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Relationship Between Factors of Online Marketing and Student Enrollment Decisions in Higher Education:
An Analysis Using Structural Modeling Techniques

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ABSTRACT

The aim of this paper is to identify the relationship between factors of online marketing and enrolment decision in higher education. With an increasing number of higher education institutions, it has become important to attract students for enrollment, and this paper will help them to do so. Through extensive literature review, the gap was identified and impact of online marketing on each step of the college selection process is tested. Data were collected from various colleges from Nepal. The sample size of the data was 800, and data were collected through structured questionnaire. SPSS and AMOS software were used to analyze the data. Factor analysis has been done, and structural equation modeling has been used in the study to show the relationship between the variables. The findings of the study showed that there is a significant relationship between factors of online marketing and enrollment decision in higher education. These findings will be helpful for higher education institutions for increasing the enrollment number.

KEYWORDS

College Choice, E-Mail, Internet, Internet Advertisement, Marketing Strategies, Parent's Expectation, Search Engine Marketing, Search Engine Optimization, Social Media

INTRODUCTION

The role of marketing has changed in today's time as it has reached to a level where everything and everyone is depended on world of internet. Digital technology has changed marketing scenario. Organization of any kind with the help of online marketing tools are coming in direct contact with their customers and are getting all the information that they want (Gielens & Steenkamp, 2019). With the increasing use of internet, students by the means of social media today look for information regarding the universities, study programs, and go through the comments of alumni of the university that interests them (Gray, Shyan Fam, & Llanes, 2003; Shields & Peruta, 2019).

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Dynamic changes that are taking place because of the digital technology has forced higher education institutions to rethink their way of marketing. Students and other stakeholders need for information and communication needs to be fulfilled by the higher education institutions. Online technologies has offered universities the tools and strategies that can be used by them to fulfill these needs (Alexa et al., 2012). In today's scenario, higher education institutions are keen on finding new ways and methods for engaging prospective students, their alumni and other stakeholders by using Twitter (Kowalik, 2011). For higher education institutions online advertising is a brilliant way to give information about their existence and tell about the advantages that one would gain from the institutions (Evans 2009). Selecting a college does not only include students but includes their parent's also. Parents also take part in the decision making process for selecting a college. Parents are involved in various stages of the college selection process. Parents have expectations from their children, which is a major influence for the students to be enrolled in a college. Students also consider their parents for fee their parents can provide them for the college. At each stage of the college selection process parents get involved (Borders et al.,2011). The purchase behavior of people has changed now. The decision making process is influenced by social media (Constantinides & Fountain, 2008). A student decision for higher education and the kind of institutions that they want is motivated by the amount of involvement their parents have in this decision (Jun & Colyar's, 2002). Parents or their family member influences school leaving student's decision for being enrolled in a higher education institution initially and later on they gather information from the outside sources (Hossler & Foley, 1995).

The study will identify various factors that affect online marketing of higher education institutions and its relationship with enrolment decision making. The study will also show the effect that online marketing factors will have on enrolment decision. Practically the study is going to present a base for higher education institutions to develop online marketing strategies. It will help the institutions to understand parents and student's role in the enrolment decision-making process and will help to increase the enrolment ratio for higher education institutions.

BACKGROUND

According to Kassandra (2019), to support student's college decision process most of the parents visit the website of the college to collect the information. Parents are an important part of the college selection. Therefore, by sending emails to parents, proper content targeting parents and student's both may increase the chance of choosing a college. According to Shields & Peruta (2018), an understanding of the way and manner a prospective student use social media may be advantageous for the colleges. Mitchall & Jaeger (2018), in their study found that parents who are involved well in the college choice process or planning of college motivates their children in the process of choosing a college, whereas their low involvement with their children planning of their college demotivates their children in the college choice process. According to Joana & Maria (2018), education institutions are investing their money on online marketing activities. With the increase use of social media around the world, it's imperative for universities to be on social media. Being on social media does not mean just having a page of the university there. The university has to be active on it. It has to allow and encourage people to communicate with each other and with the university. One things that is necessary for the universities is to update a proper content on the social networking site with right timing (Peruta & Shields, 2018). For colleges and students social media has become important as it is helping them to communicate well (Berman, 2018). Onyemaechi (2018) conducted a study to see the impact that advertising would have on enrollment rate in LPU-Laguana and found that there is a positive relationship between both. The study has taken 2016- 2017 academic year, compared it with the previous year's enrollment rate, and saw a 7% additional enrollment from that of the previous year. Social media sites like YouTube and Instagram are being used for promotion by different social media influencers. Social media influencer has positive effect on their followers as their followers believe in them and so are willing to opt for the brand that is being advertised by the influencer.

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Social media influencer are those people who have their account in the social media and have many followers following their account (Vaibhavi & Leena, 2018; Belagatti, 2017). Today internet marketing has gained lot of importance in every business. With the growing importance of internet, marketing different opportunities and challenges are being faced by the organizations. Therefore, new online marketing strategies should be made and focused on (Grubor & Jaksa, 2018). According to Ashan (2017), recently competition among higher education institutions has increased for recruiting students, which has made the universities to treat students as their customers. According to Pham et al., (2017) universities are trying to gain people trust and providing them with information by using youtube. Sathya (2017) in his study found that majority of the people feel that internet is one of the best way to get the information that they require. Consumers trusts more on electronic word of mouth. Access to internet has become easy and available because of which internet marketing has gained popularity (Lalitha &Chandra, 2016). The decision of a student for enrolment in a higher education institute is influenced by the online advertisements. The website of the university is very significant as it influences the decision of getting into a university and selecting it (Jan &Ammari, 2016). According to Brech et al., (2016) universities, which have large number of enrolled student and high reputation, enjoy more fans on Facebook than other universities. Giving right post at the right time plays a crucial part in increasing the fan base of the universities. The use of social media was not just limited to facebook and you tube by students while searching for colleges. They also depended on blogs. Giving more preference to electronic word of mouth for evaluating the colleges (Galan et at., 2015) the reviews given by other customers has an effect on the decision making process of other's interested in it (Chen et al.,2015). According to Jabeen et al., (2015), socio economic factors, personal attributes, reference group, academic factors and environment and facilities are factors that affects the student's choice for higher education. It is important for the higher education institutions to consider students as customers and make marketing strategies according to their needs and requirements. The marketing strategies should be such that it can affect different factors that are important for student's decision making for enrolment (Farjam&Hongyi, 2015). A college's reputation and its marketing tactics are significant as they influence the student's enrolment decision (Jager & Jan, 2015).

Higher education institutions have been adopting business like marketing strategies in case of traditional or internet marketing as they treat students as their customers. Website of the higher education institution is the first information provider to the students and therefore, has to be creative and presented well and also for higher education institutions information sharing the most important thing(Saichaie, & Morphew, 2014; Garwe,E.C. 2014). Social networking has brought changes in higher education institutions. Transferability of knowledge,information sharing and communication has been made easy (Kezar, 2014). Social media helps the university to target their audience (Belangar et al.,2013). According to Lubna riz 'v(2013), consumers purchasing decision can be changed and is very much relied on online word of mouth like online chats and email.Online advertising has an impact on consumer's mind. Online advertising features like content writing, creativity and ways of interaction has great influence in customer's mind(Vida,2012). The use of social media by higher education institution is at an increasing rate for college search process but it is not clear whether it influences prospective student's choice of college (Nyangau & Bado, 2012). According to Maity et al., (2012), the major role of internet in people's life is the searching information needed by them. The website design, pop ups and other online marketing strategy should be considered seriously as in today's time consumer get attracted by these. According to Greenwood (2012) these educational institutions should plan their own page in social media site, often used by the students, acquire the innovative preferences.

One of the most important thing to consider here is the content available. If the content is not satisfactory, it may make the student lose interest (Constantinides & Zinck, 2011). Omboi, B.N. (2011) stated that student's enrolment rate could be increased in the private universities by doing advertisement. Varsity Outreach (2011) stated that the admission officers and the marketing department in the colleges are using social media to look for students they require and reach out to

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them. Facebook being a major admission tool followed by twitter, youtube and blogs. Borders et al., (2011), in his study found that if a student frequently visits the page of the university created by the university on social media sites increases the chance of applying in that same university. For any latest news or information that anyone requires within seconds related to any issue the internet gives it all. Internet is spreading as major source of information (Dogruer et al., 2011). According to Joseph (2010) advertising has a positive effect on student's college choice decision. Paulsen (2010), in his study found that selecting a college for higher education is an important decision for both students and their parents. A negotiation takes place between them that is going to influence the decision of selecting a college. Parents open the door for their children to get into a college but their participation in this college going process differs according to their socio-economic status.There are various factors that influence the decision of being enrolled in a university. Universities have to market themselves according to the factors that influence student's choice of university (Kusumawati et al.,2010). Social media is the new hybrid element of promotional mix. Consumers have turned away from traditional advertising methods. They want immediate access to information (Mangold & Faulds, 2009). Through social media, universities can create their own web community to attract students (Hayes, Ranchman, & Walke, 2009). Barnes and Matson (2009), study revealed that admission officer's use of social media for attracting students kept on increasing from the previous years. Ninety one percent of the respondents agreed that for future recruitment strategy of students they would use social media. Higher education institutions with their marketing strategies starting with sending direct mail to parents consisting recruitment material, leads to discussion about that institutions with their children (Rowan-Kenyon,2008). According to Bayraktaroglu & Aykol (2008), the internet has become a main source from where consumers gather all the information. Online advertising covers the information that a student needs (Brown & Oplatka, 2006). Social media, display ads and the website of the college seems to be effective it is able to get response from most of the students. The rich stock of information about the higher education institution if supplied to the students will affect their decision of choosing a college (Briggs, 2006). According to Perna (2006), one important factor that has restricted the student's college choice process is the affordability of that college in terms of income. There is a strong relationship between choice of college for enrolment and the family income. Higher the involvement of parents in the higher education institution, higher will be the chances of student's enrolment. So, higher education institutions should know the way to involve parents (Perna, & Titus, 2005). Communication is the key for higher education institutions marketing activity. The college website is the most common and the first thing that is going to inform the students about the college (Carnevale, 2005). So much competition among the universities has given rise to so many challenges in higher education institution. Therefore, it has become very important for the universities to have enrolment numbers achieved, be financially stable, give a competitive edge and well thought academic structure (Cetin, R. 2004). The competition between higher education institutions has confused the customers mind as they are in dilemma for choosing a college. These institutions with the help of new technologies for marketing purposes can lessen this dilemma (Drummond,2004). Ramasubramanian (2003), in his study found that the images displayed on university website have a positive impact on the prospective students'. The architecture and the greenery plays an important role creating impression in front of the students. According to Paulsen (2001), the college going thinking starts from class 9 onwards, which is dependent on the grades and parents involvement. By the time one reaches class 11 and 12 the search stage starts. In colleges, the recruiting team are making attempts for recruiting and holding best and brilliant students for graduation with the expectation to see improvement in the reputation of the college and in future, overall increase in enrollment (Ogbuehi & Rogers, 1990).

There are five stages of the college selection process: the pre-search behavior, search behavior, application decision, choice decision, and matriculation decision through which students and their parents got through. The first stage, which is the pre-search behavior, is influenced by many factors. This stage is for a long period. This is the stage where at first a student recognizes the need for higher education and then look for options of various colleges. The demographic factors like, parent's

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education and income, socioeconomic status and the like may influence this stage. Then is the search stage. This behavior of students to look for colleges is said to start from high school. This stage is said to be very complex for the students. Searching for colleges means looking for all the information that they can get about colleges to choose one. They look for different sources of information not just one. Students here are influenced by their parents, relatives, friends, teachers and alumni. Many colleges are considered at this stage. The search stage stops when the student chooses colleges where they submit the application. Then is the third stage that is the application decision making. Here, students have already narrowed down their choices and have submitted the application. Students usually apply in those colleges where they have chances of being selected and which meets their criteria for the college. Financial aid is the information, which the students are not sure about at this stage. Then is the choice stage where a student has choices of set of colleges to get enrolled in. Here students have full information about the colleges and the financial aids. The choice stage ends when the students finally select a specific college. But, for some students the choice stage ends later on as they decide to postpone their admission as they might have not got their preferred college or maybe because of other factors. Then is the matriculation decision. Now, finally entering the college decision may change for some students. There can be many reasons for that like a student who was in the waiting list of his or her preferred college may get chosen there or financial or some other issue may have arisen for not getting into the selected college. This harms the college as they become from highly selective college to less selective college (Randall, 1986). According to Chapman (1981), in the past most studies are on the steps of college information search by the students and not on actual enrolment taking place, nor on the tools used by the colleges to attract the students. Printed material are most useful in attracting the students to make their college decision.

Main Focus of the Article

The existing literature of the study shows that online marketing is important for higher education institution but focuses on only one type of online marketing that being social media marketing. These literature show social media being used by higher education institutions to attract the students, to increase the enrollment rate, whereas there are various online marketing strategies that higher education institutions are using. Further, the factors of online marketing and their relationship with enrollment decision is also not known.

Development of Hypothesis

According to Shields &Peruta (2018), an understanding of the way and manner a prospective student use social media may be advantageous for the colleges. Higher education institutions have been adopting business like marketing strategies in case of traditional or internet marketing as they treat students as their customers. Varsity Outreach (2011) stated that the admission officers and the marketing department in the colleges are using social media to look for students they require and reach out to them. Facebook being a major admission tool followed by twitter, youtube and blogs. Being on social media does not mean just having a page of the university there. The university has to be active on it. It has to allow and encourage people to communicate with each other and with the university . According to Bayraktaroglu&Aykol (2008), the internet has become a main source from where consumers gather all the information. Online advertising covers the information that a student needs (Hemsley-Brown &Oplatka, 2006). Apart from social media there are other online marketing strategies that higher education institutions are using, these literatures only state the social media strategy. Therefore, the following hypothesis is developed.

H1: Online marketing factors does not have a significant impact on the pre search behavior of student's enrolment decision.

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For colleges and student's social media has become important as it is helping them to communicate well (Berman, 2018). Sathya (2017) in his study found that majority of the people feel that internet is one of the best way to get the information that they require. Consumers trusts more on electronic word of mouth. Access to internet has become easy and available because of which internet marketing has gained popularity (Lalitha & Chandra, 2016). The reviews given by other customers influence the decision making process of the customers (Chen et al.,2015). Borders et al., (2011), in his study found that if a student frequently visits the page of the university created by the university on social media sites increases the chance of applying in that same university. The above literatures have taken only social media as the source of information for decision-making. There is no study talking about the other online marketing strategies. Therefore, this hypothesis was developed:

H2: Online marketing factors does not have a significant impact on the search behavior of student's enrolment decision.

According to Ashan (2017), recently competition among higher education institutions has increased for recruiting students, which has made the universities to treat students as their customers. According to Greenwood (2012) these educational institutions should plan their own page in social media site, which are often used by the students, acquire the innovative preferences. According to Pham et al., (2017) universities are trying to gain people trust and providing them with information through the use of youtube. Borders et al., (2011), in his study found that if a student frequently visits the page of the university created by the university on social media sites increases the chance of applying in that same university. These literatures state focus only on social media effect on application decision, not about other online marketing strategies. There is no literature stating the relationship between application decision of a student for higher education institution and online marketing. Therefore, this hypothesis was developed:

H3: Online marketing factors does not have a significant impact on the application decision of student's enrolment.

According to Joseph (2010), advertising has a positive effect on student's college choice decision. The purchase behavior of people has changed now. The decision making process is influenced by social media (Constantinides& Fountain, 2008).Omboi, B.N. (2011) stated that student's enrolment rate can be increased in the private universities by doing advertisement. Social media, display ads and the website of the college seems to be effective it is able to get response from most of the students. The rich stock of information about the higher education institution if supplied to the students will affect their decision of choosing a college (Briggs, 2006). The competition between higher education institutions has confused the customers mind as they are in dilemma for choosing a college. These institutions with the help of new technologies that are now they are for marketing purpose can lessen this dilemma (Drummond, 2004). There is not enough literature showing the relationship between online marketing and choice decision, as these literatures has taken only social media for their study. Therefore, this hypothesis was developed:

H4: Online marketing factors does not have a significant impact on the choice decision of student's enrolment.

Onyemaechi (2018) conducted a study to see the impact that advertising would have on enrollment rate in LPU-Laguana and found that there is a positive relationship between both. The study has taken 2016- 2017 academic year, compared it with the previous year's enrollment rate, and saw a 7% additional enrollment from that of the previous year. Social media sites like

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YouTube and Instagram are being used for promotion by different social media influencers. Social media influencer has positive effect on their followers as their followers believe in them and so are willing to opt for the brand that is being advertised by them (Vaibhavi & Leena, 2018). The reviews given by other customers has an effect on the decision making process of other's interested in it (Chen et al.,2015). Barnes and Matson (2009), study revealed that admission officer's use of social media for attracting students kept on increasing from the previous years. The purchase behavior of people has changed now. The decision making process is influenced by social media (Constantinides& Fountain, 2008). These literature talks about adopting marketing strategies for enrolment decision and social media part in it. They do not state other online marketing strategies that are being used by the higher education institutions. Therefore, this hypothesis was developed:

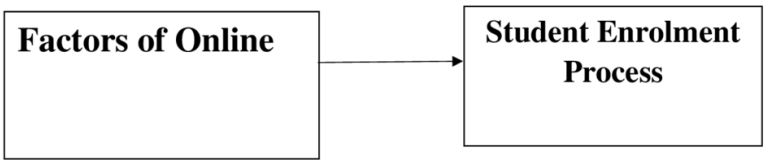
H5: Online marketing factors does not have a significant impact on the student's matriculation decision.

METHODOLOGY

The goal of this paper is to find the relationship between factors of online marketing and enrolment decision in higher education. With an extensive literature review the gap was identified and impact of online marketing on each step of college selection process[1] tested. A conceptual model was developed shown in Figure 1.

The study has used descriptive and exploratory research. Factor analysis has been done and Structural Equation Modeling has been used in the study to show the relationship between the variables. The survey done consisted of the population of those students who have yet to take admission in the college and also for those who have already taken admission in a college. A questionnaire was developed and collected from Kathmandu. The sample size of 800 was taken to obtain an effective output. For this study descriptive and exploratory research has been used. Primary and secondary data has been used in the study. The data was collected from the college-going students in Kathmandu, from the urban area, Nepal. The data was collected from 450 management students and 350 science and technology students. The data was collected from the colleges that come under the top 10 colleges in Kathmandu. Data was collected through structured questionnaire via e-mail and hardcopy. Two sets of questionnaire were developed. One consisted of 18 statements of online marketing and the other consisted of the 19 statements under enrolment decision process. These statements were made according to the study to fit them well. The study used 5- point Likert scale to collect the response, where the respondents has to rate between the ranges "strongly disagree (1)" to "strongly agree (5)". Convenience sampling was done. To test the hypothesis, we performed exploratory factor analysis, confirmatory factor analysis and structural equation modeling on the data using AMOS. SEM has been used in the present study as we intend to examine causal and effect relationship between the said constructs in the study.

Figure 1. Conceptual model (Source: Authors, 2021)



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Ethical Consideration

Research ethics is very important to complete research. This research has been done considering the research ethics. The respondent of the study will be given enough time to response so that they can fill their response correctly, according to their view and their participation will be voluntary. The purpose collecting the data will also be mentioned to them. Consent of the respondent will be taken to use their given data. Respondent confidentiality will be maintained, to ensure their data privacy. Identity of the respondent will only be disclose with their permission, if permission is not given then the identity will not be disclosed. For using secondary data, references will be given appropriately. Hence, the research ethics will be properly followed in this research.

Data Analysis and Result

The Kaiser-Mayer-Olkin (KMO) measure of testing ampleness was used for dissecting the quality of relationship among factors. The KMO sample sufficiency metric have been devised to determine the appropriateness of factor examination as it tells if factor analysis can be performed with the given information or not. The estimated KMO varies between 0 & 1 and, high esteem around 1.0, indicating that with the given information factor investigation may be attractive. For testing or for examining the KMO examining ampleness should be greater than 0.5. The value of KMO is .908 as shown in Table 1, which is adequate for testing.

Table 1 shows the loadings of 18 variables, 5 factors extracted. If the absolute value of the loading is high, factors contributing to the variable will be more. Five factors of online marketing were identified by using rotation method in factor analysis. The five factors of online marketing has been disclosed by the Principle components table and Associated variable descriptive statistics table, they are Convenience, Building Relationship, Outreach, Easy access of communication and Value creating.These factors affect the enrollment decision-making process. For investigation of the fundamental components, which are related to the 18 variable Principal component analysis, was done. This disclosed that factor 1 had 39.034% variance and similarly factor 2 had 10.100%, factor 3 had 7.489%, factor 4 had 6.413% and factor 5 had 5.851% of variance.

Confirmatory Factor Analysis: In the present study, a two stage structural equation modeling approach has been embraced. Here, those factors, which were found during EFA, are confirmed by the means of CFA (confirmatory factor analysis approach). To carry out CFA, amos software was used in the study using maximum likelihood estimation. Sample size of 800 was taken to get the effective output. Confirmatory factor analysis model has been constructed, shown and run in the study.

Figure 2 shows various variables on CFA with 18 items of online marketing (there are 5 items on Convenience, 4 items on Building Relationship, 3 items on outreach, 3 items on ease of communication and 3 items on value creating).

Table 2 shows the following results of confirmatory factor analysis and the model fit indices. CFI value ranges between 0 to 1 and if the value is greater than 0.95 it shows a good fit. Chi- square value required for good fit $0 \leq$ Chi-Square/df ≤ 2 and for acceptable fit $2 \leq$ Chi-Square/df ≤ 5. Root mean square error of approximation, which is RMSEA if < 0.05 then we reject the model. The TLI value ranges between 0 to 1 and if the value is >.9 it shows a good fit and the value of CFI will always be greater than the value of TLI. For normal fit index good fit value is, $0.95 \leq$ NFI ≤ 1.00 and acceptable is $0.90 \leq$ NFI ≤ 0.95.; CMIN: 3.995, P value =0.00, CFI (Comparative Fit Index): .943, GFI (Goodness of Fit Index): .931, AGFI (Adjusted Goodness of Fit Index), NFI (Normed Fit Index): .926, RMSEA: .061, TLI: .931. This shows the measures suitability so that they can be used in the further analysis.

Table 3 shows relationship between all the factors. To see whether there exists a relationship between the constructs or not convergent validity and discriminant validity test was done. In case of convergent validity, CR and AVE were checked. The value of CR> 0.7 is considered as good fit. In the case of AVE the value if is above 0.5 it is then acceptable. In discriminant, validity MSV and ASV is checked. In this case, the value of MSV has to be smaller than the value of ASV and

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Table 1. Factor loading and KMO Bartlett's Test of Sphericity

Attributes	Factor loading				
	Component 1 (Convenience)	Component 2 (Building Relationship)	Component 3 (Out Reach)	Component 4 (Easy Access of communication)	Component 5 (Value Creating)
Online marketing helps in two way communication	.762				
Online marketing covers wide area and saves time	.815				
Online marketing creates awareness about the existence of the higher education institution	.802				
Online marketing increases the credibility of the higher education institution	.696				
Online marketing helps to target one particular group	.723				
Online marketing helps to keep in touch regularly with the target group		.775			
Online marketing is the easiest and fastest way of sharing information		.796			
Online marketing helps in making relations with the target group		.754			
Online marketing allows you to connect to different people of the same interest and make space for discussion		.822			
Online marketing raises the interest in higher education institutions			.742		
Online marketing helps the higher education institutions to reach the national and international boundary			.783		
Online marketing makes it easy to get information about different colleges and universities.			.773		
Marketing done through online media are trustworthy				.746	
Online marketing helps to seek required information about the colleges and universities				.825	
Online marketing helps to create positive image about the higher education institutions				.665	
Online marketing is cost effective					.727
Online marketing changes the way people take decisions					.693
Online marketing facilitate quick responses and feedbacks					.778
Eigen value	7.026	1.818	1.348	1.154	1.053
Variance explained (%)	39.034	10.100	7.489	6.413	5.851
Cumulative variance explained (%)	39.034	49.133	56.623	63.036	68.887
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.	.908				
Bartlett's Test of Sphericity	Approx.Chi-Square	6700.443			
	Df	153			
	Sig.	.000			

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Figure 2. Model of relationship between factors on online marketing (Source: Authors)

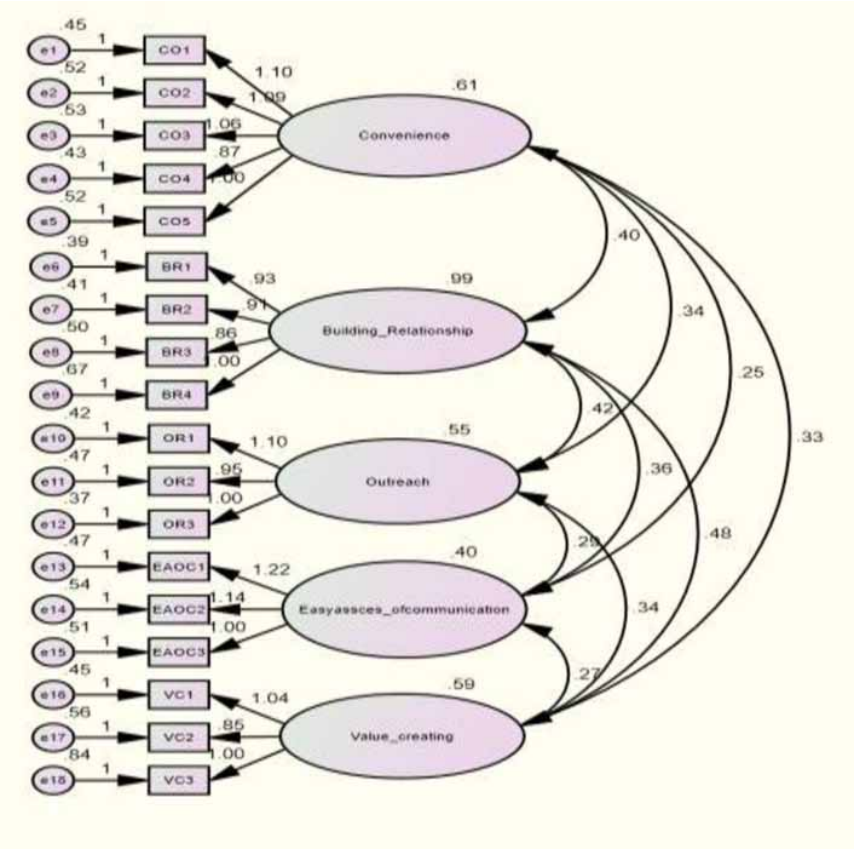


Table 2. Model fit indices

Indicator	Required for Good Fit	Required for Acceptable Fit	
CMIN (Chi-Square/df)	0 ≤ Chi-Square/df ≤ 2	2 ≤ Chi-Square/df ≤ 5	3.995
P value overall	0.05 ≤ p ≤ 1.00	0.01≤ p ≤ 0.05	.000
CFI (Comparative Fit Index)	0.97 ≤ CFI ≤ 1.00	0.95 ≤ CFI ≤ 0.97	.943
GFI (Goodness of Fit Index)	0.95 ≤ GFI ≤ 1.00	0.90 ≤ GFI ≤ 0.95	.931
AGFI (Adjusted Goodness of Fit Index)	0.90 ≤ AGFI ≤ 1.00	0.85 ≤ GFI ≤ 0.90	.905
NFI (Normed Fit Index)	0.95 ≤ NFI ≤ 1.00	0.90 ≤ NFI ≤ 0.95	.926
RMSEA	< 0.05	<.08	.061
TLI	>.9	<5.0	.931

Source: Authors

the value of ASV has to be smaller than AVE (MSV <AVE and ASV<AVE). AVE value of all the factors is acceptable as they are above 0.5 and likewise for CR, it is considered as good fit as all the values are above 0.7. Also, value of ASV and MSV is smaller than that of AVE. Therefore, there is a relationship between all the factors as can be seen in the result.

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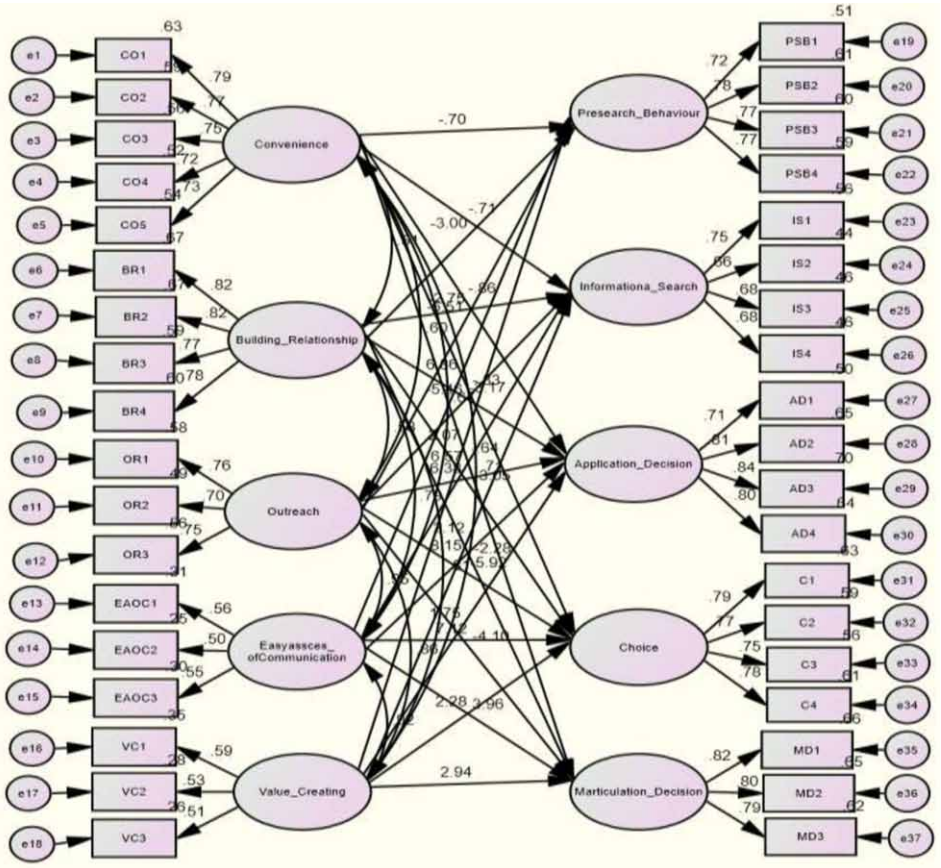
Table 3. Convergent and discriminant validity

	CR	AVE	MSV	ASV
Convenience	0.816	0.598	0.16	0.1
Building Relationship	0.843	0.574	0.17	0.17
Outreach	0.874	0.635	0.17	0.13
Easy Assces of communication	0.802	0.575	0.12	0.08
Value Creating	0.831	0.622	0.23	0.12

Source: Authors

After obtaining a successful result from confirmatory analysis, Structural Equation Model was developed as can be seen in Figure 3 and was run using software Amos. Structural equation modelling is a very effective technique that is used to test the causal relationship (Kaikire *et al*., 2016). It facilitates to see the relationship between the observed and latent variables (Ramlal, 2016).The main aim of using structure equation model was to test multiple regression equation all together. Further, it also helped us to see the relationships between factors of online marketing and enrollment decision process. Relationship between factors of online marketing on each step of the enrolment decision process has

Figure 3. Structural model of factors of online marketing impact on student enrollment decision process (Source: Authors)



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been tested through Structural equation modeling. The main aim of using structure equation model was to test multiple regression equation all together. Further, it also helped us to see the relationships between factors of online marketing and enrollment decision process. Sample size taken here was 800.

The significance of relationships between the variables has been shown in Table 4. The online factors have a significant impact on the higher education enrollment decision process. The impact of convenience on pre- search behaviour (β=.-5.890 and p=.017); Building relationship on pre- search behaviour (β=-2.293 & p= 0.14); outreach on pre- search behaviour (β= -5.895 & p=0.17); Easy access of communicationon pre- search behaviour (β = 10.032 & p= 0..19) ; Value creation on pre-search behaviour (β= 3.573 & p=.030) found significant impact on enrollment decision making. The impact of Building relation on information search (β= -2.226 & p= .019); convenience on information search (β=-5.733 & p= 0.15); Outreach on information search (β=- 6.182 & p =0.15); Easy access of communication on information search (β= 10.174 & p= .020); Value creating on information search (β= 2.790 & p= .045) found significant impact on enrollment decision making. Impact of Building relationship on application decision (β=-2.391 & p= .025); Convenience on application decision (β= -4.828 & p= .025); Outreach on application decision (β=-6.730 & p= .017); Easy access of communication on application decision (β= 11.760 & p= 0.16); Value creating on application decision (β= 2.142 & p= .027) found significant impact on enrollment decision making. Impact of Convenience on Choice (β= - 4.852 & p= .027); Building relationship on Choice (β= -2.447 & p= .017); Outreach on Choice (β= -6.652 &p= .014); Easy access of communication on Choice (β= 10.936 & p= .019); Value creating on Choice (β= 2.979 & p = .046) found significant impact on enrollment decision making. Impact of convenience on Matriculation decision (β= -5.686 & p = .037); Building relationship on Matriculation decision (β= -1.912 & p= .007); Outreach on Matriculation decision (β=- 4.799 & p= .011); Easy access of communication on Matriculation decision (β=6.341 & p= .050); Value creating on Matriculation decision (β= 4.017 & p= 0.000) found significant impact on enrollment decision making. The p value of all the variables are less than 0.05 indicating a significant relationship. This shows that there is a significant relationship between factors of online marketing and enrollment decision for higher education.

From the above statistics, it is found that our null hypothesis is rejected and the alternative hypothesis is accepted which means that online marketing factors have impact on higher education enrollment decision.

CONCLUSION

Today internet marketing has gained lot of importance in every sector. With the growing importance of internet marketing, different opportunities and challenges have occurred the organizations. Therefore, one needs to focus on new online marketing strategies (Grubor & Jaksa, 2018). Dynamic changes that is taking place because of the digital technology has forced higher education institutions to rethink their way of marketing. Students and other stakeholders need of information and communication needs to be fulfilled by the higher education institutions. Online technologies has offered universities the tools and strategies that can be used by them to fulfill these needs (Alexa et al., 2012). Therefore, it is important for higher education institutions to opt for different online marketing strategies. The study has identified five factors of online marketing for higher education institutions (Convenience, Building Relationship, Outreach, Easy access of communication and Value creating). It shows the effect of these online marketing factors on each stage of enrolment decision process. The present study has an important contribution for higher education sector. Firstly, the research has contributed to the body of knowledge by bringing into light the factors of online marketing and showing a significant relationship between them and enrolment decision process of higher education institutions, especially in the context of Kathmandu. The study has shown that online marketing has a significant impact on students' enrollment decision-making process in case of higher education institutions. The result shows clearly that a student's decision to get enrolled in a higher education is influenced by the

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Table 4. Estimates of hypothesis model

			Estimate	S.E.	C.R.	P	Label
Presearch_Behaviour	<--	Convenience	-5.890	.507	-1.360	.017	par_38
Presearch_Behaviour	<--	Building_Relationship	-2.293	.932	-2.461	.014	par_39
Presearch_Behaviour	<--	Outreach	-5.895	2.471	-2.386	.017	par_40
Presearch_Behaviour	<--	Easyassces__ofCommunication	10.032	4.257	2.357	.018	par_41
Presearch_Behaviour	<--	Value_Creating	3.573	1.358	1.895	.030	par_42
Informationa_Search	<--	Building_Relationship	-2.226	.953	-2.336	.019	par_43
Informationa_Search	<--	Convenience	-5.733	.520	-1.410	.015	par_44
Informationa_Search	<--	Outreach	-6.182	2.530	-2.444	.015	par_45
Informationa_Search	<--	Easyassces__ofCommunication	10.174	4.357	2.335	.020	par_46
Informationa_Search	<--	Value_Creating	2.790	1.391	2.005	.045	par_47
Application_Decision	<--	Building_Relationship	-2.391	1.066	-2.244	.025	par_48
Application_Decision	<--	Convenience	-4.828	.581	-1.426	.034	par_49
Application_Decision	<--	Outreach	-6.730	2.829	-2.379	.017	par_50
Application_Decision	<--	Easyassces__ofCommunication	11.760	4.875	2.412	.016	par_51
Application_Decision	<--	Value_Creating	2.142	1.550	1.383	.027	par_52
Choice	<--	Convenience	-4.852	.558	-1.526	.027	par_53
Choice	<--	Building_Relationship	-2.447	1.023	-2.391	.017	par_54
Choice	<--	Outreach	-6.652	2.716	-2.449	.014	par_55
Choice	<--	Easyassces__ofCommunication	10.936	4.677	2.338	.019	par_56
Choice	<--	Value_Creating	2.979	1.494	1.995	.046	par_57
Marticulation_Decision	<--	Convenience	-5.686	.388	-1.768	.037	par_58
Marticulation_Decision	<--	Building_Relationship	-1.912	.711	-2.687	.007	par_59
Marticulation_Decision	<--	Outreach	-4.799	1.888	-2.543	.011	par_60
Marticulation_Decision	<--	Easyassces__ofCommunication	6.341	3.244	1.955	.050	par_61
Marticulation_Decision	<--	Value_Creating	4.017	1.056	3.804	***	par_62

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online tools. The online activities or marketing strategies are appealing to the students. Impact of online marketing has been shown on each step of the student's enrollment decision making process, thus giving a deeper insight. Secondly, the models were developed and put to test by using statistical techniques, making it easy for others to look at them and get an easy and better understanding. The analysis of the Structural Equation Modelling shows that there is a significant relationship between factors of online marketing and enrollment decision process

Managerial Implication and Limitations

The study is effective for higher education policy makers. The findings of the paper is very helpful for higher education institutions especially for the marketing department to attract students. The competition between higher education is very high. This paper will also help them to understand the online marketing strategies effect more nicely. Thus, helping them to face the competition. It is important to understand that online marketing strategies are important for all the stages of the decision -making process and hence, marketing should not be limited to just few stages. Colleges and universities marketing department should make a solid online marketing strategy to attract the students. As the study shows that students decision making process for enrollment is affected by internet, therefore, marketing department needs to be updated with the internet platforms and strategies.

There are limitations in this study. Further, the models developed in this study can be used. The study does not differentiate between stages where online marketing maybe less effective than the other. Therefore, further study can be done to investigate this. Secondly, further study can be done to see the moderating effects. In addition, the study does not include parents, income, higher education institutions reputation and like. Therefore, this can also be an area for further studies.

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A Systematic Analysis of Content Structural Efficiency for Estimating Higher Educational Institution Engagement Over Facebook

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ABSTRACT

The purpose of this research article is to examine the Facebook content themes and structural efficiency of higher education institutions in order to understand how these factors influence engagement. The sample consisted of 4,703 Facebook posts from the top ten most popular Indian and global higher education institution pages. The platform engagement options were used to categorize audience reactions, while structural attributes were examined in accordance with applicable theory using a negative binomial regression model. Factor analysis and descriptive metrics were used to evaluate theme efficacy. The findings highlight the significance of developing a comprehensive assessment of content structural efficiency. The paper presents a number of evidence-based recommendations for projecting and estimating content performance. This study adds to the body of knowledge by first merging the content subject with its structural parts. Previously, the content theme was thought to be a qualitative unit.

KEYWORDS

Audience Engagement, Audience Interest, Content Structure, Content Theme, Post Interaction

1 INTRODUCTION

Websites and social media are the most regularly used digital marketing channels (Taiminen & Karjaluoto, 2015). Digital marketing extends convenience along with marketing (Lau et. al., 2018). Social media facilitates interaction, information, word-of-mouth, customization, and trendiness (Yadav and Rahman, 2019). Social media can be utilized for a variety of purposes like awareness and real-time feedback (Al-Awadhi and Al-Daihani, 2018). Social media marketing communication had a favorable impact on attitude (Duffett, 2017). Social media has a favorable impact on a brand's functional and symbolic image (Gokerik et al., 2018). Brand loyalty, brand consciousness, and value consciousness are all positively influenced by social media marketing (Ismail, 2017). Social media exposure has a great impact on buying (Karen et al., 2017). Brand image and reputation are significantly impacted by social media (Godey et al., 2016; Seo and Park, 2018).

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Napoleoncat reported (Facebook users in India, 2021) India has approximately 450 million Facebook users in August, 2021. Comparing reports for January and August, 2021 shows an increase of 14.5% user within 7 months and the user base between the ages 25-34 grew by almost 10%. Facebook's global user base grows at a yearly pace of 12% (Newberry, 2021) whereas it grows at a rate of 12-13% in India (Statista, 2021). Approximately, 23.8 percent of Facebook users are between the ages of 18 and 24, while 82 percent of college graduates are on Facebook globally (Aslam, 2021). Higher education has remained among the top ten social media-using industries for the past five years, with 6.8 posts per week (Feehan, 2021). Higher education has the third highest average engagement per post on Facebook internationally; Higher education in India is thriving and according to a forecast from brandequity (2021), digital ad expenditure across all sectors will increase by 65 percent. Higher education institutions had lower rates of engagement on Facebook in 2018 than in 2017, and this has become a Year-on-Year (YoY) phenomenon. The reported global average for Facebook interaction per post across all industries is roughly 3.28 percent, but higher education was obtained at 0.12 percent in 2019 (Feehan, 2019), which is significantly lower than the average rate. Higher education has a lower interaction rate with lower spending per click on Facebook than the global average (Irvine, 2019).

2 LITERATURE REVIEW

There is a lot of information available in the business section of the Facebook website on success stories of educational institutions implementing Facebook for lead generation and conversion in their depository. The world average engagement rate for Facebook post is 0.18% (Newberry, 2021) while average engagement rate for higher educational pages was 0.14% (Feehan, 2021) and 0.16% under low performing category (Jipa, 2021). Feehan (2021) in the report on higher education for RivalIQ (2021) obtained 0.65% average engagement for top 10 most performing higher educational institution pages, but the highest engagement rate obtained by an individual unit was just 1% (University of Iowa). It also mentioned the average engagement rate for colleges and universities declined by 20% in 2021. The marketing success cases available at the Facebook depository for higher education are not directly linked with content engagement rather it's more of deep learning of target marketing. If Feehan 2019 and 2021 reports are to be compared, the average engagement rate for higher education increased by 0.02% over three years. The engagement rate has been defined as the total engagement divided by the number of followers in the aforementioned studies. That means an increase in community size can further dilute the engagement rate. Hence this study proposes to design a model to estimate engagement based on content structure.

2.1 Determining Engagement Measure

Researchers have used platform interaction attributes to measure content flow. Interaction KPIs and metrics are significant factors for social media activity evaluation (Keegan and Rowley, 2017). Earlier researchers have studied the Facebook interactions such as: share, like, comment and reaction. They have defined these interactions as recommendation, endorsement, conversation and expression (Hennig-Thurau et al., 2004; Hennig-Thurau et al., 2010; Swani et al., 2013; Dhaoui, 2014; Alboqami et al., 2015; Liu et al., 2017). Moran et al. (2019) and Tafesse (2015) have mentioned the requirement of accessing engagement in terms of interaction. Target interaction can be defined as the interaction option of a platform that a post intends to target for engagement. As this study is about Facebook the above mentioned interactions has been used for evaluating engagement.

2.2 Content Theme (CoTh)

The study intends to categorize contents published by higher education institutions' on Facebook by 'Theme' and identify the best content structure for performance (engagement). There are few distinguishable characteristics; vividness and novelty (Tefesse, 2015) and interaction cue (Moran et al., 2019). However, this study assumes that CoTh is independent of any structure, implying that

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the content structure is always built around a specific theme. The overall content structure has been depicted in Figure 1. Specific contents allow pages to stay in touch with the community (Lipsman et al., 2012). It is more important to focus on content rather than the channel (Kelsey & Lyon, 2017, p. 51). Interactive contents with instructive elements are most engaging and informative contents act as engagement driver (Alboqami et al., 2015). Entertainment and interaction are some aspects of social media posting (Kim and Ko, 2010). The entertaining, emotional, and interesting contents highly influence social media engagement (Barger et al., 2016). Different posts have varying effects on engagement (Tsimonis and Dimitriadis, 2014; Cawsey and Rowley, 2016). Interactive posts and mix appeal receives the highest interaction on Facebook (Kusumasondjaja, 2018). Entertainment is a powerful engagement driver (Park et al., 2009; Muntinga et al., 2011). Puto and Wells (1984) suggested a two-dimensional categorization model that was based on transformation and information appeal, which was again suggested by Laskey et al. (1989). The researchers have concentrated heavily on broad categories; however, these categories are still made up of multiple sub-categories of content message call theme. Tafesse and Wien (2017) proposed a framework of twelve brand post categories depending on message design. Previous research has provided significant knowledge on the importance of content appeal but has not clearly demonstrated a method for analyzing the impact of CoTh on overall engagement.

2.3 Content Theme Measurement

The need for CoTh quantification in this study is eminent, as considering it a qualitative unit would not generate estimates for the proposed structural effectiveness model. The impact of social media community interaction over the brand image, loyalty, purchase decision, engagement development, and satisfaction has been studied and established by many earlier researches (Ramanathan et al., 2017; Pongpaew et. al., 2017; Vorah and Bhardwaj, 2019; Rossmann et al., 2015; Gruen et al., 2006; Shareef et al., 2019). Other sets of studies defined different levels of community participation over social media like; consumption, contribution, and creation (Muntinga et al., 2011; Tsai & Men 2013; Shareef et al., 2019). These studies helped in understanding the relevance of community participation towards engagement development. Hence, this study proposed to quantify the CoTh in terms of community interaction. The quantification of CoTh has been done based on metrics that can be processed with publicly available data. Keegan & Rowley's (2017) suggested metrics for evaluation should be decided based on need and possible processing. Peters et al. (2013) illustrated the importance of content measurability and network influence. As such, this study proposes two metrics: 'post interaction ratio' (PI) and 'contribution ratio' (C_{ont}) for measuring theme worth.

RQ-1. What is the most engagement effective *CoTh* based on PI and C_{ont} for higher educational institutions?

2.4 Content Structure

a. Content Type (CoTy)

According to the platform's specifications, content on Facebook follows a set format: video, album, image, text, and story. Posts with high degree of representation have a positive impact on performance (Kaplan & Haenlein, 2010). Sensory properties are influenced by post-structural factors (Fortin & Dholakia, 2005). Tafesse (2015) discussed the positive effect of post vividness (type) on audience reaction. Online content with a variety of features, such as an image and text, might provide higher-order appeal (Daugherty et al., 2008). Facebook posts with multiple structural aspects have a beneficial impact on audience response (Sabate et al., 2014). The influence of video is higher than any other

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format as per the Facebook website. Facebook believes content type is a major signal for content placement (Cooper, 2021). We propose 'Post Type' categories as Text, Link, Image, Album and Video.

b. Perceived Action (PA)

The theoretical knowledge of the impact of interaction on post-engagement is divided into two halves. Some says that brand post interactivity boosts overall engagement (de Vries et al., 2012; Sundar & Limperos, 2013; Moran et al., 2019) while others say that interactivity has a negative impact on engagement (Fortin & Dholakia, 2005; Hanna et al., 2011; Tafesse, 2015). Interactivity can have three effects on post engagement: good, negative, and neutral (Liu, 2003; Fortin & Dholakia, 2005). However, interaction has an impact on post-engagement in either direction. Interactivity is classified into three degrees based on complexity: low, moderate, and high (de Vries et al. 2012; Tefesse, 2015; Moran et al., 2019). We propose 'Perceived Action' categories as Interaction, Participation, Subscribe and Conversion.

c. User Involvement (UI)

The way content appeals to the user for interaction have been theoretically considered as involvement. User interaction influences content engagement (Moran et al., 2019; Sabate et al., 2014; Schmidt et al., 2008; Tefesse, 2015). One sort of involvement is content consumption (Calder & Malthouse, 2008, p. 253). Different types of content necessitate varying levels of user engagement. For example, an image (with or without interaction appeal) can be consumed with a simple view, whereas an online workshop or an article necessitates a higher level of involvement. Hence, the proposed factor is User Involvement whose overall structure has been built with primary data as no literature is available to define the elements. We propose 'User Involvement' categories as View, Read, Passive Participation and Active Participation.

d. Target Audience (TA)

Target audience in the context of content analysis has received little attention from researchers and theoretical information for target audience is not readily available. However, Facebook offers ample functionalities for customizing audience targeting. As a result, the target audience becomes an important part of the engagement. Several studies emphasize the relevance of the target audience for engagement based on real time data from social media marketing platforms (Newberry, 2019; Forsey, 2021; Zote, 2021). Since, the level of information required to fully implementing the target audience as an aspect of content structure is limited; the content appeal to a certain segment of the audience has been used as the categorization criterion in this study. We propose 'Target Audience' categories as Internal, External and Not Specified.

RQ 2: What is the most effective content structure for higher education institutions?

2.3.1 Proposed Content Structural Efficiency Model

The structural efficiency model is prepared with available theoretical literature on content structural models and its applicability has been demonstrated in this study.

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Figure 1. Content Structural Efficiency Model



We propose 'CoTh', 'CoTy', 'PA', 'UI', and 'TA' as elements of the overall content structure. The model to estimate structural efficiency is illustrated in figure 1. This study considers the theme to be independent of structure and dependent on 'target interaction'. Thus, once the page decides the specific interaction it aspires to target with a post; it requires estimating the most contributing theme and adjusting the structure accordingly. The following hypotheses have been formulated based on the literature review to support the significance of structural elements per 'target interaction'.

H_02_{ij}: Post type has no significant impact on Interaction
H_03_{ij}: Perceived Action has no significant impact on Interaction
H_04_{ij}: User Involvement has no significant impact on Interaction
H_05_{ij}: Target Audience has no significant impact on interaction

where 'i' is type element and 'j' is interaction.

Based on the research questions 1 and 2, the objective of this study is to *Identify most effective content structure for higher education institutions' Facebook engagement performance.*

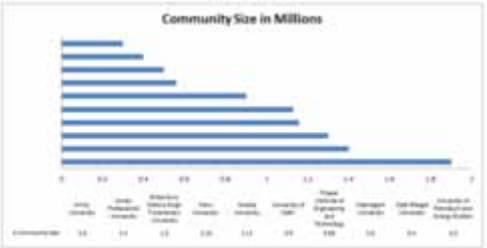
3 METHODOLOGY

The analysis is divided into two parts: global (G) and Indian (Ind). The analysis was carried out in order to reach a conclusion about the comparative scenario of both sample groups.

3.1 Sampling

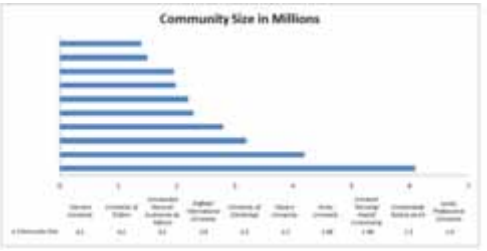
For determining the pages, the community size of a Facebook page as of January 1, 2021 was used. All the posts made between January 2021 and May 2021 by the top ten higher educational institutions' Facebook pages (both global and Indian) were extracted. Figure 2 and 3 illustrates the base samples for data extraction.

Figure 2. Indian Sample



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Figure 3. Global Sample



3.2 Data Collection

The content Meta description per post and related interaction elements have been extracted using the online data extraction tool, Fanpage Karma on May 15, 2021.

3.3 Proposed Model for Analyzing CoTh Performance

The research analytical model is divided into two parts: content theme effectiveness (RQ-1) and content structural efficiency (RQ-2).

3.3.1 Content Theme Effectiveness (RQ1)

The proposed model for CoTh effectiveness is based on the effective interest of the audience; the most successful content theme combination was determined using the Adjusted R^2 value achieved through step-wise multiple logistic regressions. The content theme mix was created using a structural equation modeling approach. The audience interest was determined using the 'post-interaction ratio' (PI). The PI equation, i.e., the ratio of effective interest was created using Munoz et al. (2017) and contribution value calculation is based on the relative measure concept suggested by Barman (2020). The following equation determined PI values:

$$PI_j = \frac{Engagement}{Exposure} \text{ ----------} \quad (1)$$

where, 'PI_j' is the PI for post j

The resulting PI values were used in factor analysis to determine the base model for the theme mix. Furthermore, the model mix is analyzed with step-wise multiple regressions considering 'post exposure' as an independent variable (X) against the dependent variable, 'engagement' (Y). All the themes obtained from the factor analysis were considered as a base model for the regression analysis and low contribution themes were removed with every successive rotation until adjusted R^2 fell below the base model. The data set extracted for this study was observed to be non-normal, hence the study employs log-log transformation regression (Benoit, 2011), as explained below:

$$log(Y_{ej}) = \alpha_e + \{\text{å}\beta_1 \, log(PE_{j1}) + \beta_2 \, log(PE_{j2}) + \ldots\ldots\ldots + \beta_n \, log(PE_{jn})\} + \varepsilon_i \text{ -------} \quad (2)$$

Where $log(Y_{ej})$ indicates the log of engagement (e) for posts 1, 2, 3, and so on. β_1, β_2, β_3, β_4, β_n indicates the regression coefficient for $log(PE_{tn})$, and $log(PE_{t1})$ represents the post exposure for post j=1,2,3,4, and so on. Based on the Adjusted R^2 value derived from the regression analysis, the best performing content mix has been estimated. The theme contribution was obtained with the following equation:

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$$\text{Content Contribution } (C_{ont}) = \frac{Theme \quad Engagemen\left(CTx\right)}{Aggregate \;\; Engagement} \text{ ----------} \tag{3}$$

3.3.2 Content Structural Efficiency (CSE)

The structural efficiency model is based on models proposed by previous researchers (Kumar et. al., 2018; Tefesse, 2015; Moran et al., 2019). A wide number of studies have employed platform interaction options as parameters for analysis (de Vries et al., 2012). It was studied independently by verifying the statistical significance of the structural elements and their total impact on 'Target Interaction' (like, comment, share, and reaction). The model testing equation is given below. Previous research (Moran, 2019; Hilbe, 2014) recommended using the Negative Binomial model to examine the effect of factors in a count data set.

Model Equation:

$$Y_{ij} = exp\,\beta\,(\text{å}\,\beta_o + \beta_{\text{Type tj}}\,X1j + \beta_{\text{Perceived Action aj}}\,X2j + \beta_{\text{Target Audience nj}}\,X3j + \beta_{\text{Involvement zj}}\,X4j).$$

Where; $Y_{ij} => Y_{1j}$ or Y_{2j} or Y_{3j} or Y_{4j} ; log of post like, share, comment and reaction respectively and exp β_{zi} is the equation regression coefficient obtained as parameter estimates (z = element).

3.3 Content Classification Approach

The classification of content for social media posts can be done either using an inductive or deductive approach (Anandarajan et al., 2019, p-16). Tafesse and Wien (2017) proposed a stepwise deductive-inductive categorization approach. Their framework was employed as an initial content assortment technique. Because it lacks categories that are specifically appropriate to higher educational institutions. Hence, the final list of categories was prepared using real-time data.

3.4. Content Classification Framework

Framework for this study has eight adjustments to the framework suggested by Tafesse and Wien (2017). Table I represents the content categories and adjustments.

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Table 1. Content (Theme) Categories

WTafesse & Wien (2017)	Study Conceptual Framework	
Proposed categories	Result	New Proposed Name
Emotional	Retained*	Emotional*
Functional	Re-adjusted**	Institutional**
Educational	Split***	Learning*** and Informative***
Brand Resonance	Retained*	Brand Resonance*
Experiential	Retained*	Experiential*
Current event	Retained*	Current event*
Personal	Re-adjusted***	Students***
Employee	Retained*	Employee*
Brand community	Retained*	Brand community*
Customer relationship	Retained*	Relationship*
Cause-related	Re-adjusted***	Social***
Sales promotion	Re-adjusted***	Promotional***
	Newly Created****	Placement****

*category scope kept as suggested by Tafesse & Wien (2017), **category definition has been changed to meet sample characteristics' in the line with original definition and a new name has been proposed, ***category has been split in to two or more categories based on sample characteristics', ****categories added based on extracted sample.

3.5 Rating of Non-Count Variable

The non-count variables are rated based on variable complexity or dimension. The rating is based on earlier researches as shown in Table II.

Table 2. Rating for Content Structural Elements

S. No	Element	Variable Type	Categories	Rating	Based on
1	Theme	Non-count	*Refer table 2*	N/A	Tafesse & Wien (2017) and author's judgment
2	Type*	Count	*Text*	1	Tefesse, 2015; Moran, 2019
			Link	2	
			Photo	3	
			Album	4	
			Video	5	
3	Perceived Action*	Count	*Click*	1	de Vries et al. 2012; Tefesse, 2015; Moran, 2019; Author Judgment
			Interaction	2	
			Subscription	3	
			Conversion	4	

Table 2 continued on next page

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Table 2 continued

S. No	Element	Variable Type	Categories	Rating	Based on
4	User Involvement*	Count	*View*	1	Calder and Malthouse, 2008; Author judgment
			Listen	2	
			Read	3	
			Participate	4	
5	Target Audience*	Count	*Internal*	1	Alboqami et al.,2015; Newberry, 2019; Author Judgment
			External	2	
			Not Specified	3	

*non-count variable converted to count

4 ANALYSIS AND FINDINGS

4.1 Interpreting Theme Mix (Base-Model)

Any variable with a communality value lesser than 0.5 may struggle to load significantly (Hair et al., 2010). Hence, the commonality measure of the data for each post were evaluated and posts with communality measures below 0.5 were excluded in the factor analysis. In total, 126 and 86 posts from the global (0.421 to 0.845) and Indian (0.321 to 0.963) samples were omitted respectively.

The factor analysis was used to determine the base model for both the sample groups. For global, 1757 posts explaining 81.126% of variance loaded onto six factors attributable to 13 themes and 1364 posts for Indian loaded onto five factors explaining 70.826% of variance attributable to 11 themes. Table III illustrates four distinct theme models for both global and Indian samples. The BM (-1) was found to be the most efficient model for both the global and Indian samples. The pattern obtained from the analysis suggests that the base model can be improved by removing low contribution themes initially. However, it tends to fall when more than one theme was removed with BM (-2). The overall analysis indicates that BM is the second best fit model for any combination which can be obtain through PCA analysis of PI values.

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Table 3. Theme Mix Model

Global					Indian			
	Base-model (BM)	BM(+n)	BM (-1)	BM(-2)	Base-model (BM)	BM(+n)	BM (-1)	BM(-2)
	Base-model (Themes obtained from PCA)	BM plus rest of the themes	BM less Brand Community	BM (-1) less Employee, Placement and Relationship	Base-model (Themes obtained from PCA)	BM plus rest of the themes	BM less Current Event and Emotional	BM(-1) less Relationship
Constant	1.245	2.145	2.169	2.144	0.721	1.972	2.156	1,846
p-value	0	0	0	0	0	0	0	0
R^2	0.55	0.77	0.81	0.74	0.47	0.75	0.77	0.72
Adjusted R^2	0.71	0.49	0.76	0.69	0.69	0.41	0.73	0.67

Note; Base-model represents the mix of themes obtain through factor analysis. Mix (+n) represents themes added to the base-model and Mix (-1,-2) represents base-model less theme category.

Table 4. Content Elements Parameter Estimates

			Global				Indian		
Parameters	**Rating**	**Like**	**Share**	**Comment**	**Reaction**	**Like**	**Share**	**Comment**	**Reaction**
Text	[Ty=1]	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***
Link	[Ty=2]	0.958*	0.762*	0.651*	-	-	0.066*	0.024*	0.167*
Image	[Ty=3]	0.9*	0.776*	0.417*	0.897*	0.049*	0.696*	0.345*	0.75*
Album	[Ty=4]	0.833*	0.687*	0.437*	1.051*	1.303*	0.206*	1.005*	0.505*
Video	[Ty=5]	1.33*	1.066*.	1.718**	0.652*	1.275*	0.284*	0.067**	0.492*
Click	[PA=1]	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***	**BL****	**BL***
Interaction	[PA=2]	0.589*	1.13*	0.73*	1.064*	0.013*	0.003*	0.091*	0.077*
Subscription	[PA=3]	0.639*	-	0.395*	1.145*	0.007*	0.002*	0.103*	0.062*
Conversion	[PA=4]	0.598*	-	0.341*	0.856*	0.004*	0.002*	0.069*	0.045**
View	[UI=1]	**BL***	**BL***	**BL****	**BL****	**BL***	**BL***	**BL***	**BL****
Listen	[UI=2]	1.143*	1.204*	1.847*	-	1.006*	-	1.975*	0.215*
Read	[UI=3]	2.406*	1.157*	2.224*	1.554*	0.083*	-	2.048*	0.222*
Participate	[UI=4]	-	1.927*	0.986*	-	0.065*	1.098*	2.015*	0.166**
Internal	[TA=1]	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***	**BL***
External	[TA=2]	1.296*	1.012*	0.772*	1.264*	0.065*	-	0.91*	0.043*
Not Specified	[TA=3]	1.335*	1.368*	1.509*	1.294*	1.237*	0.723*	0.567*	0.076*

Note: BL represents the baseline variable for each element where *Significance at 0.05 and **significance at 0.01; ($p < 0.05$, 0.01 = null hypothesis rejected)'-' represents the elements identified to be not statistically significant with the hypothesis testing

4.2 Theme Structural Efficiency

Tables IV represents the statistical significance of the elements tested at a 95% level of confidence, and the parameter estimates based on exponential beta *(exp β)*. Testing hypothesis for 'Like' suggest

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that H_0 got rejected for each of the elements at all levels except UI-4 *($H_0$5a4, p=.218, p>0.05)* for global samples and Ty-2 *($H_0$2a2, p=.878, p>0.05)* for Indian samples. For 'Share', H_0 got rejected for all elements except PA-3 *($H_0$3b3, p=.194, p>0.05),* PA-4, *($H_0$3b4, p=.555, p>0.05),* TA-2 *($H_0$2b2, p=.625, p>0.05)* for global sample. Indian sample resulted in rejection of H_0 for all elements at all level for 'Share' except UI-2 *($H_0$4b2, p=.206, p>0.05),* UI-3 *($H_0$4b3,p=.094, p>0.05*) and TA-2 *($H_0$5b2, p=.704, p>0.05*). The entire set of H_0 for 'Comment' got rejected for both global and Indian samples. Ty-5 and UI-1 for a global sample and Ty-5 and PA-1 for the Indian sample were rejected at 99% level of confidence. The entire set of H_0 for 'Reaction' got rejected for the Indian sample, where UI-1 and UI-2 were rejected at 99% level of confidence. For global sample, all H_0 has been rejected except three; Ty-2 *($H_0$2d2, p=.176, p>0.05),* UI-2 *($H_0$4d2, p=.589, p>0.05)* and UI-4 *($H_0$4d4, p=.181, p>0.05).* Thus in total, seven null hypothesis for global and four for the Indian samples were accepted. According to the findings, the Read element (UI-3) can increase Like by 2.406 times (24.6 percent approx) and Comment by 2.048 times (24.8 percent) for global and Indian samples, respectively. For both global and Indian samples, these are the highest of all parameters. Baseline estimations were considered to be one, hence no effect on engagement was observed for any given parameter. The increase (>1) in engagement for the global sample with Type has been estimated for Album (R=1.051), Video (L=1.33, S=1.066, C=1.718) and decrease (<1) has been estimated for Image and Link. Only Reaction showed an estimated increase in engagement for PA-2 and PA-3. For every given level of PA, the rest of the interactions were expected to decrease for the entire sample. Except for UI-4 for Comment, UI estimates showed an increase for all interactions at all levels (0.987). Except for the Comment at TA-2, all interactions were projected to increase for TA at all levels (0.772). The biggest TA increase was predicted for Comment at TA-3 (1.509). The highest global sample drop was predicted for Comment at UI-4 (0.986), implying that 98.6 percent of comments are likely to decline with post-expecting user participation. For the Indian sample, engagement is expected to rise for Like at Ty-4 (1.303) and Ty-5 (1.303). Ty-4 (1.005) was estimated to have a higher comment increase, whereas the rest of the Ty levels were estimated to have a lower interaction. For the Indian sample, PA at all levels tends to decrease for all types of interaction, with Comments at PA-3 (0.103) being the most affected. For example, UI-2 (1.006) is expected to rise, whilst UI-3 and UI-4 were expected to fall. Share is likely to increase for UI-4 (1.098) rests were found to be not significant. Comment has been estimated to increase at all levels of UI with the highest being estimated for UI-3 (2.048). The reaction has been estimated to decline for all UI levels. All levels of TA tends to decline for all types of interaction for Indian samples except for Like at TA-3 (1.273). The highest decline has been estimated for Comment at TA-2 (0.91).

5 DISCUSSION

The descriptive analysis indicates that a bigger community size corresponds to a higher level of engagement volume ($Global_{eng}$>$Indian_{eng}$ and $Global_{cs}$>$Indian_{cs}$). However, it leads to a falling PI ratio: {$Indian_{PI}$ (16.6%)>$Global_{PI}$ (6.8%)}. Differences in contribution score and PI rates of themes imply that different themes produce varying levels of engagement (Alboqami et al., 2015; Gruen et al., 2006). The regression analysis suggests that overall engagement efficiency can be increased by avoiding low contribution themes. The element analysis in content structure is consistent with previous research findings that demonstrate an *exp β* reduction with increasing UI and PA (Moran et al., 2019; Tefesse, 2015; Subtle, 2014). On the other hand, with a high level of vividness, engagement is likely to rise (Tefesse, 2015). According to *exp β* estimates, the most appropriate structure for a global sample is *'a Video appealing for subscription with reading involvement targeted towards a non-specified audience is most likely to increase engagement through share and like'.* The same interpretation for the Indian sample suggests that *'an album with subscription appeal and reading involvement targeted at a broader audience is most likely to increase engagement through like and comment'.* This interpretation can be applied to any form of interaction depending on the type of

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content. For example, the most 'Like' generating global theme was 'Learning,' with 0.251 contributions (25.1 percent), followed by 'Emotional,' with 0.212 for share. Now if a page wants to target 'Like' then the findings of the study suggest that it publishes *'a learning video with click interaction with reading involvement targeted towards not-specified audience'*. The technical standing for content structure is Interaction $_{\text{Like}}$ = Structure {TY=5, PA=1, UI=3, TA=3, CT=Learning}. For the Indian sample, the 'Students' post obtained 0.719 contributions for like and 0.406 for comment. The study suggests publishing *'a student-related album with conversion participation with reading involvement targeted towards not-specified audience'*. Reading involvement for 'Like' *(TA=3$_{Like}$=0.083)* has been estimated to decrease however comment is estimated to increase by 2.048 which is highest among all the exp β obtained. The technical standing is *Interaction $_{Like+comment}$ = Structure {TY=4, PA=4, UI=3, TA=3, CT=Student}*. Similarly, the study identifies the most avoidable content structures, such as *'a promotional post containing a link (TY=2$_{Like}$, 0.958) for conversion (PA=4$_{Reaction}$, 0.856) through participation (UI=4$_{Comment}$, 0.986) targeted towards the external audience (TA=2$_{Comment}$, 0.772)'*. The above statement contained structural elements with *exp β* less than one, and the theme 'Promotional' was included because it has the lowest contribution per interaction (Table V). It illustrates the contribution scores per interaction for each theme which represents the most interacted interaction. Informative contributes highest towards Comment (at 0.49) and Experiential (at 0.21) contributes highest towards Share. For Indian samples, Students dominate all contributions per interaction across all themes with a 0.71 contribution value. The analysis suggests that when a page intends to target Like from the global sample, it should use Learning and the Indian should use Students as per adjusted optimized content structure obtained from the parameter analysis.

Table 5. Theme Contribution Scores

GLOBAL					INDIAN				
Theme	**Like**	**Share**	**Comment**	**Reaction**	**Theme**	**Like**	**Share**	**Comment**	**Reaction**
Learning	**0.251629**	0.097477	0.335363	**0.391481**	Student	**0.7196**	0.299135	**0.406946**	0.082131
Informative	0.11002	0.026966	**0.495891**	0.075931	Placement	0.004186	**0.354985**	0.212917	0.366265
Experiential	0.11031	0.00401	0.048838	0.064755	Learning	0.002898	0.27934	0.182229	**0.397687**
Emotional	0.079468	**0.212376**	0.022113	0.095256	Experiential	0.060599	0.012615	0.104452	0.025297
Current Event	0.083969	0.189363	0.015873	0.09554	Institutional	0.021301	0.028644	0.045965	0.102553
Students	0.089026	0.050451	0.021346	0.105032	Brand Community	0.009287	0.015276	0.012254	0.011603
Social	0.073963	0.146455	0.016633	0.07773	Event	0.007516	0.004258	0.006394	0.003475
Institutional	0.053159	0.131444	0.012986	0.054084	Emotional	0.002511	0.001365	0.008802	0.003235
Relationship	0.062486	0.010425	0.013864	0.008995	Relationship	0.008815	0.001342	***0.00313***	0.003102
Brand Community	0.033825	0.03537	0.004656	0.010974	Brand Resonance	***0.002298***	***0.001172***	0.008535	0.002056
Brand	0.02804	0.01649	0.007319	0.017901	Employee	0.147182	0.000685	0.003344	0.001338
Employee	0.008718	0.000945	0.000735	0.001254	Informative	0.168152	0.000895	0.003077	***0.000532***
Placement	0.003011	***0.000566***	***0.000444***	***0.000476***	Promotional	0.044475	0.000289	0.001953	0.000727
Promotional	***0.000865***	0.000725	0.003938	0.000592	Social	0.018585	0.015276	0.012254	0.011603

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6 IMPLICATION

The study provides a critical understanding of content structure that can be used to project and estimate engagement. Earlier research provided sufficient structural knowledge (excluding theme) for producing engagement but did not provide a clear methodology for determining the adjustable content theme for a particular structure based on interaction. It extends the usage of previous models to the higher educational institution's Facebook contents and quantifies a theme that was previously provided as a qualitative metric. Assuming that each page has access to its own historical data, the study recommends determining the best content structure.

Implementation Methodology

a. Target Interaction

The implementation must be based on the "target interaction" concept (Moran et al., 2019). Target interaction refers to the interaction option that a page intends to target with a piece of content. Each page must determine the most likely target interaction based on contribution value and parameter estimates of content structure. Figures 3 and 4 illustrate the shift in estimates when targeting 'like' and 'share,' respectively. The curves in the figure that are less than one represent those that are likely to decline while targeting a specific interaction.

b. Impact of Target Interaction

Once a page has decided on the target interaction, it must calculate the impact of targeting. The impact of the target has been defined as the interaction opportunity cost of targeting a specific interaction. The equation below shows how to calculate the target interaction value and impact value.

$$\text{Target Interaction Value}_{(L/S/C/R)} = \sum\left(optimized\ strcutal\ units\ x\ target\ theme\ contribution\right)$$

$$\text{Impact Value} = \sum\left(non\ optimized\ structural\ units\ x\ \ target\ theme\ contribution\right)$$

*Note: Estimate ³ 1 should be considered as positive and £ 1 as negative
*Here, Impact value is the opportunity cost of engagement against a target interaction.

Table 6. Target and Impact Estimates

Elements	Target	Impact	Impact	Impact
	Like	Share	Comment	Reaction
Album	1.303	0.206	0.005	0.505
Click	1	1	1	1
Listen	1.006	0	1.975	0.215
Not Specified	1.237	0.723	0.567	0.076
	Share	Like	Comment	Reaction
Text	1	1	1	1
Click	1	1	1	1

Table 6 continued on next page

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Table 6 continued

Elements	Target	Impact	Impact	Impact
	Like	**Share**	**Comment**	**Reaction**
Participation	1.098	0.065	2.015	0.166
Internal	1	1	1	1
	Comment	Like	Share	Reaction
Album	1.005	1.303	0.206	0.505
Click	1	1	1	1
Read	2.048	0.083	0	0.222
Internal	1	1	1	1
	Reaction	Like	Share	Comment
Text	1	1	1	1
Click	1	1	1	1
View	1	1	1	1
Internal	1	1	1	1
	4	4	4	4

Figure 4. Target 'Like' for Indian Sample

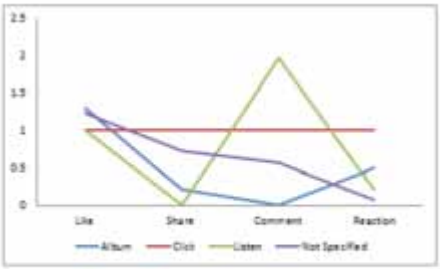


Figure 4: Target 'Like' for Indian sample

Figure 4 illustrates estimated impact on interaction for 'Target Like'. The fall for Share (Alb=0.206, Ns=0.723), Comment (Ns=0.567) and Reaction (Alb=0.505, Lis=0.215 and Ns=0.076) are shown in table VI.

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Figure 5. Target Share for Indian Sample

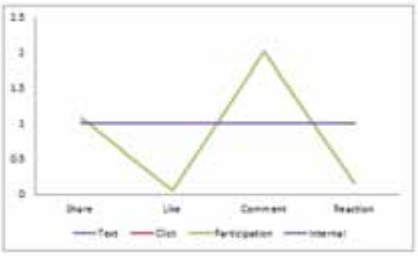


Figure 5: Target 'Share' for Indian Sample

Figure 5 illustrates the movement of estimates for 'Target Share'. The optimized structure (Table VIII) suggests majority of the contents at baseline are likely to increase Share interaction; however like and reaction are supposed to fall with UI-4.

Figure 6. Aggregate Estimates

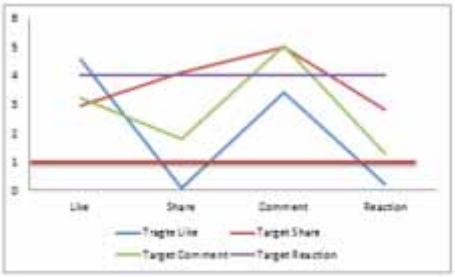


Figure 6: Aggregate Estimates

Figure 6 illustrates the aggregate estimates for each interaction, as well as their respective movement for impact interactions. It demonstrates that targeting share rather than like results in higher engagement performance, despite the fact that like is the most common interaction choice for any given content theme. The only target interaction estimate to fall below baseline has been observed with Like.

Table 7. Target Estimates

	Like	Share	Comment	Reaction
Target Like	4.5656	0.071	3.413	0.204
Target Share	2.935	4.098	5.015	2.834
Target Comment	3.22	1.794	5.053	1.273
Target Reaction	4	4	4	4

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c. Estimating Impact Gap

The Impact Gap (Figure 5) is the estimated change in volume of interaction based on target interaction calculated as follows:

$$\text{Impact Gap} = \text{Target}_{\text{interaction}} \text{ Elements} - \text{Impact}_{\text{interaction}} \text{ Elements}.$$

Figure 7 illustrates the Impact Gap for the above-mentioned target like and target shares. The curves' movement suggests that 'target like' estimations have a larger interaction opportunity cost than 'target share' estimates. Impact values (Table VIII) can be determined for each interaction based on the desired combination.

Figure 7. Impact Gap Curve

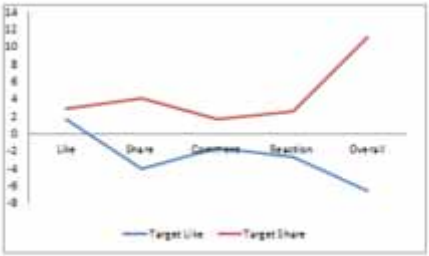


Figure 7. Impact Gap Curve

Table 8. Impact Values for Target Interaction

		Aggregate		
Target	Impact			Total
Like	Share	Comment	Reaction	
4.5656	0.071	3.413	0.204	7.684
Share	Like	Comment	Reaction	Total
4.098	2.935	5.015	2.834	14.882
Comment	Like	Share	Reaction	Total
5.053	3.22	1.794	1.273	11.34
Reaction	Like	Share	Comment	Total
4	4	4	4	16

Therefore the interaction optimization for Indian sample can be achieved with 'target share' content structure. Based on contribution score, *Placement* (0.354985) is most likely to increase share at highest rate. Hence the technical structure for content optimization at lowest impact is: $\text{Interaction}_{\text{share}}$ = (*Placement, Ty=1, PA=1, UI=1, TA=1).*

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LIMITATIONS

The study only deployed a meta-description of a post to determine content category due to the lack of a comprehensive categorization framework. Hence, future studies can develop a better model by initially developing a precise categorization framework applicable to higher educational institutions.

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Exploring the Impact of E-Marketing on Consumers' Online Cognitive and Affective Tendencies in Developing Nations:

How to Win Over Digital Consumers in the Digital Era

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ABSTRACT

Consumers have become smarter; therefore, smart techniques are required to grab their paramount interest toward a particular product or service. Since advanced information communications technologies (ICTs) have radically changed the business operations, the marketing paradigm has also dramatically transformed the promotional tactics by utilizing innovative ICTs. Corporates and researchers still believe that more work is required to unveil the role of evolving ICTs in boosting the marketing communications to attract the consumers in today's era. The study investigates the nexus between e-marketing (eM) model and consumers' impulsive buying tendencies (IBTs) using two core streams (i.e., cognitive and affective tendencies) drawing on theory of planned behavior (TPB). The key findings illustrated that eM has positive influence on consumers' IBTs (i.e., cognitive and affective). In addition, comprehensive linkages affirmed the positive connection of each dimension of eM (i.e., mobile marketing, email marketing, and internet marketing) on consumer's IBTs.

KEYWORDS

Consumer Behavior, Electronic Marketing, Emerging Economies, Structure Equation Modeling, Theory of Planned Behavior

1. INTRODUCTION

Appealing the consumers using e-marketing platforms is an evolving sphere that is obtaining intense concern of the marketers and practitioners in order to understand the insights within distinct themes across the world. Although various conventional marketing tools are already widely operational and reached at peak to target consumers over the past few decades (Waheed, Yang, Khan, Khan, & Farrukh, 2018). Nevertheless, the current competitive market scenario has changed the traditional trends and shifted towards advanced digitalized communication mediums to win over consumers

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to widely catch the consumers' attention (Waheed & Jianhua, 2018). Since the consumers are also extensively utilizing digital means of ICTs such as e-newsletters, social media, application in mobile devices, websites, and emails to collect product-specific information and try to understand the other features of the product by getting information from various digital platforms (Bawm & Nath, 2014; Walczak, Kellogg, & Gregg, 2012). However, obtaining the necessary and required information from various sources is not an easy task for the consumers since many factors involved in the information searching process on the current market scenario (Baumgartner & Steenkamp, 1996).

It is very essential for marketing professionals to identify and use advanced digital advertising tools to serve consumers in a better way (Constantinides, 2004; Constantinides et al.,, 2010). Because consumers are more attached with online platforms in today's era such as by using social media, emails, and rest of the Interne-based tools. In this context, to trigger their attention toward particular product by means of eM could play effective role in broadcasting the information (Waheed & Yang, 2018). The instinct behavior of consumers of product or service in terms of buying is sometimes unintended which is referred to as occasional or impulse buying attitude (Tuyet Mai, Jung, Lantz, & Loeb, 2003). The tendencies of impulse buying behavior are related to two elements named as affective and cognitive (Verplanken & Herabadi, 2001). Moreover, this study mainly will answer the question how utilizations of e-marketing tools could effectively enhance organizational image and may support to broadcast the information of a particular product or service to consumers particularly while triggering their impulsive behavior. According to experts, the affective aspect of impulsive behavior is related to action, while the cognitive aspect explains opposite to the individual needs of a consumer (Verplanken & Herabadi, 2001). Indeed, prior researchers and practitioners have been investigated the role of Internet, e-marketing, and utilization of digital technologies in various outlooks such as promotional events (Ashfaq et al., 2021; Gao et al., 2021; Waheed et al., 2021). Although several studies have been done to understand the insights of online platforms with respect to consumers' attention and advertisement perspectives in different industries worldwide (Canarslan & Bariş, 2022; Mostafa & Hannouf, 2022). Moreover, an exceptional work have been considered by the worldly scholars to uncover the influence of social media advertising and remainder Internet-based tools such as online gaming (Alharthey, 2021; Sadek, 2021). Likewise, another interesting study assure the insights how consumer might be engaged using Facebook platform (Al Kurdi & Alshurideh, 2021; Gupta & Sharma, 2021).

Nonetheless, our study specifically converges to observe that how the utilization of e-marketing may strengthen or weaken the tendencies of impulse tendencies of the consumers. Advertisement through e-marketing and online platforms is a key element for the whole marketing process of an organization (Alam et al., 2019; Jamil, Qayyum, & Lodhi, 2022; Ling et al., 2010), especially in case of establishing brand positioning and awareness because it can influence the attitude of consumer buying tendency (Brengman et al., 2001). Sometimes psyche of consumers is negatively hit by the advertising ads (Waheed & Jianhua, 2018; Waheed & Yang, 2017).

Pakistan is an emerging nation and is recognized as 6th most populated countries bearing around 189 million populations with a growth rate of 2.1% as mentioned in the study of Waheed and Jianhua (2018). Social culture of Pakistan is based on various values and norms whereas, the economic culture is rich and bears creative approaches. despite this, the e-marketing model of El-Gohary (2012) was currently adopted to test its influence on consumers' impulsive tendencies. El-Gohary (2012) presented five factors of e-marketing especially from the B2C domain such as mobile marketing, email marketing, internet marketing, extranet marketing, and finally intranet marketing. At the present, we adopted the most influential tools of e-marketing within the Pakistani market such as mobile marketing, email marketing, and internet marketing to examine the influence on consumers' IBTs. First, mobile marketing is grounded by the extent of utilizing the many distinct associated components of information and communication technologies such as social applications (Apps), SMS, MMS, JAVA marketing, Bluetooth marketing, and Wireless Apps. Protocol (Bauer, Reichardt, Barnes, & Neumann, 2005; Leppaniemi, Karjaluoto, Sinisalo, & Salo, 2006; Smutkupt, Krairit, & Esichaikul, 2010).

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This study endeavors to meet certain objectives as follows. First, the study attempts to explore the cumulative connection of e-marketing (eM) on consumer impulsive buying tendencies (IBTs). Second, the connection of each factor of eM will independently be examined on IBTs such as between mobile marketing (MbM) and IBTs, email marketing and IBTs, and internet marketing and IBTs, respectively. The parts of this study are arranged as follows: The first part addresses the related earlier literature of various studies as well as formations of hypotheses and research framework. The second part presents the research methodology which explains the constructs measurement, sampling procedure, and data analysis techniques. Next part emphasizes a discussion about study results and findings of hypotheses testing. Final part concludes the complete study and suggests the managerial implication, highlights the limitations and proposes future directions.

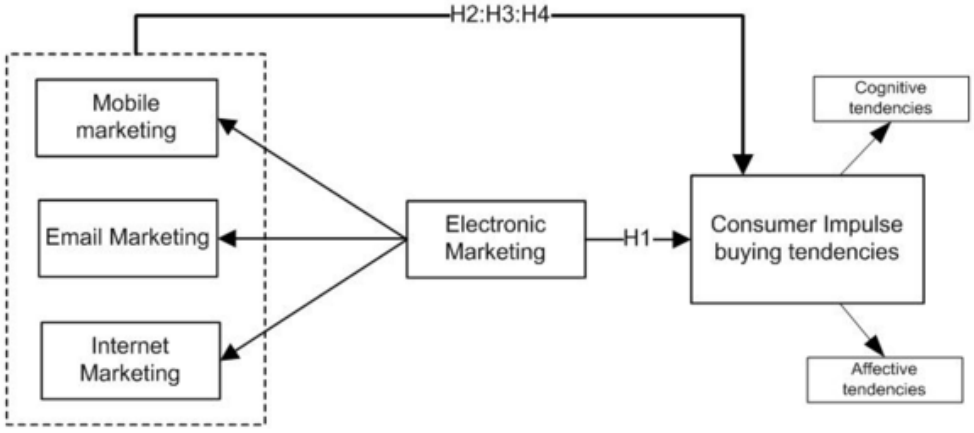
2. THEORETICAL SUPPORT AND LITERATURE REVIEW

Theory of the planned behavior (TPB) is the theory of psychological phenomena, behavior, belief of the individual based on three main dimensions, i.e., subjective norms, attitude, and behavioral control that eventually support to shape the behavioral intention (Ajzen, 2020). In the past, several theories have been used to support the ICTs and IS research such as technology acceptance model (TAM), innovation diffusion theory (IDT), and Unified theory of acceptance and use of technology (UTAUT) model (Shahid, Yang, Waheed, & Rashid, 2021). There is no empirical study available which tested the effects of mobile marketing (MbM) as well as email marketing (EmM) on instinct purchasing behavior of the consumers in Pakistan. Thus, this study primarily decides to fulfill such prevailing academic literature and research gap and secondly, to achieve core objective of this study through focusing on following two main goals. On the other hand, email marketing involves promotion of the product through email. Organizations are currently seeking to engage varied users for long-term relationships by using email marketing platforms (Hajarian, Camilleri, Díaz, & Aedo, 2021; Pavlou, 2003).

2.2. Hypotheses Formation

This study proposed four main hypotheses to understand the insights regarding e-marketing with respect to consumers' impulse buying tendencies (IBTs) as following explained.

Figure 1. Research framework and hypotheses formation



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2.2.1. E-marketing (eM) and Consumers IBTs

Em is defined the extent to which information or knowledge about a particular product or service is being transmitted using dissimilar Internet-based platforms (El-Gohary, 2011, 2012). E-marketing is a capability that presents comprehensive and unique ICTs tools to attract consumers. E-marketing has different sub-dimensions, including mobile advertising, email advertising, intranet advertising, and extranet advertising (Eid & El-Gohary, 2013; El-Gohary, 2011, 2012). Organizations can gain competitive advantages by utilizing promotions campaigns through e-marketing tools which eventually positively influence consumer behavior (Waheed & Jianhua, 2018; Waheed & Yang, 2017). E-marketing has a positive linkage on distinct kinds of consumers behavior such as exploratory buying behavior or product acquisition behavior (Waheed & Jianhua, 2018; Waheed & Yang, 2017).

E-marketing could support to win over the consumers by focusing promotional practices using online media and tactics (Brodie, Brady, Brodie, Coviello, & Winklhofer, 2008; Eid & El-Gohary, 2013). According to experts, eM support to achieve the level of effective relationships between organizations and consumers (Gilmore, Gallagher, & Henry, 2007). Consumers can get detailed information even in an easier way using internet technologies and large numbers are using Internet technologies (Mathews, Bianchi, Perks, Healy, & Wickramasekera, 2016; Villa, Ruiz, Valencia, & Picón, 2018). Based on positive connection of eM toward consumer's behavior along with support of TPB, we currently assumed the following hypotheses to test within Pakistani market in order to empirically contribute in the literature by revealing the relationships as e-marketing has a positive connection with consumers' impulse buying behavior (IBTs).

2.2.2. Mobile Marketing (MbM) and Consumers IBTS

Nowadays, it is quite easy to spread information or to advertise the product in the market anytime and anywhere due to the remarkable expansion in the usage of mobile devices (Brodie et al., 2008; Coviello, Milley, & Marcolin, 2001). By establishing a marketing campaign through mobile devices features such as MMS, SMS, and WAP is a profitable and time saver method (Barwise & Farley, 2005; Smutkupt et al., 2010). Experts claimed that MbM has a positive influence on consumer behavior if organizations embrace such tools to promote the product of service (Waheed & Jianhua, 2018). MbM is a critical tactic for advertising and seizing consumer attention because of extensive users (Waheed & Jianhua, 2018). Therefore, organizations can achieve consumers' attention and motivate them in considering the product through mobile marketing. Hence, we assumed the following relationship to test in the case of Pakistani market as MbM has a positive connection with consumers' IBTs

2.2.3. Email Marketing (EmM) and Consumers IBTs

EmM has defined the promotion of the product or service using the email channel to target a particular target market (Rettie & Chittenden, 2003; Rettie, Grandcolas, & Deakins, 2005). EmM is an effective approach to target the consumers in order to broadcast product information in a more quick and comprehensive manner (Waheed & Jianhua, 2018). EmM has a positive linkage with consumer behavior since experts have been concluded such nexus over the past years (Waheed & Jianhua, 2018). Because of the fact that email is cost-effective and easy to manage, therefore email is considered a vital player in spreading information (Sager, 2012; Zhang, Kumar, & Cosguner, 2017). EmM is the cheapest way to spread information in a short period (Rettie, 2002; Rettie & Chittenden, 2003). Emails most often play a pivotal function for information dissemination and contact management among distinct sellers and buyers, especially when it comes to marketing of business activities (Pavlou, 2003). Thus, the proposition is pretended in this study as EmM has a positive connection with consumers' IBTs.

2.2.4. Internet Marketing (InM) and Consumers IBTs

InM is often defined as the tool of marketing using distinct online platforms being used to promote a particular product or service by conveying the message to the end-users (Franch, Martini, & Inverardi,

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2003; Stewart & Zhao, 2000). InM is a unique tactic which is mostly adopted by advertising firms because firms have realized that a huge number of the population is using the internet for a different purpose. Nowadays, organizational are swaying from conventional to internet-based methods in order to convey the message (Kayumovich & Annamuradovna, 2020). Hence, InM has positive outcomes in targeting the consumers therefore using InM tools for marketing purposes may be more prolific for the organizations. Taking this notion into account, the study attempts to uncover the following assumption to validate the findings with more empirical evidence by proposing relationships as InM has a positive connection with consumers' IBTs.

3. METHODS

The surveys were developed consisting of two basic sections, i.e., demographic and basic questions related to eM. The demographic characteristics were considered on certain items (see Table 1). A total of 800 questionnaires were administered among consumers using all possible means such as emails and personal visits. Out of them, seven hundred and fifty surveys were received. subsequently, after the removal of inappropriate survey results, seven hundred surveys were presently taken into account for further statistical analysis.

3.1. Measurements

This study adopted two types of variables, i.e., independent variable (IV) and dependent variable (DV) where eM is acting as IV, while IBTs as DV. The measures for both IV and DV were adopted from prior studies. First, we picked three variables from the model of eM proposed by El-Gohary (2012). Subsequently, for the sake of measuring MbM, measures adopted from the study (Leppaniemi et al., 2006) based on three items such as SMS, WAP, and MMS. Second, for the sake of measuring EmM, measures adopted from (Sterne & Priore, 2000) also based on three items such as e-promotion, viral marketing as well as e-newsletters. Third, scales for InM were adopted from Bianchi and Mathews (Bianchi & Mathews, 2016) based on five terms. However, IBTs measures obtained from (Verplanken & Herabadi, 2001) based on 20 items scales. Verplanken and Herabadi (2001) advocated two aspects of IBTs: cognitive and affective aspects. Since this study adopted both aspects of IBTs; therefore, 10 items were linked to the cognitive aspect, while 10 items were utilized to measure the affective aspect.

3.2. Analysis Tools

In order to test hypotheses, this study used Statistical Package for Social Science (SPSS) version 21.0 with the integration of SmartPls applying structural equation modeling (SEM) being inspired from past studies (Ashfaq et al., 2021; Farah & Shahzad, 2020). As this study utilized a five-point Likert scale, nominal scale, and ordinal scales where all coded from 1-5. Thereafter, following certain statistical techniques were employed to produce outcomes. First, to calculate the demographic information of the respondents, Descriptive statistics were applied. Descriptive statistics enable the investigators to define the facts in detail. Second, to assess the validity and reliability of construct HTMT, common method biased (CMB) technique, and discriminant validity was applied. By using two modes, the convergent validity can be assessed such as factor loadings and average extracted variance where the results values for both measures must be greater than .5 as suggested by (Fornell & Larcker, 1981).

4. RESULTS

4.1. Demographic Information

In this research, four attributes of the respondent's profiles were examined where a detailed classification of a specific demographic profile is presented in following Table 1.

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Table 1. Respondents' Profiles

Characteristics	*n=700*	
	Frequencies	%
Gender (male=400; female=300)		
Qualification level Graduation Master Ms/Mphil PhD Others	240 140 120 150 050	34.29 20.00 17.14 21.43 7.14
Age level 20-30 year 31-40 year 41-50 year >51 year	100 220 280 100	14.29 31.43 40.00 14.29
Year of Internet Using 2-4 year 5-7 year 8-10 year >11 year	080 100 350 170	11.43 14.29 50.00 24.29
Note. *others sectors like semi-govt.**other associated personnel		

4.2. Construct Description

The following Table 2 represents the general metrics of the constructs in order to measure the particular latent variables such as independent and dependent constructs. The results show that seven variables of which four were independent, while one is a dependent variable.

4.3. Mean, SD, α, and AVEs

Table 3 represents the values for means, alpha, average variance extracted (AVE), and standard deviation (SD) in which all values are normal such alpha values are higher than the level of .7 and AVEs are higher than the level of .5 which vividly indicate the accuracy of the values where all outcome are normal as per recommendations.

4.4. Forbell and Larcker and HTMT Discriminate Validity

Such validity explored using two main methods such as HTMT and presented in Table 5 and Fornel and Lacker methods as shown in Table 4. All values show that there are no issues about the validity and reliability of the constructs.

Likewise, the validity is also calculated using HTMT method in order to further ensure the validity of the variables with deep understanding about reliability and validity as presented in following Table 5.

COMMON METHOD BIAS AMONG VARIABLES

Some diverse tools and techniques exist to ensure the common method bias (CMB) on the focused variables. Likewise, Podsakoff et al. (2003) suggested another technique to explore CMB using the common latent factor (CLF). We applied the CLF technique to assure the CMB of the variables such as i4.0 technologies and sustainable development. Podsakoff et al. (2003) advocated that CLF might be determined by contrasting the loadings with CLF and without CLF. To this end, the output values must be lower than 0.02 (Podsakoff et al., 2003) as currently, values are less than such criteria, which affirmed no CMB in the dataset.

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Table 2. Constructs' Descriptions

Construct	Model Type	Indicators	Predefined reliability
eM	factor	05	1.000
MbM	factor	03	1.000
EmM	factor	03	1.000
InM	factor	05	1.000
CgT	factor	10	1.000
AfT	factor	10	1.000
*Age**	factor	--	1.000
*Gnd.**	factor	--	1.000
*Ed.**	factor	--	1.000
Note. eM=e-marketing; MbM=mobile marketing; EmM=email marketing; InM=internet marketing; CfT=cognitive factor of IBTs; AfT=affective factor of IBTs *control variables			

Table 3. Mean, SD, α, and AVEs

	Latent	Mean	SD	α	AVEs
1	eM*	3.150	0.700	0.708	0.786
2	MbM*	3.254	0.726	0.826	0.706
3	EmM*	3.801	0.683	0.831	0.691
4	InM*	4.037	0.686	0.805	0.819
5	CgT**	3.920	0.640	0.874	0.700
6	AfT**	3.241	0.701	0.798	0.654
*independent variables; **dependent variables					

Table 4. Discriminate validity

	eM	MbM	EmM	InM	CgT	AfT
eM	**0.887**					
MbM	0.241	**0.840**				
EmM	0.325	0.450	**0.831**			
InM	0.015	0.235	0.241	**0.905**		
CgT	0.484	0.536	0.232	0.341	**0.837**	
AfT	0.348	0.352	0.307	0.421	0.420	**0.809**
Squared correlations; AVE in the diagonal.						

Table 5. Validity using HTMT

	eM	MbM	EmM	InM	CgT	AfT
eM						
MbM	0.120					
EmM	0.235	0.365				
InM	0.025	0.241	0.230			
CgT	0.392	0.462	0.304	0.236		
AfT	0.322	0.324	0.164	0.369	0.341	

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4.5. Pearson Correlation Model

Pearson's correlations matrix is used to explore the linkages of the variables as shown in Table 6. The values clearly represent the connections among used variables.

4.6. Validity of Latent Variables

The following Table 7 indicates the values in order to affirm the reliability and validity of the. In this analysis, the validity of the constructs was examined with the support of AVE and loadings of the constructs. The comprehensive results are reported based on mean and SD accordingly.

Moreover, the experts have been suggested that the values of composite reliability should be higher than .7 whereas the values of average extracted should be higher than .5.

4.7. SEM Model Analysis

Table 8 indicates the decision regarding proposed four hypotheses of this study using the statistical technique of SEM which guarantees the relational direction between particular variables.

5. DISCUSSION AND IMPLICATIONS

This study contributes to the literature with more empirical evidence by evaluating the relationships between e-marketing and consumers' IBTs. Moreover, comprehensive findings additionally contribute in the literature of e-marketing by confirming positive nexus of e-marketing and its each dimension such as mobile, email, and internet marketing on IBTs

From managerial viewpoint, the marketers are suggested to realize the significance of e-marketing tools such as mobile marketing, internet marketing, and email marketing to attain the intention of consumers where such a humongous market of internet and mobile users who reside in Pakistan. The marketers must know the pros and cons of a marketing plan to avoid any problem during an ongoing marketing campaign before implementation of any strategy. Marketers must understand the potential of each medium since the present study shows that mobile and email marketing is more effective than internet marketing having a strong positive connection. Similarly, the following points must be kept in mind by marketers while publicizing the information by means of e-marketing. It is immensely needed that the information which marketers are going to be publicized through such mediums should be permission-based. As the researchers and the professionals revealed fact that if the information is sent without permission, the recipient of information would either perceive a bad image regarding the product or services or would not be able to get proper attention (Barwise & Farley, 2005; Pavlou, 2003; Scharl, Dickinger, & Murphy, 2005; Schooley, Walczak, Hikmet, & Patel, 2016; Waheed & Jianhua, 2018; Waheed & Yang, 2017; Zhang et al., 2017). Therefore, the

Table 6. Correlation Matrix

	eM	MbM	EmM	InM	CgT	AfT
eM	**1.000**					
MbM	0.346	**1.000**				
EmM	0.241	0.403	**1.000**			
InM	0.246	0.491	0.456	**1.000**		
CgT	0.367	0.546	0.223	0.251	**1.000**	
AfT	0.494	0.467	0.346	0.210	0.324	**1.000**
Evaluated using suggested criteria (Taylor, 1990)						

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Table 7. Reliability and validity

Latent variables	Items	Mean	SD	Loadings	AVEs	CR
eM					0.786	0.806
	eM1	3.880	0.832	0.654		
	eM2	4.080	0.813	0.647		
	eM3	3.541	0.792	0.701		
	eM4	3.540	0.751	0.650		
	eM5	3.950	0.744	0.607		
MbM					0.706	0.828
	MbM1	3.670	0.888	0.653		
	MbM2	3.960	0.864	0.589		
	MbM3	3.870	0.906	0.654		
EmM					0.791	0.830
	EnM1	4.010	0.893	0.607		
	EnM2	3.830	0.965	0.654		
	EnM3	3.830	0.865	0.607		
InM					0.819	0.809
	InT1	4.080	0.787	0.653		
	InT2	4.050	0.783	0.589		
	InT3	3.980	0.829	0.654		
	InT4	4.120	0.742	0.607		
CgT					0.754	0.799
	CgT1	4.081	0.787	0.653		
	CgT2	3.531	0.783	0.647		
	CgT3	3.540	0.829	0.707		
	CgT4	3.940	0.842	0.650		
	CgT5	3.541	0.786	0.707		
	CgT6	3.542	0.703	0.647		
	CgT7	3.950	0.829	0.701		
	CgT8	3.641	0.740	0.655		
	CgT9	3.541	0.707	0.647		
	CgT10	3.440	0.789	0.707		
AfT					0.809	0.820
	AfT1	3.420	3.520	0.545		
	AfT2	3.541	0.687	0.607		
	AfT3	4.001	0.780	0.701		
	AfT4	3.932	0.829	0.750		
	AfT5	3.545	0.842	0.759		
	AfT6	3.520	0.788	0.640		
	AfT7	3.950	0.709	0.771		
	AfT8	3.547	0.899	0.659		
	AfT9	3.543	0.740	0.627		
	AfT10	3.450	0.787	0.743		

Note. eM=e-marketing; MbM=mobile marketing; EmM=email marketing; InM=internet marketing; CgT=cognitive factor of IBTs; AfT=affective factor of IBTs *control variables

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Table 8. Structure equation modeling (SEM)

Factors and relationships on PPV-C	β	Sig.	Hypothesis
H1: eM to BITs	±	0.389***	*Supported*
H2: MbM to BITs	±	0.268***	*Supported*
H3: EmM to BITs	±	0.365***	*Supported*
H4: InM to BITs	±	0.263***	*Supported*
***Level of Sig =0.001; level of SRMR= 0.0698			

marketers should get permission from consumers to urge IBTs among consumers and such action would attract more consumers in today's competitive environment. Hence, marketers should consider all the suggestions before carrying or forming any advertising strategy using e-marketing and each tool of e-marketing within Pakistan.

6. CONCLUSION

This study concluded that by concentrating on emerging ICTs particularly e-marketing mechanisms that may augment the communications to effectively broadcast the information to the consumers in today's business environment. Nowadays, consumers have become smarter therefore smart tactics might be employed to seize their attention. E-marketing platforms might be used to attain the highest intention of the consumers, especially to achieve IBTs in a more prolific and innovative manner. This study summarized that marketing campaigns might be improved by considering each sub-dimensions of e-marketing such as mobile marketing, internet marketing, and email marketing. Finally, it is concluded that e-marketing is a sound predictor of IBTs therefore organizations may adopt e-marketing tools in order to better embrace the attention of consumers along with the inclusion of MnM, EmM, and InM dimensions of e-marketing.

Despite the above concluding lines, discussion, and managerial implications, this research has some restraints which could be admitted in future research. First, the ultimate data analysis of the study is limited to 700 sample size within the province of Punjab and Khyber Pakhtunkhwa (KPK) of Pakistan. Therefore, further studies could increase the sample size to validate the study's validity in the whole of Pakistan and other emerging nations. Secondly, it did not incorporate any moderating variable between e-marketing and IBTs. The study could be enlarged with moderating variables like age, gender, social class, or any other factor. Lastly, the study focused only on the IBTs of consumers whereas future studies can focus on the role of such technologies on other buying behaviors within Pakistan or worldwide.

FUNDING AGENCY

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Does Personality Influence the Frequency of Online Purchase Behavior?

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ABSTRACT

Scholars have acknowledged the influence of personality traits on individuals' behavior. However, the effect of personality on (repetitive) online purchase behavior remains under-explored. This is a considerable deficit in knowledge because a marketing paradigm has shifted its interest to examine strategies for attracting new buyers as well as retaining existing customers. To further the understanding of this issue, this paper attempts to examine the relationship between the Big Five personality traits—openness, conscientiousness, extraversion, agreeableness, and neuroticism—and the frequency of online purchases among 386 university students in Malaysia. Through a structural equation modelling analysis, the authors found that extraversion, agreeableness, and neuroticism were positively related with the frequency of online purchases. In contrast, openness and conscientiousness showed no significant effect. Overall, this research suggests that marketers should target buyers with certain personality traits because they are likely to use online platforms more often for purchasing items.

KEYWORDS

Big Five Personality, Malaysia, Online Purchase, Personality, Repetitive Purchase, University Students

INTRODUCTION

Online business, commonly known as e-commerce, refers to the process of selling and buying goods or services via computer networks using protocols specifically designed for that purpose (Organization for Economic Co-operation and Development, 2011). However, e-commerce today is not limited to handling only business transactions. It also involves exchanging, sharing, and storing personal information (Dutta, 1997). Unfortunately, such personal data not only are available to the seller but also can be exploited by other parties. Hence, online purchasing has become riskier than conventional shopping, resulting in strong resistance from customers.

Research on factors explaining the willingness of individuals to use online platforms for buying items has progressed substantially in recent years. However, the existing literature focuses heavily on the supply-side factors of the websites themselves, such as perceived enjoyment (Brusch and Rappel, 2020) and ease of use (Xia et al., 2018). Although such studies merit acknowledgement, we still lack an understanding of the impact of personality on e-commerce purchases. Indeed, this represents a considerable gap in the literature because personality has long been recognized as a dominant approach for understanding the subsequent behavior of individuals (Gohary et al., 2014).

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While a few studies have considered personality as the determinant of several online shopping dimensions, such as intention to buy online (Dobre and Milovan-Ciuta, 2015) and impulsive purchases (Leong et al., 2017), the effect of personality traits on a specific online buying behavior — that of repetitive purchase — remains under-explored. To fill this gap, this study attempts to examine the effect of the Big Five personality traits (BFP) — agreeableness, neuroticism, extraversion, openness, and conscientiousness — on the frequency of online shopping.

While intention to buy online is mainly tested using predictive models of planned behaviour (Ajzen, 1991) and technology acceptance models (Lee et al., 2003), impulsive purchases involve unplanned, unreflective, and spontaneous purchasing (Piron, 1991). In this study, we focused on repetitive online purchasing as our dependent variable. To measure it, we employed the frequency of online shopping as a proxy to determine how often buyers use websites to make a purchase. The more frequently buyers purchase via the internet, the more frequently they use online platforms for shopping repeatedly. As the internet becomes an important means for conducting many activities, it is beneficial specifically for marketers to ensure that customers buy online and continue to do so in the future (Chiu et al., 2012). In fact, sustainable profits in the e-commerce market depend on how successfully businesses retain existing online shoppers (Hu et al., 2015; Wen et al., 2011).

As a contextual contribution, we tested our model amongst university students, an important social group in marketing research. Although the spending capability of students is often limited, they are well-educated and have a strong exposure to technology (Hu et al., 2009). According to the Internet Users Survey 2018 by the Malaysian Communications and Multimedia Commission, full-time students represent around 12.1 percent of all internet users in Malaysia. Also, the largest group of internet users comprises individuals in their 20s, who spend about 8 hours a day browsing the internet, half of which is spent in online shopping. Thus, university students provide an insightful context for research exploration.

PERSONALITY AND ONLINE PURCHASE BEHAVIOR

Most online business activities are handled virtually during pre-purchase (i.e., information seeking), actual purchase (i.e., payment), and post-purchase (i.e., feedback, review, and after-sales service) stages. Accordingly, the decision to purchase at online stores involves various psychosocial states including sensory reactions, perceptions, attitude formation, preferences, satisfaction evaluation, and loyalty formation (Dobre and Milovan-Ciuta, 2015). As such, repetitive online purchasing behavior and its relationship to customers' personality traits demands a comprehensive investigation.

Whilst initial purchase explains the likelihood that a potential customer purchases from an online seller for the first time at a given time, repeat purchase defines the subjective probability that the same customer will continue to buy a product using the same online channel (Davis, 1989). Therefore, the critical aspect to understand the purchasing experience occurs at the end of the initial purchase process because it greatly influences the likelihood that existing customers will return to the same online site (Ariely and Carmon, 2000).

Accordingly, both academics and practitioners acknowledge the importance of distinguishing between new and repeat customers in the online commerce industry. The literature argues that the two segments differ with regard to their motivations and activities related to the purchase of online products or services, whereas online sellers assert that the classification helps them in strategizing the market segmentation (Ryu and Han, 2011). For example, potential customers and repeat customers possess different amounts of information and use different criteria for making purchase decisions. Specifically, compared to the former, the latter are more knowledgeable in understanding and analyzing the information of an online store due to their previous experience using the platform for shopping (Kim and Gupta, 2009).

Literature in many disciplines utilizes the Big Five Model of Personality by McCrae and Costa (1990) as an underpinning framework to define the personalities of individuals (e.g. Ayob and

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Makhbul, 2020). The Big Five personality traits (BFP) consist of five dimensions: agreeableness, neuroticism, extraversion, openness to experience, and conscientiousness. Everyone possesses all five traits but at different levels (e.g., a person might be high in agreeableness and neuroticism but low in the other traits).

Prior studies in marketing have examined the effects of BFP on two common online purchase spectrums: intention to buy online (Bosnjak et al., 2007; Huang and Yang, 2010) and online impulse purchases (Badgaiyan and Verma, 2014; Thompson and Prendergast, 2015; Turkyilmaz et al., 2015). Findings from these studies consistently show that openness is positively related to the willingness to buy online, while extraversion and conscientiousness significantly influence impulse purchases.

However, research on the influence of BFP on repetitive online purchases remains scarce although scholars have long argued that personality traits are indeed an important determinant of the intention to make a future online purchase (Bosnjak et al., 2007). Hence, in order to advance literature on e-commerce, this research aims to delineate the effects of personality on (repetitive) online shopping whilst controlling for demographic factors such as gender, age, academic score, study field, university, economic status, and computer literacy.

HYPOTHESES

Openness

Openness refers to a set of traits related to creativity, curiosity, and imagination (Turkyilmaz et al., 2015). Individuals strong in this personality trait like to seek new ideas and implement novel approaches. This trait contrasts with conventionality, which involves a preference for familiarity and routine (Leong et al., 2017).

In the context of e-commerce, people who are ready to accept new technologies would more likely to shop online (Agarwal and Prasad, 1998). Also, online stores often provide customers with more varieties and choices of products or brands than the limited selection displayed at physical stores (Dabholkar, 2006). Thus, buyers with an open personality will actively purchase online in order to fulfil their desire for new things such as the latest interesting offers on the internet (Tsao and Chang, 2010).

Hypothesis 1: Openness is positively related to the frequency of online purchases.

Conscientiousness

Conscientiousness is generally characterized by an emphasis on achievement, competence, hard work, responsibility, self-dependence, and a strong will (Tsao and Chang, 2010; Turkyilmaz et al., 2015). People with this personality tend to be more risk averse and appreciate long-term relationships (Leong et al., 2017). Because e-commerce is a new means for shopping, it may be less preferable for conscientious customers who are looking for familiarity in buying products at physical stores. Also, the insecure and risky nature of online platforms explains the reluctance of conscientious customers to buy online (Tsao and Chang, 2010).

Hypothesis 2: Conscientiousness is negatively related to the frequency of online purchases.

Extraversion

In contrast to introversion, extraversion is associated with traits such as active, assertive, cheerful, energetic, outgoing, sociable, and talkative (Turkyilmaz et al., 2015). Extraverts are often dominant and actively seeking excitement (Tsao and Chang, 2010). As a result, some reasons have emerged to expect extraverts to make online purchases more often than introverted individuals. First, extraverted customers are likely to embrace a new digital platform for shopping due to their strong receptivity

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to novel ideas and experiences (Tsao and Chang, 2010). This idea is backed by Saleem et al. (2011), who found a positive relationship between extraversion and the propensity toward using a computer. Second, the unique experience of online shopping compared to traditional purchases at stores, including factors such as unlimited choices and greater discount, makes it a preferable channel for extraverted people who consistently seek to maximize pleasure (Tsao and Chang, 2010). Lastly, since e-commerce has become a global shopping trend, extraverts are more likely to adopt it as a symbol of social recognition, power, and status, compatible with their personality (Leong et al., 2017).

Hypothesis 3: Extraversion is positively related to the frequency of online purchase.

Agreeableness

Individuals high in the personality trait of agreeableness tend to exhibit altruistic, caring, courteous, forgiving, kind, respectful, and sympathetic behaviors (Saleem et al., 2011; Tsao and Chang, 2010; Turkyilmaz et al., 2015). Thus, agreeable consumers are likely to be easily influenced by persuasive ads at online stores with exciting audio and visual components, as well as to enjoy the virtual interactions with fellow online buyers and sellers (Karl et al., 2007). Because online promotions are often more aggressive and more frequently updated than physical stores, customers with a highly agreeable personality are expected to buy via the internet more frequently.

Hypothesis 4: Agreeableness is positively related to the frequency of online purchases.

Neuroticism

Neuroticism refers to poor emotional adjustment or emotional instability (Tsao and Chang, 2010). People with this personality trait possess characteristics such as anxiousness, fearfulness, depression, and sadness (Turkyilmaz et al., 2015). Socially, more neurotic people tend to prefer to be isolated or unsociable. Hence, they are reluctant to be in a situation that requires them to take control or be in charge (Judge et al., 1997; Leong et al., 2017).

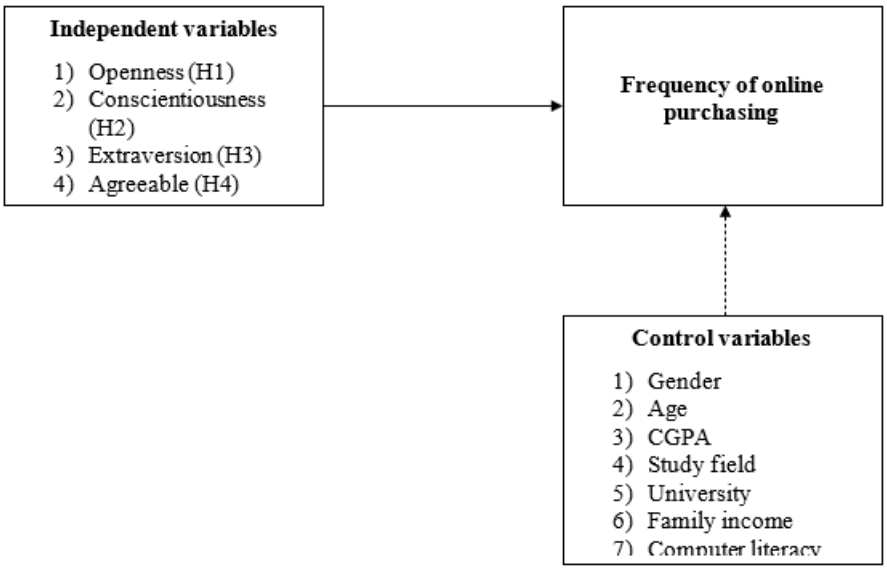
Applying this understanding to the choice of shopping mode, we expect that neurotic individuals prefer to engage in online shopping as way to avoid the attention of other people (Tsao and Chang, 2010). Instead of physically going out and purchase things from the store, neurotic buyers can hide behind their screens and do their shopping virtually. Therefore, they can avoid engaging in direct social interactions with other people since neuroticism involves a lack of social motivation (To et al., 2007).

Hypothesis 5: Neuroticism is positively related to the frequency of online purchases.

The model used in this research is shown in Figure 1.

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Figure 1. The framework of study



DATA

This research utilizes data from the Student Behavior Survey (SBS) 2019/2020, which is intended to serve as a nationally representative sample of Malaysian university students. SBS data is the most recent and comprehensive dataset providing information on many aspects of student behavior. The SBS was carried out nationwide among university students at different levels (undergraduate and postgraduate), in both public and private institutions across study fields. The SBS also collected information on personal demographics and family backgrounds. The SBS yielded responses from 466 students. However, only 386 responses were complete and useful for the analysis.

Table 1 shows the descriptive analysis of respondents in the SBS. Eighty percent of respondents were female, and respondents averaged 21 years of age. Almost 90 percent of respondents attended public universities, and more than 60 percent studied the sciences. The average academic performance, or cumulative grade point average (CGPA), was rather high at 3.35/4.00. Lastly, 60 percent of the respondents came from low-income families, 30 percent from middle-income families, and the rest from high-income families.

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Table 1. Descriptive analysis of SBS respondents

Variable	Description	Measurement
Independent variables		"Please rate each statement that according to how well it describes you" 1 = Strongly Disagree to 5 = Strongly Agree
Openness		I am curious about many different things I am an ingenious and deep thinker
Conscientiousness		I am a reliable person I do things efficiently I make plan and follow through with them
Extraversion		I am talkative I am full of energy I am outgoing and sociable
Agreeable		I am considerate and kind to almost everyone I am helpful and unselfish with others I am generally trusting
Neuroticism		I am depressed I can be moody I get nervous easily
Controls		
Gender	79.8% = Female 20.2% = Male	0 = Female, 1 = Male
Age	Mean = 21.6 years old	Continuous
CGPA	Mean = 3.35 / 4.00	Continuous, between 0.00 to 4.00
Field	61.1% = Social Science 38.9% = Science	0 = Social Science, 1 = Science
University	11.9% = Private 88.1% = Public	0 = Private, 1 = Public
Family Income	60% = Low 27% = Middle 13% = High	Dummy, middle income as a reference group
Computer Literacy	Mean = 1.88	"How good are you in using a computer and navigating its content?" 1 = Not Good at All to 5 = Extremely Good.

VARIABLES

The dependent variable is captured from a single question, "How often do you use the internet for purchasing goods/services," measured on a five-point Likert scale ranging from 1 (Never) to 5 (Always). Similarly, explanatory variables are also measured on a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree) for the statements adopted from the Big Five Personality Inventory by John and Srivastava (1999), also shown in Table 1.

Lastly, we included both demographics and additional characteristics of the individuals as controls: gender, age, CGPA, study field, institution, family income group, and level of computer literacy.

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RESULTS

Prior to the model estimation, we tested for non-response bias by comparing both characteristics and findings between early (received within one month) and late (received after one month) responses. Based on testing of the group means, there is no significant difference between the two groups. Next, we checked for common method bias through Harman's one-factor test. Since there is no single factor accounting for most of the covariance in the independent and dependent variables, we are confident that the dataset does not possess this problem (Podsakoff et al., 2003). The bivariate correlations between all variables are shown in Table 2.

Table 2. Correlational table

		1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Gender	1.000													
2	Age	0.079	1.000												
3	CGPA	0.047	-0.201	1.000											
4	Field	0.128	-0.197	-0.073	1.000										
5	University	-0.173	-0.077	-0.170	0.064	1.000									
6	Low income	-0.085	-0.088	-0.005	-0.167	-0.042	1.000								
7	High income	0.066	0.026	-0.124	0.121	0.051	-0.479	1.000							
8	Computer literacy	-0.101	-0.136	-0.063	-0.027	0.077	0.118	-0.110	1.000						
9	Online purchase	-0.020	0.009	0.008	-0.063	0.093	0.151	-0.154	0.337	1.000					
10	Openness	0.018	0.116	0.006	0.037	0.085	-0.048	0.068	-0.074	-0.169	1.000				
11	Conscientiousness	-0.005	0.040	0.061	-0.049	0.006	-0.124	-0.093	-0.126	-0.033	-	1.000			
12	Extraversion	-0.009	0.035	-0.078	-0.161	0.055	0.123	-0.017	0.038	-0.011	-	-	1.000		
13	Agreeable	0.029	-0.069	-0.101	-0.044	0.179	0.099	-0.063	0.018	0.065	-	-	-	1.000	
14	Neuroticism	-0.051	-0.197	-0.039	0.057	0.039	0.036	0.062	-0.033	-0.109	-	-	-	-	1.000

This study employed Structural Equation Modelling (SEM) to test the hypotheses on the relationships between the personality traits and the frequency of online purchases. SEM provides a conceptually engaging way to precisely test a theory regarding relationships among variables and latent constructs (Hair et al., 2018).

Assessment of Confirmatory Factor Analysis

Confirmatory Factor Analysis (CFA) is a procedure used to validate all latent variables in the model. There are two methods for executing CFA: individual-CFA and pooled-CFA. We opted to deploy the latter because it is more efficient, more accurate, and able to monitor one set of fitness indexes for all constructs in the model. Also, by using this method, all constructs are pooled and linked using double-headed arrows to assess the correlations among the constructs. The results from the CFA model for six latent variables (five independent variables and one dependent variable) range from 0.740 to 0.920. The model also shows that the correlation coefficients among the constructs range between 0.060 and 0.580, thus suggesting no multicollinearity exists among the variables.

Assessment of the Measurement Model

The first step of SEM is to test the measurement model. The results obtained from the pooled-CFA process were assessed to form the measurement model. The fit index values are Relative Chi-Square=1.986, RMSEA=0.051, CFI=0.975, TLI=0.969, and PGFI=0.658 (the full definition of

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theses abbreviations is supplied in Tables 3, 6, and 9). Therefore, our model is acceptable following the recommendation of Hair et al. (2018) that three out of four of the goodness-of-fit results must meet the requirement. The summary of model fit for the measurement model is shown in Table 3.

Table 3. Analysis for measurement model

	AFI		IFI		PFI
Fit Indices	Relative Chi Square (<5)	RMSEA (< =0.080)	CFI (> =0.900)	TLI (> =0.900)	PGFI (> =0.500)
	1.986	0.051	0.975	0.969	0.658

Notes: AFI-Absolute fit indices, IFI-Incremental fit indices, PFI-Parsimonious fit indices RMSEA- Root mean square error of approximation, CFI-Comparative fit index, TLI-Tucker-Lewis index, PGFI-parsimonious goodness of fit index

In the measurement model, we also tested Convergent Validity (CV), Discriminant Validity (DV) and Composite Reliability (CR). CV refers to a set of variables or items that are assumed to measure a construct and to share a high proportion of common variance (Hair et al., 2018). It is tested by using factor loadings and Average Variance Extracted (AVE). Both factor loadings and AVE should measure a minimum of 0.500, which indicates high CV (Hair et al., 2018). CR refers to the degree to which an instrument is measured according to the dimensions of the constructs (Hair et al., 2018). The acceptable cut-off point for CR is between 0.600 and 0.700 (Hair et al., 2018). The results are presented in Table 4.

Table 4. Analysis for convergent validity and composite reliability

Constructs	Items	Factor loadings (>0.50)	AVE (>0.500)	CR (0.600-0.700)
Openness (O)	O1	0.811	0.718	0.884
	O2	0.804		
	O3	0.922		
Conscientiousness (C)	C1	0.889	0.757	0.903
	C2	0.922		
	C3	0.795		
Extraversion (E)	E1	0.910	0.762	0.906
	E2	0.863		
	E3	0.892		
Agreeableness (A)	A1	0.872	0.697	0.873
	A2	0.814		
	A3	0.817		
Neuroticism (N)	N1	0.845	0.663	0.855
	N2	0.856		
	N3	0.737		
Frequency of online purchase (FOP)	OP1	0.870	0.789	0.918
	OP2	0.908		
	OP3	0.886		

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Discriminant validity refers to "the extent to which a construct is truly distinct from other constructs" (Hair et al., 2018). It also means that factors or items only measure one latent construct. The cut-off point for the AVE is greater than 0.500. The point of the DV of the constructs is to explain whether the items are redundant. Furthermore, as shown in Table 5, by comparing the r^2 values (correlation between tested paths) with the AVE value, findings showed that the r^2 of all variables' values are less than the AVE. Consequently, this indicates that each construct is distinct.

Table 5. Analysis for discriminant validity

Tested path	r	r^2	AVE_1	AVE_2	Result
A <--> N	0.438	0.192	0.697	0.663	Valid
A <--> E	0.362	0.131	0.697	0.762	Valid
A <--> O	0.141	0.020	0.697	0.718	Valid
A <--> C	0.121	0.015	0.697	0.757	Valid
A <--> OP	0.578	0.334	0.697	0.789	Valid
N <--> E	0.344	0.118	0.663	0.762	Valid
N <--> O	0.019	0.000	0.663	0.718	Valid
N <--> C	0.140	0.020	0.663	0.757	Valid
N <--> OP	0.711	0.506	0.663	0.789	Valid
E <--> O	0.750	0.563	0.762	0.718	Valid
E <--> C	0.559	0.312	0.762	0.757	Valid
E <--> OP	0.445	0.198	0.762	0.789	Valid
O <--> C	0.162	0.026	0.718	0.757	Valid
O <--> OP	0.058	0.003	0.718	0.789	Valid
C <--> OP	0.181	0.033	0.757	0.789	Valid

Assessment of the Structural Model

The second step in SEM is to test the structural model by examining the hypothesized relationships among latent variables. The structural model denotes one endogenous relationship linking the hypothesized model's variables. In this study, the focus of the structural model is to examine and test the interrelationship between exogenous (Big Five Personality) and endogenous (frequency of online purchase) variables. The present study adopted the Total Disaggregation Structural Model, involving only latent variables. A total of five hypotheses were tested. The fit indices values are Relative Chi-Square=3.220, RMSEA=0.076, CFI=0.942, TLI=0.929, and PGFI=0.664. Therefore, the structural model is declared to be a good fit. The summary of the model fit for the structural model is shown in Table 6 and illustrated in Figure 2.

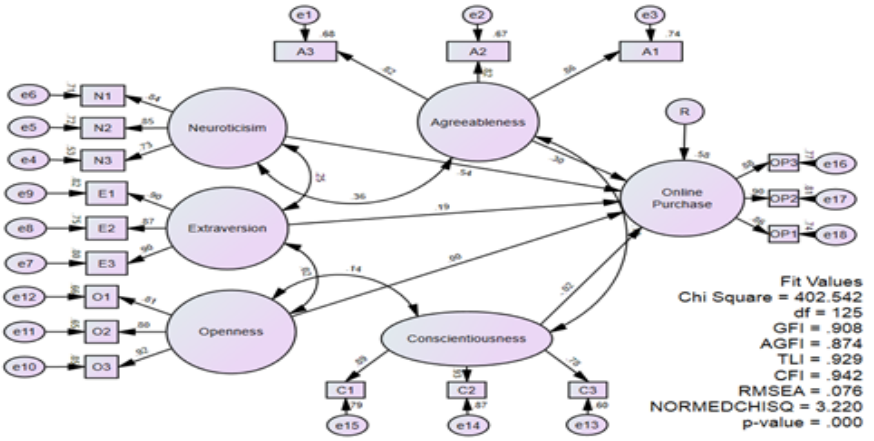
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Table 6. Analysis for structural model

	AFI		IFI		PFI
Fit Indices	Relative Chi Square (<5)	RMSEA (< =0.080)	CFI (> =0.900)	TLI (> =0.900)	PGFI (> =0.500)
	3.220	0.076	0.942	0.929	0.664

Notes: AFI-Absolute fit indices, IFI-Incremental fit indices, PFI-Parsimonious fit indices RMSEA- Root mean square error of approximation, CFI-Comparative fit index, TLI-Tucker-Lewis index, PGFI-parsimonious goodness of fit index

Figure 2. The path diagram of structural model



The aim of testing the structural model is to examine the interrelationship between exogenous (openness, conscientiousness, extraversion, agreeableness, and neuroticism) and endogenous (frequency of online purchase, FOP) variables. The results are presented in Table 7, which shows mixed results. The relationship between extraversion and FOP is positively correlated, β=0.186, p=0.000. Also, there is a significant relationship between agreeableness and FOP, β=0.297, p=0.000. Similarly, neuroticism and OP also statistically related with β=0.541, p=0.000. However, the relationships between openness and FOP, and conscientiousness and FOP are insignificant, β=0.000, p=0.993, and β=-0.020, p=0.629, respectively.

Table 7. Result of hypotheses testing

Hypotheses	Causal path			β	Estimate	S.E.	C.R.	p
H_1	Openness	→	OP	0.000	0.000	0.039	0.009	0.993
H_2	Conscientiousness	→	OP	-0.020	-0.160	0.033	-0.483	0.629
H_3	Extraversion	→	OP	0.186	0.145	0.033	4.349	0.000
H_4	Agreeableness	→	OP	0.297	0.375	0.060	6.216	0.000
H_5	Neuroticism	→	OP	0.541	0.554	0.058	9.589	0.000

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Control variables

To ensure the model we developed is stable and fits the data, we tested the control variables — namely, gender, field, age, CGPA, university, family income, and computer literacy. These variables were checked to determine whether or not they can statistically confound the effect of independent variables on the model. However, the results presented in Table 8 show that all control variables were not significant in explaining the frequency of online purchase (except CGPA, which is significant at $p<0.100$).

Although the control variables reported are insignificant, there is a slight change in the model's goodness of fit, as shown in Table 9.

Table 8. Result with control variables

Variable	Causal path		β	Estimate	S.E.	C.R.	p
Openness	→	OP	0.003	0.003	0.041	0.073	0.942
Conscientiousness	→	OP	-0.011	-0.009	0.041	-0.214	0.830
Extraversion	→	OP	0.162	0.132	0.045	2.934	0.003
Agreeableness	→	OP	0.306	0.406	0.067	6.039	0.000
Neuroticism	→	OP	0.533	0.555	0.057	9.733	0.000
Gender	→	OP	-0.044	-0.096	0.083	-1.159	0.246
Field	→	OP	0.011	0.019	0.070	0.277	0.782
Age	→	OP	0.040	0.022	0.022	1.001	0.317
CGPA	→	OP	-0.066	-0.180	0.109	-1.653	ɣ
University	→	OP	0.004	0.011	0.103	0.102	0.918
Low Income	→	OP	0.018	0.032	0.077	0.414	0.679
High Income	→	OP	-0.020	-0.049	0.109	-0.455	0.649
Computer literacy	→	OP	-0.036	-0.048	0.050	-0.944	0.345

Table 9. Modification of structural model after including the control variables

	AFI		IFI		PFI
Fit Indices	Relative Chi Square (<5)	RMSEA (< =0.080)	CFI (> =0.900)	TLI (> =0.900)	PGFI (> =0.500)
	1.565	0.038	0.976	0.963	0.578

Notes: AFI-Absolute fit indices, IFI-Incremental fit indices, PFI-Parsimonious fit indices RMSEA- Root mean square error of approximation, CFI-Comparative fit index, TLI-Tucker-Lewis index, PGFI-parsimonious goodness of fit index

DISCUSSION AND →

Due to acknowledging the benefits of e-commerce, all parties including the government, sellers, and buyers are willingly embracing the emergence of this trend. However, in the case of companies, in order to sustain business growth via the online platform, their strategies should be twofold: attracting new customers whilst at the same time ensuring existing purchasers will continuously buy products through the internet.

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To further understand what drives individuals to engage in online shopping often, this research examined the effects of personality traits on repetitive online purchase, as measured by the frequency of online shopping, among a specific customer group of university students. Drawing on BFP, we developed hypotheses and tested them through SEM using the SBS data of 386 students in Malaysia.

However, the empirical model supported only three of the five hypotheses proposed. The results showed that extraversion, agreeableness, and neuroticism were positively related to the frequency of online purchases, whereas openness, and conscientiousness showed no significant effect. Certainly, the findings verified prior studies' findings that individuals with personality traits of excitement-seeking (Tsao and Change, 2020), respectfulness (Saleem et al., 2011), and unsociability (Turkyilmaz et al., 2015), are interested in using digital platforms and are eager to do so more frequently. In other words, online shopping is proven to be a means that provides a pleasurable shopping experience for extraverts, persuasive marketing strategies for agreeable people, and the privacy preferred by neurotic buyers.

On the other hand, openness and conscientiousness were found not to explain the frequency of online shopping. There are a few possible reasons for this. Firstly, people high in openness tend to be highly curious and imaginative. Therefore, they might be less likely to stick to the same channel (i.e., online or physical store) for shopping. Rather, these buyers might be more likely to make purchases via both means interchangeably, depending on impulse, offers, or other factors. Second, although we expected reluctance among conscientious people to engage in online shopping, some of them might be interested because internet shopping is more convenient in terms of effort and time, which is suited to a personality that appreciates efficiency (Huang and Yang, 2010). In short, we cannot predict the propensity for engaging in online shopping based on individuals' personality traits of openness and conscientiousness.

Overall, this study contributes to advancing literature on online marketing by examining the effect of personality on repetitive online purchase, thus responding to the call for research focusing on repeat as opposed to new online customers (Kim and Gupta, 2009). Building on the conventional Big Five personality traits, our research sheds light on an important yet relatively neglected market segment: repetitive online customers. Therefore, we theoretically and empirically bridge the traditional understanding of personality traits with the new perspective in marketing literature. Indeed, repeat buyers are currently perceived as a preferable market segment that should receive high priority and customized marketing strategies (Ryu and Han, 2011).

This research also offers practical managerial implications. Most importantly, it suggests that marketers should customize ads and specifically target customers with certain personality traits — extraversion, agreeableness, and neuroticism — because they are likely to use online platforms more often for buying items. From the controlled variables, we would suggest that the interest in internet shopping is not correlated with gender, age, economic status, or even computer literacy. This a strong signal to both policy makers and businesses that e-commerce, especially in Malaysia, is likely to further grow in the future. Hence, all parties should grab this opportunity and earn the vast benefits.

LIMITATION AND FUTURE RESEARCH

Despite its contributions, this research is not without caveats. Mostly importantly, we did not comprehensively control for all possible confounding variables. First, we acknowledged but did not control the supply-side factors in our model such as perceived security (Alharbi et al., 2013) and ease of use of the websites (Xia et al., 2018). Second, we did not consider the socio-psychological factors, especially social influence of close circles such as the families and friends which is deemed critical in theory of social learning. Lastly, since the research was conducted in a single country, our analyses could not insert country-level variables in the model such as digital infrastructure or digital regulations in the country (Oxley and Yeung, 2001).

For future research, we hope that these limitations could be rectified particularly on the theoretical aspect by modelling a complete set of online shopping predictors through integrated psychosocial-

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technology theories. Together, the methodological aspects could also be improved by employing a robust quantitative approach across countries.

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Determinants of Influencer Credibility and Platform Credibility to Understand the Effectiveness of Indian Fashion Influencers

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ABSTRACT

The concept of influencer marketing refers to the use of selected influencers to create and promote branded content to the influencer's followers and the brand's selected target market. Fashion influencers on social media are the main transmitters of information to their network and play a successful new fashion diffuser. However, some influencers are proven to be more successful in doing so. Therefore, this paper aims to understand the factors pertaining to source credibility and platform credibility in determining influencer effectiveness on social media. A huge number of variables have been identified during the literature review for this study. The research design of the present study has been 'quantitative' and attempts to bring empirical findings. A structured questionnaire has been used to collect data from 367 respondents through a purposive sampling method and analyzed using SEM. This research study presents the findings and elaborative results in the context of India. Implications and recommendations of research are further discussed in the context of emerging markets.

KEYWORDS

Consumer Behaviour, Fashion Influencers, Indian Influences, Influencer Credibility, Influencer Marketing, Social Media, SEM

1. INTRODUCTION

The solution to a changing world is innovation. Companies have been forced to implement creative product design and marketing communication approaches because of the evolving nature of brand and customer relationships and the ever-increasing clutter. A successful communication strategy goes a long way toward ensuring the long-term viability of a brand (Brown et al., 2008; Lee & Kim, 2020). The communication networks enabled the brands to interact with their customers and involvement with the community (Hudson & Hudson, 2013).

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The marketing ethos is to create content that can communicate and react to the target market over a long period. The decline in the use of conventional media and the growth of social networks have substantially affected knowledge and artistic content accessed and viewed by people (Kumar et al., 2010). Consumers are becoming more knowledgeable and, as a result, more critical of advertising messages as a result of increased digitization. Social media use is common among Millennials and the younger generation (Gottfried and Shearer, 2016). Consequently, it is increasingly necessary to get information from social media sites.

The emergence of social media has provided an opportunity for brands to communicate with consumers more transparently and organically. The use of social media brand influences is on the rise, and it has emerged as one of the most significant marketing trends of 2017 (Talavera, 2015). Influencer marketing refers to how powerful people help to transfer ideas online and offline (Lee & Kim, 2020; Forbes.com). Social media influences are digital or web celebrities who have a large number of followers across one or more social media sites and wield significant control over their followers (Agarwal et al., 2020). These online personalities are experts in a variety of fields, including travel, food, fashion, and lifestyle, among others.

Consequently, the idea of influencer marketing refers to the use of chosen influencers to build and/or endorse advertised content to the influencer's fans and the brand's target market (Yodel, 2017). Compared to the content created by a brand itself, content produced by an influencer is regarded as more authentic and organic (Talavera, 2015). According to the Influencer Marketing Global Survey Report (2019), consumers often share the content published by the influencers with friends and family, thereby increasing the audience reach, eWOM and increasing brand awareness. Entertainers and Fashion/beauty influencers are the top two most followed influencer categories.

Marketers need to look at influencers as part of the ecosystem and should include them in the mainstream. Various authors in the past (Woods, 2016; Djafarova and Rushworth, 2017) have investigated the different factors that make influencer marketing effective. De Veirman et al., (2017) suggested that the "followers '/followees' ratio" that the influencer has and the type of product being promoted should be of prime consideration when developing an influencer marketing strategy on Instagram. Over the last few years, the influencer marketing industry has continued to expand. The impact of influencers needs to be studied keenly, especially in the context of an emerging economy. Emerging economies like India are still characterized by a need to touch and feel in a consumption context (Talavera, 2015). Given that India has one of the largest millennials, targeting them through web-based platforms has become imperative to branding success. There have been papers developed in the past that have explored the roles of influencers in the age of social media (Woods, 2016). Very few of these papers have looked at the concept of influencer marketing in an emerging economy like India. This paper analyses the factors driving the influencer's effectiveness in India, which is also quite similar to the situation in other emerging economies in Asia. The authors have also tried to present a rare conceptual model in the literature presently available on the subject.

This paper tries to understand two important variables and their roles in the effectiveness of an influencer in emerging economies. The authors, through secondary research, have understood that even though influencers are using a lot of different web-based platforms, not every platform is equally effective in creating a positive impact on the consumers. Thus, the authors have undertaken to study the impact of the credibility of platforms used for promotions on the followers and their ultimate purchase intentions. Given the premium that consumers in emerging markets put on trust and familiarity, this paper investigates the role of influencer's credibility on the consumer's psyche. The paper presents an overview of influencer marketing in terms of its concept and emergence as a new and effective tool for brand communication. The authors then present the relevant findings from the review of the literature and the concept of influencer and platform credibility. The paper then takes the readers through the problem statement and presents the hypotheses undertaken in this study. After discussing the methodology adopted for this study, the authors finally present the analysis and the relevant findings.

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2. LITERATURE REVIEW

Influencer marketing is a strategy of promotion and marketing that utilises prominent individuals or opinion leaders to drive consumers' brand awareness and/or their market-related decisions (Brown et al., 2008; Scott, 2015). The influencer's innate characteristics play a fundamental part in alluring brands and marketers to persuade them closely. Therefore, the attractiveness of influencer marketing is mounting exponentially. Influencer marketing is a form of marketing in which advertisers and brands pay influencers to create and promote their advertised content to their fans, the target market (Yodel, 2017). Influencer followers claim that influencer-created content is more organic, authentic, and meaningful than brand/firm-created marketing contact (Talavera, 2015). The influencer's inherent characteristics are crucial in convincing brands and marketers to pay close attention to them. One benefit is that marketers can choose more affordable influencers than the huge fees necessary to sign one or more renowned celebrity endorsers (Till et al., 1998).

Furthermore, most social media influencers have already developed themselves by specialising in a particular field, such as technology, fashion, or travel. This suggests that when influencers partner with brands relevant to their fields of expertise, customers are more likely to trust their opinions (Till et al., 1998). Influencer marketing yielded eleven times the ROI of conventional ads, according to the same post (Nandagiri & Philip, 2018). However, the situation is very different in emerging markets such as India and Africa, where many research findings are not produced. The benefits and drawbacks of influencer marketing have been discussed extensively, but mostly in a global context. Despite being the world's second-largest country in terms of smartphones and internet usage, India has a small share of the global influencer marketing industry. Influencers are the latest opinion-makers who create content for brands on platforms such as Instagram, YouTube, Twitter, Facebook, and other social media to gain a following and brand awareness. They also generate positive electronic word of mouth and create customer engagement (Rani & Shivaprasad, 2018; Rani et al. 2021).

Many scholars believe that in Social Media Influencer marketing, confidence and perceived credibility may impact the efficacy of the campaign (Lee & Kim, 2020). Bruns (2020) suggested a model to understand how perceived authenticity and trust are linked to purchase intent. With the advent of increased Internet infrastructure and an increased share of social media users, the state of social media marketing and influencer marketing in India is rapidly expanding (Lim, et. al, 2017). Influencers in different stages of the customer decision process in the fashion industry are examined in another study conducted in the Indian context on fashion and lifestyle items. Text-based content from influencers was the most trustworthy source of information for young Indian women (Handa & Khare, 2013; Sudha & Seena, 2017). The fake follower issue has arisen due to the rise of influencers as new opinion leaders through social media, and its implications for influencer marketing must be understood. In an empirical setting, Hu et al. (2015) find that a monopolist recruiting a limited number of influencers will reduce sales unpredictability. Non-traditional celebrities with large follower counts are considered more likeable and trustworthy, according to studies including Jin and Phua (2014), Djafarova and Rushworth (2017), and De Veirman et al. (2017). They have a huge impact on consumer product engagement and purchase intent.

3. THEORETICAL BACKGROUND

Bandura proposed social learning theory (SLT) in 1977 (Bandura & Walters: 1977), and it has since been widely used in the fields of advertising and promotional communication. People learn through observing others, according to this belief. It supports the concept that people are driven to act or behave in a specific manner as a result of their direct or indirect social contacts (Bandura & Walters: 1977). This concept has been used in previous marketing research to explore how socialisation agents such as family, friends, and celebrities impact customer consumption behaviour (Lee & Kim, 2020). This study proposes social learning theory as a contextual framework for understanding how social

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media influencers alter customer behaviour and perception and how this influence affects sales. The notion of SLT is that individuals learn new behaviour through observation from others. If individuals observe positive, desired outcomes in the observed behaviour, they are most likely to imitate and adopt the same behaviour by themselves. In this study, the influencers act as the person being observed and show positive/negative outcomes of the product through their content. They, in return, influence the behaviour of followers as they learn from the influencers they follow.

3.1 Influencer's Credibility and Social Media Platform Credibility

Previous researchers have investigated factors that contributed to the effectiveness of influencer marketing across various contexts (e.g., Colliander and Dahlen 2011; De Veirman et al., 2017; Djafarova and Rushworth 2017; Johansen et al., 2017; Woods 2016). The Bruns Intention by model concept is that Social Media Influencer Marketing is driven to purchase intention when consumers perceive an Influencer as authentic. The literature suggested that some of the key factors were influencer credibility and trust in the influencer. The model also describes how consumers gain confidence in Influencers when they believe them to be genuine and how trust contributes to purchase intent. The study investigated the effects of perceived source confidence, experience, product match-up, and brand meaning transfer on customer attitude and purchasing intention.

With the emergence of technology that helps create similar offerings, brands are likely to lose out on the uniqueness of their offering. The same is true not only for physical products but also in the case of services today. As more corporations are equipped with emerging technologies that help them create similar offerings, brands and services are commoditized. In such situations, it becomes imperative for brands to invest in marketing campaigns that help create a psychological difference in the consumer's mind (Xiao et al., 2018; Lee & Kim, 2020). Trust and credibility play an important role in brand creation. Companies are thus exploring avenues to establish a credible and long-lasting relationship with the target market. Companies have started forging relationships with their customers through celebrities whom the customers follow. The emergence of social media has created an opportunity for brands to connect with their customers by investing in a relationship with popular social media influencers. As the literature review has revealed, customers tend to trust messages and communication from their liked and followed social media personalities than directly from the company's side (Xiao et al., 2018; Lee & Kim, 2020). Hence it becomes important to analyze the factors that influence such marketing success. It becomes important to explore why certain influencers are more successful and what factors help deliver a message to a successful and long-lasting member. After undertaking an in-depth literature review, this paper presents a model to unearth the elements that ensure both the influencer's and platform's credibility, impacting the customer's purchase intentions and knowledge about the brand. The review of existing publications revealed that influencers have the highest impact in the fashion segment as many consumers like to copy the lifestyle of the influencers by indulging in things that these influencers seem to use as a part of their life.

Evans (2017) discovered that endorsers outside of their respective expertise fields could indirectly damage consumers' perceptions, resulting in negative purchasing intentions. Second, the attractiveness of social media influencers as a source of information failed to affect consumers' purchasing intent. The findings backed up a previous study by Colliander and Dahlen (2011), which found that physical attractiveness influences consumers' perceptions of source legitimacy. Till et al. (1998), argued that attractiveness is not a powerful dimension for generating buying activity because there is a poor logical relation between an attractive endorser and purchasing intent.

The product match-up of social media influencers was found to be important with purchasing intention and customer attitude. The product match-up hypothesis established that an endorser's product-related messages must be consistent to achieve a successful advertising outcome (Lim et al., 2017). Furthermore, various endorsement works of literature have highlighted the value of the product or brand congruence with its endorser as a primary criterion for advertising success (Brown et al., 2008;

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Carrillat et al., 2013). In this report, millennials' perceptions of product congruency with social media influencers significantly affected their purchasing intent. Chakraborty and Bhat (2018) looked at the relationship between source credibility and brand awareness indirectly, finding that online reviews' source credibility and review quality were significant predictors of consumers' perceived review credibility, which influenced brand knowledge and purchase intentions. The current study wants to see whether influencer reputation directly impacts brand awareness and purchasing intent in this review.

According to literature, social media influencers' value transfer positively impacts consumers who connect the symbolism associated with endorsers and the endorsed brands. These consumers are more likely to perceive and pass brand meanings, resulting in increased purchases of the endorsed brands (Sudha & Seena, 2017). According to this study, respondents were more likely to embrace meanings from brands supported by social media influencers they saw as similar to themselves or whom they respected. Hence, the authors suggest that influencer's personal/source characteristics improve Influencer's overall effectiveness:

H1: Influencer's credibility has a significant positive relation with Influencer's effectiveness.

Several studies, including those by Cortland (2017) and Phua et al. (2017) found that social media users are wary of marketing ads in their feeds because the site was designed to foster relationships rather than sell goods. Furthermore, it will be important to learn how social media as a medium affects influencer marketing reputation. In the context of social media influencers, where attitude plays a significant role in mediating the fit between a product and influencers, this phenomenon makes sense (Lim et al., 2017). Factors influencing YouTube influencer marketing credibility: a heuristic-systematic model was established where the insightful importance of influencer-generated material, influencer's trustworthiness, attractiveness, and similarity to followers positively affect followers' trust influencers' branded posts, which in turn influence brand awareness and purchase intentions. YouTube is a perfect media outlet for implementing influencer marketing campaigns due to the rise of influencer marketing. The findings show that user perceptions of information credibility on YouTube are influenced by trustworthiness, social impact, claim consistency, and information participation. The authors suggest the following hypotheses, keeping in mind the significance of the influencer's network complexities that improve the Influencer's overall effectiveness:

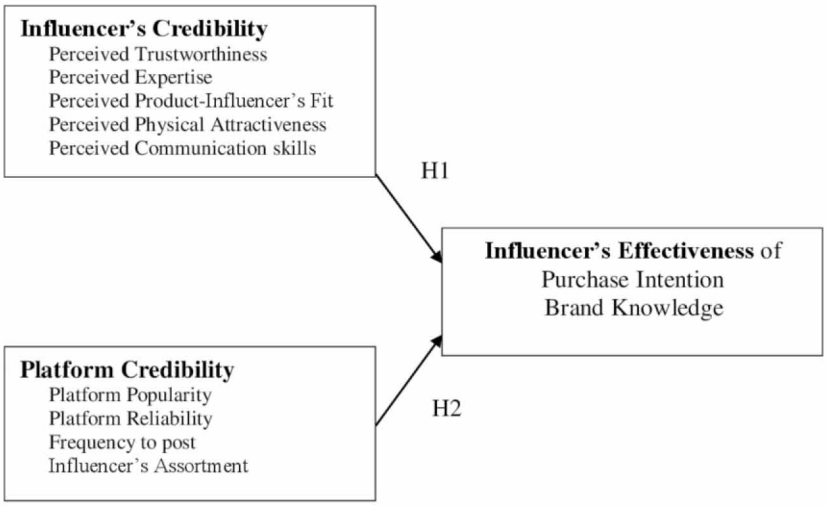
H2: Platform Credibility has a significant positive relationship with the Influencer's effectiveness.

Influencers have altered the fundamentals of marketing communication and how consumers process knowledge. Identifying the factors that affect the source reputation of fashion product influencers and the network credibility of social media platforms could raise awareness of influencer marketing in India As a result, the analysis will investigate the relationship between the following independent variables and influencer effectiveness. The proposed hypotheses are summarised in Fig. 1.

The hypothesis of a positive relationship between consumer attitude and purchasing intent was shown to be correct. The ultimate goal of influencing a person's decision to buy a certain product is highly subjective to that person's beliefs (Hudson and Hudson, 2013). According to the findings of this report, respondents who have a positive attitude toward social media influencers are more likely to want to buy the influencers' Similarly, this result corroborated previous research by Hudson and Hudson (2013) and Chakraborty et al. (2018), which found that perceptions have a positive effect on purchasing intent. Finally, customer attitude is significantly mediated by the relationship between source attractiveness, product match-up, and meaning transfer. The following proposed conceptual model for the study will be tested using various statical tools and applications.

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Figure 1. Conceptual Model of Influencer's effectiveness



4. RESEARCH METHODOLOGY

The study employs a systematic research design method to present empirical results on influencer credibility measures. A standardised questionnaire was used for this research study since quantitative research was designed to collect data using the e-survey process. The items on the questionnaire were taken from previous research studies that measured the credibility of influencers' sources and platforms. The items for the scale of source credibility are taken from McCroskey & Teven (1999); Bhattacherjee & Sanford (2006) and modified in the present context. Few items were adopted from Dedeoglu et al. (2019). The items of platform credibility are adopted from the previous study by Zha et al. (2018). The instrument used in this study had 24 items, and they were measured on a 5-point Likert scale to capture the responses from participants. The study measures the five dimensions of influencer's credibility: Perceived Trustworthiness, Perceived Expertise, Perceived Product-Influencer's Fit, Perceived Physical Attractiveness, and Perceived Communication skills of the influencer. Also, there are four dimensions of platform credibility i.e. popularity of platform/site, reliability of Platform/site, frequency to post, and presence of other popular influencers, included in the study for measurement. The Dependent variables are Purchase intention and Brand knowledge which are indicated in constructed instruments of measurement.

The said measurement scale pooled items from the previous existing scale, and therefore a preliminary approach to testing this scale is executed with the help of exploratory factor analysis (EFA) and further confirmatory factor analysis (CFA). The EFA examination using principal component analysis supports the constructs of the measurement model proposed three iterations (eigenvalue >1). In the final results, 2 items in the questionnaire were dropped due to cross-loadings. Then, CFA was performed to assess the scale properties of the measurement model and again obtained satisfactory results to establish the model's validity. Factor analysis revealed a three-factor solution accounting for 69.1% of the total variance. Each indicator loaded on the expected factor.

4.1 Data Collection

There are over 462 million internet users in India, with 200 million of them using social media. India is the world's second-largest online economy, after China. 66% of Internet users use social media sites daily in urban India. Users of social media are often subjected to ads from social media influencers. Influencer-created content is preferred over ads because it is detailed and

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objective-oriented (Nandagiri & Philip, 2018). The research samples are "any Indian internet users on social media who follow one or more fashion influencers". Surveying this population has not been easy as Indian social media users were 310 million in 2018 (statista.com). The 216 million users on social media are interested in fashion products, and approximately 1 out of 3 users follow influencers on any social media platform. Therefore, the sampling frame is 72 million Indians on social media who follow one or more fashion influencers, mostly micro-influencers (Żyminkowska, 2019).

Further, the structured questionnaire for this study has been circulated through social media posts to the potential samples. The questionnaire was distributed using a purposive sampling technique to reach the right respondents. The famous fashion influencer's/micro-influencer information was obtained from the official report (Pragati, 2019) and then the email address of followers through the comment box was taken. Further, the survey instrument has been shared via email and mobile numbers. A total of 381 respondents' data of qualified participants have been received over two weeks, out of which 367 respondents' data has been further considered for analysis.

Further, the multivariate analysis will be applied to understand the relationship between latent and measured variables of the measurement model. This method is preferred for the study because estimating multiple interrelated independent variables can be understood as the dependent variable in a single analysis. Therefore, the study has analyzed the data by Confirmatory Factor Analysis (CFA) and Structural Equation Modelling (SEM) to test the developed hypotheses. The SPSS software of version 22.0 has been used for all the basic analysis, and AMOS 21.0 version has been used for CFA and SEM Models.

5. RESULT AND DISCUSSION

The theoretical foundation of this research leads to hypotheses and data has been collected to examine the proposed hypothesis of the study. First, data preparation has been undertaken by identifying missing data, analyzing outliers, calculating standard deviation and descriptive statistics of individual items & individual responses. The test of Kaiser-Meyer-olkin (KMO) test and Bartlett's test of Sphericity has been performed to measure sample adequacy. The KMO value (0.81) and a significant chi-square value indicated that factor analysis has been appropriate for analysis.

Once the data has been adequately prepared for analysis, the model evaluation has been carried out. All the exogenous variables have been measured using a 5-point Likert scale, whereas a different 5-point Likert scale is used to measure endogenous variables. During the common method variance analysis, the largest variance explained by the first factor was 45.73% of the total variance explained. This result signifies no common method bias (CMB) in the data set; therefore, the analysis can be further taken (Podsakoff et al., 2003; Podsakoff and Organ, 1986). The models were compared via chi-square tests to control CMB, which demonstrated that the former was superior to the later. Therefore, common method bias was not a concern for this study, further supporting the two-dimensional structure.

The data sets which have been retained for the final analysis of the proposed model are 367. The table below represents the demographic profile of respondents. From Table 1, it is evident that the majority of respondents are Female (57.2%) of the age group 16-25 (45.2%), having a Graduation degree (45.5) as their dominant educational qualification. The data also indicate that the most popular social media platform to follow fashion influencers is Instagram and Facebook followed by youtube. This demographic data helps the researcher understand that this study can be generalised on which category of the population.

The demographic measure constituted the first section of the research instrument, followed by the proposed Independent variables and additional variables. The latent variables have been measured based on a previously validated scale. Therefore the Exploratory Factor Analysis (EFA) has been conducted to know the total variance extracted. The items of Influencer's credibility, Platform

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Table 1. Demographic Profile of Respondent

Demography		Frequency	Percentage
Gender	Male	157	42.7
	Female	210	57.2
Age (in years)	16-25	166	45.2
	26-35	128	34.9
	36 & above	73	19.9
Educational qualification	Under-Graduate	117	31.9
	Graduate	167	45.5
	Post-Graduate or above	83	22.6
Which social media Platform do you follow fashion product influencers (Select multiple as applied)	Instagram	242	65.9
	Facebook	132	36.0
	YouTube	216	58.9
	Blog Post	124	33.8
	Others	31	8.4
Which type of review do you prefer to follow	Detailed Video	99	27.0
	Short Video	146	39.8
	Picture and text description	102	27.8
	Plain Text reviews	18	4.9
	Audio	2	0.5

Credibility, and Influencer's Effectiveness fall into three extractions, explaining the total variables extracted as 69.1, which is acceptable for further CFA & SEM.

Before the model estimates, each sub-construct undergoes a CFA analysis. Many of the items with a high P-value were investigated further for model estimates. The results of SEM were obtained using AMOS software and are presented below. In this analysis, the two independent constructs 'Influencer's source credibility (SC)' and 'Platform/Site Credibility (PIC)', as well as the dependent variable 'Influencer's Effectiveness (IE),' met all of the criteria, with all loadings exceeding 0.5 and adequate composite reliability (CR). Average Variance Exacted (AVE) has been reported greater than 0.5 which is suitable for the given model (Hair et al., 2009). This allows an understanding of the correlation for the indicator variable of a construct. As a result, all constructs met the reliability and convergent validity requirements. Table 2 represents the data systematically.

The fitness of the model has been tested and indicated on different values. The final model for source credibility exhibits the chi-square divided by the degree of freedom ratio, giving the value of 2.50, which is a good acceptable fit. Also, the absolute fit parameter value, like root mean square error of approximation (RMSEA), is 0.064, which is below the recommended cut-off level but acceptable in range. The incremental fit parameters, namely the comparative fit index (CFI), give 0.98, which Bentler, 1990 recommends. The normal fit index (NFI) is 0.96. The values are given in Table 3 exhibit fitness indicators that imply the conceptual framework. Most of the observed data set values are ranged with model fit recommended values and provide sufficient evidence that the fit between the measurement model and data is acceptable.

The structure of the model under consideration is represented in the Figure 2. All the standardized values are represented in the structural model. This indicates that the model under consideration is significant.

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Table 2. Internal consistency and Reliability of the study construct

Construct	Indicator	Item Name	Loading	AVE	CR	Cronbach's Alpha
Influencer's Credibility (SC)	Perceived Trustworthiness	PT	0.706	0.612	0.83	0.873
	Perceived Expertise	PE	0.739			
	Perceived Product-Influencer's Fit	PF	0.598			
	Perceived Physical Attractiveness	PA	0.633			
	Perceived Communication skills	PC	0.631			
Platform Credibility (PIC)	Popularity of Platform/ site	PP	0.502	0.591	0.81	0.882
	Reliability of Platform/ site	PR	0.977			
	Frequency to post	FP	0.498			
	Presence of another popular influencer	IA	0.986			
Influencer's Effectiveness (IE)	Purchase Intention	CPI	0.855	0.724	0.79	0.85
	Brand Knowledge	CBK	0.886			

Table 3. Model Fit Indices

Statistics	Suggested Values (Gefen et al. 2000)	Value achieved
Chi-square/d.f	< 5.0	2.502
RMSEA	<= 0.60	0.64
NFI	> 0.90	0.96
TLI	>= 0.95	0.97
CFI	>= 0.95	0.98
GFI	> 0.80	0.95
AGFI	> 0.80	0.92

Table 4 represents the regression weights of all the variables is represented along with P values. The regression values stand significant in all cases. The regression value of SC (Source Credibility) on IE (Influencer's effectiveness) is significant ($p < .001$). PIC (Platform Credibility) on IE (Influencer's effectiveness) is also significant ($p < .001$). The variables of source credibility that is PT (Perceived Trustworthiness), PE (Perceived Expertise), PF (Perceived Product-Influencer's Fit), PA (Perceived Physical Attractiveness), PC (Perceived Communication skills) estimates stand significant stating positive relation between endogenous variables and exogenous variables. The factors of Platform credibility also positively signify the relationship with the PP (Platform Popularity), PR (Platform Reliability), FP (Frequency to post) & IA (Influencer's Assortment).

The level of R square (co-efficient of determination) has been assessed in the case of hypotheses testing. 'Influencer's credibility' construct accounts for 42% of the variance in explaining Influencer's

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Figure 2. SEM Structure of Model

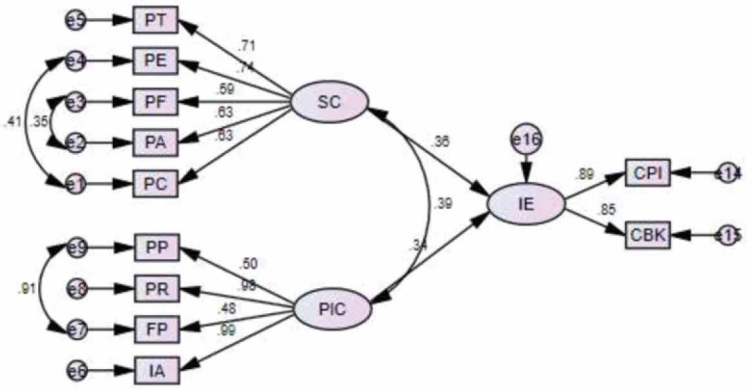


Table 4. The Regression Weightage of Model

Regression Output			Estimate	S.E.	C.R.	P Value
IE	<---	SC	0.417	0.079	5.287	***
IE	<---	PIC	0.262	0.043	6.148	***
PC	<---	SC	1			
PA	<---	SC	0.997	0.115	8.669	***
PF	<---	SC	0.987	0.12	8.224	***
PE	<---	SC	1.13	0.085	13.28	***
PT	<---	SC	1.028	0.111	9.257	***
IA	<---	PIC	1			
FP	<---	PIC	0.442	0.043	10.219	***
PR	<---	PIC	0.992	0.023	43.476	***
PP	<---	PIC	0.458	0.042	10.879	***
CPI	<---	IE	1			
CBK	<---	IE	1.086	0.081	13.419	***

Note: IE (Influencer's effectiveness), SC (Source Credibility), PIC (Platform Credibility), PT (Perceived Trustworthiness), PE (Perceived Expertise), PF (Perceived Product-Influencer's Fit), PA (Perceived Physical Attractiveness), PC (Perceived Communication skills), PP (Platform Popularity), PR (Platform Reliability), FP (Frequency to post) & IA (Influencer's Assortment)

effectiveness. Whereas, the 'Platform credibility' construct accounts for 27% of the variance in explaining the Influencer's effectiveness. Together, they account for 69% variance in explaining the dependent variable, i.e Influencer's effectiveness. The value presented in the table below is sufficient to support the hypotheses of this research paper. Finally, this presentation demonstrates that the data analysis is sufficient to prove that the Influencer's credibility and Platform credibility have been significant in predicting the Consumer's Purchase intention in the context of the Fashion industry's influence.

The study revealed a significant positive effect of Perceived Trustworthiness, Perceived Expertise, Perceived Product-Influencer Fit, Perceived communication skills and Perceived Physical Attractiveness on source credibility. There is significant evidence even to conclude that Popularity of

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Table 5. Hypotheses Testing Results

Hypothesis	Relationship	Effect	Decision
H1	Influencer's credibility has a significant positive relation with Consumer's Purchase intention.	0.417	Supported
H2	Platform Credibility has a significant positive relation with Consumer's Purchase intention.	0.262	Supported

Platform/site, Reliability of Platform/site, Frequency to post and Presence of another famous influencer, shape-up the platform credibility for influencer's marketing. Overall, the source and platform credibility directly affect the 'Purchase intention' and 'Brand knowledge of consumer'. Therefore, it can be concluded that influencers should exhibit the qualities of Trust, Expertise, good-fit with the product, commendable communication skills, & attractive physical appearance, which indicate credibility in the source for a fashion product. The finding also indicates that consumers prefer the popularity & reliability of social media sites along with the frequency to post by many influencers' as credible platforms for influencers. Finally, perceived source credibility and perceived platform credibility signify purchasing intention and enhance the consumer's brand knowledge.

6. DISCUSSION AND IMPLICATIONS OF STUDY

This study is the first to elucidate the effect of source credibility and platform credibility in the context of influencers of emerging markets like India. By examining the influencers market from a comprehensive perspective, this study has extended the application of considering the factors pertaining to source credibility and platform credibility, which influence the purchase intention of consumers. The proposed hypothesized model investigated the potentials of Trust, Expertise, good-fit with the product, admirable communication skills & attractive physical appearance of influencers on the effectiveness of influencer's marketing. The result of this study suggests that influencers generated post's informative value and their attractiveness ensures credibility. Our findings explore the attributes of trust and expertise as the most impactful variable to establish the source dimension within an influencer's credibility model. Our findings showed that influencers' trustworthiness, attractiveness, and perceived product-fit positively affect their followers follower's belief in influencer branded posts which are in line with the findings of perceived studies like Djafarova & Rushworth (2017) & Lou & Yuan (2019). This study also contradicts the results of Lee & Koo (2015), which examines the relationship of source credibility on social media influencers with purchase intention and found the relationship insignificant for all the variables of source credibility. It is noteworthy that the physical attractiveness of influencers was a non-significant factor in most of the previous examinations. However, in the context of fashion influencers, the factor of physical attractiveness is important to shape the credibility of influencers.

The finding indicates the dimension of platform credibility through the variables pertinent to the platform's reliability, the popularity of the platform, frequency of influencer's posts and an assortment of many similar influencers. There are many social media platforms present in India for Influencers. But the study indicates that the sites that are more popular among consumers, perceived as reliable, many other influencers are present to choose from and frequently do post about fashion products are considered superior to see influencer's posts like Instagram (Rebelo, 2017). It reinforces the research findings of other researchers in this field (Chou et. al, 2015; Lou & Yuan, 2019; Zha et. al, 2018). The social learning theory often strengthens the field of influencer effectiveness. Human behaviours are learned from the environment through the observation learning process. Influencers are powerful in creating an effect on consumers who follow them. This paper is one of the rare few

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attempts to put source and platform credibility together to determine its relationship with purchase intention and brand knowledge.

6.1 Implications of the Study

The investigation has theoretical as well as practical implications. The study's findings, in theory, expand existing literature on influencer marketing by investigating how various aspects such as influencer credibility and platform credibility may influence the success of this brand's marketing by impacting purchase intention. The Social learning theory adoption proves that customers learn new behaviour through observation from influencers and as influencer's credibility increases positive outcomes in the observed behaviour of those who follow them, it implies imitating and adopting the purchase intention. Furthermore, the study's findings provide theoretical insight for the persuasive knowledge of the social learning model by demonstrating how influencers help followers learn about fashion products.

This study offers marketers a few practical implications in selecting social media influencers for their marketing program from the managerial perspective. The limitations of modern marketing communication and ineffective mass media have given rise to unreliability among brand-endorsed communication. Consumers are looking for those sources of communication that they find independent and reliable at this point in time. Influencers are those opinion leaders on social media who are extremely effective in brand knowledge and positive attitude formation. Entrepreneurs and practitioners can utilize this framework to tap the potential of social media influencers. Particularly, the firms selling less-differentiated brands can target the micro-influencers to reach their target market segment. These influencers have also been taken as an alternative to avoid costly celebrity endorsement and help to deeply connect local and niche markets to approach product diffusion in urban and rural markets. While considering an influencer's marketing program for fashion products in India, marketers can consider influencers who are good fit for the product, perceived as trustworthy, knowledgeable, attractive and possess good communication skills to gain a competitive advantage. Besides, managers can choose the social media platform that is more popular, reliable and encompasses an assortment of other fashion influencers to gain more effectiveness in their influencer marketing program. This study can also be useful for micro-influents who want to expand their followers and look for sustainable growth as fashion influencers on social media.

Some limitations of this research work are highlighted in this section. First, although this study considers several factors related to influencer and platform credibility, other factors such as message credibility and content quality can be found and included in future research. According to studies, other variables such as message credibility, receiver motivation, and receiver personality characteristics may all play a role in an influencer's effectiveness. Second, fashion influencers on social media sites are taken into account in evaluating influencer effectiveness. Limiting the scope was important to keep the current study focused, but it also made it difficult to parse the results to other forms of influencers. Physical appearance, for example, is a major factor in increasing source credibility, but it may not be as important in the cases of other influencers. As a result, this study can be applied to other product categories and different cultural settings. Finally, this research aims to develop a basic theoretical model that can explain the factors that affect influencer effectiveness. Influencers and their followers are becoming more prevalent. Influencer marketing is a field of study that will need more time and effort from potential researchers to provide valuable information and aid in developing a focused strategy for influencers in the future. Future researchers can study the combined effect of influencers with advertising messages in impacting product purchase intention.

7. CONCLUSION

The principal aim of the research paper was to investigate the effect of Influencer's credibility and platform's credibility on consumer purchase intention in the context of fashion influencers. Influencers

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play the role of catalysts in the marketing program's effectiveness; however, this effect depends on the consumer's perceived credibility of the influencer. The result shows that an influencer's credibility and platform credibility are significant in purchasing intention and following a fashion influencer on social media. These findings add to the literature in three ways. First, the analysis of fashion influencers on social media complements knowledge by providing information on factors influencing influencers' effectiveness. Second, the analysis of these relationships between source credibility, platform credibility, and Consumer purchase intention can be strategically utilized by marketers as an innovative tool to market fashion products in India. Third, the study provides the foundation for potential research in the field of study. Overall, it is virtually impossible to ignore the collective wisdom in social media-originated by influencers of all types that's why influencers marketing can be considered a cost-effective and more efficient way to market products in the fashion sector.

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Investigating Marketing Communication Mix on Brand Performance Indicators:

Evidence From the Financial Services Sector in Ghana

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ABSTRACT

In today's world of business, an increased number of brands are competing in order to gain market share. As a result, companies' promotion has become more significant in order to reach out to target customers. Effective financial service (FS) delivery is driven by the strategic application of each element of the marketing communication mix (MCM) in the organization. The population of this study was employees in Ghanaian organizations where the MCM tools are known to be frequently used and their customers. Confirmatory factor analysis (CFA), fit statistics for the measurement, and structural models were used. The first four models were the measurement models of MCM, brand quality, brand loyalty, and brand performance respectively. The fifth model is the structural model through which all relationships and hypotheses were tested simultaneously. The study finds that the first four models did not produce a good fit because they failed to meet the following criteria: p-value > 0.05; RMSEA < .08; TLI > .90; GFI > .80; and AGFI > .90.

KEYWORDS

Brand Quality Perception, Confirmatory Factor Analysis, Financial Service, Marketing Communication Mix, Service Quality Perception

INTRODUCTION

The services sector is the fastest growing sector in the Ghanaian economy and represents around 56.2 percent of GDP in 2017 (the budget statement and Economic Policy of the government of Ghana, 2018). In this report, again the financial service sector is the main contributors to the quota services.

Banks and financial services are brands that aim to provide the highest product and therefore consumer value services. Traditionally, advertising, marketing activities, direct marketing, sales promotion, internet marketing, public relations and personal sales form marketing methods. These instruments are frequently called the integrated communication marketing mix (Saeed et al., 2013; Lekhanya, 2015). Extant literature indicates that companies often use one or more of these instruments to communicate with clients and the public. Sometimes, the choice of such instruments by a company depends on which of its financial conditions and communication needs are better adapted to one or more.

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Some studies have established the positive association between brand quality and unique consumer communication tools (Okyere et al., 2011; Porcu et al., 2012). This evidence shows that the quality of brands improves as the degree of use of appropriate consumer communication tools increases. Unfortunately, the effect on brand quality (i.e. the consumer contact mix) of all market communications methods is poorly seen (Porcu et al., 2012; Frimpong, 2014).This situation poses a literature gap, considering the significant investment of financial service providers on market communication (Idris et al., 2012); Takalani, 2015) and the fact that the goal of market communication is to maximize brand quality.

Touching on the methodological problem, there is substantial evidence on the association market communication tools and brand quality, customer satisfaction, brand loyalty, and brand performance (Madhavaram et al., 2005; Ismail et al., 2012; Lekhanya, 2015). This notwithstanding, very few of these studies have considered all market communication tools in testing the foregoing association thus researchers' emphasis on a few or specific tools predominates in the literature. In the financial services sector in Ghana, studies examining market communication mix tools and the above association failed to apply appropriate statistical methods in validating measurement scales used (Frimpong, 2014). When the tools of consumer communication are linked to brand quality, brand loyalty and brand performance, the relationship between a market communication tool and brand performance is possible to mediate brand loyalty entirely or in part.

The value for improving organizational performance in these subsectors of concepts like MCM and IMC would ultimately represent a major impact on the overall GDP of a developing country such as Ghana.

LITERATURE REVIEW

Communication activities in marketing are an important task for companies as their income and survival are inevitably linked to the needs and desires of consumers. In this age of diverse and competitive market conditions, changing demand also depends on consumer knowledge, retention, persuasion and information about existing services (Okyere, Agyapong, & Nyarku, 2011). This led to the use by financial services organizations of many marketing communications strategies for their success in this type of competition.

Marketing Communication Mix

Marketing communications assist to define an organization's associations with consumers not just by the kind of messages exchanged, as well as by the choice of medium and occurrence to go with their customers' preference (O'Guinn *et al*., 2006).

The main purpose of marketing communication according to Vodak et al. (2016) is the transmission of the message between the sender and the recipient. The amount of the actual customers willing to pay for the service delivered can be proved for the contribution of a good message transferred.

Marketing communication mix tools are intended to inform and persuade an audience with a view to influencing the behavior of that audience (Abubakar, 2014). These tools go a long way to contributing in the growth of business in many aspects if practiced efficiently like increase in volume of sales, amount in return on investment and maintain goodwill (Gassmann, Kausch and Enkel, 2005).

In order to engage with their target audience, financial service organizations, according to Manisha (2012), rely heavily on effective communication mix tools between themselves and their customers. Service companies have identified absolute focus on existing customers as a primary success strategy in achieving customer satisfaction (Rahim & Malik, 2010; Hashem, 2010), which is the basis of customer retention, in today's business environment characterized by rapid change, knowledge explosion, technological advancement, and where many organizations are striving to increase their customers.

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The blend of marketing communications, according to Kotler & Keller (2011), is a complex mixture of six components, including advertisement, product promotion, events and experiences, public relations, direct marketing, and personal selling.

Advertising

Advertising deals with any paid form of non-personal presentation and promotion of ideas, goods, or services by an identified sponsor. The paid aspect is significant since the space for the advertisement is bought. The organisation has various options of advertising, but it must endeavour to use of the most cost-effective and customer-cantered option in the marketing of services (Kotler & Armstrong, 2010; Schultz et al. 2007; Manisha, 2012). It includes mass media (such as TV, radio, and magazines). By paying for the advertising space, a firm decides on what to say and, to some extent, the message to be conveyed. It is one of the elements of the promotion mix which financial service firms consider as prominent in the overall marketing mix and the most visible element of the communications mix because it makes use of the mass media, i.e. newspapers, television, radio, magazines and billboards (Abubakar, 2014).

According to Aliata et al. (2012), the amount spent on advertising a good or service has a statistically important connection with profit efficiency. However, his research did not show that advertisement strategy is successful in increasing product sales volume.

Personal Selling

Another element of the marketing communication mix (MCM) is personal selling, which is the personal presentation by the firm's sales force to make sales and build customer relationships (Porcu et al. 2012; Kotler & Armstrong, 2010; Rawal, 2013). According to Manisha (2012), personal selling is suitable to winning new customers, though it is equally suitable for delivering services to the expectation of existing customers.

Okyere et al. (2011) also argued that this two-way contact practice enables the seller to recognise and customize the sales presentation in light of this experience to identify the unique needs and challenges of the customers. It links the needs of the company to the wishes of its clients, hence completing the transaction is a personal obligation to sell (Kotler and Keller, 2012; Dmour et al., 2013).

Event Marketing

Event marketing has grown popular in recent decades and has been an experiential promotional vehicle. Schmitt (1999) posit that it is an experiential marketing method that treat rational and emotionally driven consumption as an integral experience. Combining a range of promotional activities aimed at communicating brand values and improving the value of consumers' experience, event marketing provides opportunities for consumers to participate with the company, their brands and the community (Close et al. 2006). Researchers have pointed out that events must not be regarded as an independent marketing tool, but as an integral part of the marketing communications in order to achieve their aims. When integrated into different forms of communication, events should achieve optimal brand effects. ((Gupta 2003, Väänänen 2008)

Sales Promotion

Ofosu-Boateng (2020), posits that, marketers use sales promotion tools to draw a stronger and quicker buyer response, including short-run effects such as highlighting product offers and boosting sales. Sales promotions are instruments which stimulate the sales of products and brands, especially in consumer-packaged goods. Sales promotions is the short-term promotional offers to promote selling or purchase of a product (Okyere, Agypong & Nyarku, 2011) including free presents, cash-off vouchers, product samples, and so on (2012). According to Aliata et al. (2012) there is statistically important linkage between promotion, sales and performance. They confirm that the promotion of

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sales increased brand love, thus improving brand performance and high profitability. Nevertheless, research has not shown how sales promotion strategies can obviously improve sales performance.

Direct Marketing

Direct marketing is defined as direct connections with targeted individual customers to obtain both an immediate response and cultivate long-lasting customer relationships. It is often referred to as direct mail or as dialogue marketing, personal marketing and database marketing (Bird, 1989; Kotler and Armstrong, 2012). According to Sellahvarzi, Mirabi & Parizi, (2014), due to the rapid growth in both communications technology and the introduction of direct marketing, the overall nature of marketing communication changed.

Public Relation

According to Ngwenya (2015), public relations have existed since the 20th century, and the main job is to inform, persuade and integrate the public into the early society. Mubushar *et al.* (2013), were of the view that, an increasing number of organisations are using public relation to communicate with their audiences about the organization and brands. Public relation involves building good relations with the company's numerous publics by gaining favourable publicity, building up a good corporate image and handling unfavourable rumours, stories and events It includes press releases, sponsorships, special events and other forms (Belch and Belch, 2007; Kotler and Keller, 2012).

Internet Marketing

The internet has become wildly more readily accessible globally in the past ten years. As people spend a great deal of time on internet, it has consequentially become an essential place for marketing as well. Perry (2018) posits that email marketing campaigns can be used to search and effectively boost long-term and short-term sales, but are to be seen as part of an all-encompassing communication strategy aimed at building longer-term relations with the recipient of communications.

Social Media as a Marketing Communication Mix Tool

The social media has also been academically integrated and a broad range of study has been produced into social media marketing and associated issues including online word of mouth (WOM) and online networks. Despite what academics and professionals have researched and learned about the issue in the previous 15 to 20 years, the future of social communication in marketing may not be just continuity, because of the fast-paced and ever-changing nature of social media and the way in which consumers use it. According Gordon et al. (2019), the value of advertising on the social media continues to be investigated, and how it interacts with other formats of the media, such as television (Fossen and Schweidel, 2019). Some current studies have examined social media from the point of view of WOM, which are the impact that WOM transfers have on others (Herhausen et al. 2019), which impact the type of WOM content shared on the behavior of others (Villarroel Ordenes et al. 2018), and which motivates consumers to make their posts on the social media (Grewal et al. 2019)

Integrated Marketing Communication (IMC)

There is growing evidence that IMC results in a strong positive impact on brand and financial performance (Luxton et al, 2015; Porcu, et al 2017).IMC is defined as a strategic approach to management of marketing communications activities in an organization by the Chartered Institute of Marketing (CIM, 2012). Lamb et al (2012) also argue that IMC is the systematic coordination of all sales programs, primarily in media, advertising, promotion of sales and personal selling and public relations as well as other forms of communications. Porcu et al. (2012) on the other hand, defines it as a concept that directs and coordinates the process of planning, implementing and supervising brand messages by which brand-customer relationship is build.

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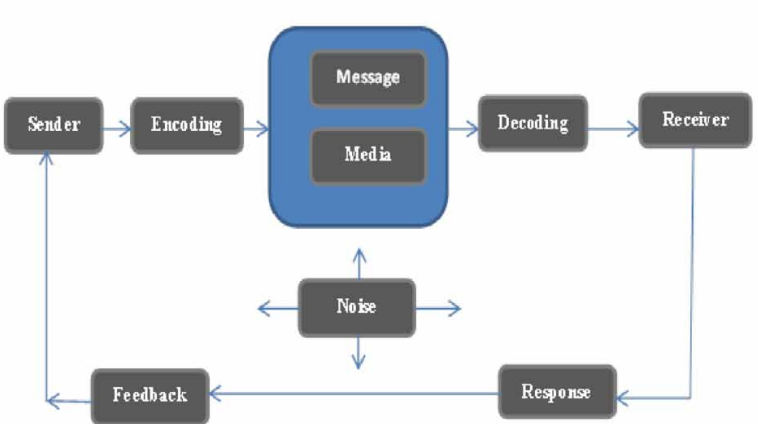
In today's turbulent world, Eagle et al (2015) see integrated promotion mix as essential. Using expert promotional integration, companies invest in corporate image advertising in order to increase brand awareness through a single advertisement slot. Keller (2016) identified the challenges facing customers as a result of the ever growing choice of brand contact points, and suggested further customer / agency relationships

THEORETICAL REVIEW

An Overview of Marketing Communication Mix Models

Over the years, the concept of market communication has been an active subject of academic debate. The meaning of market communication has not changed during the history of this debate (Jerman & Zavrsniky, 2012), yet the theoretical foundations of the concepts are continually evolving. However, marketers need to understand how communication works to communicate effectively. The nine elements of communication in Figure 1 are described (Kotler & Armstrong, 2010; Ekhlassi et al. 2012). Two of these elements are the major parties in a communication: the sender and receiver (Kotler & Armstrong, 2010; Wellman & Molander, 2008).

Figure 1. The Communication Model



(Source: Adapted from Kotler & Armstrong (2010); Akerlund, (2004) and Ekhlassi et al. (2012)

This model points out several key factors in good communication. Senders need to know what audiences they wish to reach and what responses they want. They must be good at encoding messages that take into account how the target audience decodes them. They must send messages through media that reach target audiences, and they must develop feedback channels so that they can assess the audience's response to the message. For a message to be effective, the sender's encoding process must mesh with the receiver's decoding process (Kotler & Armstrong, 2010). Therefore, the best messages consist of words and other symbols that are familiar to the receiver. The more the sender's field of experience overlaps with that of the receiver, the more effective the message is likely to be (Ekhlassi et al. 2012).

Market Communication Mix and Brand Loyalty

Brand loyalty is essentially a term for repeated purchases. Chen et al. (2019) characterize brand product loyalty as a driver of repurchase quantity and brand appeal and of memorable brand experience to

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enhance branch awareness and appreciated value. These purchases are required to generate revenue and achieve optimal performance in a very competitive situation. It is the result of Porter's (1985) theory of competitive strategy's competitively advantage, and capacity. Brand is an instrument of strategic marketing and is associated with value-based pricing strategy, according to Nadanyiova et al. (2018), Kicova, & Nadanyiova (2017). Their analysis found that the frequency of dairy products purchases depended on the satisfaction of customers with their product quality.

Janoskova & Krizanova (2017) argue that "brand identification with consumers creates advantages that can lead to higher prices for the same product and higher profit margins, growth, and a strong net value for the same product." A successful marking differentiation has been gained globally (Singh & Singla, 2018), and the assessment of delivered quality brand products from the consumer point of view is necessary for the companies

From the Social Exchange (SET') perspective, people would go for a service or product which has a positive 'worth' or results a term used to describe the benefit of a product or service that is more than cost (Porcu et al., 2012). The theory also argues that people have a stronger attachment to the most valuable services or products. Therefore customers compare several alternatives to know, which has the greatest value in a competitive situation. In this regard, the relationship between individuals and brands depends on the value or customer value of these brands.

The ability of a company to offer customer value on a competitive market depends on the capacity and use of available resources (Hanaysha & Hillman, 2015).If service providers provide customer value on a sustainable level, customers are invited to repeated purchases. Since the ability in time to meet customer needs parallels the capacity to deploy market communication channels, brand loyalty can be connected to market communication from a competitive standpoint.

This is because the capabilities and resource bases of companies vary. Therefore, the impact of market communication channels on repeated purchases would be dependent on the company being able to deliver customer value from the SET perspective and whether or not the company has the resources and the ability to apply a correct mix of communication channels to sensitize it to its ability to offer customer value consistently.

AIDA Model

AIDA is one of the communication models used by companies to facilitate their product and service delivery. The model was an early attempt to explain how the potential audience of publicity could pass a number of steps and the main one was attention. It was developed in 1925 by Elie St. Elme Lewis who was a publicity and sales pioneer and thought it was effective if an ad contained the four qualities. Palmer (2005) assumes that the message is delivered in line with an easy model: promoting prospective attention; interest prospects for a product; prompt desire prospects, and action from the perspective.

The model suggests that customers react in cognitive, affective, and conative sequences to sales information. The sales officer first takes care of the salespeople, welcomes them, visits them or promotes them through sales. Then, reliable marketing, display or publicity created by this product by showing how the products can satisfy the desires of the customer. Finally, it is possible to use a single sales strategy to encourage purchases (Lamb et al, 2012).

The Hierarchy of Effects Theory (HET)

Brand awareness is defined as the extent to which a product or service can be recalled. If a person is unable to recall a product or service, it is impossible for him to buy it. This argument is in parallel to the premise of the Hierarchy of the Theory of Effects (HET). Lavidge & Steiner, founders of the brand HET, (1961), said buying a brand is the last of six steps in which customers decide to consume the brand.

The HET contends that being aware of the service or product is the foundation of a purchase. Madhavaram badrinarayanan and Mcdonald (2005) have opined that this kind of awareness might

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not necessarily be 'being aware about the specific service'. Rather, it could be as good as being aware of the service provider such as a bank.

For the HET, "knowledge" of a product is of greater significance than "awareness" of that product. The reason being that having knowledge of a product takes the individual closer to making a purchase; thus, knowledge of the product is associated with a deeper understanding of brand characteristics. It is on the basis of this knowledge or understanding that the customer begins to 'like' the service or brand after gaining knowledge of it.

The HET is probably most logical and important when 'preferences' come into play in its fourth stage. This stage describes the fact that customers compare the value of one brand to another because they understand the value of this brand. In a competitive situation, many more brands can be compared, and it is reasonable for the one known to give the best value to their customers.

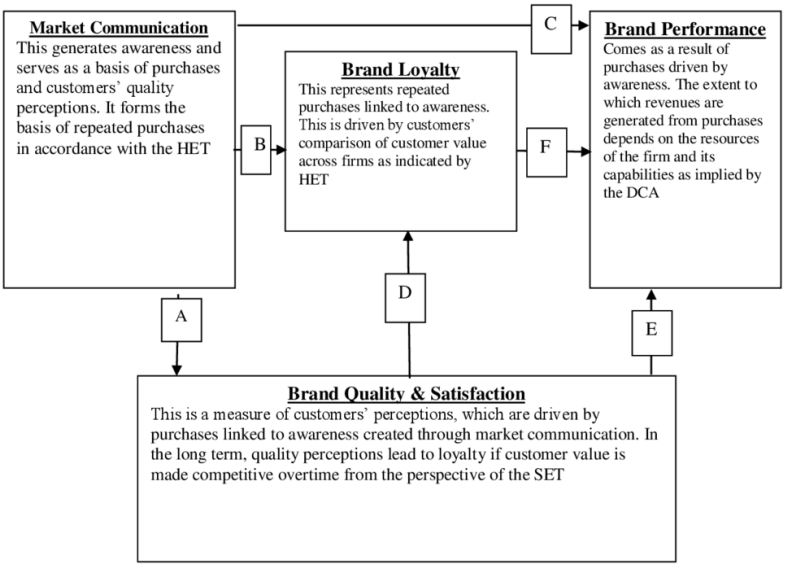
The final stage of the HET before purchases is "conviction". At this stage, individuals demonstrate the desire to purchase the service or product. They do so either by expressing willingness to acquire it if need be or sampling it. Sampling the service enables the individual to verify its value as communicated by the firm.

The last phase of HET before purchase is 'conviction.' At this point, people show the will to purchase the service or product. Either through their willingness to acquire it or by sampling it, if necessary. Sampling the service enables the person to verify their value as reported by the company.

Purchases are the primary ways customers make brand quality perceptions. The HET explains that awareness is the beginning of the process in which customers make a purchase. Market communication is the ultimate process used by financial services firms to generate brand awareness. If awareness is the basis of purchases and customers' brand quality perceptions, each communication channel used in the firm can be positively associated with brand quality as shown in Figure 1 So the potential linkage between a market communication channel and brand quality is founded on the argument of HET that awareness leads to purchases on which brand quality perceptions are made.

KEY: A-F represents the linkages or relationships among the variables.

Figure 2.The Theoretical Model



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Source: The Researcher's Construct adapted from the HET, SET, and DCA

It is necessary for a firm to ensure that its market communication channels are effective. For a communication channel to be effective, it must be suitable. Thus, the right channel must be used to communicate on the right subject. For instance, a TV advert is more suitable for visualising the physical attributes of a firm, which the SERQUAL model recognises as 'Tangibles'. This form of advertising can hardly demonstrate 'empathy', which is a dimension of the SERVQUAL model that emanates from customers' behaviour and emotional competencies. Hence, the strength of the relationship between a market communication channel and brand quality would be determined by whether or not this channel has been used to communicate the right subject. From this perspective, each channel would often make a different effect on each domain of brand quality.

The SET explanation by Saeed et al.'s (2013) suggests that customers would always like a positive value or customer value in their consumption of a service. The benefits from the service should therefore exceed the expenditure of customers on the service. The result of a customer's needs is satisfaction. The satisfaction of a customer's needs means that the service provider accounts for its positive value. Quality perceptions are arguably opinions that show a service provider's ability to meet customer needs. Brand quality can be linked with customer satisfaction from this perspective.

EMPIRICAL REVIEW

Several studies examined the effectiveness of different marketing techniques employed by financial service companies to improve customers' patronage and thus brand performance. Empirical research shows that advertising influences consumer selection decisions (Joshi & Hanssens, 2010; Akanbi & Adeyeye, 2011).

According to a research conducted by Ashkan (2016) on the role of selected MCM tools including advertisement in sales increase and promotion in Hamadan Province Insurance Companies. It was revealed that advertising has an impact on sales increase.

Additionally, Njawa (2015) conducted a research on the effects of advertising on organizational performance: Case study of Tigo telecommunication network. The results indicate that advertising significantly influences the performance of the organization.

Festus (2016) further studies on the impact of sales promotion on Guinness Ghana Breweries' organizational performance study showed that sales promotion had a positive and substantial relationship with performance.

Moreover, a study by Aliata et al. (2012) found that sales promotion and performance are statistically important links. They confirmed that promotion improved brand preference, thus improving brand performance and high income. Despite this, study showed no way to evidently improve sales promotion strategies.

Mohammadi and Sohrabi (2018) studied the Effect of Marketing Mix Elements on Customer Satisfaction with Mediating Role of Electronic Customer Relationship Management, using customer satisfaction as a dependent variable and Marketing Mix Elements as independent variable. The results confirm that the presence of the intermediary variable CRM has a positive and significant impact on customer satisfaction. Therefore marketing Mix elements do not have a direct influence on electronic stores customer satisfaction.

CONCEPTUAL FRAMEWORK

The marketing communication mix is a set of marketing tools or activities used to facilitate the delivery of products and/or services to customers. Drawing from all tools, it is argued that these tools share significant covariance. Thus each pair of these tools are correlated. To illustrate, a firm's advertising options may be deployed via the internet. In recent years, viral marketing campaigns have been made and fashioned in the form of advertising through the internet media. Particularly in Ghana, newspaper

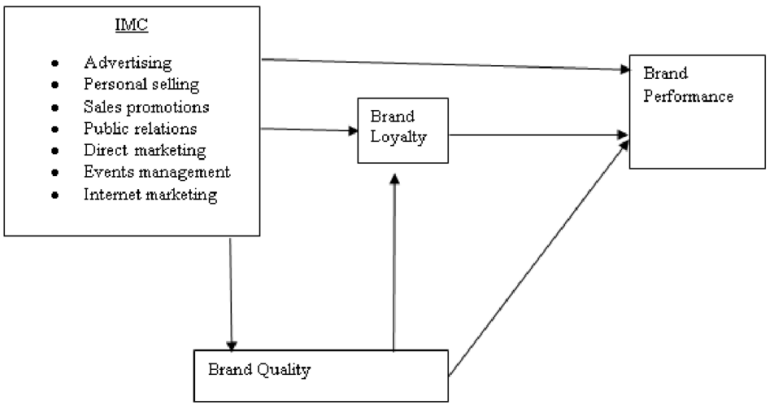
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advertising, which has become a common platform for many service firms (Okyere *et al.,* 2011), is channeled through the internet. On the basis of this argument, it could be said that internet marketing and advertising are correlated and therefore share significant covariance.

In addition, many firms have to advertise their events to achieve the desired impact. The newspaper, internet and billboard are common stages often used by Ghanaian firms in creating awareness on their events. So many researchers (Porcu *et al.,* 2012; Ismail *et al.,* 2012) have contended that without creating awareness on events using advertising, a firm's event management endeavor may prove futile. From this point of view, it is arguable that events marketing and advertising is correlated. The covariance of internet marketing and events marketing is justified by the fact that the promotion of events can be carried out through internet marketing.

Personal selling are events marketing platforms for many service firms (Idris *et al.,* 2012). Particularly in the insurance sector in Ghana, personal selling is associated with events and promotional selling (Okyere *et al.,* 2012). In this vein, events marketing, personal selling and sales promotion are seen as a composite tool for facilitating service delivery and customer patronage. By implication, personal selling, sales promotion and events marketing can share significant covariance. Therefore, these three marketing communication tools can correlate with advertising.

Figure 3. Conceptual Framework



Source: Frimpong 2015: A Conceptual Framework Developed from this Study to investigate into the Theoretical Relationship between MCM and BP

In direct marketing, firms give priority to communicating with customers without any third-party influence or barrier. In recent years, businesses, especially financial institutions, have used their websites and other internet platforms to engage in direct communication with customers (Al Khattab *et al.,* 2015). Moreover, events and promotional programs are direct marketing platforms. For instance, an event may be used as a way to disseminate information to customers and to take their views. Hence direct marketing is likely to make the strongest relationship with events marketing and sales promotion. This notwithstanding, direct marketing can co-vary with other marketing communication tools.

These seven tools of marketing communication drive the course of service delivery, enabling service firms to create awareness on products and to improve patronage. From the point of view of the communication model, these tools, depending on how they are applied, influence customers'

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perceptions about the firm. On the basis of these perceptions, the IMC construct can influence brand quality. Coupled with evidences from other studies (Idris *et al.,* 2012; Takalani, 2015) the IMC construct is expected to make a significant effect on brand quality (BQ) as indicated in Figure 1.

IMC tools, depending on how they are implemented, may impress customers to develop positive or favorable impressions about a business. These arguments explain the basis for the potential effect of integrated marketing communication tools on brand loyalty. Brand quality serves as a mediating variable in the relationship between the IMC construct and brand loyalty (BL).

Within the framework represented by Figure 1, the main dependent variable is brand performance, which is a measure of the performance of service firms surveyed. The basic predictor or independent variable is IMC, with brand quality and brand loyalty serving as mediating variables. Through the use of a Confirmatory Factor Analysis (CFA), the mediating effects of BQ and BL are tested in assessing the relationship between IMC and BP. The importance of using CFA in the analysis is expressed by the need to identify the correlation between pairs of the IMC tools. On the bases of these conceptions, the following primary alternative hypotheses are tested in this study:

H_{1} - Financial services sector in Ghana applied all the identified marketing communication mix
H2 - MCM tools have effects on brand quality
H_{3} – MCM tools have effects on brand loyalty
H_{4} – MCM tools have effect on brand performance
H_{5} – Brand quality mediates the relationship between each MCM tool and brand performance.
H6- Brand loyalty mediates the relationship between each MCM tool and brand performance.

RESEARCH METHODOLOGY

To a small extent, the exploratory research was employed in this study in the face of the main descriptive research chosen. The exploratory research was employed at the level of literature review and content analysis, where Creswell et al (2017) classify literature review and content analysis as types of exploratory research. Considering the main research design of this study which is descriptive research, it is worth saying that the quantitative research design was dominant in this study. This is because an examination of relationships in descriptive research is done in the context of hypothesis testing (Kichenham and Pfleeger, 2002; Creswell, 2008; Williams, 2011), where hypothesis testing is done in quantitative research (Kichenham and Pfleeger, 2002), while the qualitative content analysis was also used as a supporting research design.

The Population

The population of this study was employees in Ghanaian organisations where the IMC tools are known to be frequently used. Owing to how large it is, the target population was potentially heterogeneous. To ensure that a representative sample was reached, a total of 3,012 participants were in the target population with respect to all characteristics. Depending on which type is used, sampling is carried out in the face of assumptions and other considerations. Considering the nature of the target population, the simple random and stratified sampling techniques were the specific probabilistic sampling methods employed.

Sample Sizes by Strata

The sample size was 978. 792 of them were complete and returned. After taking out poorly completed questionnaires, a total of 757 questionnaires were analysed.

The sample size of each stratum was determined using the following formula of Krejcie and Morgan (1970):

$$s = X^2NP(N-1)/d^2(N-1) + X^2(1-P)$$

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s = required sample size
X^2 = the table value of chi-square for 1 degree of freedom at the desired confidence level (3.841)
P = Population proportion (assumed to be 5% since this will provide the maximum sample size)
d = the degree of accuracy expressed as a proportion (.05).

The names of all members of the list of each category were identified with numbers from 1 to the population size of the category. These numbers were entered into Ms Excel 2010 and transported to SPSS version 21 for each category. The random sampling procedure in SPSS was used to select the sample sizes.

Description of Data

In this study, data used involves nominal, ordinal and ratio data type. The third type of data used in this study is the interval data type, which include scores given to the main variables and constructs (i.e. IMC tools, brand quality, etc.). This type of interval data is also discrete. Data used in this study can also be classified into secondary and primary.

Variables and Measurement

In this study, four main constructs were measured, namely IMC tools or MCM, Brand Quality (BQ), Brand Loyalty (BL) and Brand Performance (BP). MCM was measured using 7 items of integrated marketing communication in accordance with Lekhanya (2015). Basic independent variable (IV) is MCM whereas the dependent variable is BP. BQ and BL are conceptually mediating variables. Moreover, measurement of these constructs was made based on a combination of a five-point Likert scale which has the following levels:

Strongly disagree (1); Disagree (2); Not sure (3); Agree (4); and Strongly agree (5).

MCM and BP were measured using items borrowed from the study of Lekhanya (2015). Brand Quality and BL were measured using the SERVQUAL model items and brand loyalty scales respectively. Evidently this study involved two major variables, namely latent variables and manifest variables. The latent variables include the 4 constructs identified earlier and the manifest variables are individual items of the constructs. A latent variable is that variable which cannot be directly measured (e.g. MCM, BQ, etc.) but is measured using a series of other variables called manifest variables (Creswell, 2008; Bebell *et al.* 2004). Meanwhile, variables used in this study are dependent (e.g. BQ, BL and BP), independent (i.e. MCM) and mediating variables (BQ and BL).

DATA ANALYSIS

Data analysis was commenced with data screening in which outliers and other anomalous data items were removed from data entered into the statistical software used. Confirmatory Factor Analysis (CFA) was used fit four measurement models in an effort to assess the validity and reliability of the measurement scales and to test all hypotheses concurrently, thereby maximining statistical validity and internal validity through the reduction of type 1 error.

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RESULT

Demographic Characteristics of Customers

Table 1. Shows the summary statistics of customers who participated in the study on the basis of gender, education service experience and firm category.

Variable	Level	Frequency(n)	Percent (%)
Gender	Male	521	69%
	Female	236	31%
	Total	757	100%
Education	Basic/secondary	189	25%
	Diploma	132	17%
	First degree	213	28%
	Master's degree	114	15%
	PhD or equivalent	109	14%
	Total	757	100%
Service experience	Less than 3 years	197	26%
	3-5 years	321	42%
	6-10 years	132	17%
	Above 10 years	107	14%
	Total	757	100%
Firm category	Bank	598	79%
	Non-bank	159	21%
	Total	757	100%

In Table 1, 69% (n = 521) of all customers who responded were males; and 31% (n = 236) of all customers who responded were females. Apparently, most participating customers were males. Moreover, 25% (n = 189) of all customers who participated had basic/secondary education; 17% (n = 132) of all customers who participated had a diploma; 28% (n = 213) of all customers who participated had a first degree; 15% (n = 114) of all customers who participated had a master's degree; and 14% (n = 109) of all customers who participated had PhD or an equivalent qualification. Therefore, the largest proportion of customers who responded were holders of a first degree. In addition, 26% (n = 197) of all responding customers had been baking with their current bank for less than 3 years; 42% (n = 321) of all employees had worked in their current company for between 3 and 5 years; 17% (n = 132) of all customers had been banking with their current bank for between 6 and 10 years; and 14% (n = 107) of all customers had worked been banking with their current bank for more than 10 years. Hence, the largest proportion of customers had banked with their current bank for between 3 and 5 years. Finally, 79% (n = 598) of all customers who responded were customers of banks; and 21% (n = 159) of all responding customers were customers of non-banks. Table 2 shows the results of the sampling adequacy test (conducted via Exploratory Factor Analysis) on all measurement scales applied. As the table depicts, the scales tested apply to marketing communication mix (MCM), brand quality, brand loyalty, and brand performance.

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Table 2. Sampling Adequacy Tests

Construct	Kaiser-Meyer-Olkin (KMO) measure of Sampling adequacy	Bartlett's test		
		Chi-square	df	P
MCM	0.872	1431.21	675	***
Brand quality	0.763	82.101	321	***
Brand loyalty	0.801	401.99	342	***
Brand performance	0.799	399.11	289	***

***p<.001

In Table 2, MCM accounts for a KMO value of 0.872, which is the largest value in the table. All the scales accounted for a KMO value that is greater than 0.5 as recommended in the literature (Morse, 2002). Moreover, every scale produced a significant Bartlett's test at 5% significance level as recommended in the literature (Morse, 2002; Drost, 2011). As a result, the sampling adequacy requirement of the scales were met.

Tables 3 and 4 show the results of the CFA used to assess the psychometric properties of the scales used. The CFA is a continuation of the EFA in which all items of each scale were retained on the basis of the *communality > 0.5* criterion. That is, no item was removed from the original scales because all items met the foregoing criterion.

Table 3. Psychometric Properties of Scales

Construct	Domain	APVE	CA	CR	AVE	MSV	ASV
Marketing communication tools	Advertising	23.11	0.922	0.768	0.591	0.112	0.051
	Personal Selling	14.91	0.921	0.768	0.590	0.112	0.051
	Sales promotion	13.92	0.901	0.751	0.578	0.110	0.049
	Public relations	11.02	0.932	0.777	0.597	0.114	0.051
	Direct marketing	10.71	0.927	0.773	0.594	0.113	0.051
	Events management	9.22	0.901	0.751	0.578	0.110	0.049
	Internet marketing	8.81	0.900	0.750	0.577	0.110	0.049
Brand quality	Factor 1	22.91	0.955	0.796	0.612	0.116	0.052
	Factor 2	18.82	0.899	0.749	0.576	0.110	0.049
	Factor 3	16.21	0.809	0.674	0.519	0.099	0.044
	Factor 4	9.33	0.904	0.753	0.579	0.110	0.050
	Factor 5	7.19	0.911	0.759	0.584	0.111	0.050
Brand loyalty	Factor 1	43.99	0.932	0.777	0.597	0.114	0.051
	Factor 2	24.11	0.905	0.754	0.580	0.110	0.050
Brand Performance	Factor 1	39.01	0.912	0.760	0.585	0.111	0.050
	Factor 2	23.12	0.905	0.754	0.580	0.110	0.050
	Factor 3	11.98	0.954	0.795	0.612	0.116	0.052

Note: APVE = Average Percent of Variance Extracted; CA = Cronbach's alpha; CR = composite reliability; AVE = Average Variance Estimate; MSV = Mean Shared Squared Variance; ASV = Average Shared Squared Variance

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With reference to Table 3, the Average Variance Estimate (AVE), Maximum Shared Squared Variance (MSV), and Average Shared Squared Variance (AVS) are statistics used to evaluate the validity of the scales. Cronbach alpha (CA) also assesses the reliability of the measurement scales. It can be seen that each factor of the scales has a CA greater than or equal to 0.7 as recommended elsewhere (Drost, 2011; Asiamah et al., 2018). In addition, the AVE ^{3} 0.5, CR > AVE, MSV < AVE, and ASV < AVE criteria applied in the literature (Kelava, 2016; Asiamah et al., 2018) are met. Moreover, each construct produced an average percent variance extracted (APVE) of at least 70%, which according to Kelava (2016) indicates a good reliability and fit and fit of the scales. According to Asiamah and colleagues, the following conditions satisfy validity and validity:

i. reliability – CA/CR > 0.7;
ii. convergent validity – CR > AVE, AVE > 0.5; and
iii. Discriminant validity – MSV < AVE, ASV < AVE.

The above being the case, validity and reliability of the scales are confirmed or supported by the data. Table 4 shows CFA fit statistics for 5 models fitted.

Table 4. CFA Fit Statistics for the Measurement and Structural Models

Model	Chi-square	p-value	RMSEA	TLI	GFI	AGFI
1 (Measurement MCM)	1321.432	0.002	0.321	0.803	0.714	0.806
2(Measurement BQ)	184.09	0.041	0.072	0.899	0.867	0.901
3(Measurement BL)	187.09	0.000	0.401	0.798	0.778	0.799
4(Measurement BP)	109.11	0.000	0.195	0.900	0.898	0.905
5(Structural)	99.02	0.321	0.031	0.988	0.979	0.999

Note: RMSEA = Random Mean Square Error of Approximation; TLI = Tucker-Lewis Index; GFI = goodness of fit; AGFI = adjusted goodness of fit index.

In Table 4, the first four models are the measurement models of MCM, brand quality, brand loyalty, and brand performance respectively. The fifth model is the structural model through which all relationships and hypotheses were tested simultaneously. In Table 4.5, the first four models did not produce a good fit because they failed to meet the following criteria: p-value > 0.05; RMSEA < .08; TLI > .90; GFI > .80; and AGFI > .90.

Table 5. Univariate and Multivariate Normality Assessment

Variable	z-score	Univariate normality	
		Shapiro-Wilk's	p
Advertising	2.34	0.342	0.091
Personal Selling	2.44	0.943	0.543
Sales promotion	-2.43	0.423	0.132
Public relations	2.01	0.522	0.201
Direct marketing	2.91	0.432	0.138
Events management	2.03	0.522	0.202
Internet marketing	-2.01	0.401	0.109

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Table 5 continued

Variable	z-score	Univariate normality	
		Shapiro-Wilk's	p
Brand quality	-2.33	0.501	0.199
Brand loyalty	2.01	0.843	0.401
Brand performance	2.22	0.899	0.422

Note: each p2 value of the Mahalanobis distance test in the main structural CFA model satisfies the condition p>0.05, with the smallest of the values being 0.195.

The z-score in Table 5 is an indicator of the distribution of data associated with a variable. The z-score signifies a normal distribution when it is falls within the range -3 to +3 on a real number line (Sawilowsky, 2005). It can be seen in the table that each score produced in the table falls within this range, which confirms a high likelihood that each variable's data was normally distributed. Sometimes however, a variable's z-score can fall within the above range but its data would not be normal enough to support a parametric test. So, the only way to confirm univariate normality is to conduct the Shapiro-Wilk's test for a sample size of 2000 or less. Data is normally distributed in this assessment when each variable accounts for a p-value greater than 0.05, $p > .05$ (Asiamah & Mensah, 2017). In Table 5, this criterion is met by each variable. Multivariate normality is a requirement for testing complex relationships using CFA, and this assumption is tested using the structural model through which all relationships are tested (Asiamah, 2017). Multivariate normality was therefore assessed through the fifth CFA model, which produced p2 values of the Mahalanobis distance test that satisfy the condition $p>0.05$, with the smallest of the values being 0.195. That being so, multivariate normality was met, for which a basis is established for conducting CFA.

DISCUSSION AND FINDINGS

With respect to the framework established, brand quality and loyalty are mediators of the effect of the marketing communication tools or Marketing Communication Mix (MCM) on brand performance. Since this mediation effect is partial, the current study implies that the communication tools can significantly influence brand performance without the mediation of brand quality and brand performance. In practice, this suggests that brand performance can be better driven by gearing market communication towards achieving brand quality and loyalty. It also suggests that brand quality and loyalty are not a necessary route towards the realisation of brand performance goals. This evidence is supported by previous studies (Opoku et al., 2014; Opuni et al., 2014) which have found that market communication makes a direct positive effect on brand performance even if brand quality and loyalty are not achieved. That is, a customer can patronise a service not because he or she is satisfied by this service but is merely aware of it as a result of market communication.

The result of this study relating to the effect of the MCM tools on brand quality is not at variance with evidences from previous studies (Ekhlassi et al., 2012; Aghaei et al., 2014). Moreover, Akerlund (2004), Manisha (2012) and some studies buttress the significant effect of the hypothesised MCM tools on brand/brand quality in both developed and developing country contexts. This study's result is however at a minor variance with the study of Opoku et al. (2014) and Frimpong (2014). The variation has to do with the fact that Frimpong (2014) and Opoku et al. (2014) used brand quality instead of brand quality as the dependent variable to the MCM tools.

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Result shows that brand performance is positively correlated with each of the seven market communication programs at 1% significance level. Advertising is the strongest predictor of brand performance among the communication tools. Increasing the level of market communication in terms of advertising and other communication tools (that significantly predict brand performance) can increase the performance of the financial services providers. This has a bearing with the stance of Opoku et al. (2014), who posited that sales promotions and personal selling are MCM tools directly used in generating sales/revenues and therefore make a stronger effect on brand performance.

Generally, the findings suggest that, the effects of advertising, personal selling and events management on brand performance are fully mediated by brand quality. However, brand quality fails to mediate the effect of internet marketing on brand performance though internet marketing makes a direct effect on brand quality. The study reveals that, brand quality makes a positive effect on brand performance and loyalty at 1% significance level. This implies that increasing brand quality can enhance brand loyalty and brand performance.

The partial mediating role of brand quality in the relationship between the MCM tools and brand performance is also corroborated to some extent in marketing literature. Firstly, the study of Danquah (2014) supports the positive mediation of brand quality on business performance in terms of ROI in the banking sector, where brand quality is a measure of brand quality. However, the mediating role of brand quality in the study of Danquah (2014) is not traced to MCM. But Rawal (2013) and Danquah (2014) are of the opinion that brand quality, regardless of whatever determines it, can mediate business performance and growth.

Finally, brand loyalty fails to mediate the effect of sales promotion and internet marketing on brand performance, though internet marketing makes a direct effect on it. By inference, the MCM tools are relevant to brand performance but their effect on brand performance is maximised when loyalty is achieved. So, it is always a better option to achieve brand loyalty as a way of achieving brand performance.

It can further be seen that, brand loyalty makes a positive effect on brand performance at 1% significance level, which is an indication that increasing brand loyalty can increase brand performance.

The mediation of MCM and business performance in this study is supported by the studies of Frimpong (2014) and Opoku et al. (2014). Impressively, these studies have been conducted based on data employed from the banking sector of Ghana, a situation that confirms the validity of this study's results and the consistency of its findings.

IMPLICATION

In Africa, few studies have been conducted to reflect the role of MCM in marketing (Opoku et al., 2014). However, quite a number of studies have been focused on individual tools or a section of the MCM tools in an African perspective (Saeed et al., 2013; Opoku et al., 2014). This study is therefore among few research indicating the level of application of the MCM tools from an African point of view.

The findings infer the need for the promotional planners to know the roles that advertising and other marketing communication mix elements will play in the overall marketing plan. Marketing communication planners must focus on information in the marketing plan that is relevant to the promotional strategy. This study unveils the need for the financial institutions to prioritise some MCM tools in order to minimise cost.

Being the fastest growing sector in the economy, the service sector has gained increased attention to prioritise some MCM tools in order to minimise cost. The research findings suggest that financial services must make their internal marketing orientation effective if the impact of the various marketing communication mix tools on brand performance should be maximised. This orientation can be improved and maximised in the firms by creating a balance between investment in people and strategies, and technologies and raw materials.

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The result relating to the effect of the MCM tools on brand quality is not at variance with evidences from previous studies (Ekhlassi et al., 2012; Aghaei et al., 2014). Moreover, Akerlund (2004), Manisha (2012) and some studies buttress the significant effect of the hypothesised MCM tools on brand/ brand quality in both developed and developing country contexts. In a Ghanaian context, Opoku et al. (2014) found that each of these MCM tools influences brand quality in the banking sector both at the level of correlation coefficients and regression analysis.

Theoretically, the study validates and uniquely integrates the theory of reasoned action (TRA) social exchange theory (SET) to explain the positive effect of MCM on BG and BL. In this vain, quality perception and repeated purchases are seen as facets of customer behaviour (from the point of view of the theory of reason action) resulting from brand awareness which conveys knowledge about the value or benefit.

CONCLUSION

These significant results suggest that brand quality makes a positive effect on brand performance and loyalty at 1% significance level. This outcome confirms that increasing brand quality can enhance brand loyalty and brand performance. Apparently, all hypotheses (about direct effects) of the study have been established. Brand loyalty increases if each of the seven marketing communication mix tools increases. In this case, the strongest predictor of brand loyalty among the seven communication programs is public relations. By implication, the marketing communication tools are relevant to brand performance but their effect on brand performance is maximised when brand quality and loyalty are achieved.

The effect made by each of the MCM tools on brand performance is powered by internal marketing orientation. This is in view of the evidence that the effect of each of the MCM tools on brand performance significantly reduces when internal marketing orientation is controlled for. This means that the effect made on brand performance by each of the MCM tools and the MCM construct is significantly contributed by internal marketing orientation. Research findings imply that management of these financial firms should see BQ attainment as a necessary way to maximize BL through MC.

Finally, the study reveals a theoretical extension of integrated model by treating the Resource-Based view (RBV) and Dynamic Capability Approach (DCA) as explanations of the antecedents of effective market communication and brand performance.

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Effect of Social Media Personalization on Brand Strength:
A Study of the Brand Advertised on Facebook

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ABSTRACT

Personalization is now used at every corner and position of social media sites. Using personalization in marketing communication is an effective communication strategy for many businesses to improve customer satisfaction and increase sales in a reduced advertising budget. This research is an attempt to measure the impact of personalized advertisement on customers' social media account. This empirical research develops and tests a conceptual model using the structural modeling technique of the data collected from Facebook users, who have seen personalized advertisements of the preferred brand on their Facebook account. Results suggest that personalized advertisement of the brand in customers' social media accounts is positively related to brand experience. Brand experience is also positively related to brand equity and mediates the relationship between perceived personalization and customer-based brand equity. Both brand experience and brand equity collectively enhance the brand strength of the brand advertised on Facebook.

KEYWORDS

Brand Equity, Brand Experience, Facebook Advertisement, Personalization, Social Media Advertisement

INTRODUCTION

The progressive growth of technology, digital devices, and communication channels has provided businesses so many opportunities. By the strategical utilization of changing, technology firms can provide more value to their customers, increase experiences, satisfaction, and loyalty (Villarejo-Ramos & Sanchez-Franco, 2005). Many businesses use social media sites for personalized marketing to create a more tailored customer brand experience and encourage impulse buying by offering products for individual needs aligned with their tastes and preferences. Personalized advertisement is a technique to provide endless customer experience and an effective way to strengthen brand trust (Chinomona, 2013).

Social media has revolutionized marketing communication; it is a significant and rising area of research that attracts both practitioners and academicians. Personalized advertisements on social media sites with Facebook have gotten much consideration over the previous studies. Despite broad exploration on the use of individual personal information in online marketing activities (Tran et

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al., 2020), recent studies show that more understanding is required to know how personalized advertisements are well aligned with customer interest. This includes enhance customer experience and limit advertisement evasion (Tran et al., 2020), (Boerman et al., 2017), (Stojanovic & Andrreu, 2017). There is also an absence of studies in the Indian context which investigates the outcome of social media personalization (Yadav and Rahman, 2018). Providing personalization in e-shopping is the most popular marketing trend, and social media consistently stays in vogue because of its highly interactive nature (Shanahan, Tran, & Taylor, 2018), (Blasco-Lorena et al., 2016). So, it is necessary to study personalization in the social media context (Growth from Knowledge, 2019), (Khan et al.,2017).

Customers in social media no longer behave as passive recipients of entire marketing activities like conventional media; instead, they assume an active user in creating and sharing brand-related content. This content-sharing opportunity motivates social media users to share information online to get the word out about the characteristics, experiences, and reasons they support the brand (Schivinski and Dabrowski,2015). Facebook advertising is being used in this work since it is a popular social media advertising channel. Facebook surpassed 1 billion registered accounts and currently has 2.45 billion monthly and 1.49 billion daily active users (Aboulhosn,2020). Facebook gives a chance to users to make their accounts where they can post their personal information like pictures, educational & job history, interests, and hobbies. Each user spends nearly 20 mins per visit, making Facebook quite popular among users and companies uses social media for communication purposes (Dehghani and Tumer,2015), (Yadav et al. 2017b). According to the survey results, millions of retail websites have integrated with Facebook, spending on digital marketing tools and platforms has increased approximately two and half times compared to the last decade (Survey CMO, 2019, (Aboulhosn,2020). Social media increases the visibility of the brand through its high accessibility and reachability as compared to traditional media like television, indoor, and outdoor display (Romaniuk, 2013)

Using personalization in marketing communication is an effective communication strategy for many firms to increase customer interest and sales in a reduced advertising budget (Ansari & Mela, 2003). Previous studies on personalization have shown that addressing people by their names increases response rate and helping behaviour both in an online and offline environment (Maslowska et al., 2016), (Fatkin,2017*)*. (Heerwegh, 2005) confirmed that personalizing messages in traditional media, for example, e-mails make individuals more special, unique, and valued; consequently, they are bound to conform to the request. Personalization effects are defined in two ways., first, is perceived personalization which occurs when a sender of a message intentionally modifies and delivers the message to recipients that were design based on previously collected data from them. Moreover, the recipient perceives that a particular message fits their need, whereas the second is actual personalization. The recipients receive the messages from the sender looks very general common for all, and the customers may or may not show interest in that message (Li, 2016). Today, the firms use personalized advertisements in social media based on consumer online behaviour, recent searches, and purchase histories on the internet (Li, 2016). These advertisements on social media account create interest and curiosity in consumers. The attractive images and video of the brand are advertised in consumer's social media account to increase their experiences and interest in shopping online (Ho & Bodoff, 2014). Some research work on personalization also says that personalized advertisement reduces customers, ad avoidance raises brand awareness, increases brand credibility and sales (Baek and Morimoto, 2012), (Oberoi, Patel, & Haon, 2017).

Several researchers have given their views on the effectiveness of personalized advertisements on traditional media are well documented such as direct mail (Baek & Morimoto, 2012), telemarketing (Yu & Cude, 2009), mobile messaging (Xu, 2006), and website ad personalization. However, the effect of personalization (Awad & Krishnan, 2010) on social media accounts like Facebook and Twitter is very few may be due to its recent disruptiveness, and emergence is still under observation (Tran et al., 2020), (Khan et al.,2017), (Shanahan, Tran, & Taylor, 2018). This research attempts to measure the outcome of a personalized advertisement on a customer's social media account (Oberoi, Patel, & Haon, 2017). In this study, the researcher tries to develop and test a conceptual model measuring

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the effect of the personalized ad on consumer-based brand equity, which further develops the brand strength of the band advertised on customer social media account (Facebook). This study also examines the mediation effect of consumer brand experience, which has not been discussed in previous studies. The study considers the three forces of brand equity: brand awareness, perceived quality, and perceived value, which play a major role in developing brand strength. The model proposed in the study will be helpful for the firms dealing with online business using social media environment and want to strengthen their brands with having resource constraints. This analysis will guide to find out the differences in the use of personalization in social media. In this study, Facebook is being used as a social media platform because it provides the most economical and targeted form of advertising to small businesses.

The rest of the paper discussed the theoretical background of the research followed by hypotheses development, methodology verifies both measurement and structural model and at the end, the study concluded with theoretical and managerial implications.

THEORETICAL BACKGROUND

Perceived Personalization

Personalization considers as a very important psychological component in developing social media marketing strategies. Personalized advertisement delivered to customers depends on their individual preferences. That exclusive preference is captured by many firms to create personalized offers and to retain their customers (Montgomery & Smith, 2009). Personalization is an effective marketing strategy in terms of identity and delivering marketing mix to the individual based on their preferences (Khan, Lewis, & Singh, 2009). Previous studies on personalization (Li, 2016) suggested that personalization is of two types: pull personalization generally deliver personalized service when customer request it and can customize in terms of pre-determined or pre-specified product/service characteristics, on the contrary, push personalization provide all the information to the customer of their preferred based on their past activities, past preferences, and purchases and recommend the product of services related to their past searches (Maslowska et al., 2016). Perceived personalization has persuasive power. It increases the user's attention and motivates them towards the content of the message; it exerts persuasive effects by being recognized as "for me" and evoking the feeling of "me-ness" (Kalyanaraman & Sundar, 2006), (De Keyzer et al., 2015).

Personalization is the opportunity for firms to lead in the current competitive online environment. Businesses have acquired a massive amount of data related to consumer shopping behaviour, web browsing, consumer preferences, consumer buying habits, etc (Lavie, Sela, Oppenheim, Inbar, & Meyer, 2010). Personalized advertisement based on customer demographic data, search patterns, and past online buying helps customers get products they may need. Hence, personalized advertisement is perceived as very significant and trustworthy positively influences customer interest for the brand (Baek and Morimoto, 2012). The potency of personalized content depends on its ability to influence consumer behaviour, such as attention, intelligence, attitude towards the advertisement, and behavioral intention. Personalized advertising is a powerful tool that improves advertising relevance for users and increases ROI for advertisers. It provides an improved experience for users and advertisers. However, the firm should comply with the ethical guidelines as per the culture and the country. When dealing with personal information firms have to be clear about the sensitivity of the information and the ethical issues. In Asian culture, people are especially sensitive about sharing data on sex, health, and finances. People are less comfortable with other people knowing it. In India collecting information from people under 13 is not considered ethical. Indian is a culturally diversified country and the advertiser has to respect religious sentiments as people are very sensitive about it. Some content is legally restricted in the country it should not be a part of customized advertising. In the country relation hardships like divorce and the customized ads targeted to that are also discouraged. Thereby

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personalized advertisement should follow the personal, religious and cultural boundaries to provide a satisfying customer experience.

In their research (Dehghani & Tumer, 2015) suggested that advertising on social media, especially on Facebook, set a new platform for advertising and promotion; it provides a more personal and intimate connection with users, which ultimately enhances the brand image and brand equity. This study well defined personalized ads posted by Online retailers on Facebook based on customer previous activities and search behaviour on the internet. Personalization benefits both the firm and the customer as both can directly interact with each other and share plenty of personal information. Marketers effectively utilize those details to create customized offers for them and advertising to a great extent that increases brand awareness, customer satisfaction, and loyalty (Logan, Bright, & Gangradharbatla, 2012). (Stojanovic, Andreu, & Curras, 2017) Used multidimensional brand equity approach to analyze social media communication influences on brand image and loyalty. Previous Personalization studies generally show both positive and negative outcomes, adverse in terms because some users did not respond to the advertisement or avoid the same responses. Some Facebook users perceive the ad as credible and personally made for them and have a positive thought and favorable attitude towards it. However, some Facebook users negatively respond to advertising they see and fail to care for their personal preferences (White, Zahay, Thorjorson, & Shavitt, 2008). Marketers need to understand what personalized means for a customer and perceive the personalization process (Tran & Trang, 2017). Here, one challenge for marketers always comes to an understanding of what an individual perceives as personalized (Tam & Ho, 2006). In the study of (Shanahan, Tran, & Taylor, 2018), we have seen the impact of social media personalization on enhancing brand engagement and loyalty. Recent advances in technology personalization practices implemented in the online environment allow the marketer to extend their customer services. It is integrated and pops up to every corner of the website. Finding a relevant Place for personalized advertisement on the website where it creates maximum visibility and attracts customer attention is also the biggest challenge for the marketers (Van den Broeck et al., 2020), (Tran, 2017)

The S–O–R Model

The S–O–R model (Mehrabian and Russell, 1974) is used as a theoretical framework to justify the conceptual framework developed in the present research study. The S-O-R model follows that specific stimuli or environment cues (personalized advertisement) create consumers' experiential states both affective and cognitive (brand experience). These emotional experiential states produce behavioral responses (brand equity and brand strength) (Donovanand Rossiter, 1982). In the online retailing context, the stimuli relate to the characteristics of the online retail environment (Eroglu et al., 2003). The customer's emotional and cognitive process determines the inner states, bounding their experiences, knowledge, and evaluations (Iglesia et al.,2011), (Jiang et al., 2011). The behavioral responses or consequences in the model represent customer buying behavior and customer loyalty (Sautter et al., 2004). Many previous studies adopted this S–O–R model (Roy, et al.,2016), (Koo and Ju,2010), (Shim et al. 2001) to know the influence of atmospheric cues on customer's internal affective and cognitive experiential states and subsequent online purchase intention. Our conceptual model consists of three types of variable behavior antecedent variables, OCE component, and outcome variables. Furthermore, the structured technique adopted by the S–O–R model to test the effect of personalized ads as environmental stimuli on customer brand experience to measure consumer behavior (brand equity and brand strength) as a response.

CONCEPTUAL FRAMEWORK AND HYPOTHESES DEVELOPMENT

The Relationship Between Perceived Personalization and Brand Experience

The marketer uses Personalized ads as an effective marketing communication tool to reach their customers and provide them positive brand experience. Every interaction between the customer and

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the brand leaves some impression, either positive or negative (Brakus, Schmitt, & Zarantonello, 2009). Customers experience the brand at every moment of contact they have with the brand. A firm should provide a seamless experience across the customer's entire journey of brand touchpoints along with pre, during, and post-purchase situations (Homburg, Danijel, & Kuehnl, 2015). Marketing Communication is pre-purchase brand stimuli anticipated through advertising and the level of value they receive from individual treatment (Alloza, 2008). Consumer experience with the brand is subjective, internal consumer responses by brand-related stimuli such as product design, identity, packaging, marketing communication, distribution, environment, etc. (Ramaseshan & Stein, 2014). (Brakus, Schmitt, and Zarantonello, 2009) conceptualized and empirically established that brand experience is a multidimensional construct consisting of four dimensions— sensory, affective, behavioral, and cognitive.

Sensory brand experience appeals to all the five senses (sight, touch, sound, smell, and taste), affective experience appeals to the sentiments and innermost feelings of a consumer, and Intellectual experience stimulates consumers analytical and imaginative thought that engage consumers creatively (Brakus, Schmitt, & Zarantonello, 2009). Companies use personalized advertising by using age, gender, location, past search behaviour, and buying history of consumers to create their interest in a brand and provide them positive brand experience (Baek and Morimoto, 2012). The customer feels happy to see the personalized advertisement of the brand they love and searches a lot on the internet. Social media provides a unique and focused environment for marketers to perform various marketing activities. Online marketers can more effectively approach and attract their target audience by using personalization activities on social media. An online marketer can give value to their customers by treating personally individual users and their requirements, reducing their efforts to search for products, and feeling them a special and valued customer. Companies are now boosting their advertising budget to customize the offer to their customers and give them a good brand experience. The personalized ad should be interesting and beautifully designed; it also considers customers' personal preferences. The more personalized consumers perceive an ad, the more they feel a good experience with the brand.

H1: Consumer Perceived Personalized ad has a positive impact on their brand experience.

The Relationship Between Brand Experience and Consumer Based Brand Equity

Experiential communication is a medium to pass a brand value to their target customers. It delivers required and relevant information and gives effective pleasure and attachment of a brand to consumers (Schmitt B., 2003). Research on experience suggested that good experience always determines brand choices and can create brand Equity (Blasco & Velazquez, 2017). (Keller K., 1993) defines brand equity as consumer reaction to the brand's marketing, focusing on the two components of brand knowledge: brand awareness and image. The study of (Aaker D., 1996) provides the most accepted brand equity or brand value phenomena. A set of assets and liabilities linked to the brand. It provides an opportunity for online marketers to directly interact with their customers, strengthen their communication, and provide extreme values to their customers. Social media provide prompt and regular, instant value and increase brand equity (Yadav and Rahman, 2018). Three forces of brand equity brand awareness, value, and quality are used to determine the effectiveness of personalization on social media.

The construct of brand awareness comes from a consumer's knowledge about the brand and "the ability to recognize/identify or recall a brand to its product category" (Yoo, Donthu, & N, 2000). (Aaker D., 1996) describe "brand awareness as the strength of brand presence in the consumer minds". Personalization is helpful for the firm if it successfully provides knowledge to customers about all the customized offers related to their preferred brand. Social media provides a rich platform for advertisers and customers to make a direct personal connection. Both can share a vast amount of information to enhance knowledge. The marketer uses this knowledge to make a more personalized marketing mix for customers, and customers receive this knowledge to more study the brand. Customers having a

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good experience with a brand can easily identify and become more familiar (Iglesia, Singh, & Batista-Foguet, 2011). Personalization increases customer's brand experience if the brand delivers the right message to the right customers at the right time. Experience leaves an impression in consumers' minds, which helps them in taking further brand-related decisions (Iglesia et al.,2011). Brands provide personalized services to their customers, increase their experiences, and leaves an impression, which keeps reminding them about the brand (Hutter, Hautz, et al., 2013). Brands post personalized Advertisements on social media with colorful images, well-designed videos, meaningful content that creates a sensory impression and develops a positive brand image. These visual stimuli create brand identity increases customer awareness and knowledge about the brand. The more memorable experiences provided by the brand, the more customers get aware of the brand.

H2a. Customer Brand Experience has a positive impact on brand awareness.

The researcher also reported that there is a relationship between advertising spend and perceived quality. Consumers generally use different communication tools like advertising to judge or measure the quality of the product. They perceived the brand which has high investment should be high in quality. Therefore, the consumer perceives the highly advertised brands should be high in Quality (Villarejo-Ramos & Sanchez-Franco, 2005).

Similarly, consumers judge the brand's quality (Shanahan et al.,2018) through the effectiveness and efficiency of the personalized ad. Customers generally think that advertisements in social media quickly develop a strong connection with consumers (Baird & Parasnis, 2011). It creates a positive image that provides a good experience and engages the customer's mind. User-generated content in social media like a product review, comments, suggestions, and feedback is also an important information source, enhancing consumer attraction and interest towards the brand (Stojanovic, Andreu, & Curras, 2017). The brands are trying to associate with their customers through personalized advertisement. The amount of personalization increases customer's involvement and attachment with the brand, which helps determine the quality and reliability of the brand (Shanahan, Tran, & Taylor, 2018). Personalized advertisement on users Facebook account always try to increase the customer brand experience, customer highly connected and engaged with the brand and perceived it as a quality brand (Shanahan, Tran, & Taylor, 2018).

H2b. Customer Brand Experience has a positive and impact on Perceived Quality.

In general, the term consumer perceived value (CPV) is the trade-off between the benefits consumer have from the product or services and the sacrifice made in terms of cost, effort, time paid, etc. (Zeithamal, 1988), (Dodds & Monroe, 1985). It assumes that consumers got the benefit when personalized advertisements treated them as unique and special customers (Brakus, Schmitt, & Zarantonello, 2009). Brands look valuable for customers if it provides customized offers that fit their needs, reduces efforts to search relevant offers, and saves time making a purchase decision. The researcher suggests that customer perceived value can be conceptualized along four dimensions: financial aspects such as price, discount, and value for money) (Holbrook, 2006). Functional value (quality and uniqueness) means that offers should provide functional benefits like high performance, social value (such as social recognition, status, and prestige), finally the individual value (self-assessment and self-recognition/identity). The consumer should be treated with respect and feel valuable (Verhoef et al., 2009). Brands advertised on social media try to make a personal connection with the brand by providing customize offers that increase customer experience with a brand which helps to measure the value brand provides.

H2c.The Customer Brand Experience has a positive impact on Perceived Brand Value.

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The relationship Between Brand Experience And Brand Strength.

The most acceptable customer value is positively related to customer satisfaction, trust, and commitment to the brand (Bakanauskas & Jakutis, 2010). Individual commitment, trust, and satisfaction towards the brand are responsible for strengthening consumers' relationship with the brand, which further increases brand loyalty. The set of associations, affection, and love of consumers for a brand are obvious to build the strength of the brands (Srivastava & Shocker, 1991). Brand strength can be calculated by how many consumers will buy the brand after seeing the personal advertisement of the brand on their social media account and recommend it to their friends. Firms use advertisements of their brands on consumer's social media accounts by using data analytics. Companies easily get all the information about customers, their preferred brand, previous purchase, and past search results. After getting all the track records of customer internet activities, they promote their brand on their personal social media account. Seeing the advertisement of their most search brand on the internet in their social media account will give the customer pleasure and produce a good experience with a brand. Maximum people have their accounts on social media, so personalized advertisement on a customer's social media account is a good branding strategy to promote the product and develop a strong connection between the customer and the brand. Firms now using modern marketing communication methods and techniques for building strong brands (Keller, 2009). Brand strength can include affective, cognitive, and behavioral components. In detail cognitive component addresses individual attention towards the evaluation of the brand; the affective component measures the personal feelings and sentiment towards the brand (Wiedmann, Hennings, Schmidt, & Wijstefeld, 2011), while the behavior component refers to the satisfaction, loyalty, repurchase intention towards the brand. A customer's positive experience with a brand creates a good impression on their sight and helps in building brand strength.

H3: The Customer Brand experience has a positive impact on brand strength.

The Relationship Between Customer-Based Brand Equity And Brand Strength:

Brand equity is always considered an important tool to measure brand value. Ideally, as the relationship between the customer and the brand increases, consumer experience and satisfaction with the brand also increase, ultimately driving consumer purchase intention (Beig & Khan, 2018). Band equity can be evaluated in two different ways; firm-based focuses on the value brand provided to the firm and customer-based brand equity is considered the valued customer receives from the brand (Simon & Sullivan, 1993). Firms use social media communication tools to get deeper interaction with their customers. Now in the modern communication environment, firms get opportunities to easily connect with loyal customers, share knowledge and influence their perception about products and services provided. So, the intensity of brand interaction on social media influences the customer-brand relationship, which helps consumers make brand strength (Wiedmann et al.,2017). Advertising encourages customers to become more familiar and loyal to the brand (Dehghani & Tumer, 2015). Previous studies found that the amount of spending on advertising positively related to brand loyalty with moderating role play by brand image, perceived quality, and customer satisfaction (Ha, Swinder Janda, & Muthaly, 2011). Customers perceive advertising as a standard tool to judge the quality of the product. A customer's positive evaluation of the brand's quality influences their purchase intention (Stojanovic, Andreu, & Curras, 2017). A customer generally makes the purchase decision based on what they received and what is given (Zeithamal, 1988), and why do they select to purchase and reject the particular product? Consumers choose those products from which they receive certain benefits and have some value (Dodds & Monroe, 1985).

H4a: Brand awareness has a positive impact on brand strength
H4b: Perceived brand quality has a positive impact on brand strength
H4c: Brand value has a positive impact on brand strength

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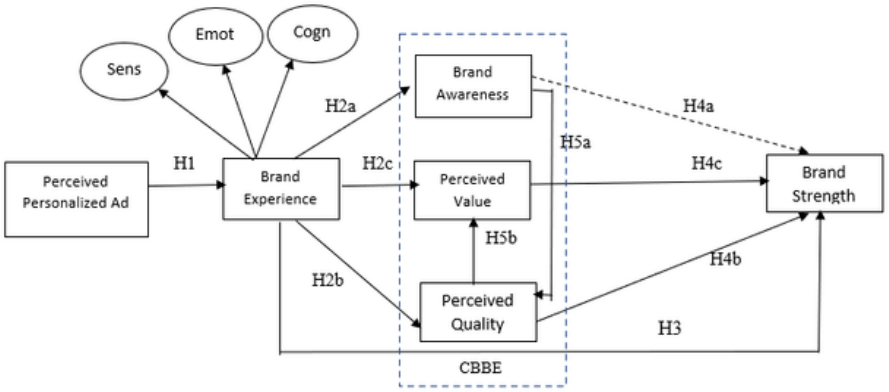
The Relationship Between the CBBE Dimension:

This research represents the standard hierarchy of model to find the effective order among the dimension of consumer-based brand equity. In this research, three practical components of brand equity are used brand awareness, perceived quality, and value (Aaker D., 1991). Brand equity is developed by increasing brand awareness in the consumer's mind. Such memories, retained in the consumer's mind, help to determine the quality of the brand. Social media provides the direct interaction of the consumer with a firm which helps to increase the knowledge and determine the quality of the brand (Aakar, Yoo, Donthu, & Lee, 2000). The brand is perceived as high quality, high performance at minimum cost gives some value to the customer. Thus, the following hypothesis is designed:

H5a: Brand Awareness Positively Influences the Perceived Quality
H5b: Perceived Quality Positively Influences the Perceived Value

Based on the above relationships between constructs identified in the present study, the conceptual model is presented in figure 1.

Figure 1. Conceptual model



Note: CBBE: consumer- based brand equity, Sens: sensory, Emot: emotional, Cogn: cognition

RESEARCH METHODOLOGY

Data Collection and Procedure

This study aims to measure the outcome of a personalized advertisement on a customer's social media account. Facebook was chosen for this study because many reports indicated that India has a great number of Facebook users and has effectively adopted online searching and buying. This makes Facebook users a perfect target sample for the study (Yadav and Rahman, 2018). Also, there is a lack of studies in the Indian context which examine the outcome of social media personalization (Yadav and Rahman, 2017a). Here participant was selected who had a Facebook account could proceed here, asked a question to the respondent do you have an account on Facebook? Suppose the respondent has their accounts on Facebook. In that case, they can answer the question have you see any personalization advertisement of the brand (product and service) of your choice on your Facebook account? Only those who answered yes are eligible for the survey. Here respondents can randomly

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select the brand (products and services) of their choice and then answer the questions; all the responses are collected for further study. An online survey was conducted, the questionnaire was distributed through an invitation link. The link was forwarded to respondents on their E-mail -Id, social media accounts and through text messaging. The convenience sampling technique was employed to collect data. In this study, university students were selected as target respondents, because they are proficient in technology and have sufficient exposure to the internet medium (Islam and Rahman, 2017), Also, students are frequent users and active contributors of social media (Kim and Ko, 2012), (Bolton et al., 2013). The questionnaire has two sections. The first section asked about the respondents' demographic details and the second section was designed to be set up on the "five-point Likert scale started from "strongly disagree, neutral, to agree strongly".

Sample Demographic Characteristics:

Data collection was performed from the different age groups, qualifications, and gender; the characteristics of the sample are presented in table 1. The participants are generally young, and the gender distribution is almost equal.

Table 1 Demographic profile of the sample

Table 1 Demographic profile of the sample		
Variable	characteristics	percent%
Age	below 18	1.49
	18-25	48.25
	26-35	39.3
	36-47	6.96
	46 above	3.98
Gender	Female	52.73%
	Male	47.26%
Respondent type	student	54.72
	Non-Student	45.28
Time accessing Facebook per day	less than 1 hour	45.77
	1-3 hour	39.3
	3-5 hour	10.94
	5-7 hour	3.98
Highest qualification	Higher Secondary	0.99
	Senior Secondary	2.48
	Graduation	35.82
	Post-graduation	58.7
	Doctorate	1.99

Measurement Scale:

The factors identified in this study and factor loading mention in table 2. Mostly Scales used in the present study are reflective and taken from prior studies. The 4-item measured in a scale of perceived personalization has been taken from (Shanahan et al.,2018), (Srinivasan, Anderson, & Pannavolu,

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2002). To measure the item of four constructs used to create a brand experience sensory, emotional, intellectual, and behavioral, adapted the scales created by (Brakus, Schmitt, & Zarantonello, 2009)). The 4-item scale of brand strength is adapted from (Hieke, 2010). Brand awareness measured by the 4-item scale slightly updated the scale given by (Aakar, Yoo, Donthu, & Lee, 2000). Brand value measure with 5-item scale adapted from (Boo, Busser, & Baloglu, 2009), and for perceived quality adapted 4-item scale given by (Yoo & Donthu, 2001). Here, consumer brand experience is a second-order construct consisting of three zero-order constructs: sensory, emotional, and intellectual (Brakus, Schmitt, & Zarantonello, 2009). Other constructs, brand awareness, perceived quality, perceived value, and brand strength, are zero-order constructs. All factor loadings mentioned in the table are in the range of .548 to .879 level, as given by (Costello & Osborne, 2005). Here, the range of alpha lies between .89 to .95, all the results shown in table 2.

Table 2 Constructs, Items and Factor loading

Brand Experience	
Images and videos of personalized ad make a strong impression on my sense	0.836
Personalized ad appeals very attractive in a sensory way	0.830
Personalized ad creates interest to my senses	0.715
Personalized ad engages me to think about the brand	0.688
Personalized ad generates curiosity on me	0.585
Personalized ad induces feelings and sentiments towards the brand	0.656
Creates personal connection and interaction with brand	0.551
Perceived Value	
It shows that brand high in performance	0.869
It treats me as special and valued customer	0.857
Personalized offers provide functional benefits to me	0.823
It provides good value for money	0.764
Personalized offers are worth for their price	0.762
Brand Awareness	
It made me to think first about the brand	0.801
It provides more knowledge about the band	0.785
It helps me to easily identify the brand	0.780
It makes me more familiar with brand	0.692
Perceived Personalization	
It provides me purchase recommendation	0.825
Personalized advertisement appeal like it is customize fit with my needs	0.800
It treats me like I am unique customer	0.776
It informs me regularly about the offers as per my preferences	0.691
Perceived Quality	
Information receive is transparent I can trust on that	0.760
It gives me detailed and quality information	0.697

Table 2 continued on next page

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Table 2 continued

Brand Experience	
Get effective response of all queries and doubts about the brand	0.623
It provides reliable and clear information	0.579
Brand Strength	
It impulse me to purchase the brand	0.750
It invokes me to recommend this brand to my friends	0.744
It creates positive attitude towards the brand	0.689
Personalized offers give me pleasure	0.636
Personalized ad save time to take purchase decision	0.426

Measurement Model

To test the conceptual model first measurement model and then the structural model was tested (Anderson & Gerbing, 1988). To check the reliability and validity of the construct used in the study confirmatory factor analysis with the maximum likelihood method was performed.

To test the measurement model, measure all the latent variables to validate the scales used in this research. For this, draw the all dependent and independent latent variable in the CFA Model Next checked the validity (both convergent & discriminant) of the constructs used in the model so for this, need to measure the following: Composite reliability (C.R.), Here C.R. value Range from .780 to .960 which should be greater than .70 (Bagozzi & Yi, 1988), the average variance extracted (AVE) ranging from .510 to .832 which should be greater than the recommended value .50, results showed that the value of C.R. and AVE values of all construct is satisfactory (Fornell & Larcker, 1981). All the values of constructs reliability and validity mention in table no.3. So, the requirement of convergent validity has been completed; Discriminant validity of the construct was validated if the value of AVE were higher than the value of ASV and MSV (Hair, Black, Babin, & Anderson, 2010), which is also satisfactory. The result of reliability and validity (both convergent & discriminant) are represented in table 3.

Table 3. Constructs Reliability and Validity

Constructs	Alpha	CR	AVE	ASV	MSV	Mean	S.D.
Perceived Personalisation	0.82	0.849	0.7726	0.584	0.5266	3.4	0.67
Customer Brand Experience	0.93					3.29	0.75
Average score of items of sensory factor		0.924	0.804	0.6642	0.369		
Average score of items of emotional factor		0.8574	0.6674	0.565	0.433		
Average score of items of intellectual factor		0.78	0.51	0.4	0.32		
Brand Awareness	0.96	0.906	0.73	0.4788	0.348	3.6	0.65
Perceived Quality	0.92	0.845	0.578	0.4019	0.396	3.36	0.76
Brand Value	0.86	0.96	0.832	0.564	0.25	3.24	0.65
Brand Strength	0.94	0.905	0.7626	0.659	0.5	3.6	0.71

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Now to verify measurement model fitness, check the (Cmin/df); if its value is less than 3, then the model is fit. Here (Cmin/pdf) value 1.51, comparative index .959 value is Tucker-Lewis's index is .952 AGFI .809 means the model yields a good result. The root means square (RMSEA) .051 value shows that the model is a good fit; all the values come within the defined range (Hair, Black, Babin, & Anderson, 2010).

Structure Model

To test all the relationships defined in the conceptual model, check the fitness of the structural model. The value of Cmin/df was 1.524, CFI .957, TFI and AGFI .810 was the root mean square value RMSEA .051, and the RMR value is .060. All measured values come under the threshold value. It shows that the model is a good fit. The assessment of the model measures the R square value of brand experience (.421), quality (.593), awareness (.441), value (.344), and brand strength (.680), which represent that the perceived personalization is the good predictor. The result of hypothesis testing is shown in table 4. The findings of the research indicate that perceived personalization is positively related to consumer brand experience (B=0.61, p< .01) and brand experience mediate the relationship between personalized advertisement and the three-component of the brand equity Brand Awareness (B=.564, p<.01), perceived Quality (B=.331, p<.01) and value (B=.490, p<.01). Consumer brand experience has a significant effect on brand strength (B=.524, p<.01). Brand Awareness (B= .013, P>.05) has not any significant impact on brand strength but the perceived value (b=.172 p<.01) and perceived Quality positively (B=.14, p <.01) impact on brand strength. Finally, brand awareness is positively related to Perceived Quality (B=.483, p<.01), and perceived quality positively influences brand value (B=.205, p<.05). Brand awareness has an indirect effect on brand strength through perceived quality and perceived value. Table 4 presents to show the parameter estimates of the structural model.

Table 4. Testing of hypothesis

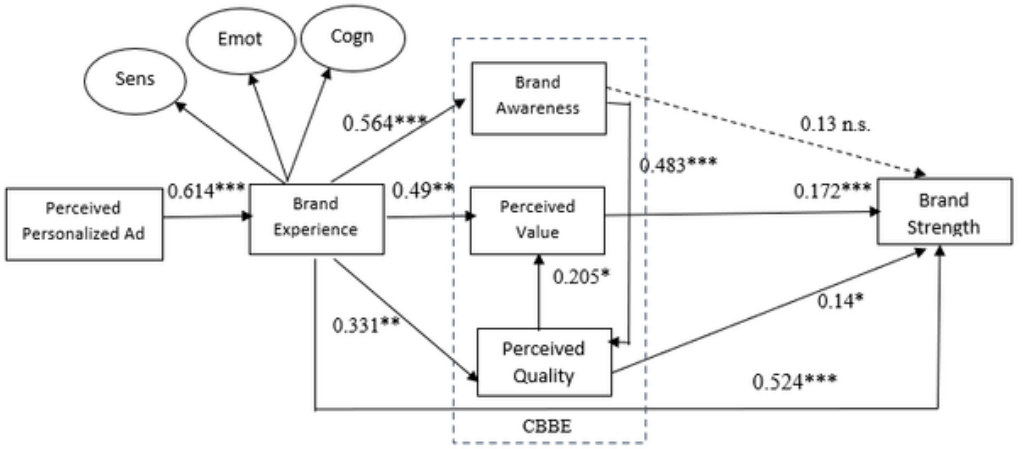
Table No.4 Testing of Hypothesis

	Estimate	S.E	C.R.	P	Hypothesis
PP -> BE	0.614	0.085	7.255	***	H1: Accepted
BE -> BAW	0.564	0.101	5.582	***	H2a: Accepted
BE -> PQ	0.331	0.127	2.608	0.009	H2b: Accepted
BE-> PVAL	0.49	0.166	2.954	0.003	H2c: Accepted
BE -> BSTREN	0.524	0.12	4.328	***	H3: Accepted
BAW -> BSTREN	0.013	0.1	0.135	0.893	H4a: Not Accepted
PQ -> BSTREN	0.14	0.82	0.144	0.01	H4b: Accepted
PVAL -> BSTREN	0.172	0.046	3.428	***	H4c: Accepted
BAW -> PQ	0.483	0.116	4.166	***	H5a: Accepted
PQ -> PVAL	0.205	0.101	2.029	0.02	H5b: Accepted

Note: PP-perceived personalization, BE-Brand experience, BAW-Brand awareness, PQ-Perceived brand quality, PVAL-Perceived brand value, BSTREN-Brand strength

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Figure 2. Structural model



Note: ***ρ≤.001, **ρ≤.01, *ρ≤.05, n.s-not significant, CBBE: consumer- based brand equity, Sens: sensory, Emot: emotional, Cogn: cognition

MEDIATION ANALYSIS

Mediation analysis was performed using direct, indirect, and total effects of the variable analysis. All three effects are statistically tested by performing a bootstrap at 2000 and a 95% significance level. Here brand experience mediates the relationship between personalized ads and the three elements of brand equity brand awareness, perceived value, and quality. Brand experience also mediates the relationship between perceived personalization and brand strength. Although no hypothesis was developed to check this relationship. The results exhibit that brand experience fully mediated the relationship between personalization ad and brand awareness but partially mediated with perceived quality, value, and brand strength. The result shows in table 5.

Table 5. Mediation Analysis Using Direct & Indirect Method

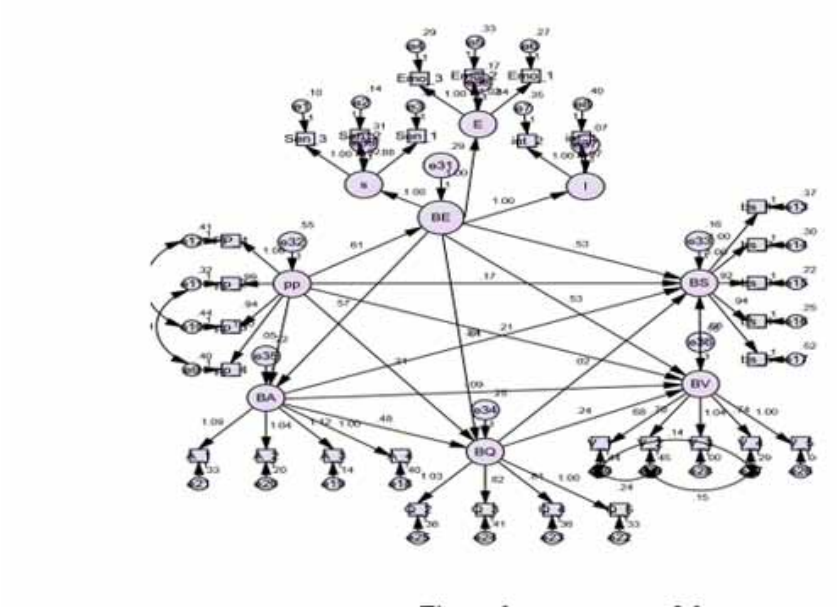
	Direct effect	Indirect effect	Total effect
	Loadings		
PP->BE	0.61***		
BE->BAW	0.629***		
PP->BE->BAW	0.056 n.s.	0.408*	0.464***
BE->PQ	0.29**		
PP->BE->PQ	0.204**	0.375*	0.579***
BE->PV	0.369***		
PP->BE->PV	0.15**	0.321**	0.475***
PP->BS	0.39***		
PP->BE->BS	0.61**	0.52***	0.748***

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DISCUSSION

This work is considered concerning the recent calls for further research to find the effect of personalized advertisements on social media. This study finds that personalized advertisement positively impacts brand-related outcomes such as brand experience, brand equity, and brand strength. Data were collected from the customers who have Facebook accounts and have seen the personalized advertisement of the brand advertised on their Facebook account. Customers prefer all these brands that they already searched for or buy online. The study provides important implications for academics and practitioners discussed below:

Figure 3. Structural model



THEORETICAL IMPLICATIONS

The results of the present study considerably contribute to the marketing literature in various manners. First, even though the study has found different effects of personalization, the outcomes stay uncertain. Some studies described positive effects that include increased customer satisfaction, purchase intention, engagement, brand loyalty, brand awareness, and customer retention (Blasco-Arcas et al., 2016) ;(Shanahan, Tran, and Taylor, 2018). Although negative effects are also reported in some studies covering customer avoidance, reactance for a personalized ad, and discuss privacy issues (Aguirre et al., 2016), (Chen et al., 2019). So, it is very much needed to know why these irregularities of thoughts exist. Even though various types of personalization have been defined in various studies, this research focuses on a particular kind made after a customer saw the advertisement of the brand in their social media account, which they searched on the internet. In country like India people are very sensitive about sharing personal details, so marketer should follow the cultural and religious boundaries to provide customers satisfying experiences. The outcome of this research proves that its effects are positive and supported the hypothesis. This paper reveals insight into the process of defining the role of a personalized advertisement on Facebook. If online marketers efficiently implemented this

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communication strategy, it can become a powerful tool for their businesses. Personalization, created through matching message content with customers' needs, makes the message relevant and improves customer experiences.

Second, buyers looking for a brand on the internet with which they want to use and have the good past experience itself is appropriate to conduct the research. (Tran et al.,2020). When buyers feel an association between their needs and the characteristics of the brand, a positive experience is established. This impact improves customer experience with the brands and creates brand awareness and provides brand value. Consumers find personalized messages to be more pertinent, more persuasive, better remembered, which improves brand familiarity; thus, consumers spend more time processing these messages. The outcome of this study confirms the finding of the previous studies that prove a positive relationship between brand experience and brand equity (Khan and Fatima 2017) (Lee and Jeong 2014), Expanding on personalization and brand equity literature, the present research established the connection between the customer experience with a personalized advertisement of the brand advertised on Facebook and customer-based brand equity after searching that brand online. To sustain in today's highly competitive market, firms need to develop up-to-date advertising strategies to always be informed and aware of their customers about the brand. The study already confirms that personalization is part of successful advertising and marketing tactics. The positive attitude of consumers towards personalized communication is because it provides several benefits to customers, creates values, and improves customer connection with brands.

Third, brands provide benefits to customers and try to create long-running good relations with them (Shanahan et al., 2019). (Tho, Trang, and Olsen, 2016). Companies have invested a lot of time and energy to investigate the principal brand equity (Kim and Ko,2012). Brand equity is driven by brand image, awareness, brand value, quality, and experience (Being and Nika,2019), (Khan and Fatima,2017). It leads to a greater probability of repurchase and a less inclination to switch brands (Shanahan et al., 2019). The finding of this research goes above and beyond by identifying the effect of brand equity on brand strength in the social media context. This study tried to explain the missing link between how perceived personalization in social media affects brand strength. Here study proved that personalization is a critical driver of brand equity, which increases the strength of the brand advertised on social media. To gain and sustain brand strength depends on the consumer attachment with a brand, experience provided by the brand, and the management of brand value and benefits individuals receive (Wiedmann et al.,2017). This analysis will guide us to find out the differences in the use of personalization in social media and how it helps develop brand strength.

Fourth, in the present research, we did not develop any hypothesis to explain the mediation effect. The paper proposes the drivers of brand experience (perceived personalized ad) and outcomes (brand equity and brand strength). However, it also shows a mediation process that improves the impact of perceived personalization on brand equity. Mediation analysis reveals that brand experience positively mediates the relationship between the three elements of consumer-based brand equity (BA, PQ, BV) and perceived personalization. Brand experience also positively mediates the between personalization ad and brand strength. Results show that personalized advertisements on social media positively affect brand equity and strength if it successfully increases consumer experience. Therefore, firms should focus on the practical implementation of personalization content to provide a good brand experience.

Lastly, this research also provides the interconnection between the various components of brand equity. The three components of CBBE used in this research are brand awareness, value, and quality. The study's outcome shows that brand awareness impacts the brand's perceived quality, and perceived quality further impacts brand value. Previous studies also developed and confirmed the same results in the social media context (Aaker et al.,2000), (Keller & Lehmann, 2003).

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MANAGERIAL IMPLICATIONS

This work provides significant implications for firms using Facebook advertising as a critical advertising strategy. As discussed, the social media platform is highly interactive than other traditional media. Personalization is more effective in social media than in other media environments. Firms should keep a bird's eye on customer preferences and provide more customized offers that fit their needs. Social media marketers acknowledge the massive amount of information customers share on different social media platforms and communities and personalize their marketing practices. For the marketer, creating a personalized offer is the biggest challenge, but the effective implementation of personalization will help increase their profit margin. IBM took the first initiative to make the team with Facebook and fuel the advertising technology. The attractive advertisement, eye-catching Images, Meaningful videos attract customer attention and give them an excellent sensory impression. A firm should be honest with all advertisements published on their account so the customer can rely on them and easily access their quality and value. This study might guide well to the firm dealing with online business and using social media environment. Brand managers can use social media personalization as the most vital communication tool to engage their customers. Customers' engagement with the brand increases their familiarity and results in sustainable relationships.

Although the study shows the positive results of personalized advertising, some studies criticize the effect of personalization. (Tucker, 2014), (Chen et al., 2019). Especially, (Hoffman et al.,2010) contended that social media revalorized the entire communication system by transfer the power and control of brands from the companies to the customers, who are more actively engaged in brand communication through interactions, comments, and recommendations. This study exhorted that online marketer are cautious while picking social networking sites to advertise their brands. The adequacy of a personalized advertisement on social media is problematic because the number of fake Facebook accounts is increasing very fast (Krombholz et al., 2012). This incredulity can be managed if firms understand how to customize their social media advertisements. The right implementation of personalization could improve customer attitude or brand image promoted online, as confirmed by the outcomes of the present study.

Social media should provide customers with complete control of the entire online activities, like what they want to see or not on their personal social media account. Companies can also benefit from this policy because companies will not waste their budget and time promoting ads to customers who do not want to. These resources can be further utilized efficiently to enhance the quality of personalization activities.

To take advantage of the benefits of social media, companies invested billions of dollars on social media to do social commerce (Liang et al., 2011), (Rudolph, 2015). that encourage customers to buy and sell products and services on a social media platform. The study results also support that social commerce company can use personalized advertisements to reach their marketing goals. This communication strategy can improve customer perception about the brand, improves the customer-brand relationship, and develop positive brand strength. This is a more cost-effective communication strategy than other communication tools (Comscore, 2019).

New advanced technology provided many conveniences for online shopping and brought them closer to the companies. Presently customers can purchase any product from any brand available online after just a few clicks. These conveniences can create huge competition and critical problems for the marketers, for example, customer retaliation. How can companies create a strong bond with their customers to prevent customers from switching to other brands? This is the big question for online marketers. This research can provide a possible solution to the online marketers – if the company develops their personalization communication strategy that encourages a customer to consistently associated with a brand can retain their most profitable customers. Previous advertising literature confirmed that personalization improved customer perception about the brand and reduced consumer ad avoidance behavior (Shanahan, Tran, & Taylor, 2018), (Tran, 2017). Also, these results are

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limited only to a specific social networking site, Facebook. The marketer should also implement this personalization approach to other social networking sites like Instagram, Twitter, etc. If the outcome of personalization acts in the same way as it on Facebook, companies have the chance of creating this personalization advertising strategy on those media also.

Brand managers need to build brand communities on social media to strengthen their brand and have a better conversation with their customers. The brand communities reassemble the most loyal customers in one place, which is a massive win for any brand. This community should have to give personal attention to their customers and serve them in a better way. Companies can also strengthen their brand through posts brand content, videos, and images on the brand fan page. The number of likes, shares, and comments on these posts helps managers of brands that operate brand fan pages to collect their customer's requirements. In this research, we explained personalized advertisement development based on past search behaviour and online activities. They can also check how many customers switch to a website from clicking advertisement in their Facebook account. Brand managers need to closely monitor all these observations to create a personalized advertisement and develop high customers responses to the brand.

LIMITATIONS AND FUTURE RESEARCH

Although our research is empirically tested, it has a specific limitation. The first and essential limitation is the people gap in understanding the advertisement as personalized or general. Here the method chooses to explain the personalization is not ideal. Future studies should use different techniques like experimental design by giving respondents different conditions and randomly selecting the particular condition. The comparison between different conditions should develop for the best outcome. This can also set a good explanation of the effect of the perceived personalization of the brand advertised on social media.

The effect of personalization in traditional media we already tested and in the given research model is given to measure the effect of personalization on social media. As we know, the social media environment is highly distinct, unique, and interactive as we already discussed, there is a difference in perception between the social media context with opposed to another context. For future research, there should be a difference between the perception of social media and the other context in a single study. Finally, in this study, we using data collected from Facebook users in a future study; we can collect the data from other social media platforms like Instagram and Twitter to compare the result, test the model and check the validity of the construct used.

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Factors Affecting the Behavioural Intentions of Indian Millennials:
An Analysis of Online Food Delivery Services

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ABSTRACT

Previous studies have examined consumer attitudes toward online services, while few studies have explored factors affecting the behavioural intention of online food delivery services. The purpose of this study is to explore factors that affect the adoption of online food delivery services of Indian millennials by examining the structural relationship between hedonic motivation, utilitarian motivation, price-saving orientation, time-saving orientation, attitude, and behavioural intentions using the theory of planned behaviour. Data from 328 Indian millennials were collected for the study by a self-administered and structured questionnaire. The proposed model was investigated empirically using exploratory factor analysis (EFA) for the validation of scale and then using confirmatory factor analysis (CFA) and structured equation modeling (SEM). The study's results showed that only the relationship of time-saving orientation with the attitude towards was proven insignificant, while others were significant.

KEYWORDS

Dining Out, India, Millennials, Online Food Delivery Services, SEM

INTRODUCTION

Online food delivery (OFD), defined by Li et al. (2020), is a process through which food is ordered online, prepared, and delivered to the consumer. These services have seen a rise in demand over the past few years and are deemed to grow further in the coming years steadily. In numbers, the revenue generated by these platforms globally was estimated to be $50 billion in 2018 to 107.4 billion in 2019 and are projected to exceed to $200 billion by 2027 (Statista, 2020). These changes have helped the food markets innovate new ways as it was reaching its saturation point (Cho et al., 2019). The market in India is in its nascent stage, with only a handful of companies operating in the region. However, given the size of the country and its market, the revenue numbers have been on the rise touching $370 million in 2017 at an exponential rate, made possible due to the food-tech start-ups

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founded in the country like Zomato and Swiggy (The Economics Times, 2018; Technavio, 2019). With their market penetrating strategies, these companies have been able to catch the imagination of the Indian consumers. Food delivery is not a novel concept for Indian consumers. For example, Mumbai dabbawallas have been doing it since the 1890s (Baindur & Macário, 2013). However, as online food delivery gained popularity in the 2000s in other parts of the globe, in India, it was only introduced a decade ago (Pigatto et al., 2017). All these advancements have also resulted in the food buying behaviours of the consumers and are expected to be widespread in times to come (Ciro et al., 2020; Alalwan, 2020). Furthermore, as a result, compared to other counties like Indonesia (Elvandari et al., 2018; Suhartanto et al., 2019), Columbia (Correa et al., 2019), China (He et al., 2019), and many more, the literature suggests that are only a few studies that have been conducted in the country's context for studying the main drivers of adoption of this technology (e.g., Bilgihan et al., 2014; Levin & Taylor, 2014; Vernhoef et al., 2015; Cho et al., 2019; Gunden et al., 2020; Yeo et al., 2017).

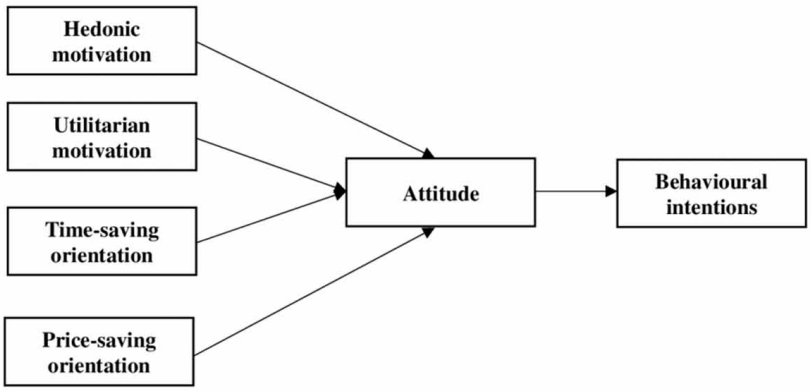
Hence, the current study focuses on the Indian market only. As demographics is one of the critical predictors for online buying behaviour (Li et al., 1999), the study focuses on the millennial consumers in the country who are considered to be tech-savvy and aware of these services better in comparison to other generations (Norum, 2003; Jackson et al., 2011; Wolburg & Pokrywczynski, 2001; Business.com, 2020). Furthermore, adoption of such technology is subject to favourability of the relationship between attitude and behavioural intention with complex decision-making process based on many factors like values, social and personal (Kimes, 2011; Littler & Melanthiou, 2006; Saarijärvi et al., 2014; Bisogni et al., 2005; Fishbein & Ajzen, 1975). Hence, this study mainly focuses on four such factors that have been tested in previous studies, like hedonic and utilitarian motivations defining online food delivery characteristics (Nejati & Moghaddam, 2013; Yeo et al., 2017). Moreover, food belongs to a low-involvement product group making it a price-sensitive and time-saving action defined by price-saving and time-saving orientations (Saunders et al., 2012; Yeo et al., 2017). The proposed model is shown in figure 1.

LITERATURE REVIEW

Theoretical Background

Online food delivery services have innovated ways of purchasing food by the consumers, and these innovations have been able to meet their personal and social needs. The study takes the attitude-behaviour relationship approach as it is known and proven for measuring the intentions of consumers (Madden et al., 1992). Previous researchers like Venkatesh et al. (2003), De Pelsmaeker et al. (2017),

Figure 1. Proposed conceptual model



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Wang & Somogyi (2018), and Piroth et al. (2020) have investigated the factors that affect attitude and the behavioural intention to adopt new technology through extending or diffusing the behavioural models like technology adoption model (TAM), theory of planned behaviour (TPB) and theory of reasoned action (TRA) (Davis et al., 1989; Ajzen, 1991; Ajzen & Fishbein, 1980). The theories primarily investigate the attitude-behaviour relationship where behavioural intentions are defined as the propensity to try or not to try something (Brown & Venkatesh, 2005), and attitude defined is as the favourability of one's evaluation of the behaviour in question that may incline people to adopt that particular behaviour (Rezaei et al., 2016). For example, a study by Hansen et al. (2004) found that consumers' attitude toward online grocery shopping is a prominent predictor of their behavioural intention toward online shopping. Furthermore, the studies related to food purchase have also affirmed strongly over the relationship while investigating driving factors for the usage of OFD applications (Cho et al., 2019; Lee et al., 2017; Yeo et al., 2017). Some of these studies have focused on characteristics of the mobile applications (Pigatto et al., 2017; Cho et al., 2019) and some behavioural models (Hansen et al., 2004; Alagoz & Hekimoglu, 2012). The present study also moves a step forward in studying a behavioural model by diffusing the theory of planned behaviour model by adding and exploring the effects of four exogenous variables (hedonic motivation, utilitarian motivation, time-saving orientation, and price-saving orientation) in the context of OFDs used by Indian millennials as many predictors have been proposed in prior studies developing the theory for multicomponent approach (Conner & Armitage, 1998; Conner & Sparks, 2005; Ajzen, 2011; Conner, 2015).

Hedonic Motivation

Hedonism affects a consumer's purchase and consumption patterns while deciding about a product (Rezaei & Ismail, 2014). Hedonic consumption relates to fantasy, the senses, and the experiential interaction of various interest controls (Musnaini et al., 2017). Hence hedonic motivation is the internal incentive one receives for using a technology that may include fun, enjoyment, or pleasure (Venkatesh et al., 2012). In this era of technology, Richard & Habibi (2016) investigated hedonism and observed that different websites vary in the level of hedonism. Other studies like Rezaei & Ghodsi (2014) and Yeo et al. (2017) stated that hedonic websites motivate enjoying the process giving an experiential view of hedonism a broader approach for that particular process. Also, hedonism affects cognitive-rational and problem-solving information processing as emotional arousal (Rodríguez & Trujillo, 2013; Miranda, 2009). Hence, hedonism or hedonic motivation affects the e-commerce buying enjoyment of the consumers as well as their attitude towards using them (Childers et al., 2002; Bilgiham, 2016; Wagner et al., 2016; Nath et al., 2020). Therefore, based on the arguments above, a user's attitude is believed to be affected by hedonic motivation while using OFD services. Therefore, the following hypothesis is proposed:

H1: Hedonic motivation significantly affects on Indian millennials' attitude towards OFD services.

Utilitarian Motivation

Utilitarian motivations are extended by hedonic motivations (Parson, 2002). It is also more valued and important in the e-commerce environment, as indicated in a study by Bridges & Florsheim (2008). These motivations are product-based and are related to rational, decision-effective, critical, and goal-oriented tasks (Arnold & Reynolds, 2003; Cai et al., 2018). Also, they start with a goal or task that receives perceived benefits when done effectively throughout the process (To et al., 2007). Hence, utilitarian consumers with an incentive to use brand-specific social media platforms are concerned with finding a suitable content that is appropriate and useful for that particular purpose, like the ease of use and satisfying results (Derbaix & Pham, 1998; Bridges & Florsheim, 2008; Dabbous & Barakat, 2020). Previous studies investigated the utilitarian aspects, whereas others have focused on hedonic aspects for various reasons. A study by Holbrook and Hirschman (1982) shifted the shopping

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perspective from being hedonically oriented to utilitarian oriented. Concerning food consumption, a study conducted by Nystrand & Olsen (2020) indicated that the utilitarian eating values outdid the hedonic eating values while investigating the consumers' attitude towards functional foods. Therefore, based on the above arguments, a user's attitude is believed to be affected by utilitarian motivation while using OFD services. Therefore, the following hypothesis is proposed:

H2: Utilitarian motivation significantly affects Indian millennials' attitude towards OFD services.

Time-saving Orientation

Modern societies have evolved in a way that has become time-sensitive (Zuzanek, 1998). Online shopping has been able to curtail this feeling by saving travel time to and from a retail store (Morganosky & Cude, 2000; Punj, 2012). One of the primary and prominent characteristics of fast and takeaway food is convenience, which saves time (Bashir et al., 2019). Similarly, OFD services have been able to help eliminate this feeling by reducing waiting time at a crowded restaurant, also allowing them to save the payments for later and lets them check the previous orders for efficient checkout to save time (Berdychevsky & Gibson, 2012; Statista, 2020). Also, time-saving orientation defines such characteristics involving online shopping that saves time for the consumers (Jensen, 2012). They prefer their food to be delivered as fast as possible (Yeo et al., 2017). Also, time-saving orientation is a significant predictor of the consumers' attitude and intention to use these services (Yeo et al., 2017). Hence, making time-saving orientation one of the prominent factors that affect the attitude of the people purchasing online (Khalil, 2014). Therefore, based on the arguments above, a user's attitude is believed to be affected by time-saving orientation while using OFD services. Therefore, the following hypothesis is proposed:

H3: Time-saving orientation has a significantly affects Indian millennials' attitude towards OFD services.

Price-Saving Orientation

Price is the monetary value one gives in exchange for a product or service in a purchase agreement (Nagle et al., 2010). The price-saving orientation refers to the fiscal paybacks that a consumer gains from technology (Pitchay et al., 2021). For OFD services, price-saving promotions in the form of free delivery or promotional incentives often serve as a marketing tool (Choi et al., 2021; Ray et al., 2019). Consumer look for a price-saving opportunity via these discounts and a study by Choi et al. (2021) have suggested that OFD services provide these advantages as online consumers can compare the prices of different companies (Gentry & Calantone, 2002; Eriksson & Nilsson, 2007; Chiu et al., 2014). However, price-saving orientation considers not only the monetary savings but also the perspective of not incurring any additional costs to purchase a product or using a service, making low prices attractive to the managers as well, while food consumers tend to opt for different levels of food quality and price (DelVecchio & Puligadda, 2012; Rodríguez & Trujillo, 2014). Therefore, based on the arguments above, a user's attitude is believed to be affected by price-saving orientation while using OFD services. Therefore, the following hypothesis is proposed:

H4: Price-saving orientation has a significantly affects Indian millennials' attitude towards OFD services.

Attitude and Behavioural Intention in OFD Service Setting

Attitude is a predictor of behaviour defined as the degree of favourability towards a particular behaviour in question (Ajzen, 1991). It is also the representation of the act of the individual's behavioural

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intentions is positively or negatively valued (Ajzen,2005). As reasoned by Ajzen & Fishbein (1977), the attitude has a positive effect on behaviour. Authors like Rezaei et al. (2016) have also suggested that a positive attitude leads to high motivation to do a particular act. Moreover, concerning the adoption of technology, a seminal paper by Davis (1989) has concluded that the adoption of such services highly depends on the behavioural intention and the individuals' attitudes. Examples include participation in virtual communities, e-learning (Liaw, 2008; Liaw et al., 2007), Internet-based learning (Lee et al., 2005), personal digital assistant in the healthcare industry (Mun et al., 2006), search engines (Liaw & Huang, 2003), technology in the hotel industry (Lam et al., 2007), blogs (Hsu & Lin, 2008), and internet banking (Shih & Fang, 2004). Therefore, based on the arguments above, a user's behavioural attitude is believed to be affected by the user's attitude while using OFD services. Therefore, the following hypothesis is proposed:

H5: Indian millennials' attitude towards OFD services significantly affects their behavioural intention.

METHODOLOGY

Survey Instrument

The questionnaire included two sections. The first section included the screening question to identify the consumers of OFD services and their demographic profile. The second section included the construct of the study to be measured for the study. The constructs for the study were acquired and adapted from the existing literature to make it relevant to the current context. All the items were measured on a 5-point Likert scale ranging from 5 for strongly agree to 1 for strongly disagree. Furthermore, they are shown in the Table 1.

Data Collection and Sample

Before the main study, the questionnaire was distributed among the final year students pursuing a master's program in business administration, research scholars, and academics of the department to reflect upon the clarity on the readability of the instrument. The pilot group responded to all the scale items, and a few minor modifications were made to the wording of items. Their suggestions were incorporated by changing the wording. Then the main study was conducted online using Google forms sent across the social media platforms using purposive or judgmental sampling techniques among the Indian millennials using OFD services. The questionnaire received was 328, out of which only 320 were used for the final analysis, and their demographic profile is depicted in table 2. The sample was adequate for the recommended 10:1 item-to-response ratio suggested by Hair et al. (2014). Also, G*Power 3 (Faul et al., 2007; Faul et al., 2009) software was used for a post hoc analysis that

Table 1. Scale and their sources

Scale	Items	Source
Hedonic motivation	5	Rodriguez and Trujillo, 2013; Khajehzadeh et al., 2014
Utilitarian motivation	4	Voss et al., 2013
Price-saving orientation	3	Rodriguez and Trujillo, 2013
Time-saving orientation	4	Alreck and Settle, 2002
Attitude	4	Childers et al., 2001
Behavioural intention	3	Rodriguez and Trujillo, 2013; Kuo and Yen; Taylor and Todd, 1997

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Table 2. Demographic profile of the respondents

	Category	Frequency	Percentage
Gender	Male	226	70.6
	Female	94	29.4
Age	Between 18-25 Years	153	47.8
	Between 25-40 Years	167	52.2
Education	Intermediate	19	5.9
	Graduate	101	31.6
	Post Graduate	200	62.5
Occupation	Student	232	72.5
	Working Professional	84	26.3
	Home Maker	4	1.3
Family's Monthly Income	INR 0- 25,000	84	26.3
	INR 25,000- 50,000	106	33.1
	INR 50,000-1,00,000	65	20.3
	Above INR 1,00,000	65	20.3

resulted in a power of test more than the suggested 0.8 by Cohen (1988) that confirmed the sample was adequate for further analysis.

DATA ANALYSIS

The data analysis for the study was done using Statistical Package for Social Science (SPSS) version 25 and Analysis of Moment Structure version 24, used in many studies in the marketing and management field.

Measurement Model

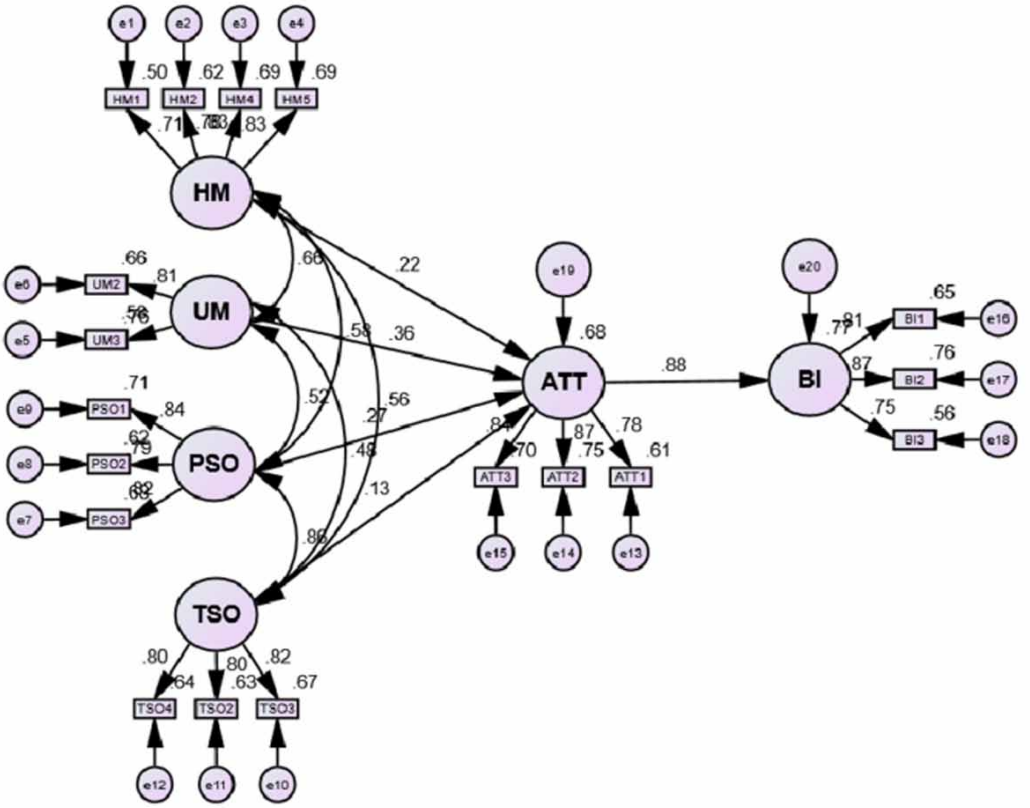
For the evaluation of the measurement model, the reliability and validity of the items and the corresponding constructs were conducted using Cronbach alpha and Principal Component Analysis (PCA). The values of Cronbach alpha ranged from 0.785 to 0.864 for the scales shown in table 3, which is above the acceptable threshold of 0.7 suggested by Nunnally (1978). Second, the variance inflation factor was measured to check multicollinearity among the independent variables that came to less than 3. Also, the composite reliability ranged from 0.760 to 0.876, which was above the threshold of acceptance (Hair et al., 2014). After this, the validity was tested using PCA on 22 items with orthogonal rotation (Varimax) that led to the exclusion of 5 items, namely HM1, UM1, UM4, TSO1, and ATT4. Furthermore, the Kaiser-Meyer-Olkin measures of adequacy and Bartlett's test for Sphericity for these items were checked after the exclusion of 5 items, and it was found to be 0.924 and significant at $p<.001$. The measurement model is shown in figure 2.

Common Method Bias

Self-reported data are vulnerable to common method bias. To investigate whether our data was free from such bias, we conducted Harman's post hoc one-factor analysis. The highest noted variance

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Figure 2. Measurement Model



explained was 14.62 percent, less than 50 percent, confirming the absence of common method bias (Liang et al., 2007).

Hypotheses Testing

The final evaluation of the model fit showed an acceptable fit of the model to the data (Goodness-of-fit statistics: $\chi 2$=360.644, df=124, $\chi 2$/df=2.908, $p < .001$, NFI=0.909, CFI=0.938, TLI=0.923, and RMSEA=0.076), regression values given on the arrows in the model. The standard factor loadings of all the variables fell in the range of 0.716 to 0.885, and they loaded to their related latent constructs at $p < .001$. The detailed standardized factor loadings of the items are shown in table 3. The utilitarian motivation was retained with two observed variables as they were highly correlated and had a factor loading of more than 0.7 (Worthington & Whittaker, 2006). The table depicts the calculated average variance extracted (AVE) ranging from 0.613 to 0.703, better than the accepted value of 0.500 (Hair et al., 2014). Furthermore, table 4 shows the discriminant validity where the diagonals indicate the square root of AVE and the off-diagonal depicts the correlation. The diagonal's values are greater than the off-diagonals inferring the existence of discriminant validity conferring to the Fornell Larcker criterion (Fornell & Larcker, 1981). Finally, the hypotheses testing for H_1 to H_5 in the proposed model were made employing path analysis via structural equation modeling (SEM). Where H_1 showing relationship of hedonic motivations with attitude ($\beta = 0.235$, t-value = 3.123, $0.05 < p$), H_2 showing the relationship of utilitarian motivation with attitude ($\beta = 0.372$, t-value = 4.815, $0.05 < p$), H_4

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Table 3. Factor loadings, Composite Reliability, AVE and Cronbach Alpha values

Constructs	Items	Factor Loadings	CR	AVE	Cronbach Alpha
Hedonic motivation	**HM2**	0.736	0.866	0.619	0.842
	HM4	0.848			
	HM5	0.782			
Utilitarian motivation	**UM2**	0.786	0.760	0.613	0.785
	UM3	0.828			
Price-saving orientation	**PSO1**	0.767	0.854	0.661	0.853
	PSO2	0.749			
	PSO3	0.724			
Time-saving orientation	**TSO2**	0.626	0.838	0.633	0.853
	TSO3	0.803			
	TSO4	0.827			
Attitude	**ATT1**	0.729	0.876	0.703	0.864
	ATT2	0.772			
	ATT3	0.784			
Behavioural intention	**BI1**	0.693	0.849	0.653	0.844
	BI2	0.710			
	BI3	0.689			

Table 4. The square root of AVE and factor correlation coefficients

	Attitude	Hedonic motivation	Time-saving orientation	Price-saving orientation	Behavioural intention	Utilitarian motivation
Attitude	**0.838**					
Hedonic motivation	0.608***	**0.786**				
Time-saving orientation	0.609***	0.544***	**0.796**			
Price-saving orientation	0.652***	0.562***	0.849***	**0.813**		
Behavioural intention	0.822***	0.771***	0.625***	0.660***	**0.808**	

Note: The diagonals represent the square root of AVE and the off-diagonals represent the correlation. Correlation is significant at the *** p < 0.001 (2-tailed), Items in bold are the square root of AVE.

showing relationship of price saving orientation with attitude (β = 0.223, t-value = 2.338, 0.05< p), and H_5 showing relationship between attitude and the behavioural intentions (β = 0 .940, t-value = 13.383, 0.05< p), were all supported. Only hypothesis H_3 showing the relationship of times saving orientation with attitude (β = 0.106, t-value = 1.169, 0.05< p) was not supported. Moreover, the standardized regression weights with the other estimates for the relationships are given in table 5.

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Table 5. Hypotheses test results

Hypothesis	Beta-coefficient	SD error	t-value	Significance
H_1	.235	.075	3.123	.002**
H_2	.372	.077	4.815	***
H_3	.223	.095	2.338	.019**
H_4	.106	.090	1.169	.242
H_5	.947	.072	13.138	***

Note: (1) *** p < 0.001 has a significant effect and **p <0.01 probability level, (2) The model is found to be statistically significant at 0.05 level of significance; however, the certain relationships marked in bold in this table are found to be statistically insignificant at 0.05 level of significance.

DISCUSSION AND CONCLUSION

The current study explored the factors affecting Indian millennial consumers' behavioural intentions while using OFD services. The hypotheses H_1 to H_5 tested the relationships of these exogeneous variables with the attitude leading to their behaviour. While hypotheses H_1 and H_2 tested the relationship of hedonic and utilitarian motivations with the attitude that was found to be significant in this study coherent to the studies by Babin et al. (1994), Eroglu et al. (2005), Ryu et al. (2010) and Chang et al. (2016) showcasing that the consumers were affected by these two motivations while using OFD services and the current population's consumption happen due to utilitarian reasons (i.e., to curb hunger) and hedonic gratification (i.e., good taste) (Dhar & Wertenbroch, 2002; Batra an& Ahtola, 1991). Furthermore, the H_3 tested the significance of price-saving orientation affecting the attitude of consumers came to be significant indicating that Indian millennial consumers are price and value-conscious while using the OFD services (Jin & Suh, 2005) and also incoherent to a study by Dazmin & Ho (2019) that price is not an important factor influencing OFD services decisions making. Furthermore, unsurprisingly, hypothesis H_5 tested the consumers' attitude affected their behavioural intention was also significant and fell in line with a previous study by Liang & Lim (2011) on buying food online. Lastly, the only hypothesis H_4 testing time-saving orientation affecting consumers' attitudes was insignificant. This distinguished the Indian millennial consumers from other consumers and was incoherent to previous studies like Kedah et al. (2015) that highlighted that time is an important reason for the e-commerce environment. Thus, it can be concluded from this result is that Indian millennial consumers are inconsiderate of the delivery timings, as delays may affect the attitude of the consumers and their satisfaction and can be bettered in the future as it fluctuates the demand for the services (Bateson et al., 1992; Lovelock et al., 1999; Dholakia & Zhao, 2010).

IMPLICATIONS

The proposed model of the study validated the factors that affect the decision-making of Indian millennial consumers while using OFD services. The present study explored the factors using the attitude-behaviour model and found most of them significant, providing contextual knowledge of consumers' behavioural intention for using OFD services in India. This study provides insight into the Indian market and studies its most tech-savvy population using a novel technology, as the current market is yet in its nascent stage and yet to evolve in the coming future. Furthermore, the body of knowledge created provides a forum for further creation of theoretically guided frameworks to explore other user behavioural constructs for online food delivery services. The current study also has managerial implications, such as the variables studied could potentially be looked at and improved upon by the OFD services providers and their partnered restaurants. The improvements can be implied as increasing the hedonic motivations. There can be new innovative ways of plating

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the food for delivery that can be taken care of by the chefs in the restaurants while better packaging for maintaining the food temperature.

In contrast, transportation of the food being delivered can be handled by these OFD services providers, which will increase the hedonic pleasure. Then, for the utilitarian motivations, the quality of the food served by these restaurants can prove to be a chance to match the quality of home-cooked food. As for the price saving aspect of the orders, it is important to give reasonable pricing on the food more competitive to the price offered by the offline restaurants. Lastly, the time-saving aspect of the food delivered by OFD services providers can use artificial intelligence to track the whole process of booking the food orders and provide the best prices based upon their frequency of using their services. Furthermore, it shall also help in tracking the order streamlining the real-time food preparation and delivery information. Artificial intelligence can be integrated as a chat-bot section of the applications of these OFD services, making it easier for the consumer to place orders and track. It will also help in amending the errors of misplacing the orders due to technical glitches as incorporated artificial intelligence in these chat-bots will work with more precision and accuracy.

LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

The current study suffers from a few limitations due to its research design, as the cross-sectional design is vulnerable to methodological bias. Secondly, the population only focused on the Indian millennials' consumer perspective and not on the other age groups in the market, limiting its generalizability. However, this study has indicated ways for future scholars and researchers to undertake their studies. First, future research should involve consumers of various age groups across the different geographical areas of online delivery services. Second, longitudinal and experimental designs may be used in future examinations to examine and explain relations over time and assess causality in their links. Lastly, as the world has witnessed a pandemic and is still coping with its effects while this paper was being written, the data presented in this study was from the pre-pandemic era. Hence, making this study a pre-pandemic study, and thus the studies carried out in the post-pandemic market can be compared to measure the changes that happened due to the pandemic on the market dynamics and consumers' behavioural intentions.

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Impact of Social Media Marketing Activities on Purchase Intentions via Customer Equity Drivers:
An Empirical Case Study on Smartphone Brands

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ABSTRACT

In the light of growing interest and usage of social networking sites, smartphones, and internet availability, this study aims to analyze the impact of social media marketing activities on the purchase intentions of the customer via customer equity drivers—relationship equity, value equity, and brand equity—in the context of the smartphone industry. An online survey was conducted with a total of 343 respondents who were active users of social media. The collected data were analyzed using structural equation modeling. The findings of the study revealed that interactivity was the most important dimension of social media marketing activities (SMMAs) and personalization with e-WOM being second. SMMAs were found to have a significant impact on customer equity drivers, and value equity and brand equity were positively related to the purchase intentions of the customers. The findings of the study will help managers to develop SMMA strategies and boosting future sales by influencing the purchase intentions of the customers through perceived SMMA.

KEYWORDS

Brand Equity, Customer Equity Drivers, Purchase Intention, Relationship Equity, Social Media Marketing Activities, Social Networking Sites, Value Equity

INTRODUCTION

The usage of Social Networking Sites (SNSs) e.g., Facebook, Twitter, Instagram, etc. has been growing enormously over the past few years. Facebook is the most popular network in the world, with over 2.6 billion monthly active users as of the first quarter of 2020. (Clement, 2020). In 2020, there were above 3.6 billion social media users worldwide which is expected to increase to 4.41 billion by 2025 (Tankovska, 2021). On average social networking usage of internet users worldwide amounted to 144 minutes per day as of 2019 (Clement, 2020). The digital population of India is increasing rapidly with more than 680 million users active on the internet. It is forecasted that with the growing ease of internet access available to the population of India, there will be almost 423 million social network users in India from a little over 326 million users in 2018 (Keelery, 2020). This clearly demonstrates that with the increase in internet penetration, the digital population of India is growing fast and people are becoming accustomed to using the internet. This rapid increase in digital convergence

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has also given rise to the digital advertising industry. Companies have started recognizing the need of incorporating social media into their businesses and therefore, today, India's digital advertising industry is worth over 160 billion Indian rupees, and it is estimated to reach 560 billion rupees by 2023 (Keelery, 2020).

Social media marketing (SMM) refers to the promotional activities of the companies to promote their products and services on SNSs to reach a huge customer base online. Social media allows the exchange of various types of content in digital networks. Users create profiles on social media platforms or in a smartphone application designed and maintained by social media companies. User profiles can connect with other users' profiles and brands and create networks (Obar & Wildman, 2015). Using social media, brands can interact and communicate better with their consumers and intensify their association with them, can improve customer services, maintain customer relationships, inform about new products, offers, etc. Facebook, Instagram, Youtube, and Twitter are the most popular SNSs that are widely used by companies and celebrities in promoting themselves and their brands. Social media marketing is also more genuine in its interactions with people, attempting to reveal rather than manipulate the brand's profile (Karamian et al., 2015). Many businesses are turning to social media for marketing, customer service, external promotions, and internal employee communications as it becomes more convenient and relevant (Seo & Park, 2018). Even though social media marketing has not completely replaced traditional marketing, it is definitely an alternative tactic. Companies are increasingly using social media as a more cost-effective and efficient alternative marketing practice that directly communicates with customers, rather than traditional advertisement channels (Castronovo & Huang, 2012). The major difference between traditional media and social media is customer engagement. On average, users spend almost two to three hours on social media sites. Television ads nowadays do not play a much effective role in attracting customers due to easy channel swapping. This makes social media an effective and attractive platform for brands to promote their product and attract current and future prospective customers, build strong customer relationships, and ultimately build customer equity.

The ways of shopping and information generation of Indian consumers are changing rapidly and there has been a gradual shift from offline shopping to online modes of shopping. This has led to the emergent use of Social Media Marketing in India which provides a developing vision and strong association with online customers making it necessary for the companies to analyze its actual impact on the purchase intentions of the customers. If the brands in India want to grow in both the online and offline domains, the latest factors they should be concerned about are the effects of their social media presence on customers, how much value and brand loyalty they are able to create through their Social Media Marketing Activities (SMMAs) and the relationship they are able to build with the customers. These factors would help companies in building up their digital marketing strategies to not only retain existing customers but also increase their customer base by turning potential customers into active ones.

Many smartphone brands, in order to be competitive and with a perspective to reaching their current and future prospective customers directly and in a cost-effective manner have increased their presence on SNSs by creating their Twitter account, Facebook pages, YouTube channels, Instagram accounts, etc. On analyzing social media marketing activities of top smartphone brands it was analyzed that Apple having millions of followers on social media sites, has created a popular hashtag #shotoniphone campaign where it interacts with its customers by asking them to upload their pictures clicked with iPhone using this hashtag and the selected pictures and videos are posted on the official Instagram account of Apple creating a sense of belongingness in the customers and also taking continuous feedback and promoting its camera quality. Apple Music tweets daily about the latest collections and the Apple support page continuously guides its customers about data protection measures and the new features added.

Samsung, on the other hand having 161 million followers on Facebook, regularly post audio-video and written content on their social media accounts about the new products launched with the

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information on new features added, continuously updating its customers about promotional offers (cashback, EMIs, discounts). It also provides a direct link to purchase below these contents. It entertains and entices its customers to buy its smartphone by uploading amusing pictures shot by different models of Samsung smartphones. Samsung also posts creative ads on its Instagram account giving social messages and showing a brilliant example of building brand and relationship equity.

Likewise, another popular brand OnePlus also has the #shotononeplus campaign which helps them in promoting their excellent camera quality and editing features available. It gives updated and newest information about new product launches. OnePlus has aired the world's first-ever augmented reality (AR) launch of its new smartphone, OnePlus Nord, in July 2020 which it identifies as "in your house, on your palm!" The AR invites were sold to the fans at the price of 99 INR which gave its fans a privilege to experience the launch even before the official launch. Fans purchasing AR invites also had a chance to participate in the Launch Day Lottery with assured gifts.

Oppo, another leading brand in the smartphone industry attracts a huge customer base and engages its customers on its social media accounts by offering a chance to win an Oppo smartphone by sharing their experience with previous Oppo smartphones. It has the #QuizAlert and #SaleAlert campaign where customers are being enticed to make purchases at sale value or win an Oppo mobile by participating in its quiz. The strategy helps Oppo not only to attract customers' attention but also facilitate e-word-of-mouth about its different models of smartphones.

Social media marketing has been a prominent research topic for the past decade. This research contributes substantially to the extant literature in the field of social media marketing. There are a lot of researches that focus on a firm's perception of the effects of SMM; the outcomes of SMMAs by way of comments, likes, tags, etc. on sales value but very few research studies focus on the customer's perception about SMMAs of their favorite brands and the impact of the same on their purchase intentions. Also, those few kinds of research focusing on customer's perspectives are done mainly in the context of luxury fashion brands (Kim & Ko, 2012), the airline's industry (Seo & Park, 2018), banking (Chahal & Bala, 2017), and supermarket, and e-commerce industry (Yadav & Rahman, 2018), with very few or no study focusing on electronic gadgets or smartphones. There has been a paucity of attention given towards investigating the impact of SMMAs on various types of consumer responses in the extant literature (Yadav & Rahman, 2018; Kim et al., 2010). Though, the previous literature confirms the impact of SMMAs on customer loyalty via customer equity drivers (Yadav & Rahman, 2018); impact of SMMAs on customer responses via brand equity alone (Seo & Park, 2018; Godey et al., 2016); impact of the benefits of social media marketing on SNSs and brand trust on consumer behaviors (Kananukul et al., 2015); impact of SMMAs on customer equity via customer equity drivers (Kim & Ko, 2012); impact of customer equity drivers on future sales (Vogel et al., 2008), but the impact of SMMAs on purchase intention via customer equity drivers is not well analyzed in the recent literature of social media marketing. This research aims to bridge this gap by investigating the impact of SMMAs on purchase intention via customer equity drivers. According to the theory of planned behavior and reasoned action, it is very important to predict purchase intention as it is highly correlated with actual purchase behavior (Ajzen, 1991). Also, the literature has revealed that very few studies have been conducted on the concept of customer equity drivers in the Indian context (Chahal & Bala, 2017; Yadav & Rahman, 2017).

This study makes a substantial contribution to the existing body of knowledge of SMM by adapting the S-O-R model which links social media marketing activities to the purchase intention of the customers via customer equity drivers. The S-O-R model helps to understand the reason behind how people behave given environmental stimuli. In the current study, SMMAs act as external environmental stimuli which trigger the emotional and cognitive state of consumers (here, customer equity drivers) which ultimately brings out certain responses, here, purchase intention of the customers (Yadav & Rahman, 2018).

The value of one customer is not limited to what he or she initially spends but is worth a lot more than that. Thus, it becomes necessary for businesses and brands to measure the true value of

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customers and the influence of social media marketing activities on them (Kim & Ko, 2012). Also, the validity of the perceived SMMAs scale needs to be confirmed in diverse industry contexts (Yadav & Rahman, 2018). Thus, this research aims to study the impact of social media marketing activities on customer equity drivers (value equity, brand equity, and relationship equity) and purchase intentions of Indian customers in the context of the smartphone industry.

BACKGROUND

The S-O-R Model

The S-O-R model was developed by Mehrabian and Russell, (1974) by studying consumer behavior and further developed by Jacoby, (2002). The current study adapts the S-O-R model which links social media marketing activities to the purchase intention of the customers via customer equity drivers. Here, SMMAs act as external environmental stimuli which prompt the emotional and cognitive state of consumers (here, customer equity drivers) which ultimately brings out certain responses (here, purchase intention of the customers) (Yadav & Rahman, 2018). Prior studies applied the S-O-R model majorly in the e-commerce industry where characteristics of e-commerce environment were studied as stimuli (Yadav & Rahman, 2018); the inner emotional and cognitive states of the customers which encompass their experiences and insights (Jiang et al., 2011) were taken as organism and the responses were studied by predicting consumer behavior in terms of their purchase behavior, customer loyalty, and e-WOM in the e-commerce (Sautter et al., 2004). We justify the application of the S-O-R model in the present study with the help of various prior studies. Yadav & Rahman, (2017) found the impact of perceived social media marketing activities on customer loyalty via customer equity drivers using the S-O-R model in the e-commerce industry. Zhang et al., (2014) adapted the S–O–R model to investigate the impact of technological social commerce characteristics on the customers' virtual experience and social commerce intention. The findings of the prior studies reveal the significance of the S-O-R model in predicting the responses of the customers in terms of their purchase behavior, customer loyalty, and behavioral intentions.

Social Media Marketing Activities as Environmental Stimuli (S)

According to (Kaplan & Haenlein, 2010) "Social Media is a group of Internet-based applications that build on the ideological and technological foundations of Web 2.0, and that allow the creation and exchange of User Generated Content." They take numerous forms like weblogs, social blogs, micro-blogging, wikis, podcasts, videos, photos, ranking, and social bookmarking. Social media has revolutionized the way brands create their content, communicate with their customers and has shifted the power from marketers to consumers to build brand image (Tsai & Men, 2013). Brands and businesses are actively using social media for marketing and advertising purposes. Social Media Marketing (SMM) is defined as a process through which companies develop, communicate and furnish online marketing offerings through social networking platforms to develop and retain stakeholder's relationships that enhance the value of stakeholders by encouraging engagement, sharing knowledge, providing personalized buying recommendations and creating word of mouth among stakeholders about current and trending offerings (Yadav & Rahman, 2018). Businesses may use SMM to establish relationships with loyal customers, influencing their perception of the product, publishing their information, and even learning about and from their customers (Karamian et al., 2015). The different components of SMMAs have been identified and used by various researchers in their studies in different settings. Initially, (Kim & Ko, 2012) identified five components of perceived SMMAs in their study and applied them to luxury fashion brands. Ural & Yuksel, (2015) adopted the same components of perceived SMMAs and tested them on the Skoda brand. Algharabat, (2017) in their research investigated the link between the same five perceived SMMAs and brand love. Yadav & Rahman, (2018) in their research paper focused on the same five perceived SMMAs concerning

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the e-commerce industry. Godey et al., (2016) applied the same components of SMMAs to test their relation with brand equity and customer response. Seo & Park, (2018) in their research paper modified the SMMAs developed by Kim & Ko, (2012) to entertainment, interaction, trendiness, customization, and perceived risk and investigated their effects on customer response and brand equity in the airline industry. Further, prior studies like Yadav and Rahman, (2018) and Zhang and Benyoucef (2016), have also perceived SMMA to be an environmental stimulus in the S–O–R model. This paper will focus on the six dimensions of social media marketing activities- Entertainment, Trendiness, Informativeness, Interaction, e-Word of mouth, and Personalization as developed by Kim & Ko, (2012) and validated by many other researchers.

Entertainment: The literature reveals that the hedonistic value of advertising like perceived entertainment plays a vital role in predicting consumers' responses to advertising (Shin & Lin, 2016). Entertainment refers to the fun that the advertising media provides to the users (Xu et al., 2009). It is considered to be the crucial component of social media that generates user engagement, positive emotion, and usage intentions of the users (Kang, 2005).

Trendiness: Trendiness refers to the dissemination of trendy and latest information (Godey et al., 2016). It is the perception of the customers about the extent to which social media provides the latest content (Yadav & Rahman, 2018). Trendiness is one of the important components of social media that provides the customers with the latest and trendy content about the products and brands.

Informativeness: Informativeness refers to the informational enrichment created by advertising content which increases the understanding of the products and services being offered (Ducoffe, 1996). It is the perception of the customers regarding the information provided on social media being accurate, useful, and comprehensive. Customers make informed decisions regarding the purchase by considering the information present on the internet and social media sites in the form of comments, reviews, ratings, and features (Yadav & Rahman, 2018).

Interaction: Godey et al., (2016) defines interaction as the exchange of information and opinion with one another. Brands are no longer the sole source of brand communications with customers, rather, social media provides customers with the platform to talk to other users of the products (Yazdanparast et al., 2016). According to Muntinga et al., (2011), social interaction refers to individuals who engage in brand-related social media platforms to meet like-minded people, connect with them, and discuss specific products/brands.

e-Word of mouth: Any positive or negative brand-related information made by consumers or passed on among other consumers via the Internet is referred to as e-WOM (Chu & Sung, 2015). In social media usage, e-WOM refers to consumer-to-consumer communications about brands (Muntinga et al., 2011). Social media is an ideal platform for customers to produce and share brand-related information with their friends, peers, and other contacts without any restrictions (Godey et al., 2016).

Personalization: Personalization refers to providing tailored information to customers based on their demographics, preferences, and geographic details which are being tracked automatically by the system (Lee et al., 2015; Xu, 2006). Yadav & Rahman, (2018) define personalization as the perception of the customers about the degree to which social media provides tailored services to satisfy customers' needs and preferences.

Social Media Marketing Activities of Smartphone Brands

According to research conducted by Socially Powerful in 2018, almost all smartphone brands have an active presence on social media mainly Instagram, YouTube, and Facebook, and also have a scope for enhancement. The top smartphone brands demonstrated the best quality of interaction with customers through Instagram stories, innovative content sharing, polls, creative use of hashtags, etc. The brands that do not focus on their social media marketing like not posting regularly, no hashtags, and interactions lose their followers on SNSs. The top brands also collaborate with YouTube tech influencers MKBHD and Unbox Therapy during launch phases to maximize their engagement across all platforms.

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Customer Equity and its Drivers as Customers' Internal States (O)

Inner states refer to the emotional and cognitive feelings of customers which encompass their experiences and insights (Jiang et al., 2011). The cognitive states relate to the mental process concerning retention, acquisition, processing, and retrieval of information whereas the emotional states refer to the sentiments like pleasure and arousal (Eroglu et al., 2001) which consumers attain from environmental stimuli. Rust et al., (2004) define customer equity as, "the total of the discounted lifetime values summed over all of the firm's current and potential customer". They also claimed that customers and customer equity have more weightage to many firms than the brands and brand equity and there has been a visible shift from product-centered thinking to customer-centered thinking. The worth of the customer to a firm is not just the revenue from each transaction but the cumulative gain that the customer will provide to the firm for the span of his relationship with the firm (Kumar & George, 2007). Customer equity amounts to both financial and non-financial benefits to a company being invested by a customer during the span of an exchange relationship (Chae & Ko, 2016). Customer equity is a behavioral variable (Kim & Ko, 2012)

Customer equity has been classified into three key drivers- brand equity, relationship equity, and value equity (Lemon et al., 2001). These drivers have been recognized in several studies as the RLZ approach i.e., Rust, Lemon, and Zeithaml approach. The RLZ approach proposed that "to maximize customer equity, customer-level evaluations of a company and/or its product(s) can be categorized into three dimensions: the value of the product or service, brand issues of the product, and relational aspects of the product" (Kumar & George, 2007). The RLZ approach has been used by Lemon et al., (2001) in the airline industry. The same approach has been used by various authors for luxury brands in the fashion industry in the context of SMM (Godey et al., 2016; Kim & Ko, 2012; Kim et al., 2010). Further, the individual drivers of customer equity have also been investigated by various authors in the context of SMM-brand equity in the e-commerce industry (Yadav & Rahman, 2018) brand equity, and customer response in the airlines industry (Seo & Park, 2018). The RLZ approach has also been used by Vogel et al., (2008) to study the effects of the three equity drivers on consumers' loyalty intentions and their future purchase behavior.

Value equity (VE) is described as the objective evaluation of the brand's worth by the customer which relies on their perception of what is given up and what is being received (Lemon et al., 2001). It is based on the customer's benefit-cost assessment (Vogel et al., 2008). Cost involves efforts, time, and the money spent to acquire benefits like quality, convenience, satisfaction, and worth.

Brand equity (BE) is described as an intangible, subjective, and emotional assessment of the brand that reaches beyond its objectively perceived worth (Lemon et al., 2001; Vogel et al., 2008). Brand equity is the most vital element of customer equity and has been studied by various researchers individually and with other drivers as well. A strong level of brand equity offers high-quality services to customers (Mokha, 2021) and thereby contributes to building stronger customer equity. According to Aaker (1996), "brand equity consists of five dimensions namely perceived quality, brand loyalty, brand awareness, brand association, and other propriety assets".All these dimensions considerably reflect the strength and value of the brand and contribute to customer equity.

Relationship equity (RE) or retention equity is described as the propensity of customers to remain in a relationship with the brand which is beyond the subjective and objective evaluation of the brand (Vogel et al., 2008). It includes key elements such as special recognition, loyalty programs, community and knowledge-building programs (Lemon et al., 2001). It becomes necessary to build powerful customer relationship equity as the brand might grow weaker than before because of the availability of alternatives (Kim & Ko, 2012). However, as the marketing approach shifts from transactional to relationship-based marketing, having a strong BE and achieving VE is no longer enough to retain customers. The current situation requires the building of a strong foundation of relationships among customers and businesses, and this foundation is RE (Yadav & Rahman, 2018).

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Purchase Intentions as Customers' Response (R)

The response component of the S-O-R model exhibits the consequences of environmental stimuli via internal states in the form of consumer behavior. Here, the outcome of environmental stimuli is taken as the purchase intentions of the customers. Purchase intentions refer to the combination of consumer's interest, engagement, and the likelihood of buying a product (Kim et al, 2010). It is based on the consumer's perception and attitude towards the brand assuming to be the customer's future behavior based on their attitude (Kim & Ko, 2012). Purchase intention is an attitudinal variable that accounts for future purchases whereas customer equity is a behavioral variable that accounts for actual purchasing records (Kim & Ko, 2012). According to the theory of planned behavior, which postulates that consumers' intentions are a very strong predictor of their actual behavior and there exists a strong connection between the consumer's attitude to their actual behavior. Thus, purchase intentions can be used to estimate the future purchase behavior of the customers.

HYPOTHESES DEVELOPMENT

Impact of SMMA on Customer Equity Drivers

As of today, communication with the customers and promotion of the products and services by the brands has been shifted from traditional marketing to social media advertising. The presence of brands on social media attracts consumers' attention towards the brands and tends to increase sales and profits of the brands. Entertainment, Interaction, Trendiness, Word of mouth, and Personalization are the key elements of SMMA which are expected to have a significant positive impact on customer equity drivers (value equity, relationship equity, and brand equity) as supported by the literature. Yadav & Rahman, (2018) in their study found the significant and positive impact of perceived SMMAs of the e-commerce industry on all the drivers of customer equity. Kim & Ko, (2012) examined the impact of SMMAs of luxury fashion brands and found their significant positive impact on customer equity drivers and purchase intention. Ural & Yuksel, (2015) found the mediating roles of customer equity drivers between social media marketing efforts and purchase intention.

The majority of customers are value-conscious and demand products that are available at a reasonable price, with minimum effort, and without compromising quality and features. They frequently visit SNSs and other social commerce platforms in search of goods delivering high adequate customer value (Yadav & Rahman, 2018). Social media communications strongly enhance brand equity (Godey et al., 2016). Seo and Park, (2018) found the significant positive impact of SMMAs of the airline industry on two types of brand equity: brand awareness and brand image. Karamian et al., (2015) established the effects of SMMAs on brand equity formation from the customer's perspective. A company can enjoy powerful brand and value by exceeding customer expectations, thereby, retaining present customers and attracting future prospective customers (Yadav & Rahman, 2018). Businesses are using social media marketing to build relationships with customers and influence their perceptions of the product (Karamian et al., 2015). Previous researchers also studied the impact of SMMA on customer equity and found a significant impact of SMMA on value equity, relationship equity, and brand equity in the case of luxury fashion brands (Kim & Ko, 2012). Godey et al., (2016) established the significant positive impact of SMM efforts on customer equity drivers.

Therefore, we hypothesize:

H1: Perceived social media marketing activities have a significant positive effect on value equity.
H2: Perceived social media marketing activities have a significant positive effect on relationship equity.
H3: Perceived social media marketing activities have a significant positive effect on brand equity.

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Impact of Customer Equity Drivers on Purchase Intention

Further, it is necessary to evaluate the purchase intentions of the customers to predict future sales and it has already been proved in the literature that intentions are a strong predictor of customers' actual behavior. Also, it has already been established in the theory of planned behavior that attitude precedes behavior and here customer equity drivers are attitudinal variables whereas purchase intention is a behavioral variable. Kim and Ko (2012) reveal that SMMA is positively related to future purchase behavior via customer equity drivers. Yadav and Rahman,(2018) also found the impact of SMMAs on consumer loyalty via customer equity drivers. Ural and Yuksel, (2015) established the relationship between social media marketing efforts and purchase intention via customer equity drivers. Therefrom the literature, the following hypotheses have been drawn.

H4: Value equity relates positively to purchase intentions.
H5: Relationship equity relates positively to purchase intentions.
H6: Brand equity relates positively to purchase intentions.

METHODOLOGY

Data Collection

To test the hypotheses of this research study, the population of the study comprised of active social media users from India who were following the smartphone brands on different SNSs. Since, this research study focused on smartphone brands, respondents were restricted to those having a smartphone of value above Rs. 10,000 to eliminate cheap smartphone brands. The data was collected by way of an online survey which was floated on SNSs like Facebook and LinkedIn and was also sent personally to the followers of different smartphone brand's on SNSs. The sampling technique used was simple random sampling. Data was collected on an online survey questionnaire from 11th October 2020 till December 2020. The sample size for the study is determined using the measure suggested by Bentler and Chou (1987) i.e., 5 observations for every item is considered to be an adequate sample size in the case of Structural Equation Modelling. There are 37 items in this study making a requirement of a minimum of 185 respondents. 343 valid responses were finally received which is more than a sufficient sample size as suggested by Bentler and Chou (1987) with 182 male respondents and 161 female respondents with no missing value.

Measures

For the purpose of this research study, the questionnaire items were developed based on a literature review and were modified by expert opinion and pilot testing. The questionnaire surveyed smartphone brand's SMMAs with six elements- Entertainment, Personalization, Informativeness, Interaction, Trendiness, and Word of mouth covered by 21 items (3-4 each) as developed by Kim and Ko, (2012). Customer equity drivers- value equity (4 items), relationship equity (4 items), and brand equity (5 items) were used from the scale developed by Rust et al., (2004); Vogel et al., (2008) and Kim and Ko, (2012). A 3-item purchase intentions scale was adopted from Maxham et al. (2001). All these items were measured using a 5-point Likert scale which ranged the responses from 5 ("strongly agree") to 1 ("strongly disagree").

Pre-testing and Pilot testing were also performed before proceeding to the final data collection. Pre-testing was done on 20 participants to get their opinion on the questionnaire's design, sequence, language, and content. The respondents proposed a few improvements to the questionnaire's layout, which the researchers have implemented. Thereafter, pilot study was undertaken on 30 respondents to check the internal consistency of items using Cronbach's alpha. The value of Cronbach's alpha was more than the suggested value of 0.70 meaning thereby, the items were reliable and further research could be carried out. To reduce non-response bias, the questionnaire designed was made in

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the English language; short but comprehensive and with a description of the purpose of the research to stimulate respondents to fill the questionnaire.

DATA ANALYSIS

SPSS 23.0 and AMOS 23.0 statistics package programs were used to analyze the collected primary data and to test the hypotheses. With SPSS 23.0, descriptive analysis was performed to analyze the results of the preliminary test and to analyze the demographic characteristics of the sample. The reliability of the instrument was tested using Cronbach's alpha. Using AMOS 23.0, confirmatory factor analysis was conducted to test the validity of each construct, while the structural model was developed to test hypotheses framed in the research study.

Demographic Analysis

Based on the results of frequency analysis, among the total of 343 respondents, the sample consists of 182 males (53.1 percent) and 161 females (46.9 percent). The total percentage of respondents belonging to the age group of up to 25 years was 54.8 percent, whereas 42 percent belonged to the age group of 25 to 40 years and only 3.2 percent of the respondents were above 45 years of age. This shows that the results of this study can be generalized to age groups of 18 to 40 years of age. Concerning the educational qualifications of the respondents, 56.6 percent were postgraduates and 40.8 percent were graduates. Regarding average household income levels, 32.9 percent of incomes were 10 lakhs INR and above; 30 percent were between 5 lakhs INR to 10 lakhs INR; 21.6 percent of the incomes were between 3 lakhs INR to 5 lakhs INR and the remaining 15.5 percent of incomes were up to 3 lakhs INR. The employment status of the respondents shows that 50.1 percent of the respondents were working professionals and 48.4 percent of them were students and the remaining 1.5 percent of them were homemakers.

Reliability Test

Internal consistency is determined using the Cronbach alpha measurement to assess the reliability of the different items included in the survey (Streiner, 2003). Cronbach alpha is a measure that is derived from a pairwise correlation between the items selected. The value of Cronbach's alpha varies between 0 and 1 and a construct should have a Cronbach's alpha score of at least 0.6 to 0.7 to have a satisfactory degree of reliability and more than 0.7 to have a good degree of reliability (Taber, 2018). The reliability of an instrument is required to estimate scale validity and also to access the quality of the research instrument. All the constructs represented in the measurement model qualify the reliability test and have internal consistency with the decent level of Cronbach's alpha values as shown in Table 1 with 0.926 for "SMMA"; 0.895 for "Brand Equity"; 0.832 for "Relationship Equity"; 0.895 for "value equity" and 0.951 for "Purchase Intentions". All the Cronbach's alpha values for different constructs are more than 0.8 and 0.9 implying a very good and extraordinary degree of internal consistency.

Confirmatory Factor Analysis

The proposed measurement model was evaluated by examining convergent and discriminant validity with the help of CFA using AMOS 23. Before examining the inter-relationships of the constructs in the structural model, the aim of evaluating the measurement model is to confirm the existence of the required degree of constructs' reliability and validity (Fornell & Larcker, 1981). The finalized measurement model comprised of six dimensions of SMMA namely Entertainment with 3 items; Personalization with 4 items; Informativeness with 4 items; Interaction with 4 items; Trendiness with 3 items and e-Word of mouth with 3 items. The other constructs are the drivers of customer equity-Brand equity, Relationship equity, Value equity, and finally, the purchase intentions of the customers.

Composite reliability (CR) offers a more reflective method of estimating the overall reliability of the instrument and also ascertaining convergent validity (Hair et al., 2010). For a scale to be

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sufficiently reliable, the value of CR should be more than 0.7 (Nunnally & Bernstein, 1994; Fornell & Larcker, 1981). The CR values as shown in Table 1 indicate CR of "SMMA" is 0.849; "Brand equity" is 0.898; "Relationship equity" is 0.835; "Value Equity" is 0.898 and "Purchase intentions" is 0.952. This implies that the constructs in the measurement model are considerably reliable as the CR values of all the constructs have more than satisfactory CR values.

The convergent validity is used to determine the degree to which the items of the construct that are theoretically related are actually related. The convergent validity can be estimated through standardized construct loadings and average variance extracted (AVE). The recommended values for convergent validity are more than 0.50 for standardized construct loadings to their observed variables as well as for AVE and more than 0.70 for CR of all the constructs (Hair et al., 2010). Table 1 shows that the loadings of the observed variables are within the range of 0.7 to 0.95 except for four items. However, the average of standardized construct loadings for each item came out to be more than 0.7. Thus, all the loadings are considered to be within the range. AVEs, as shown in Table 1 for each construct, are more than 0.5 and the CR values of all the constructs are more than 0.7. Therefore, the presence of significant convergent validity can be assured.

The discriminant validity of a construct is measured by how distinct it is from other constructs (Hair et al., 2010). The scale can be said to have adequate discriminant validity if the square root of the AVE for each construct is greater than the correlation between the constructs and all the values of AVE should be greater than maximum shared variance (MSV). Table 1 shows that all the values of AVE for each construct are more than the values of MSV for each construct. Further, Table 2 shows that the square roots of AVE marked in bold are higher for diagonal constructs as compared to non-diagonal constructs. These findings suggest that each construct is highly interrelated with its items as compared to other constructs in the measurement model. Therefore, the discriminant validity of the scale can be assured.

Finally, the overall validity of the model was ascertained through model fit indices such as goodness of fit index (GFI); normed fit index (NFI); comparative fit index (CFI) and Tucker-Lewis Index (TLI), and root mean square of error approximation (RMSEA) (Hair et al., 2010). The test statistics revealed a satisfactory fit with the data. The model fit indices were $\chi 2/df = 1.709$; CFI=0.95; GFI=0.86; TLI=0.946 NFI= 0.89 and RMSEA= 0.046. The acceptable values for goodness-of-fit indices are $\chi 2/df < 3$; CFI, GFI, TLI, NFI > 0.9 and RMSEA < 0.08 CMIN/DF < 5 (Gefen et al., 2000; Gefen and Keil, 1998). All the values are within the acceptable range except GFI and NFI. The value of NFI is below the threshold value but it is very close to the threshold value, so it still represents a satisfactory model fitness. Also, GFI is below the threshold value (0.9), but it still meets the requirement suggested by Baumgartner and Homburg (1995): the value is acceptable if above 0.8. Therefore, Table 3 represents that the measurement model is a good fit.

Common Method Bias

To address the concern of survey-based research, common method biasness test was applied following the work of Serrano et al., (2018). The researchers employed the common latent factor (CLF) method to extract the common variance from all of the individual items of the model by connecting them to a common latent factor. If the difference between the standardized regression weights of the CFA model without the CLF method and the standardized regression weights of the CFA model with the CLF method is less than 0.20, then it represents that the data is free from common method biases. The result of this study revealed that the difference between the two was less than 0.20. thus, it can be said that the data is not contaminated with common method biasness.

Structural Model

To test the hypotheses of the study, the structural model was developed and tested according to Anderson and Gerbing (1988). The model was tested by standardized coefficients and other fit statistics. The model fit was satisfactory with the results shown in Table 3. The Chi-square statistics

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Table 1. Reliability and Item loadings

Construct	Items	Standard Factor Loading	Cronbach's Alpha	Composite Reliability (CR)	Average Variance Extracted (AVE)	Maximum Shared Variance
Social media marketing Activities			0.926	0.849	0.510	0.288
¨Entertainment	ENT1	0.72	0.833			
	ENT2	0.91				
	ENT3	0.76				
¨Personalization	PER1	0.88	0.877			
	PER2	0.92				
	PER3	0.62				
	PER4	0.78				
¨Informativeness	INF1	0.88	0.937			
	INF2	0.94				
	INF3	0.88				
	INF4	0.85				
¨Interactivity	INT1	0.61	0.863			
	INT2	0.78				
	INT3	0.91				
	INT4	0.84				
¨Trendiness	TRE1	0.81	0.866			
	TRE2	0.82				
	TRE3	0.86				
¨e-WOM	eWOM1	0.89	0.857			
	eWOM2	0.79				
	eWOM3	0.77				
Brand Equity	BE1	0.85	0.895	0.898	0.638	0.073
	BE2	0.86				
	BE3	0.78				
	BE4	0.74				
	BE5	0.76				
Relationship Equity	RE1	0.56	0.832	0.835	0.568	0.085
	RE2	0.62				
	RE3	0.91				
	RE4	0.86				
Value Equity	VE1	0.8	0.895	0.898	0.687	0.286
	VE2	0.83				
	VE3	0.78				
	VE4	0.89				
Purchase Intentions	PE1	0.95	0.951	0.952	0.868	0.288
	PE2	0.95				
	PE3	0.89				

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Table 2. Discriminant Validity

	Value Equity	Social media marketing activities	Brand Equity	Relationship Equity	Purchase intentions
Value Equity	**0.829***				
Social media marketing activities	0.397	**0.697***			
Brand Equity	0.102	0.220	**0.799***		
Relationship Equity	0.201	0.291	0.106	**0.754***	
Purchase intentions	0.535	0.537	0.271	0.242	**0.932***

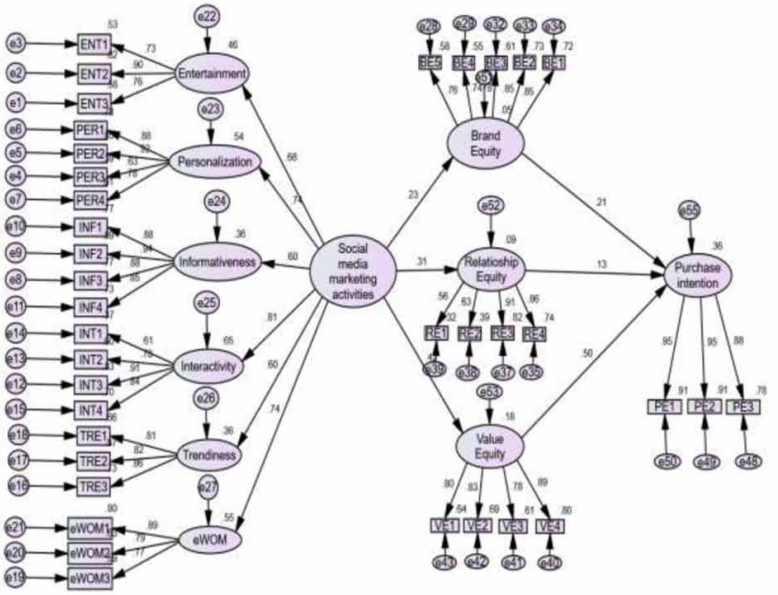
Table 3 Summary of goodness-of-fit indices

Model Fit Index	Chi-square/ Degree of freedom	CFI	GFI	TLI	NFI	RMSEA
Measurement Model	1.709	0.95	0.860	0.946	0.891	0.046
Structural Model	1.758	0.947	0.885	0.942	0.89	0.047

(χ2= 1084.6 and df= 617) was at a significant level (p=.000) and the other fit indices were within or very close to acceptable standards (χ2/df= 1.758; CFI=0.947; GFI=0.885; TLI=0.942 NFI= 0.89 and RMSEA= 0.047) as per the ranges suggested by Hair et al. (2014). After achieving satisfactory model fitness, hypotheses were tested using SEM. Out of the six hypotheses, five were significant at p<0.01 as shown in Table 4. All the paths between SMMA and customer equity drivers were significant at p<0.01. Table 4 shows that SMMA of smartphone brands have significant positive effects on Brand Equity (β= .231, p< .01); Relationship Equity (β= 0.306,, p< .01) and Value Equity (β= 0.422,, p< .01). In other words, SMMA like Entertainment, Trendiness, Informativeness, Interaction, Word of mouth, and Personalization positively influence the customer equity drivers. Concerning the relationship between customer equity drivers and purchase intentions, brand equity shows significant positive effects on purchase intentions with β= 0.214 and p< .01; value equity shows significant positive effects on purchase intentions with β= 0.501 and p< .01. However, we found no significant relationship between relationship equity and purchase intentions of the customers.

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Figure 1. Results of structural equation model



$\chi 2$= 1084.6, df= 617, p=0.000

$\chi 2$/df= 1.758; CFI=0.947; GFI=0.885; TLI=0.942 NFI= 0.89 and RMSEA= 0.047

Table 4 Hypotheses assessment results

Hypothesis	β	p values	Test Results
Brand Equity<-- SMMA	0.231	0.005	Supported
Relationship Equity<-- SMMA	0.306	0.007	Supported
Value Equity<-- SMMA	0.422	0.005	Supported
Purchase Intentions<-- Brand Equity	0.214	0.006	Supported
Purchase Intentions<-- Relationship Equity	0.133	0.07	Not Supported
Purchase Intentions<-- Value Equity	0.501	0.007	Supported

CONCLUSIONS AND IMPLICATIONS

The study here aims to analyze the effects of social media marketing activities (entertainment, trendiness, informativeness, interaction, word of mouth, and personalization) on customer equity drivers (brand equity, relationship equity, and value equity) and the purchase intentions of the customers in case of smartphone brands. The findings of the study contribute to prior literature by providing a holistic framework that demonstrates how SMMAs influence customer equity drivers and finally purchase intentions of the customers. Firstly, the relationship between SMMAs and customer equity drivers was examined and the findings of the study revealed that SMMAs positively influence all the customer equity drivers (brand equity, relationship equity, and value equity). Even though the previous

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study proposed by Ural & Yuksel, (2015) revealed that there is no significant impact of SMMAs on brand equity, this study revealed a significant impact of SMMAs on brand equity, relationship equity, and value equity which was found to be consistent with the previous literature (Godey et al., 2016; A. J. Kim & Ko, 2012; Yadav & Rahman, 2018). SMMAs provide customers with unique and distinctive value equity that is not being delivered by traditional advertising mediums. The customers enjoy personalized, trendy, and reliable information at minimum cost, time, and effort. SMMAs constantly interact with their customers by binding them with new trends, entertaining advertisements provide their loyal customers with huge discounts and offers. SMM provides customers with a huge virtual space where they can communicate within themselves and with the brand. The authentic reviews about the products are readily available on the social media pages of the brands by way of comments, posts, forums, etc. This clearly shows that SMMA influences value equity, relationship equity, and brand equity. Thus, hypotheses (H1, H2, and H3) were accepted.

Secondly, SMMA of smartphone brands comprises of six elements namely entertainment, trendiness, informativeness, interaction, word of mouth, and personalization. However, in the previous study proposed by Kim and Ko, (2012) and Godey et al., (2016) SMMAs comprised of five dimensions namely entertainment, interaction, customization, trendiness, and word of mouth. In another study by Yadav and Rahman, (2018) SMMAs comprised another five dimensions namely interactivity, informativeness, word of mouth, personalization, and trendiness. In this study, all the six elements studied are found to have a wholesome impact on customers' purchase intentions as they provide superior value to the customers by providing them with the latest, trendy, and authentic information about new product launches, discounts, and offers; entertainment by launching innovative advertisements; interactivity by providing them with a huge platform where customers can directly communicate with the brand; benefits of personalization and also facilitates e-WOM by allowing customers to read and write reviews and recommendations of each other on social media platforms. The study also revealed that interactivity was the most important element of social media marketing activities and personalization and e-WOM were the second most important elements.

Finally, out of the three customer equity drivers, only two were found to have a significant positive relationship with purchase intentions. There exists a significant positive association between value equity and purchase intentions of the customer as the reduced cost, time, and efforts facilitated by SMM help to boost the purchase intentions of the customers. Thus, hypothesis (H4) was accepted. Also, brand equity significantly influences the purchase intentions of the customers. Constant and innovative interaction of the brands with their customers by way of social media platforms plays a huge role in influencing brand equity and finally purchase intentions of the customers. Even though the previous study proposed by Ural & Yuksel, (2015) revealed that there is no significant impact of brand equity on the purchase intention of the customers, this study revealed significant impact of brand equity on the purchase intention of the customers which is consistent with the results of previous literature (Kim & Ko, 2012). Thus, hypothesis (H6) was also accepted. Surprisingly, the study found no significant positive relationship between the purchase intention of the customers and relationship equity which is different from the results of Ural and Yuksel, (2015) but confirms the results of Kim and Ko, (2012). Therefore, hypothesis (H5) could not be accepted.

The study has several implications for practitioners. SMMA has a significant impact on customer equity drivers as well as purchase intentions. The cost of losing one customer is way more than the cost of attracting one new customer. Thus, managers should focus on retaining the existing customer base as well by increasing the activities that enhance customer equity drivers. Our research demonstrates the impact of social media marketing activities on customer equity drivers. SMMA creates value for the customers by updating them with timely information at a low cost and with minimum time and effort. It builds a strong relationship with the customers by constantly interacting with them through social media pages about new launches, offers, discounts, exchange offers, etc. SMMA builds a great amount of brand equity as well by boosting brand awareness and recognition. All these equity drivers increase the purchase intentions of the customers. Thus, managers should keenly take the opportunity

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of these social media websites and ensure their presence on them. They should post good content on their pages and keep reminding customers about their brand. Managers should entice customers by creating interesting conversations about their brand. Managers should make efforts to make the customer follow their brand on social media sites. The posts and contents created by the brands should have a link below to facilitate an easy purchase mechanism which will also boost the sales of the product. Managers should use trendy measures like hashtags (#) and ad campaigns to gain popularity and create traffic. Managers can start building conversations with their customers to know their opinions and expectations about the brand. All these entertaining and interesting SMMA would help the managers create brand equity, value equity, relationship equity, and purchase intentions on the minds of their customers about their product which would help them to boost their sales.

LIMITATIONS AND FUTURE RESEARCH

The limitations and future research directions are as follows. First, this research focuses on the smartphone industry only and hence cannot be generalized to other industries. Future research should validate the present study in another industrial context. Second, this study cannot be generalized as it considers only Indian customers. India is still a developing country where internet availability and smartphone users are increasing at a high pace. However, still, a majority of the Indian population does not own a smartphone or the internet. Also, Indian customers are more privacy-sensitive. Future research should validate the results of this research in other countries as well. Third, the present study focused on the impact of SMMA on the purchase intentions of the customers. Future studies should examine the impact of SMMA on actual sales and actual customer equity value.

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Social Media Influencer Endorsement:
How Attitude Towards Endorsement Affects Brand Attitude

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ABSTRACT

This research aims to examine the effect of social media influencer endorsements on brand attitude. Further, this research investigates the mediation effect of attitude towards endorsement on brand attitude. Based on previous studies of endorsements, factors that affect the brand attitude were chosen. The factors are trust and expertise from the source credibility model, follower ratio, brand congruency with the influencer personality, and sponsorship disclosure. Primary data was collected using a structured survey of 220 respondents. The analysis was performed using Preacher and Hayes Mediation Model 4 and multiple linear regression. From the results of the path and mediation analysis, it is concluded that there is a mediation effect on brand attitude by expertise, brand congruency, and follower ratio. The research helps digital marketers to choose collaborations with social media influencers more effectively to create a positive brand attitude.

KEYWORDS

Attitude Towards Endorsement, Brand Attitude, Brand Congruency, Expertise, Influencer Follower Ratio, Social Media Influencer Endorsements, Source Credibility, Sponsorship Disclosure, Trustworthiness

1. INTRODUCTION

Social Media has a reach to the world's population with no demographic and physiographic limitations (Alrubaian et al., 2019). Millions of people use social media platforms to connect, learn, share, and grow. In India from 2015, there has been a significant jump in the number of social media users from 142.23 million social media users to 351.4 million users in 2019. Influencer marketing is said to grow to an exponential $9.7 billion in the year 2020, and companies are shifting their strategy from mega influencer to micro-influencers (Influencer marketing hub, 2020). Globally, consumers are adapting, which represents a challenge to adapt communication strategies in this changing digital world. Marketers focus their campaigns around an intersection at a digital, local, and social level.

The evolution of social media started with Wikipedia's launch in 2001, after which Napster was introduced in 2002, which was a Kuala Lumpur, based social gaming site. MySpace was introduced in 2003. One of the most famous social media platforms, Facebook was introduced in 2004 at the same time, Orkut was launched, which was more popular among the Brazil and Indian population. With currently 2 billion users of Youtube, the video streaming platform was established in 2005.

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Twitter, the microblogging and social networking site, was established in 2006 with currently 321 million active users. Instagram was introduced in 2010 as a photo-sharing site and is now the most popular social networking (Irani, 2016).

Marketers have used these high rates of social media usage to their advantage to build positive brand evaluation using influencer marketing. Influencer marketing is a form of endorsement in which companies utilize the users on social media with a high following who are known as influencers to promote their products and brand (Kádeková & Holienčinová, 2018). This concept is then used by marketers to rope in Social Media Influencers (SMIs) for collaborations and sponsored posts shared by the influencer. These SMIs significantly impact brand evaluations and brand attitude (De Veirman et al., 2017). With the rise of influencers, the question also arises about "whom to believe in an environment where anyone can say anything about anyone" these mainly occur due to problems such as fake followers and fake news (Van Der Heide & Lim, 2016). This leads us to the concept of source credibility; this concept plays a very crucial role in influencer marketing, which helps analyze the attitudes towards the endorsement, which mediates the attitude towards the brand (Seiler & Kucza, 2017). An SMI should also fit well with the brand and the reason being an influencer who does not portray an image on the same lines as the brand might lead to a negative impact on the brand, so along with source credibility, brand congruence must be also tested. The research focuses on sponsorship disclosure (i.e.) if a consumer might react differently to a paid endorsement compared to an unpaid one. This research also focuses on social media influencers who have a large following through content creation on Instagram. This emerging trend of influencer marketing has shifted customer's trust from celebrity endorsers who read from scripted messages to SMIs with a genuine audience (Schouten et al., 2020). The research gap of a combination of crucial social credibility cues such as SMI follower count (Dhanesh et al., 2019), sponsorship disclosure and SMI-brand congruency (Breves et al., 2019) and primary dimensions of the source credibility model (expertise and trustworthiness) are taken into account and how they affect attitude towards brand while mediated by the attitude towards endorsement. Understanding the mediating effect of attitude towards endorsement via the factors mentioned above would help marketers understand and analyze an influencer before bringing them on board.

2. LITERATURE REVIEW

2.1 Endorsements

According to the Federal Trade Commission guidelines, an endorsement is an advertising message that consumers will be more likely to believe as it comes from someone other than the sponsoring party (Tobin, 1973). Endorsements are among the most popular strategies in which marketers have promoted their brands, services, and products. Celebrity endorsements are used to transfer the impact and influence a celebrity has onto the brand, which would improve the customer's intent to use or purchase the product endorsed (Atkin and Block, 1983). It has been found that celebrity endorsements play a significant role in positively impacting brand attitude, the credibility of the advertisement, and purchase intentions (Erdogan, 1999) and improving brand recall and recognition (Misra & Beatty, 1990). However, time and again, researchers have cautioned us regarding how endorsements can be a double-edged sword; that is, if a celebrity does not have brand fit, then this kind of pairing can prove to be very detrimental (Erdogan, 1999). Based on research studies with the new generations, it has been published that consumers are leaning more towards SMI endorsements rather than celebrity endorsements (Schouten et al., 2020).

2.2 Social Media Influencer Endorsements

Social Media Influencer can be defined as a kind of micro-celebrity who uses self-presentation on social media platforms by creating an online image, which is used to attract and create engagement with a massive number of followers (Khamis et al., 2016). The influencers range from artists, fashion

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lovers, fitness, musicians, photographers. Influencers are categorized broadly into four types based on the number of followers: mega influencers (more than 1 million followers), macro-influencers (1,00,000 to 1 million followers), micro-influencers (1000 to 1,00,000 followers), Nano Influencers (1000 and below). With personalized feeds and information, these influencers are usually perceived to be more relatable and accessible. They have a niche set of followers who are deeply engaged and dedicated to the influencer (Wissman, 2018). A comparative study measuring the effectiveness of celebrity endorsements and influencer endorsements based on the role of credibility and product-endorsers' role fit impact on brand attitude and purchase intentions (Schouten et al., 2020). In addition to using celebrities such as supermodels, actors, and athletes, marketers are also increasingly using Social Media Influencer endorsements to add value to their brands (Marwick, 2015). Social Media Influencers are increasingly being utilized by brands and organizations to gather an audience on social media through influencer marketing. Recent technologies have made it easier for brands to identify which influencer would be the best fit for their organization through applications like IBM Watson AI Personality (Dhanesh & Duthler, 2019). In short, an SMI is someone who builds and maintains relationships with followers on social media platforms by providing them with informative, entertaining content through personal branding, which helps influence consumer's attitudes and behaviors towards a brand. Organizations try to use this personal branding to their advantage by using SMI's in their marketing strategies (Brown & Hayes, 2008). Endorsers are usually paid through a barter system (i.e.) by providing them products and, if not, then through payments. While a micro-influencer might get somewhere around $150, a mega influencer with more than 1 million followers can earn up to $15,000 per promotion (Crain, 2018). There is a fine balance that needs to be maintained between commerce and credibility for SMIs and sponsoring brands (Abidin & Ots, 2015). In exchange for payment or barter, SMI's produce content on social media platforms in the form of their personal story and connect with the brand and their personal opinions, which would favor the brand to help consumers form a positive attitude towards the brand.

2.3 Brand Attitude

Brand attitude is an individual's internal evaluation of a brand (Mitchell and Olson, 1981). A brand attitude is defined as a customer's continuous preference and trend to choose one specific brand over others (Wu & Wang, 2011). While brand attitude defines the consumer's continuous preference for the brand, it also defines the positive or negative evaluations a consumer makes towards the brand, which ultimately leads to their habitual use of the particular brand (Aaker and Keller, 1990). Studies on endorsements have proven that if there is a positive impact on the consumers' endorsement, it leads to a positive brand attitude (Bergkvist et al., 2016).

2.4 Source Credibility

Influencers usually possess knowledge in a specific subject based on which they attain followers (Lou & Yuan, 2018). The knowledge in this subject enables an influencer to gain followers with similar ideologies and thus helps build better and meaningful relationships with the audiences who evaluate the brand based on the endorsement from the influencers (Childers et al., 2018). This is termed as source credibility in this study. Previous research has demonstrated that source credibility impacts attitude towards an ad, attitude towards the brand (Mitchell and Olson, 1981; Atkin and Block, 1983; Lafferty et al., 2004). An influencer and peer endorser are similar in ways as they both rely on the source's trust and authenticity. The credibility of an endorser can be measured using the 4-dimensional conceptualization, which includes expertise, trustworthiness, attractiveness, and similarity (either ideologically or demographically) (Munnukka et al., 2016) out of which the two primary dimensions are considered in this paper based on previous research where attractiveness and similarity did not have a significant effect (Bhatt et al., 2013).

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2.4.1 Social Media Influencer Expertise

Expertise by an endorser is defined as an endorser's capability to make claims, which are valid as perceived by the consumer (Hovland et al., 1953). An endorser is most likely to be believed if the endorser possesses any relevant competency, skills, and knowledge (Homer and Khale, 1990). Based on this, endorsers can be split into expert endorsers and consumer endorsers. While expert endorsers can be used for products with high involvement and consumer endorsers can be used for low involvement products (Friedman et al., 1976). It is suggested that perceived expertise leads to a positive attitude towards the endorsements (Xiao et al., 2018). Even though there is an inconsistency among the research for expertise where some studies have found that expertise is not an important factor to be considered for credibility while performing brand evaluation while other studies have suggested that it is essential to consider product involvement when considering expertise (Seiler & Kucza, 2017). Thus the study proposes the following hypotheses:

H1: Perceived Expertise of the social media influencer will have a positive and significant impact on brand attitude.

2.4.2 Social Media Influencer Trustworthiness

While expertise addresses how equipped an SMI endorser is in a particular field, trustworthiness is when the consumer can accept the message based on the consumer's perception of whether the endorser will communicate information accurately (Sertoglu et al., 2014). While the endorser may have other positive qualities, which may lead to a good brand attitude, a lack of trustworthiness will have a negative impact (Erdogan, 1999). In addition to two primary dimensions of the source credibility model: expertise and trustworthiness, other credibility cues are to be considered to understand how the attitude towards the brand is mediated by the attitude towards endorsement based on the research gap identified. Thus forming the following hypotheses:

H2: Perceived Trustworthiness of the social media influencer will have a positive and significant impact on brand attitude.

2.5 Social Media Influencer- Brand Congruence

Endorser- Brand Congruence has been the focus of many studies and is called fit, similarity, and relevance. Endorser-brand congruence is nothing but the degree of similarity of consistency between a brand and the brand's extensions, such as an endorser (Aaker and Keller 1990), in this case, an SMI. Endorsement studies particular to influencers have shown the importance of impact on the brand congruence on having a positive brand attitude (Breves et al., 2019). Social Adaption theory has been used in celebrity endorsement based research to understand the impact of celebrity product match on consumer attitudes (Mishra et al., 2015). Brand congruence is an important factor in influencer endorsements as any perceived lack of fit between an endorser and a brand can have a negative impact on the brand evaluation, which finally affects the purchase intention (Lafferty, 2009). Volvo was criticized heavily for its poor influencer strategy in collaboration with the influencer Chriselle Lim, who, after the collaborations, posted a video of apology (Curalate, 2016). Based on the above discussion, the following hypotheses are proposed:

H3: Under conditions of Social media influencer personality–brand personality congruence will have a positive and significant impact on brand attitude.

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2.6 Sponsorship Disclosure

One of the main aims of disclosure is to discourage any attempts of deception, which thereby activates persuasion knowledge (Cain, 2011). Knowledge about persuasion and how to cope with attempts of persuasion is persuasion knowledge (Friestad & Wright, 1994). Sponsorship disclosure affects the persuasion knowledge of an influencer and has an impact on the credibility of the influencer (Carr & Hayes, 2014). An SMI must disclose potential advertisements using language such as "paid advertisement" or "sponsored" (Evans et al., 2017). It was also found that an implicit disclosure of the brands' involvement decreased the credibility of the SMI, while an explicit disclosure of the brands' disclosure increased the credibility of the SMI. Audiences must be aware of the influencer's potential biases through full disclosures (Carr & Hayes, 2014). Thus forming the following hypotheses:

H4: Sponsorship Disclosure of the endorsement by the social media influencer will have a positive and significant impact on brand attitude.

2.7 Influencer Follower Ratio

Previously performed research regarding endorsement and how they affect brand attitude have failed to account for important markers used by users to assess social media credibility, which is the number of followers or otherwise known as follower ratio (Sundar, 2008). The bandwagon Heuristic explains that when other individuals also find a source that is popular, trustworthy, or reliable, then they find it reliable too (De Veirman et al., 2017). Thus it is essential to consider that bandwagon effects might have a large impact on credibility. It has been determined in various previous studies that these cues are considered important in social media to classify if the source is credible or not (Metzger et al., 2010; Sundar, 2008; De Veirman et al., 2017). But there is also the threat of influencers acquiring fake followers to inflate the follower count on display. Thus, the following hypotheses are proposed:

H5: Influencer follower ratio of the social media influencer will have a positive and significant impact on brand attitude.

2.8 Attitude Towards Endorsements

Based on findings related in areas of research of endorsements, customer relationship management (CRM), and sponsorships that affect factors such as SMI- brand congruence, sponsorship disclosure, SMI expertise, and SMI follower relationship would have an effect on the mediating factor of attitude towards SMI endorsements rather than directly on attitude towards the brand. It has been found in previous studies of CRM that consumers' attitude towards the alliance between the non-profit organizations and brands mediates the effect on attitude towards the brand (Lafferty et al., 2004). Based on attitude theory, attitude towards an endorsement can be defined as a positive or negative evaluation of an alliance between an endorser and a brand. Based on research, it is expected that the cognitions that are endorsement related influence attitude on endorsements the same way that ad-related cognitions influence attitude towards an advertisement (MacKenzie et al., 1986). Similarly, attitude towards the brand in a celebrity endorsement is also mediated by the attitude towards celebrity endorsement, which is affected by factors such as attitude towards celebrity, celebrity brand fit, celebrity expertise, and celebrity motive (Bergkvist et al., 2016). Thus forming the following hypotheses:

H6: Attitude towards endorsement mediates the effects of expertise (H1), trustworthiness (H2), brand congruence (H3), sponsorship disclosure (H4), follower ratio (H5), on attitude towards the brand.

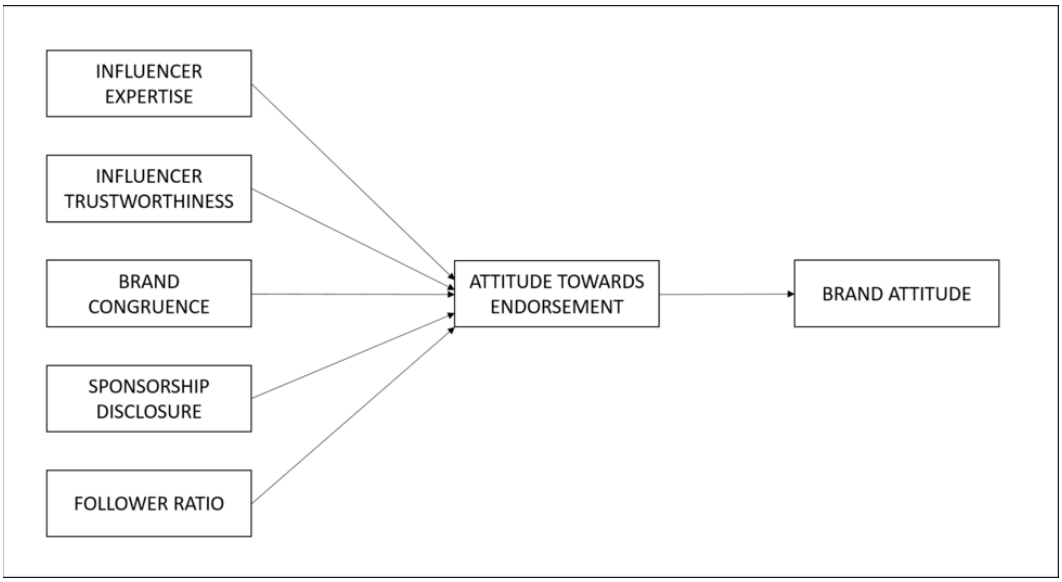
While celebrity endorsement in the context of mass media has been studied extensively over the span of last 20 years, there is a significant gap in understanding SMI Endorsements in digital

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media context. In addition, while the previous research papers have studied the effect of individual factors they have not considered the impact of 'Source Credibility Model' in combination with the Attribution Theory and Social Adaption Theory. Social Adapation thory suggests that an endorser whose personality is congruent with a brand's personality would be construed as an effective new source of information. Attribution Theory presents an argument that when the personality of an endorser matches the endorsed brand personality it leads to internal attributions. This is because of the fact that the customers are likely to think that the celebrity is endorsing the brand as it matches their personality instead of the financial benefit they receive from it. The importance of considering sponsorship disclosure stems from the above argument, it is necessary to understand if a customer's attitude is impacted by the explicit knowledge of a sponsorship which is beneficial to the endorser. The rationalization on the part of the consumer will have a significant impact on the customer's attitude towards the brand mediated by attitude towards endorsement (Mishra et al., 2015).

Based on the above discussion, the study proposes the following conceptual model of attitude formation of SMI endorsements (refer Figure 1):

Figure 1. Conceptual model of the effects of factors affecting brand attitude mediated by the attitude towards the endorsement



3. METHODOLOGY

The gaps identified from the literature review are the absence of substantial research to identify the factors affecting brand attitude mediated by the attitude towards social media influencer endorsement. It is also worth mentioning that there has not been sufficient research done in the Indian context to study Social Media Influencer Endorsement. Based on the gaps identified in the study, the objectives are:

- To establish the significant factors that influence the attitude towards the brand based on social media influencer endorsements
- To analyze the mediating effect of attitude towards endorsement on attitude towards the brand

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This study is performed using a descriptive research technique. Primary data for this research is collected through a structured online survey questionnaire. A structured questionnaire was floated, which measures respondents' attitude towards the brand, the endorsement itself, the social media influencer's expertise, trustworthiness, brand congruence with social media influencer's personality, sponsorship disclosure. The questionnaire comprises two parts, where the first part consisted of questions related to the demographics of the respondents and screening questions. The screening questions checked if the respondent was familiar with social platforms like Instagram. A visual cue was provided (product, service, and an event endorsed by an influencer) to help check if the respondent was familiar with influencers. The second part had questions related to the dependent and independent variables. All the seven constructs with a total of 29 items are measured with a 5 point Likert scale adapted from previous literature. A 5 point Likert scale is used instead of a 7 or 9 point Likert scale due to a higher number of respondents. (Shing-On Leung, 2011)

A total of 708 e-mail addresses of urban Indian college students were chosen as respondents as they are most active on social media platforms (Smith & Anderson, 2018; Baek and Morimoto, 2012). A homogenous student sample can help in reducing error variance thereby facilitating better theory-testing (Baek and Morimoto, 2012). An invitation to participate e-mail containing the survey URL was sent to each of the potential participants. Based on various researchers recommendations for appropriate sample size in multiple linear regression with mediating analysis (Dwivedi et al., 2015; Boerman et al., 2012; Bergkvist et al., 2016), a total of 283 responses were collected, out of which 220 usable responses were taken for analysis. Chronbach alpha is calculated to check the reliability, unidimensionality, and validity of each construct. Factor Analysis is performed with SPSS to reduce the individual independent factors into fewer dimensions. Path analysis is done using multiple linear regression analysis. The linear autocorrelation is checked using the Durbin Watson values, and the multicollinearity among the variables are tested using the variance inflation factor (VIF) and tolerance values.

4. DATA ANALYSIS

Out of the 220 responses, 53.4% of the recorded responses were male and 46.6% from females. Some 73.3% of the respondents were students, while 19.1% were employed full time, and the rest were employed part-time and looking for opportunities. All the respondents are exposed to social media platforms, and youtube has the highest usage rate, followed by Instagram and Linkedin then Facebook and then twitter.

Common Method Bias (CMB) occurs when the estimates in a relationship between two or more constructs are biased because they are measured with same method (Podsakoff and Organ, 1986). To assess the impact of common method bias in this study, Harmon one factor test (Podsakoff and Organ, 1986) was conducted on the 7 crucial variables in our conceptual model. Results from this test showed that the most covariance explained by one factor is 32.26%. As the covariance is less than the indicated 50% it shows that this study is not impacted by Common Method Bias (Liang et al., 2007).

Reliability Analysis

All measure items loaded on the constructs respectively have scored above 0.70, indicating good reliability of constructs. Cronbach's alpha of the constructs is given in Table 1. As a rule of thumb, researchers consider a measure to have adequate inter-item reliability if Cronbach's alpha coefficient exceeds .70 (Leary, 2007). Based on this theory, the results showed that all the study's measurement constructs have achieved good reliability (refer to Table 1).

Factor Analysis

The Kaiser-Meyer-Olkin (KMO) and the Barlett's test of sphericity are used for analyzing the sampling adequacy as well as the test for sphericity, which were both applied to all the measurement constructs. A principle-component analysis with Varimax rotation was conducted using the software tool SPSS

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Table 1 Reliability analysis of the constructs

Measurement Constructs	Number of Items	Cronbach Alpha
Expertise	4	0.785
Trust	4	0.804
Brand Congruence	2	0.769
Sponsorship Disclosure	3	0.826
Follower Ratio	4	0.824
Attitude towards the Endorsement	5	0.702
Brand Attitude	4	0.737

yielding seven factors. The results are given in Table 2. KMO is used to determine sampling adequacy and is used to ensure that the constructs that are measuring a particular concept are measuring the concept indeed. A calculated value of KMO between 0.5 and 1.0 is an indicator of sampling adequacy. The KMO measures of sampling adequacy values for the constructs were indicating normal distribution in the data and appropriate sample size. The chi-square value of Bartlett's test of sphericity compares an identity matrix with an inter-item correlation matrix. These results indicate that the p-value (Sig.) of .000 < 0.05 makes the factor analysis valid.

Table 2 Results from Factor Analysis

Measurement Constructs	KMO Measure of Sampling adequacy	Bartlett's Test of Sphericity Approx. Chi-square	Sig.	Total Variance Explained Cumulative %
Expertise	0.707	277.842	0.000*	61.142
Trustworthiness	0.742	356.578	0.000*	63.922
Brand Congruence	0.500	107.351	0.000*	81.207
Sponsorship Disclosure	0.721	239.170	0.000*	74.210
Follower Ratio	0.797	304.283	0.000*	65.569
Attitude towards the Endorsement	0.763	215.877	0.000*	57.878
Brand Attitude	0.738	189.502	0.000*	56.191

Note: p < 0.05*

Path Analysis

Path analyses are performed by applying multiple linear regression on the data with alpha as 0.05. Based on the results in this model, the paths from expertise (H1), brand congruence (H2), and follower ratio (H5) are statistically significant with significance p<0.05. The analysis also revealed two factors estimates were not statistically significant (i.e., *p* - 0.244 for trustworthiness, p- 0.777 for sponsorship disclosure). A possible cause of the offending estimate may be the relatively small sampled-size of our study *is-a`-vis* the observed variables (Kline, 2015). Figure 2 gives the results for the path analyses. The variance inflation factor (VIF) value of the regression model is between 1.119 and 1.843, which were below 10, so there was no "collinearity" in this regression model (Hair

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et al., 1998). The regression model explained a substantial amount of the variance in attitude towards endorsement (0.458). However, the variance explained was lower for the brand attitude (0.431). This might be because other factors influence attitude towards the brand, such as previous experience with the brand, word of mouth.

Figure 2 A path analysis of the effects of independent variables on brand attitude while mediated by the attitude towards endorsement.

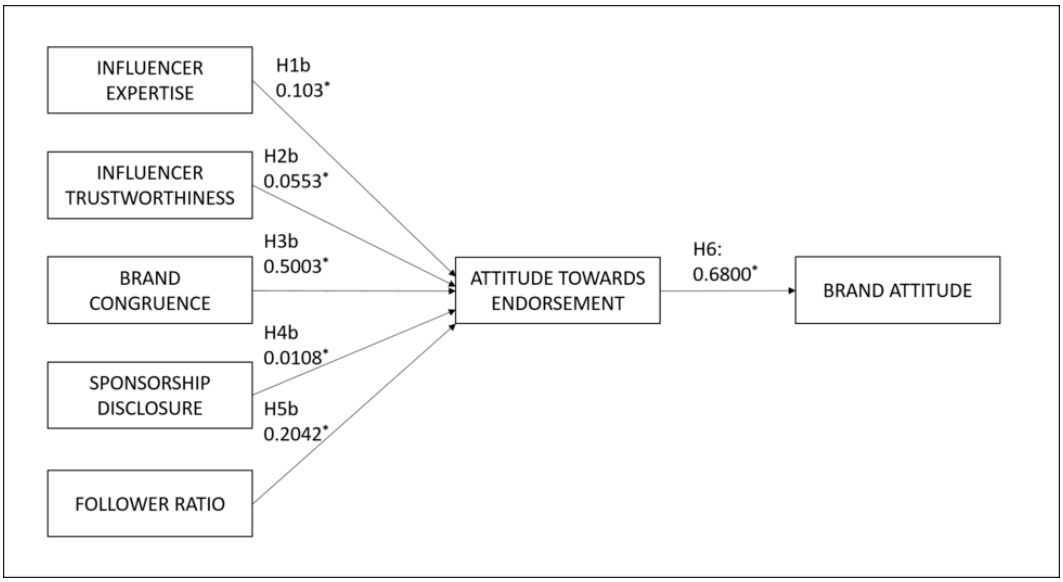


Table 3. Results from mediation analysis (dependent variable: Brand attitude; mediator: attitude towards Endorsement

Independent Variable	Mean Indirect effect	95% CONFIDENCE INTERVAL	
		Lower limit	Upper Limit
Expertise	0.0675	0.0039	0.1405
Trustworthiness	0.0376	-0.0252	0.1015
Brand congruence	0.3402	0.2469	0.4433
Sponsorship Disclosure	0.0073	-0.0439	0.0590
Follower Ratio	0.01389	0.0749	0.2109

Mediation Analysis-

A mediating bootstrapping analysis (using 5,000 sub-samples and 95 percent confidence intervals [CI]) was performed using the SPSS macro indirect (Preacher and Hayes 2008) model 4. The results of the mediation analyses with the indirect effect of the independent variables on Brand attitude with the attitude towards the endorsement as the mediating variable lends support to H1, H3, H5. Three of the independent variables of the five considered in this research had perfect mediation, which

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indicates that the mean indirect effect was significant, and the confidence intervals did not include a zero. However, trustworthiness and sponsorship disclosure around the offending estimate includes a 0 point, which indicates that model-misspecification is *not* a likely cause of the offending estimate (Dillon *et al.*, 1987) and as explained, a possible cause of the offending estimate may be the relatively small sample size. Table 3 gives the results from the mediation analysis.

5. DISCUSSION AND CONCLUSION:

From the above analysis, it can be inferred that while measuring the direct effect through regression and the indirect effect through SPSS Macro model 4 (Preacher & Hayes). Hypotheses H1, H3, and H5 are accepted, and the hypotheses H2 and H4 are rejected. It is seen that three independent variables-expertise, brand congruence, and follower ratio have a significant effect on brand attitude mediated by the attitude towards endorsement.

The mean score of brand congruency with the influencer personality is approximately four, and also the beta coefficient of brand congruency has a value of 0.505, which shows that it has a more significant impact on brand attitude. The average sentiment of consumers' attitudes towards a social media influencer endorsement is highly influenced by brand congruency with the influencer's personality. Factors such as expertise and follower ratios mean range approximately between 3.25 to 4 while having an unstandardized beta coefficient of 0.103 and 0.2042. The scores indicate that while expertise is a crucial factor to be considered, sometimes collaborations with influencers occur based on the influencer's reach and engagement more than the expertise. This explains how for each social media influencer, the ratio of the impact of each of the factors such as expertise, brand congruence, and follower ratio might vary, but they do impact brand attitude.

This study extends previous research on influencer endorsements. In line with previous research, the study shows that expertise, brand congruence, and follower ratio positively impact brand attitude. As implicated in previous studies drawbacks that follower ratio was an important factor which was not considered while measuring the impact of attitude towards brand along with Brand congruence (Mishra et al., 2015) and considering it as a factor in this model has shown that it indeed does have a positive impact. The study's uniqueness is in analyzing the impact of the mediating variable attitude towards social media influencer endorsements. The effects of the influencer-related factors included herein are wholly or partly mediated by attitude towards endorsement, rather than primarily having direct and indirect impact on brand attitude.

This study has investigated the factors of social media influencer endorsements affecting brand attitude in the Indian context while mediated by the attitude towards endorsement. The proposed model extends the source credibility model by checking the influence of factors like brand congruence with influencer personality, sponsorship disclosure, and follower ratio for the influencer. This study highlights that it is imperative to examine multiple components as they are proving to be indispensable in improving the effectiveness of social media influencer endorsements. The construct helps one to understand the effects of social media influencer endorsement on brand attitude in influencer marketing, which incorporates the value of an influencers' credibility popularity as well as match up with the brand and lays the groundwork for a more comprehensive model in the future.

6. LIMITATIONS AND FUTURE SCOPE OF RESEARCH

The study focuses on social media influencers and the credibility of influencer endorsements. It is, however, conceivable that social media users' beliefs and attitudes may change across platforms because of the message modality or influencers' content variations across different platforms. Moreover, the insignificant impact of trustworthiness and sponsorship disclosure may be studied in future studies. Future research may not only investigate the roles of interactive platforms' affordances and consumers' differences in social media influencer endorsements but also identify critical boundary conditions or mechanisms of its effects on brand building and consumer behaviors.

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It would be worth exploring the role of cultural factors in social media. Further, future studies can also study the construct with other research designs such as experimental designs or study causal relationships or replicate reported results using cross-platform social media endorsements focussed on specified brands or high performing influencers.

7. THEORETICAL AND MANAGERIAL IMPLICATIONS

This study makes significant contributions to the social media marketing literature. It offers a theoretical contribution by providing the factors impacting Indian consumer's attitude towards brand and how attitude towards endorsement acts as a mediator which helps researchers to further build a robust model.

In line with the conclusion in this study regarding sponsorship disclosure, in July 2021 the Advertising Standards Council of India ("ASCI") had released the Guidelines for Influencer Advertising in Digital Media. These guidelines are to enable consumers to identify when influencers are deriving some benefit for promoting a product, and for influencing opinions and behaviours. These Guidelines prescribe stringent disclosure requirements for influencers, placing obligations on both influencers and advertisers thus ensuring that consumers are made aware of the financial benefits of the endorsement for the influencer.

The findings in this study help provide insights to the digital marketing managers for drafting various advertising strategies on how to increase the favorable brand attitude in consumers' minds. As an effort to create a favorable brand attitude in the minds of the consumers, the suggested areas are first ensuring that the influencers' collaboration as a source of endorsement is trustworthy to be perceived well by the user. Second, ensuring that there is a brand congruence with the influencer's personality would ensure that the endorsement the consumers are exposed to helps build a better connection with the brand. One of the crucial managerial implications of this study's findings is that the credibility of the influencer in an endorsement can positively impact the endorsed brand equity, which is represented by four aspects- brand associations, brand loyalty, quality, and brand awareness. This suggests that marketers should recruit and collaborate with influencers with good credibility while making sure there is a brand congruence with the influencer's personality so that the endorsements deliver the desired meaning. A combination of such knowledge could help brands implement potentially more effective influencer marketing campaigns. In line with the above discussion, advertisers and marketers can use the social media influencer endorsements to develop a strong and positive brand attitude and attitude towards the endorsement.

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The Effect of Self-Service Technologies on Customer Experiences in Banking: The Case of Ghana

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ABSTRACT

The surge in self-service technologies (SSTs) adoption in the banking industries across the globe has changed customers' banking experiences. Customers have developed a passion for SSTs, which has affects their intentions to save with a bank. The positive effects of employing SSTs in the banking industries are evident in literature; however, little is known on how SSTs affect customer experiences in the banking industries of developing countries like Ghana. The study involved a sample of 384 banking customers in Ghana. The study confirms the positive effects of SST adoption on customer experiences. The study establishes that customers' adoption of SSTs is influenced by their perception of SST usefulness and ease of use, which increases their satisfaction level. The study finds out that when SSTs are deployed individually, they have no effect on customer experiences in the banking sector. However, when the individual SSTs are combined in usage, they collectively produce a positive effect on customer banking experience.

KEYWORDS

Affective Experience, Customer Experience, Internet Banking, Internet of Things, Mobile Applications, Physical Experience, Self-Service Technologies, Sensory Experience, USSD, Web Technologies

INTRODUCTION

The growing interest in the use of self-service technologies (e.g., web applications, internet of things, unstructured supplementary service data technology, mobile applications, automated teller machines, etc.) for business operations across the globe is gradually altering consumer behaviors and experiences (Shahid Iqbal, Ul Hassan & Habibah, 2018). Self-service technologies (SSTs) are defined as technologies that customers use independently without interaction or assistance from a firm's employees (Meuter *et al.*, 2000). The adoption of SSTs in the area of banking undoubtedly has reduced customer waiting times, and queuing (De Leon, Atienza & Susilo, 2020). This trend has resulted in customers experiencing more responsive and equally effective banking services. The manifestation of these experiences is seen around the world as customers perform transactions in the

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comfort of their rooms (e.g., transfer between bank accounts and mobile money wallets), and update personal banking details via the internet.

Customer experience (CE) is a fundamental concept that is embraced in banking industries across the globe. CE encompasses every aspect of a firm's offering, stemming from the nature or quality of customer care to packaging, products and services, ease of use and reliability (Meyer & Schwager, 2007; Schmitt, Joško Brakus & Zarantonello, 2015). The economic climate and business environments require more than just innovative and low-priced products. Firms rely also on customer experiences for survival (Voorhees *et al.*, 2017). Customer decisions, intention to purchase or rely on the services of a firm such as a bank is greatly influenced by their experiences (Tannahill & Jamshidi, 2014) as their satisfaction level depends on their experiences (Schmitt, Joško Brakus & Zarantonello, 2015). This has resulted in banks paying attention to customer experience-based systems (Kim *et al.*, 2011) which SSTs offers.

The positive effects of employing SSTs in the banking industries across the globe is evident in literature (Nijssen, Schepers & Belanche, 2016; Shahid Iqbal, Ul Hassan & Habibah, 2018). Contemporary studies have focused on personal traits, technology adoption, customer purchase intention, and customer intention to continually use a firm's services, which is customer retention oriented (Ho & Ko, 2008; Foroudi *et al.*, 2018). Little is known about the individual and combined effect of the different types of SSTs on customer experience in the banking industries. Studies in this area can help banks to understand how different types of SSTs can be combined to obtain maximum customer experiences. This study seeks to fill the knowledge gap on the individual and combined effect of different SSTs on customer experience in the banking industry using Ghana as a case site. Specifically, the study will examine the types of SSTs used by customers, the experiences they gain, the factors that affect SST adoption and acceptance as well as the individual and combined effect of the different types of SSTs on customer experiences in the banking industry of Ghana. This study will contribute to literature and serve as a lens for banks to determine the type of SST to invest in to achieve maximum customer experience especially in developing countries where banks have limited investment resources.

BACKGROUND

Self-Service Technologies (SSTs)

SSTs are technological interfaces through which customers can attain services without the direct involvement of service firm employees (Meuter *et al.*, 2000). SSTs are replacing many face-to-face service interactions with the intention to make service transactions more accurate, convenient, and faster (Foroudi *et al.*, 2018). The introduction of the internet paved the way for the application of SSTs including mobile applications, unstructured supplementary service data technology (USSD), Automated Teller Machines (ATM), debit and credit card and mobile money in banking industries (Shaikh & Karjaluoto, 2015). In the Ghanaian banking environment, internet banking (web applications), USSD, mobile applications and ATMs are the most adopted and patronized amongst SSTs (PwC Ghana, 2020) and this influenced the choice of focusing on these four SSTs.

Internet Banking (Web Applications)

Internet banking (Web applications) involves the use of a banking website for the performance of different banking services, ranging from accessing account information and general information on a bank's products and services, bill payment and the making of an investment (Hosein, 2011). Except for cash withdrawal, internet banking offers customers the opportunity to access almost any type of banking transaction at a click. In essence, internet banking provides universal accessibility from any internet linked computer, irrespective of the location worldwide (Shahzad, Xiu & Shahbaz, 2017; Yoon & Steeg, 2013)

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Mobile Applications

Over the last few years, the mobile and wireless market has been one of the fastest growing markets in the world and it is still growing at a rapid pace (Hanif *et al.*, 2020). The convenience of mobile banking is that banks undertake the banking transaction outside of the working hours and is accessible from anywhere and has indeed become customer preference (Mullan, Bradley & Loane, 2017). Since mobile banking was introduced, consumers have been able to do banking services 24 hours a day using their mobile phones without having to visit the traditional bank branches or to find a computer with broadband connection for personal transactions. This has led to the ease of use of mobile banking since the availability of the system whenever they need it will increase their intention to adopt the system and become their preference as compared to the traditional banking system.

Unstructured Supplementary Service Data (USSD)

Most people in developing countries like Ghana do not have reliable internet access, also not everyone can afford to patronize broadband for internet banking or mobile banking, therefore USSD technology is gaining prominence (Fletcher & Owusu-Sekyere, 2018). Almost 85% of the Ghanaian population are conversant with USSD due to its use for loading mobile credit and mobile money transactions. USSD is a menu-driven technology that improves user experience by unleashing the ability to easily, quickly and cost effectively send essential alerts and information via short codes (e.g. *170#) (Fletcher & Owusu-Sekyere, 2018).

Automatic Teller Machines (ATM)

Although SST services including USSD, mobile applications, web applications allow for convenience and easy banking experiences by means of updating personal banking details, transfer of funds, payment of bills without the direct assistance of the bank's employees, it lacks the ability to withdraw and deposit physical cash. ATM customizes service offerings, reduces waiting time for customers, serves as an alternative channel for service delivery and provides vital information needed by customers in the shortest possible time (Lovelock & Yip, 1996). "ATMs combine a computer terminal, record-keeping system and cash vault in one unit, permitting customers to enter the bank's bookkeeping system with a plastic card containing a personal identification number (PIN) or by punching a special code number into the computer terminal linked to the bank's computerized records 24 hours a day" (Giddens, 2008). They were introduced first to function as cash dispensing machines but currently they have the ability to accept cash deposits and others thus giving customers a more fulfilling experience.

Customer Experiences (CE)

Customers interact with firms through staff, SSTs and service environments to obtain some level of satisfaction (Firdous & Farooqi, 2017). The outcome of such interaction is described as customer experience. In the last three decades customer experience has been mentioned, discussed, and theorized by scholars and practitioners. Early scholars studying experience introduced the term in a general framework and described experience as playful leisure activities, sensory pleasures, aesthetic enjoyment and emotional response (Sirapracha, 2012). However, it is important to realize that actual experiences are distinct from services (Pine & Gilmore, 1999). The authors specifically quoted that "when a person buys a service, he purchases a set of intangible activities carried out on his behalf. But when he buys an experience, he pays to spend time enjoying a series of memorable events in a personal way" (Pine & Gilmore, 1999, p. 2). Schmitt (2010) identified five different types of experiences that firms can create for customers - sensory experience, affective experience, cognitive experience, physical experience and social experience. The economic climate and business environments require more than just innovative and low-priced products but on customer experiences for survival (Voorhees & Fombelle, 2017; Grewal & Roggeveen, 2020). Thus, for a firm in the banking sector to survive, the collective effect of these five types of experiences cannot be underestimated.

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Sensory Experience

Sensory experience relates to sight, hearing, touch, taste, and smell experiences and how they arouse aesthetic pleasure, excitement, satisfaction and a sense of beauty (Schmitt, 2010). It is a component of the CE whose stimulation affects the senses; an offering, whose aim is to provide good sensorial experiences, can address sight, hearing, touch, taste and smell so as to arouse aesthetic pleasure, excitement, satisfaction, and a sense of beauty. For example, banks utilizing SSTs inculcate biometric features which offer a safe banking environment for customers and reinforces their sensory experience through touch and eyesight related biometric authentication (Nagaraju & Parthiban, 2015).

Affective Experience

Affective experience is conceived as central to how people use, understand and interact with designed objects (Chahal & Dutta, 2015). It's a component of the Customer Experience which involves one's affective system through the generation of moods, feelings, emotions; an offering can generate emotional experience in order to create an affective relation with the company, its brand or products. In offering a better customer experience, banking firms regularly send messages which are targeted towards making customers feel the need to stay connected to their bank through various SSTs (Bilisbekov *et al.*, 2021). This creates an affection between customers and the bank.

Cognitive Experience

Cognitive experience relates to thinking and conscious mental processes to get customers to use their creativity or problem-solving skills so that they revise assumptions about a product (Chahal & Dutta, 2015; Schmitt, 2010). It is a component of the Customer Experience connected with thinking or conscious mental processes; an offering may engage customers in using their creativity or in situations of problem solving; furthermore, a company can lead consumers to revise the usual idea of a product or some common mental assumptions. Cognitive experience has been indicated to be essential in customers' intention to adopt internet banking as well as mobile banking (Sharma, 2019).

Physical Experience

Physical experience provides customers with different ways of doing things (Chahal & Dutta, 2015; Schmitt, 2010) . It is a component of the Customer Experience coming from the practical act of doing something; in this sense the pragmatic component includes but is not exhausted by the concept of usability. It does not only refer to the use of the product in the post purchase stage, but it extends to all the product life-cycle stages. ATM usage to deposit and withdraw cash or carry out a banking process at the ATM is a refreshing physical experience which reinforces a customer's intention to reuse the service (Abd, Aziz & Hussien, 2014).

Social Experience

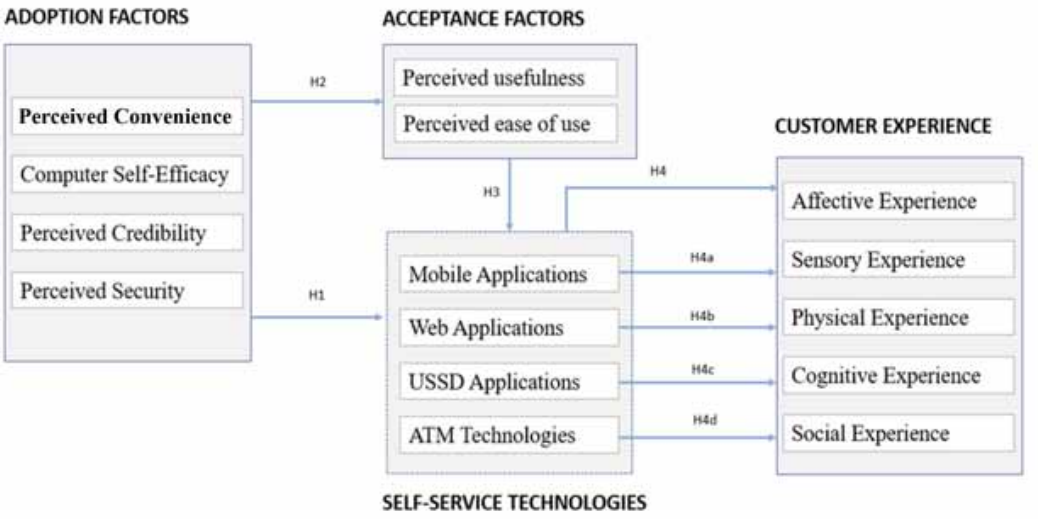
Social experience widens the private feelings of the individual by relating to something outside of the individual's closed condition thus creating a sense of self – improvement (Chahal & Dutta, 2015; Schmitt, 2010) . It is a component of the CE that involves the person and, beyond, his/her social context and relationship with other people or also with his/her ideal self. Social experience suggests a means by which a customer encourages the use/consumption of a product together with other people. This togetherness in customer use of products is the core of a common passion that may eventually lead to the creation of a community or still a tribe of fans. The social experience in customer use of a product can also be a means of affirmation of a social identity, inducing a sense of belonging or of distinction from a social group; in this case the link with the lifestyle component is very relevant. In recent times the use of mobile money through USSD in most African countries has created a social experience which has led to the informal sector being more active as a community in banking (Aborogu et al., 2016).

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RESEARCH MODEL AND HYPOTHESIS FORMULATION

The adoption and utilization of SSTs are influenced by a couple of factors which have been highlighted in contemporary literature. Literature has identified SSTs adoption factors including convenience (Nielsen, 2005), security (Nasri, 2011), computer self-efficacy (Samar, Ghani & Alnaser, 2017), credibility (Lakshmi & Ganesan, 2010) and perceived convenience (Collier & Kimes, 2013).

Figure 1 Research Model



Studies conducted to understand the relationship between SST characteristics (perceived risk, perceived ease of use, and perceived usefulness), consumer technology readiness, social pressures (coercive, normative, and mimetic), and SSTs adoption revealed that SSTs characteristics, consumer technology readiness and social pressures were crucial determinants of SST adoption (Yang, Liu & Ding, 2012; Kaushik & Rahman, 2015).

Other studies conducted to examine whether SSTs influence customer value and customer readiness revealed that, SST characteristics or adoption factors including ease of use, usefulness, and cost-saving positively influence customer values and customer readiness which then influences customers desire to adopt or use internet banking (web applications) (Boon-itt, 2015; Ho & Ko, 2008). It can therefore be inferred from literature that, SSTs like web technologies adoption are influenced by factors which include cost-savings strengthened by SSTs ease of use and usefulness. The unavailability of literature generalization of the relationship between the broad overview of SSTs adoption factors and its adoption makes it relevant to measure how the SSTs adoption factors (convenience, security, computer self-efficacy, and credibility) informs customers SSTs adoption decision and how that relationship is strengthened by acceptance factors (ease of use and usefulness). This leads to the us hypothesizing:

H1: SSTs adoption among customers in the banking industry is driven by the adoption factors, namely, convenience, security, computer self-efficacy and credibility.
H2: Adoption factors positively associate with SSTs acceptance factors namely, ease of use and usefulness in the banking industry.

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H3: Acceptance factors positively associate with SSTs adoption in the banking industry.

Carbone (1994) defined experience as ''a take away impression formed by customers' encounters with products, services and businesses- a perception produced when humans consolidate sensory information''. Customer experience is an important driver to the survival of any business organization (Grewal & Roggeveen, 2020; Voorhees & Fombelle, 2017) and this has led to the recent surge in customer experience research studies.

Foroudi, Gupta & Sivarajah (2018) conducted an exploratory study to investigate the effects of smart technology on customer dynamics and customer experience. The study sampled 330 retail stores in London and used a covariance-based structural equation modelling (AMOS) approach for the analyses and validation of results. The results showed that there is a direct relationship between customer dynamics, customer experience and customers' participation in the adoption of self-service technologies in a retail setting. The relationship was found to be driven by customers' willingness and the ability to learn.

Scherer & Wünderlich (2015) also conducted a study to examine how the ratio of self-services and personal services influences customer retention over time. The study revealed that the ratio of self-service to personal service usage affects customer retention in a u-shaped manner and that the intermediate levels of both self-service and personal services usage had the lowest likelihood of defection (Scherer & Wünderlich, 2015). The study suggested that firms should not shift customers from personal service towards self-service completely. The study explained when and how self-service technologies create valuable customer experiences. The study left a gap by not explaining the extent to which self-service and personal service should be combined to achieve a higher level of customer satisfaction. Moreover, the influences of self-services were not measured independently to assess their contribution to customer satisfaction which could serve as an insight to the banking industry in determining the type of SSTs to deploy whether individually or in a combination. This leads to the hypothesis:

H4: Customer experience is positively influenced by self-service technologies namely mobile applications, web applications, USSD applications and ATM usage.
H4a: There is a positive relationship between mobile applications usage and customer experience.
H4b: There is a positive relationship between web applications usage and customer experience.
H4c: There is a positive relationship between USSD applications usage and customer experience.
H4d: There is a positive relationship between ATM usage and customer experience.

RESEARCH METHODOLOGY

Sampling and Data Collection

The study employed a quantitative approach to objectively examine the effect of self-service technology adoption on customer experience in the banking industry of Ghana. Quantitative methods use formal processes in which numerical data are used to measure the phenomenon to arrive at a conclusion, which best fits the objectives of this study (Harrison, Reilly & Creswell, 2020).

In this study, the population comprised of customers in the banking industry of Ghana. Random sampling technique was adopted to select a sample of 384 out of the population following the mathematical formula of Bartlett, Kotrlik & Higgins (2001).

Sample size where 'n' is the required sample size, 'p' is the percentage occurrence of a state or condition (0.5), 'e' is the percentage maximum error required (0.05) and 'z' is the value corresponding to level of confidence (1.96).

Printed and electronic questionnaires were used for the data collection based on a 5-point Likert scale ranging from 1- strongly disagree to 5- strongly agree. The electronic questionnaire was created

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using google forms and distributed to the study participants. Printed questionnaires were printed for participants not conversant with electronic questionnaire.

Methods of Data Analysis

Descriptive statistics and structural equation modelling were used as the methods for data analysis. The data obtained was cleaned and analyzed with IBM SPSS Statistics 25 and SmartPLS version 3.3.2. The statistical software IBM SPSS was used for the descriptive statistics, frequency, and distribution of the data whiles the variance-based structural equation modelling software SmartPLS was used for the structural model. It estimated the various measurement model validity and reliability, the parameters of the structural equation model and other statistical measurements considered appropriate to answer the research questions. The Partial Least Squares (PLS), bootstrapping, blindfolding and PLS Predict algorithms of SmartPLS were used for the analysis of the structural equation model. In assessing the validity of the constructs and evaluating the model, four major test tools were employed. These included internal consistency tests (Cronbach's alpha and Composite Reliability), convergent validity (Average Variance Extracted), discriminant validity and collinearity assessment.

Measurement Instrument

Drawing on existing literature, the measuring instrument for the study was designed using the constructs in the research model. Adoption factors were measured with four sub constructs namely perceived convenience, computer self-efficacy, perceived security, and perceived credibility. Measuring items from scales developed by Collier & Kimes (2013), Samar (2017), Rose et al. (2012) and Wang et al. (2003) were adapted to measure the constructs. Acceptance factors in the research model were measured using two sub-constructs namely perceived usefulness and perceived ease of use. Measuring items for the two constructs were adapted from Samar (2017). In exploring the type of self-service technology adopted in the banking industry, the respondents were asked to select from among a set of options. These options were Web Technology (E-banking, online banking, or internet banking), USSD Technology (short code example *555#), Mobile App Technology and Internet of Things Technology (ATM Machines). Customer experience was operationalized by measuring five types of experiences namely sensory experience (sense), affective experience (feel), cognitive experience (think), physical experience (act) and social identity experience (relate). Measuring items for physical experience (act) and social experience (relate) were drawn from Schmitt (2015) whiles items used to measure affective experience (feel) and cognitive experience (think) were drawn from Rose et al. (2012) and Novak, Hoffman, and Yung (2000). Finally, sensory experience (sense) was measured with items drawn from S Rose et al. 2012 and Gentile, Spiller, and Noci (2007). Table 2 shows the measuring instrument for the constructs with their respective Cronbach's alpha and other reliability metrics establishing the internal consistency of the adapted measures.

DATA ANALYSIS AND FINDINGS

Demographics

A total of 374 responses were obtained for the analysis, out of which 213 (57%) were males and 161 (43%) were females. A greater proportion of the respondents were high school graduates; 191(51.1%) and 252 (67%) were employed. 118 (31.6%) have worked for five years and over. A summary of the demographic statistics is tabulated in Table 1 below:

Structural Model Estimation and Evaluation

The variance-based structural equation modelling software SmartPLS was used for the model estimation and evaluation. Internal consistency tests, convergent validity and discriminant validity were adopted to evaluate the model constructs while a collinearity measure was adopted for the structural model assessment.

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Table 1 Demographic Statistics

	Description	Frequency	Ratio
Gender	Male	213	57.0
	Female	161	43.0
Age	18-28	207	55.3
	29-38	136	36.4
	39-48	22	5.9
	49 -58	9	2.4
Level Of Education	JHS	2	.5
	SHS	21	5.6
	Degree	191	51.1
	Masters	151	40.4
	PHD	4	1.1
	Certificate	5	1.3
Employment Status	Unemployed	122	32.6
	Employed - Public sector	84	22.5
	Employed - Private sector	135	36.1
	Self-employed	33	8.8
Number of years of working	Below 1 year	81	21.7
	1-2 years	64	17.1
	2-4 years	111	29.7
	5 years and above	118	31.6

Source: (Authors construct, 2021)

Internal Consistency Tests

Internal consistency measures how well a set of indicators measures a single latent variable. Cronbach's alpha and composite reliability were adopted for the internal consistency test. According to (Nunnally, 1994), an alpha value of 0.7 at the early stages of a research is accepted. The results of the analysis in Table 2 indicated that all the constructs resulted in a Cronbach alpha greater than 0.7 confirming that the constructs met the reliability test. Composite reliability measures the extent to which a set of indicators are reliable for the study. It also considers a threshold of 0.7 at the initial stages of the research as acceptable. The result of the analysis of the data indicated that all the constructs resulted in a composite reliability value greater than 0.7 confirming that the constructs met the composite reliability test.

Convergent Validity Tests

Convergent validity is the extent to which a measure correlates positively with an alternative measure of the same construct. According to (Bagozzi & Yi, 1988; Fornell & Larcker, 1981), if an outer loading is greater than 0.708 then the latent variable can explain at least 50% of its indicator's variance. The result of the analysis on convergent validity indicated that the outer loadings SExp4 and PExp4 were less than .708, indicating that they did not meet the convergent validity acceptance level and hence were removed from subsequent analysis.

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Table 2 Construct Measurement and Model reliability metrics

Construct	Item Code	Reference	Cronbach's Alpha	rho_A	Composite Reliability	AVE
Affective Experience (AE)	AE1	Rose et al. (2012), Novak, Hoffman, and Yung (2000)	0.843	0.847	0.896	0.683
	AE2					
	AE3					
	AE4					
Cognitive Experience (CE)	CE1	Rose et al. (2012), Novak, Hoffman, and Yung (2000)	0.798	0.803	0.868	0.621
	CE2					
	CE3					
	CE4					
Computer Self Efficacy (CSE)	CSE1	Samar (2017)	0.848	0.847	0.898	0.687
	CSE2					
	CSE3					
	CSE4					
Perceived Convenience (PCon)	PCon1	Collier & Kimes (2013)	0.854	0.854	0.901	0.696
	PCon2					
	PCon3					
	PCon4					
Perceived Credibility (PCred)	PCred1	Wang et al. (2003)	0.836	0.837	0.891	0.671
	PCred2					
	PCred3					
	PCred4					
Perceived Ease of Use (PEU)	PEU1	Samar (2017)	0.867	0.868	0.909	0.716
	PEU2					
	PEU3					
	PEU4					
Physical Experience (PExp)	PExp1	Schmitt (2015)	0.711	0.724	0.823	0.539
	PExp2					
	PExp3					
	PExp4					
Perceived Security (PSec)	PSec1	Samar (2017), Rose et al. (2012)	0.854	0.863	0.901	0.696
	PSec2					
	PSec3					
	PSec4					
Perceived Usefulness (PU)	PU1	Samar (2017)	0.879	0.880	0.917	0.735
	PU2					
	PU3					
	PU4					
Sensory Experience (SExp)	SExp1	Gentile, Spiller, and Noci (2007), Rose et al. (2012)	0.738	0.754	0.837	0.565
	SExp2					
	SExp3					
	SExp4					

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Table 2. Continued

Construct	Item Code	Reference	Cronbach's Alpha	rho_A	Composite Reliability	AVE
Self-Service Technology (SST)	SST_WEB	PwC Ghana (2020)	1.000	1.000	1.000	1.000
	SST_USSD					
	SST_Mobile					
	SST_ATM					
Social Experience (SoExp)	SoExp1	Schmitt (2015)	0.897	0.904	0.928	0.763
	SoExp2					
	SoExp3					
	SoExp4					

Source: (Author's construct, 2021)

Discriminant Validity Tests

Discriminant validity is a measure of the extent to which a construct is truly distinct from other constructs. Cross loadings, Fornell and Larcker (F&L) criteria and heterotrait-monotrait (HTMT) were used to test for discriminant validity. According to (Chin, 2010; Fornell & Larcker, 1981), the square root of AVE should be greater than the highest correlation coefficient of each construct of the latent variable which was supported. Moreover, the loadings of an indicator on its assigned latent variable should be higher than its loadings on all other latent variables to meet the validity criterion. The report of the cross-loading validity analysis showed that all the loadings on the latent variables were greater than each of the loadings on the corresponding constructs, signifying that the cross-loading criterion has been fulfilled. An estimation of the correlations between the constructs in the model based on the average of heterotrait-monotrait correlation suggested by (Henseler, Ringle & Sarstedt, 2015) was adopted. According to (Ab Hamid, Sami & Sidek, 2017), 0.90 is a good threshold for HTMT validity. The results of the analysis of the HTMT validity indicated that all of the constructs were smaller than 0.9 explaining that the discriminant validity criterion has been met.

Table 3. Discriminant Validity and Collinearity Values

Constructs	Loadings	VIF	Constructs	Loadings	VIF	Constructs	Loadings	VIF	Constructs	Loadings	VIF
AE1	0.814	2.048	PCon1	0.805	1.757	PExp1	0.785	1.518	SExp1	0.834	1.790
AE2	0.875	2.665	PCon2	0.847	2.136	PExp2	0.732	1.371	SExp2	0.813	1.735
AE3	0.879	2.434	PCon3	0.879	2.498	PExp3	0.794	1.529	SExp3	0.703	1.313
AE4	0.731	1.532	PCon4	0.804	1.760	PExp4	0.612	1.192	SExp4	0.639	1.209
CE1	0.792	1.632	PCred1	0.827	1.877	PSec1	0.842	2.216	SST_ATM	0.475	1.017
CE2	0.797	1.689	PCred2	0.830	1.934	PSec2	0.886	3.103	SST_Mobile	0.465	1.013
CE3	0.780	1.664	PCred3	0.800	1.762	PSec3	0.784	1.783	SST_USSD	0.778	1.041
CE4	0.784	1.467	PCred4	0.819	1.860	PSec4	0.822	1.899	SST_WEB	0.376	1.019
CSE1	0.850	2.269	PEU1	0.846	2.593	PU1	0.870	2.423	SoExp1	0.846	2.249
CSE2	0.847	2.145	PEU2	0.885	2.909	PU2	0.872	2.471	SoExp2	0.898	3.324
CSE3	0.851	2.136	PEU3	0.871	2.433	PU3	0.856	2.239	SoExp3	0.905	3.664
CSE4	0.766	1.531	PEU4	0.779	1.882	PU4	0.829	1.955	SoExp4	0.843	2.452

Source: (Author's Construct, 2020)

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Table 4. Summary of Hypothesis Results

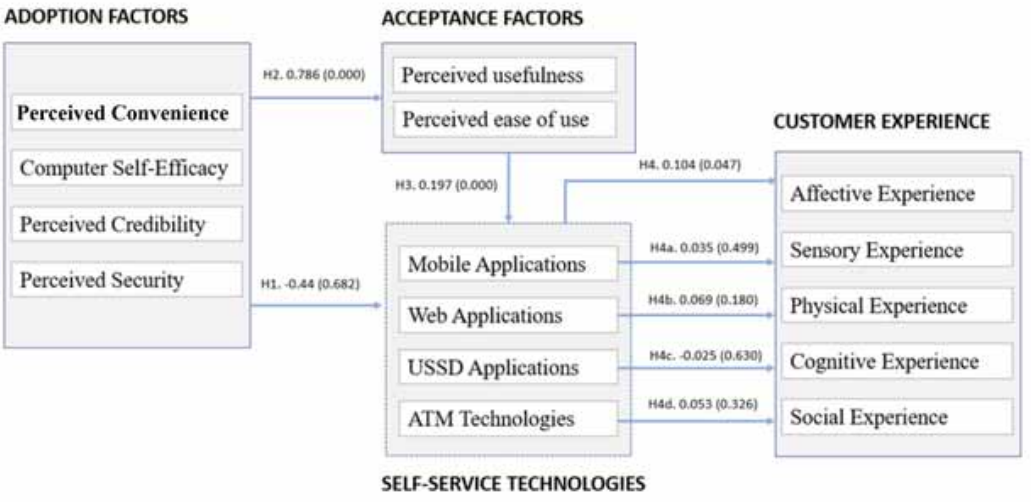
Hypothesis		Estimate	SD	T Statistics	P Values	Hypothesis
Adoption Factors -> Self-Service Technology	H1	-0.044	0.107	0.410	0.682	Not Supported
Adoption Factors -> Acceptance Factors	H2	0.786	0.025	31.840	0.000	Supported
Acceptance Factors -> Self-Service Technology	H3	0.197	0.044	4.527	0.000	Supported
Self-Service Technology -> Customer Experience	H4	0.104	0.053	1.985	0.047	Supported
Mobile App -> Customer Experience	H4a	0.035	0.052	0.677	0.499	Not Supported
Web App -> Customer Experience	H4b	0.069	0.052	1.343	0.180	Not Supported
USSD App -> Customer Experience	H4c	-0.025	0.052	0.483	0.630	Not Supported
ATM -> Customer Experience	H4d	0.053	0.053	0.984	0.326	Not Supported

**SD = standard deviation
Source: Author's Construct, (2020)

Collinearity Assessment

When two indicators in the model are highly correlated, it results in the problem of collinearity. The problem of collinearity is assessed using the Variance Inflated Factor measure (VIF). As some authors considers that there exists the problem of collinearity when VIF is equal to or greater than 5 (Hair, Ringle & Sarstedt, 2011), authors including (Diamantopoulos & Siguaw, 2006) argues that there exists the problem of collinearity when VIF is greater than or equal to 3. This study assumes a VIF threshold of 3. The results of the collinearity assessment indicated that all the indicators resulted in VIF less than 3 indicating that there was no problem of collinearity in the constructs except SoExp2, and SoExp3.

Figure 2. Validated structural model



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Results on Hypothesis Testing

The testing of the hypothesis was done using regression analysis carried out with the Smart PLS software and its bootstrapping functionality. The study presented seven hypotheses. At a 5% level of significance, three(3) of the hypotheses were supported whiles four(4) were not supported. *H2*, *H3* and *H4* were supported whiles *H1, H4a, H4b, H4c* and *H4d* were not supported. Table 4 shows a summary of the results of the hypothesis testing whiles Figure 2 shows the validated structural model based on the results of the hypothesis testing.

DISCUSSION

The study aimed at examining how different types of self-service technologies including ATM, mobile applications, web applications and USSDs individually or collectively influence customer experience in the banking industry with Ghana as a case. Specifically, the study examined the different types of SSTs used by customers, the experiences they gain from using SSTs, factors that affect SSTs adoption and the effect of SSTs adoption on customer experiences in the banking industry of Ghana. The result from the randomly sampled banking customers in Ghana revealed that customers' adoption of SSTs is influenced by their perception of SSTs usefulness and ease of use which increases their satisfaction level. Also, the combined usage of SSTs was identified to be resulting in customer experiences while the individual SSTs wasn't.

Banking customers adoption of the self-service technologies under study (mobile applications, web applications, USSD applications and ATM technologies) in the banking industry of Ghana was identified to be not directly influenced by the adoption factors (computer self-efficacy, perceived credibility, perceived security and perceived convenience) (H1: =-0.044, t = 0.410, p = 0.682, not supported). The adoption of SSTs by customers in the banking industry of Ghana does not depend on (1) how socially influential they may feel after using SSTs; (2) how secured they perceive the SSTs employed by their banks are (3) how good they are with using computer technologies and (4) how credible they perceive the SSTs adopted by their banks. This finding about the adoption factors directly influencing SSTs is not consistent with previous research (Yang, Liu & Ding, 2012; Kaushik & Rahman, 2015). Perhaps, these factors may directly influence banking customers' adoption of SSTs elsewhere due to demographic difference or perhaps the level of development. However, this is not the case in Ghana, following the empirical results of this study.

Further analysis revealed that, customers in the banking industry of Ghana prefer to use SSTs employed by their banks when they consider it useful and as well easy to use (H2: =-0.786, t = 31.840, p = 0.000, supported; H3: = 0.197, t = 4.527, p = 0.000, supported). This explains that customers in the banking industry of Ghana adopt to use SSTs employed by their banks when they perceive some level of usefulness and easy to operate qualities with them. This finding establishes that SST adoption factors are necessary but are relevant when they influence acceptance factors such perceived ease of use and perceived usefulness. This is in consonance with existing literature (Boon-itt, 2015; Ho & Ko, 2008).

More interestingly, there was a positive relationship between SSTs adoption and customer experiences when these SSTs are all available for use (H4: =0.104, t = 1.985, p = 0.047, supported), which was not the case when at least one SSTs was not available (H4a: = 0.035, t = 0.677, p = 0.499, not supported; H4b: = 0.069, t = 1.343, p = 0.180, not supported; H4c: = -0.025, t = 0.483, p = 0.630, not supported; H4d: = 0.053, t = 0.984, p = 0.326, not supported). This explains that, in the banks' quest to realize customer experiences in Ghana, they should employ all possible combinations of SSTs in their daily operations. The need for USSD technologies is justifiable by the fact that most Ghanaians do not have reliable internet access, also not everyone can afford to patronize broadband for internet banking (Asante & Hayford-Acquah, 2017). In such situations, resort to USSD technologies for transacting businesses. The need for ATM technologies can also be aligned with reasons such as the long waiting times in queues (Meuter *et al.*, 2000). Queuing in Ghanaian banks is very common and a major problem which goes against customer satisfaction most often. Also, the proliferation

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of mobile phone usage and internet facilities (though expensive in Ghana) makes it quite justifiable for the need of internet banking facilities to help customers transact business over the internet with their smartphone and or computers. Drawing from this finding, the customer satisfaction goal of the banking industry of Ghana can be achieved upon the implementation of all possible banking SSTs in order to account for every single customer's banking need with and without internet facilities.

IMPLICATIONS FOR RESEARCH

This study makes contributions to contemporary literature in the area of self-service technology, customer satisfaction and banking. The conceptual model highlights the type of self-service technologies employed and adopted by customers in the banking industries of developing countries such as Ghana (i.e. web applications, mobile applications, USSD applications and ATM technologies). It further shows the experiences (affective, sensory, physical, cognitive and social experiences) and factors that influence the adoption of SSTs (perceived usefulness and ease of use). The empirical evidence adds to extant literature by highlighting the positive effect of SSTs on customer experience. More interestingly, the study points out the relevance of deploying all possible combinations of SSTs or individual SSTs in the banking industry of developing countries such as Ghana as it influences customer experience. This study has developed a new potential research trajectory for exploration into future studies of whether banks should deploy all available SSTs or choose some SSTs over others to influence customer experience positively.

IMPLICATIONS FOR PRACTICE

In practice, this study has presented an insight to the banking industry by establishing a relationship between SSTs and customer experiences, the type of SSTs employed by the customers of the banking industry in Ghana and the factors that influence their adoption. This provides insight to bankers in deploying all possible combinations of the major SSTs (web applications, mobile applications, USSD applications and ATM technologies) used in Ghana. More specifically, the study established that, there was a statistically significant relationship between the combined effect of SSTs and customer experience while there existed an insignificant relationship between each of the individual SST and customer experiences. The Ghanaian banking industry must notice that technology is expensive, and that appropriate SSTs should be deployed where necessary. As established earlier, the Ghanaian community suffers high cost of internet and sometimes complete unavailability and in such cases, SSTs including ATM technologies and USSD applications should be deployed to reduce the cost of deploying SSTs at "wrong" geographical locations.

Practitioners need to understand that factors including customers' perceived convenience, computer self-efficacy, perceived credibility and perceived security does not directly result in customers' intention to adopt SSTs in the banking industry. And that, there is the need to deploy more useful and easier to use graphical user interface applications or technologies to customers, as they are the main influencing factors to customer SSTs adoption which tend to increase their banking experience.

CONCLUSIONS

The objective of the study was to examine the effect of self-service technologies adoption on customer experience in the banking industry of Ghana. Specifically, the study examined the types of SSTs used by customers, the experiences they gain from using SSTs, factors that affect SSTs adoption and the effect of SSTs adoption on customer experiences in the banking industry of Ghana. The analysis of the data revealed that customers in the banking industry of Ghana use web technologies, mobile applications, ATM technologies and USSD applications as SSTs. SSTs adoption by customers was found to be influenced by perception of SST's ease of use and usefulness as acceptance factors. A more thrilling revelation was that there was a significant relationship between the combined SSTs (web applications, mobile applications, ATM technologies and USSD applications) and customer

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experience; however, the relationship between each SST and customer experience turned out not significant.

LIMITATIONS AND FUTURE RESEARCH DIRECTION

The study did not peruse the effect of different SSTs (i.e. web applications, mobile applications, USSD applications and ATM technologies) on the various types of customer experiences (i.e. affective experience, cognitive experience, physical experience, social experience and sensory experience). Future research on this topic should test these effects which may explain why the individual effect of SSTs do not affect customer experience, but the combined effect affects customer experience. Furthermore, the study used data from only Ghana and that could have influenced the results, hence there is the need for future studies on this topic in another country.

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Determinants of Online Purchase Intention of Apparel Products in Lebanon

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ABSTRACT

This study investigates factors (perceived usefulness, perceived ease of use, privacy concerns, trust, innovativeness, and perceived benefits) affecting Lebanese consumers' attitudes toward online purchasing and their intention to purchase online. Data collected from 416 online shoppers were analyzed using regression analysis. All the factors affect attitude towards online purchasing except trust and privacy concerns which were not supported. In addition, results show that attitude toward online purchasing affect online purchase intention. This study is the first to empirically examine factors influencing online purchase intention towards apparel products in Lebanon. The study provides insight from an overlooked emerging country, Lebanon.

KEYWORDS

Apparel Products, Attitude Toward Online Purchasing, Lebanon, Online Purchase Intention

1 INTRODUCTION

The acceptance and usage of advanced technologies became a universal practice as people are keener to adopt a novel technology in their everyday lives (Natrajan, Balasubramanian, & Kasilingam, 2018). As a consequence of the technological advances in the marketplace during the past decades, online shopping has recorded enormous growth (Rubin, et al., 2020). This spectacular change in the consumer purchasing landscape has marked the naissance of online shopping (Faqih, 2016). The growth of online shopping has increased as businesses prefer to have an online presence (Rahman, et al., 2018). This act supports them to gain a competitive advantage relative to other companies in the market (Adnan, 2014). The new method of shopping helps consumers to purchase at any time, and wherever, to save money, and to search and access more information about the product (Al-Debei, et al., 2015).

Despite the benefits of online shopping, transactions over the internet are followed by some concerns which could discourage people and result in their unwillingness to engage in any form of online transactions (Beldad, et al., 2010). The perceived lack of human contact, lack of financial security and privacy, intangible products, failure of technology, and unfamiliar service encounters are some of the main reasons why consumers do not use online shopping (Aldhmour, 2016). When companies increasingly rely on the internet to deliver their goods and services, they should know how to maintain and develop their online presence (Beldad, et al., 2010). The key drivers that motivate customers to make an online transaction should be identified by the companies so that they can formulate strategies to enhance purchase intention (Adnan, 2014). A full understanding of the factors that motivate and deter consumers to buy online may provide novel insights that result in the growth of B2C e-commerce (Rahman, et al., 2018).

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For years, many businesses in Lebanon have started to operate online by offering several kinds of products via websites or social media pages as the online environment is cheaper. The popularity of online shopping in Lebanon as a developing country has not received much acceptance from local consumers. The new format of business conducted over the internet has not received much reputation and acceptance because perhaps consumers in developing countries are not quite set yet to admit this technology as it seems risky and insecure (Faqih, 2016). Moreover, since it is crucial to comprehend how consumers react to online shopping based on determining factors; consumers' attitude should be explored as it helps in the formation of a favorable direction, or not, toward online shopping. Based on this, our study aims to develop an apprehension of the factors that affect Lebanese consumers' attitude and to detect the influence of that attitude toward online purchase intention of Lebanese consumers to broaden the knowledge of companies in order to better develop their marketing strategies and secure their online presence from failure. Accordingly, there is a need to analyze the factors behind the shift of consumers' behavior toward using online markets (Aref & Okasha, 2020).

Our study draws on the commonly applied Technology Acceptance Model (TAM) (Davis, 1989), originally introduced in the organizational behavior domain, and applied widely in the online behavior (Aldhmour, 2016; Ben Mansour, 2016; Chiu, et al., 2005; Faqih, 2016). The two central elements of the TAM, namely perceived usefulness and perceived ease of use are included in our research model to study their influence on Lebanese consumers' attitudes. The modification of the TAM is mandatory because the model was established for a broad description of the factors influencing computer acceptance rather than for the peculiar behavioral intentions of online purchasing (Chiu, et al., 2005). Many information systems researchers have agreed on the validity of TAM in predicting the person's acceptance of several corporate information technologies but the constructs of TAM do not entirely represent the particular effects of technological and usage aspects that could shape the user acceptance of such technology (Ben Mansour, 2016). Consequently, we also employed the IDT by studying the influence of innovativeness on consumers' attitudes as suggested by previous research (Chiu, et al., 2005; Crespo & Del Bosque, 2008b; Huseynov & Yildirim, 2015). The compatibility between the TAM and the IDT helps to better explain the adoption of users for a specific information system because such a combination may be proficient to offer a stronger model rather than standing alone (Cheng, 2013). The research model includes also privacy concerns, trust, and perceived benefits to study their impact on consumers' attitudes. Previous literature asserts the importance of the impact of those elements on consumers' attitudes (Agag & El-Masry, 2016; Al-Debei, et al., 2015; Anic, et al., 2019; Faqih, 2016; Rahman, et al., 2018). These aforementioned factors are vital in influencing consumers' attitudes towards online purchasing. However, there is a lack of research applied on the Lebanese virtual market concerning this topic. Also, the effect of Lebanese consumers' attitude on Lebanese consumers' purchase intention is examined in our study to uncover the importance of attitude in affecting purchase intention over the internet.

Finally, apparel products are selected for the scope of this study. The attitude toward online shopping for sensory experiential products, such as apparel, is less strong than cognitive products, such as books and computer software (Kim & Kim, 2004). Most apparel shoppers are reluctant to make purchases online and prefer to shop in physical stores (Kim & Forsythe, 2007). By identifying which attributes are essential to online shoppers, retailers can develop better effective strategies for apparel products that make customers highly rate them better against not only competitors but also physical stores with no online presence (Kim & Kim, 2004). Accordingly, the current study is conducted in Lebanon and a model is developed to test factors affecting Lebanese consumer's attitudes and their effect on online purchase intention in the case of apparel products, specifically, clothes and footwear. Those factors are perceived usefulness, perceived ease of use, privacy concerns, trust, innovativeness, and perceived benefits.

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2 LITERATURE REVIEW

2.1 Technology Acceptance Model (TAM)

TAM was driven by cognitive psychology theories, namely, the theory of planned behavior (TPB) and the theory of reasoned action (TRA) (Ajzen, 1991). Developed by Fred Davis, it is considered the most accepted and used model to examine the acceptance and user's usage of specific information system applications, for example, internet banking, mobile applications, e-commerce systems, and online shopping websites. Initially, the model was established to describe the adoption of technology with an organization, but the constructs of TAM help also to explain the adoption of technologies by consumers used in their everyday lives (Natrajan, et al., 2018). The TAM has been built on two main building blocks which are perceived usefulness as well as perceived ease of use. According to Davis (1989, p. 320), perceived ease of use is "the degree to which a person believes that using a particular system would be free of physical and mental effort", and perceived usefulness is "the degree to which a person believes that using a particular system would enhance his/her job performance".

TAM is considered a valuable model in predicting consumer online shopping behavior (Cho, 2016). It explains the attitude toward information systems as well as predicting the use intention (Bigne-Alcaniz, et al., 2008). An individual attitude toward using an innovative system is mainly shaped by the ease of use and usefulness of the system (Aref & Okasha, 2020).

Regardless of the broad use of the TAM model in the field of research about the adoption of information technology, TAM was criticized for providing only general information on a person's views of innovative technology without taking into account individual characteristics (Agarwal & Prasad, 1999). Accordingly, there is a need for a more in-depth identification of the factors that cause the acceptance of the internet as a shopping medium (Bigne-Alcaniz, et al., 2008).

2.2 Innovation Diffusion Theory (IDT)

The IDT, introduced by Rogers (1962), is considered one of the most widely used theories to justify technological innovation. According to Rogers (1983, p. 5), an innovation is "an idea, practice, or object that is perceived as new by an individual or unit of adoption". In addition, diffusion is the process of "communicating a new technology through certain channels within a social system" (Rogers, 1983, p. 5). According to Johnson et al. (2003), diffusion indicates how innovations are transferred among people within a society and how they adapt or reject it over time (Johnson, et al., 2003). The theory has been used to examine different innovative contexts ranging from agricultural activities to high-tech products (Yi, et al., 2006).

According to the theory, the rate of adoptions depends on the perceived attributes and features of innovation, which referred to innovation characteristics and suggests many attributes possibly essential within different innovation adoption realms (Yi, et al., 2006). Furthermore, the IDT explains how attitude is formulated, how it is translated into the ultimate adoption or refusal decision, and how innovation features match this process (Karahanna, et al., 1999). The IDT claimed that prospective users build their decisions to reject or adopt an innovation according to beliefs formed about the innovation (Agarwal & Prasad, 1999). Substantial empirical support to the IDT has been received in justifying consumer acceptance in various disciplines, notably online shopping (Agag & El-Masry, 2016).

2.3 Apparel Products Over the Internet

The market share of online apparel retailing is growing considerably (Nasibov, et al., 2019). The profound diffusion of the internet caused by technological advancement and low-cost data rates led to growth in online apparel purchases (Kaushik & Dhir, 2019). Despite this, high purchases from physical stores indicate that apparel websites are not supplying consumers with the required product descriptions and presentations to aid them in their decision-making (Boardman & Mccormick, 2019). In online retail, the role of touching the product is diminished which causes uncertainties that reside in the consumers' minds (Rathee & Rajain, 2019). Besides this, clothing is categorized as an experienced

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good that differs extensively according to quality, body fit, price, and taste (Hansen & Jan, 2009). The inability to test the product in addition to the limited description available on the website may increase the uncertainty of the outcome associated with in-home shopping, specifically; for apparel products (Khakimdjanova & Park, 2005). Accordingly, consumers began to shop via online and offline channels together with the aim to eradicate the lack of physical experience (Nasibov, et al., 2019).

The available data about apparel products over the internet in Lebanon is limited or non-official. No previous official statistics were published about the percentages of sold apparel products online in Lebanon. A single report is published by BLOM Invest Bank in Lebanon (2015) which indicates that the top three purchased products over the internet in Lebanon are clothing, airline tickets, and hotel booking. According to the report, clothing products scored 38% of the products purchased online in Lebanon, followed by airline tickets (11%), and hotel booking (11%) (Daou & Mikhael, 2015).

2.4 Perceived Usefulness

Based on Davis (1989), perceived usefulness is "the degree to which a person believes that using a particular system would enhance his/her job performance" (Davis, 1989, p. 320). Usefulness deeply influences a user's tendency to use a specific tool of technology (Natrajan, et al., 2018). Perceived usefulness reflects consumer's perception that using a website for shopping will enhance the outcome of the shopping process. In other words, comparing prices and placing tracking orders will boost the effectiveness and efficiency of the process of shopping (Aref & Okasha, 2020). Those perceptions affect consumers' attitudes toward online shopping (Monsuwe, et al., 2004).

The usefulness of the system can influence consumers' attitudes and the decision whether they select an online store for purchasing or not (Al-maghrabi, et al., 2011). A simplified process and fewer steps involved would encourage consumers to use and to accept the internet to shop online (Aldhmour, 2016). According to Agag and El-Masry (2016), there is a positive relationship between perceived usefulness and consumers' attitude (Agag & El-Masry, 2016). Perceived usefulness of the electronic medium for shopping is positively correlated with attitude towards online shopping (Vijayasarathy, 2004). Therefore, the succeeding hypothesis is suggested:

H1: Perceived usefulness has a positive effect on Lebanese consumer's attitudes towards online purchasing.

2.5 Perceived Ease of Use

Perceived ease of use represents "the degree to which a person believes that using a system is free from physical and mental effort" (Davis, 1989, p. 320). Ease of use represents the non-complexity and friendliness of technology and the degree to which the internet is seen as effortless (Aldhmour, 2016). If the process was easy and does not require extra effort, shoppers are likely to use online websites to shop (Aref & Okasha, 2020). As much as a web facilitates the transaction and is easy to operate, consumers are more expected to accept it (Pavlou, 2003). Perceived ease of use depends on the navigation structure of the website, which is represented by the product indices, site maps, search functions, and the design and organization of the website (Beldad, et al., 2010).

In assessing attitude towards online shopping, perceived ease of use is highly related to that attitude (Vijayasarathy, 2004). Perceived ease of use has a positive impact on the attitude of consumers because if the process of purchasing online was simple, the evaluation of online shopping by the consumer will be positive (Agag & El-Masry, 2016). A significant positive association exists between perceived ease of use and consumers' attitude (Bigne-Alcaniz, et al., 2008). A system characterized by ease of use would generate a positive attitude (Aref & Okasha, 2020). On the other side, the results of Crespo and Del Bosque (2008a) documented that perceived ease of use does not affect attitude. Accordingly, the subsequent hypothesis is suggested:

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H2: Perceived ease of use has a positive effect on Lebanese consumer's attitudes towards online purchasing.

2.6 Privacy Concerns

Privacy concerns represent a serious barrier in front of consumers to participate in an online transaction such as online shopping, the reason behind this is that consumers have to disclose personal information (for example, home telephone number, credit card information, date of birth, social security number, etc.) (Gurung & Raja, 2016). Online shoppers consider that companies are responsible to safeguard their personal information and are most sensitive when companies fail to do so (Anic et al., 2019). Privacy concerns represent the fear of losing privacy as a result of entering personal information to accomplish an online operation (Dinev & Hart, 2005). Due to issues concerning privacy in the e-commerce setting, some consumers are reluctant to purchase online (Huseynov & Yildirim, 2015). With the unauthorized use of personal information and the limited ability of consumers to monitor any kind of those actions, consumers are considered vulnerable and there is an inherent hesitation to engage in an online transaction (Gurung & Raja, 2016).

Successful implementation of the features that permit consumers to manage their privacy within the online website is highly needed (Al-Debei, et al., 2015). Privacy concerns can result in severe problems to the safety and wellbeing of online activities, and it is necessary to be reduced to enhance attitude toward online purchasing (Faqih, 2016). High privacy concerns are deemed to negatively influence consumer's attitudes towards online shopping (Rahman, et al., 2018). Online privacy concerns negatively affect attitudes toward online purchases (Anic, et al., 2019). Therefore, the next hypothesis is suggested:

H3: Privacy concern has a negative effect on Lebanese consumer's attitudes towards online purchasing.

2.7 Trust

Trust in the website is a major matter in order to reduce the uncertainty of the online environment (Weisberg, et al., 2011). Trust is the degree of trustworthiness that an individual has about others and can be determined by their perceived benevolence, integrity, and competence (Lin, 2011). In the online environment, online companies are considered the trustee parties, whereas consumers are the trustor party (Gurung & Raja, 2016). The acceptance of an online transaction relies on the customer's trust in the online transaction, in the other partner of an online transaction, and the technology used to complete the transaction (Beldad, et al., 2010). Online companies are responsible for developing consumers' trust by being fair and keep the consumers' best interests in mind while performing an electronic transaction (Gurung & Raja, 2016).

The lack of buyer's trust is considered as a hitch faced by sellers to do business online (Akhlaq & Ahmed, 2015). Consumers' attitudes and trust are positively related in order to shop online (Monsuwe, et al., 2004). Trust remains a significant factor in modeling the attitude of consumers (Gurung & Raja, 2016). When consumers trust an online retailer, they will likely hold a positive attitude toward this retailer (Agag & El-Masry, 2016). Violation of trust in an online environment, in terms of misusing personal information, will negatively affect attitude toward online shopping and deter consumers to buy online (Monsuwe, et al., 2004). Therefore, the next hypothesis is suggested:

H4: Trust has a positive effect on Lebanese consumer's attitudes towards online purchasing.

2.8 Innovativeness

Innovativeness represents the "degree to which an individual is relatively earlier in adopting new ideas than other members of a system" (Rogers, 1983, p. 241). Innovativeness in information technology is

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seen as a symbol of risk-taking propensity that characterizes some persons (Lu, 2014). Persons with personal innovativeness characteristics can handle high levels of improbability, as well as dealing with uncertainty since they are more ready to accept and try new technology (Rogers, 1983). Information technology innovativeness is essential in determining the adoption of innovation because it captures the natural inclination of an individual to try novel technologies (Lu, 2014).

Innovativeness has a strong influence on consumers' decision to adopt new technologies so that consumers become more aware of new technologies and their adoption of them depends on personal traits such as personal innovativeness (Dewi, et al., 2019). Moreover, Huseynov and Yildirim (2015) suggested that there is a positive association between innovativeness in new technology and attitude toward online shopping. Innovativeness is a personality characteristic that can facilitate the formation of a favorable attitude toward online shopping (Limayem, et al., 2000). On the other hand, Crespo and Del Bosque (2008b) in their study concluded an insignificant relationship between innovativeness and attitudes toward internet shopping. Following the recent studies (Dewi, et al., 2019; Huseynov & Yildirim, 2015), the following hypothesis is suggested:

H5: Innovativeness has a positive effect on Lebanese consumer's attitudes towards online purchasing.

2.9 Perceived Benefits

Consumers strive to maximize their perceived benefits derived from an online deal associated with high risk and uncertainty (Forsythe, et al., 2006). Perceived benefits are "the sum of advantages that meet a consumer's needs or wants" (Wu, 2003, p. 40). The perceived benefits of online shopping could make the switch of a consumer from a physical store to online shopping (Akroush & Al-Debei, 2015). When selecting an online retailer to purchase from, consumers pursue the benefits that they will receive (Forsythe, et al., 2006).

Perceived benefits are expressed in terms of time-saving, economic advantage, browsing products with minimal efforts, obtaining knowledge about company and brands, the ease to compare products features, and that internet shopping provides anonymity when purchasing sensitive products (Al-Debei, et al., 2015; Eastin, 2002; Monsuwe, et al., 2004)

It was found that the perceived benefits of online shopping have an impact on consumers' attitudes and their intention to buy online (Adnan, 2014). The more the perceived benefits of a certain website, the more favorable attitude toward online shopping becomes (Al-Debei, et al., 2015). The favorable effect of perceived benefits on attitude was supported by the study of Agag & El-Masry (2016). In addition, Zhou, et al. (2007) was found that shopper attitude partially mediates the association between perceived benefits and online shopping intention. Therefore, the following hypothesis is proposed:

H6: Perceived benefit has a positive effect on Lebanese consumer's attitudes towards online purchasing.

2.10 Attitude and Intention in the Online Purchase Setting

Attitude is deemed as one of the key indicators that contributed significantly to the prediction of intention (Aldhmour, 2016). Attitude refers to an individual's relatively steady evaluations, feelings, and propensities toward an object or idea (Wu, 2003). As for Anic et al. (2019), attitudes towards online shopping indicate consumers' evaluations containing beliefs and psychological states about the perceived outcome of online purchases (Anic, et al., 2019). To increase consumers' intention to shop online, companies should work hard to increase consumer's positive attitudes towards online shopping so that sales of products via the internet will increase (Aldhmour, 2016).

Consumer purchase intention is considered to be fundamental in forecasting consumer behavior (Ariffin, et al., 2018). Intentions are the degree of how much effort people are willing to exert; they are the indications of how people are planning to try as well (Ajzen, 1991). If a consumer feels "good" about online activity, then he/she is more likely to shape intention to use it for shopping (Al-maghrabi,

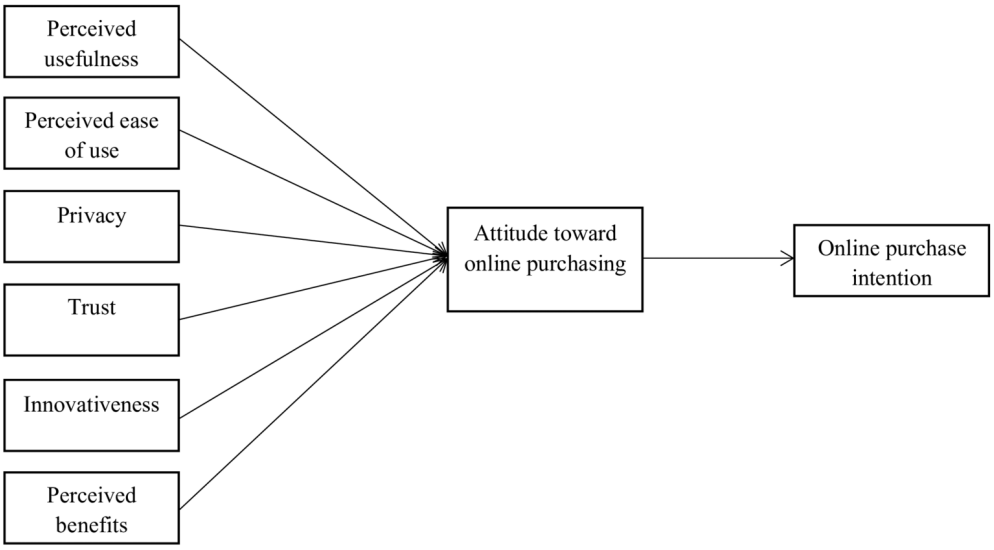
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et al., 2011). A positive attitude toward online shopping will result in an intention to use the internet for purchasing (Kim & Park, 2005). Attitude toward online shopping contributes significantly to the intention to shop online (Limayem, et al., 2000). The measurement of attitudes is important as it is considered an influencer factor as well as having a positive impact on online purchase intention (Akroush & Al-Debei, 2015). Therefore, the following hypothesis is suggested:

H7: Lebanese consumer's attitude towards online purchasing has a positive effect on their intentions to purchase online.

Based on the aforementioned literature, our conceptual framework is proposed (figure 1) to depict the relationship between antecedents and consequences of Lebanese consumer's attitude toward online shopping in the apparel industry.

Figure 1. Conceptual Framework



3 METHODOLOGY

3.1 Sample and Data Collection

The data were collected via a questionnaire survey using Google forms. The questionnaire was sent via electronic mediums such as emails and social media channels. It includes two parts; the first is designed for demographic information (namely, age, gender, income, educational level, occupation, income), in addition to questions about how much they purchased online and the duration of the purchasing process. The second part includes thirty-four statements related to the eight variables, in which respondents were asked to specify their degree of agreement or disagreement following each statement. The target population was Lebanese consumers. The selected participants must be Lebanese citizens, above eighteen years old, and have made an online purchase of an apparel product. Following those conditions, a non-probability sampling technique was used which is the purposive

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sampling technique to choose the respondents of the sample. The purposive or judgmental sampling technique conforms to a set of specific criteria or conditions (Adams, et al., 2007).

To ensure the readability and clarity of the questionnaire, a pilot study was conducted on 20 respondents. The results of the pilot study suggested minor modifications should be made to the questionnaire. The questionnaire was then distributed by email and social media channels. The desired sample size for a judgmental or purposive sample depends on the purpose of the study and the cases are selected in a way to fulfill the objectives of the study (Sahu, 2013). After looking at similar studies, the sample sizes are often more than 300 respondents. As for Dang and Pham (2018), 336 cases were selected, 486 cases were selected in the study of Al-Debei et al. (2015), 350 cases selected in the study of Ariffin et al. (2018), and 400 cases selected in the study of Faqih (2016). Accordingly, in our study, a total number of 506 questionnaires were distributed to the respondents and 416 questionnaires were usable recording a usable rate of 82.2%.

3.2 Measurement

All statements of the questionnaire are adopted from previous research studies and modified as per the requirements of the current study. All the variables were measured using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). There is a total of thirty-four items adapted to measure the eight variables of the model. Table 1 presents a summary of the number of items for each variable and the selected sources. In addition, table 2 shows the factor loading of each item on its corresponding construct.

Table 1. Measurement scales

Constructs	*Number of items*	*Sources*
PU	4	(Faqih, 2016)
PEOU	5	(Faqih, 2016)
PR	4	(Anic et al., 2019)
TR	4	(Al-Debei et al., 2015)
INOV	5	(Chiu et al., 2005)
PB	4	(Forsythe et al., 2006)
ATT	4	(Crespo & Del Bosque, 2008b)
OPI	4	(Crespo & Del Bosque, 2008b)

4 DATA ANALYSIS

The quantitative data was carried forward for analyses using the Statistical Package for Social Scientists (SPSS) version 20. This software is used by many studies in the marketing and business research fields.

4.1 Measurement Model

Evaluation of the measurement model entails scrutinizing the reliability and validity of the items and the corresponding constructs to ensure the used constructs' measures are satisfactorily valid and reliable before testing the nature of correlation in the overall model. Table 2 presents the factor loading for all the items. The factor loadings more than 0.50 are accepted. Only item OPI4 has recorded a factor loading of 0.354 less than 0.50, and as a result, it was deleted.

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Table 2. Factor loading

Construct	*Items*	*Factor Loadings*
Perceived usefulness	PU1	0.874
	PU2	0.823
	PU3	0.880
	PU4	0.862
Perceived ease of use	PEOU1	0.862
	PEOU2	0.864
	PEOU3	0.882
	PEOU4	0.512
	PEOU5	0.823
Privacy concerns	PR1	0.897
	PR2	0.939
	PR3	0.928
	PR4	0.910
Trust	TR1	0.779
	TR2	0.909
	TR3	0.892
	TR4	0.803
Innovativeness	INOV1	0.886
	INOV2	0.894
	INOV3	0.839
	INOV4	0.920
	INOV5	0.882
Perceived benefits	PB1	0.793
	PB2	0.884
	PB3	0.884
	PB4	0.854
Attitude toward online purchasing	ATT1	0.856
	ATT2	0.924
	ATT3	0.929
	ATT4	0.922
Online purchase intention	OPI1	0.932
	OPI2	0.933
	OPI3	0.901
	OPI4	0.354

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Table 3 depicts the Cronbach's alpha for all the variables. The alpha coefficients of perceived usefulness, perceived ease of use, privacy concerns, trust, innovativeness, perceived benefits, attitude toward online purchasing, and online purchase intention ranged between 0.804 and 0.938 (i.e., higher than 0.5). Thus, the results indicate that the items have reasonably high internal consistency.

Table 3. Results of Cronbach's Alpha

Constructs	*Number of items*	*Cronbach's alpha*
PU	4	0.882
PEOU	5	0.851
PR	4	0.938
TR	4	0.868
INOV	5	0.927
PB	4	0.876
ATT	4	0.929
OPI	4	0.804

The measures of variables were chosen based on an in-depth review of the existing literature. Two tests were used to examine the validity and suitability of data, namely, Kaiser-Meyer-Olkin measure of sampling adequacy (KMO) that must be higher than 0.50, and Bartlett's Test of Sphericity that must be significant. Table 4 shows that KMO values for the study variables can be considered satisfactory. Bartlett's Test of Sphericity was significant at $p<.001$. Accordingly, the data outputs are satisfactory to examine factor analysis.

Table 4. Validity test KMO and Bartlett's

Constructs	*KMO*	*Bartlett's*	*Significance (p-value)*
PU	0.813	644.545	0.000
PEOU	0.846	927.424	0.000
PR	0.770	747.715	0.000
TR	0.766	693.354	0.000
INOV	0.870	1091.040	0.000
PB	0.775	695.420	0.000
ATT	0.856	1260.383	0.000
OPI	0.761	875.112	0.000

4.2 Hypothesis Testing

In this section, multiple linear regression was employed to test the effect of the independent variables (perceived ease of use, privacy concerns, perceived usefulness, trust, innovativeness, and perceived benefits) on the attitude toward online purchasing (H1 to H6). In addition, a single regression model

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is applied to test the effect of attitude toward online purchasing and online purchase intention (H7). Table 5 shows the outcomes of hypothesis testing from H1 to H7.

Table 5. Results of hypothesis testing

Hypotheses	*Beta coefficient*	*SD error*	*t-value*	*Sig.*	*Decision*
H1	0.272	0.041	6.168	0.000	Supported
H2	0.132	0.028	3.093	0.002	Supported
H3	0.064	0.030	1.592	0.112	Not supported
H4	0.056	0.036	1.374	0.170	Not supported
H5	0.102	0.024	2.610	0.009	Supported
H6	0.456	0.043	11.864	0.000	Supported
H7	0.617	0.038	15.944	0.000	Supported

Five hypotheses out of seven were supported, while two hypotheses were not supported. The statistics presented in Table 5 show the support of H1 (β=0.272, t=6.168, p=0.000<0.05), thus, the positive relationship between perceived usefulness and attitude toward online purchasing was supported. The second hypothesis (H2) was also supported (β=0.132, t=3.093, p=0.002<0.05), which means perceived ease of use has a positive impact on attitude toward online purchasing. The positive relationship between innovativeness and attitude toward online purchasing was found to be significant (β=0.102, t=2.610, p=0.009<0.05), thus, supporting H5. Moreover, H6 was supported (β=0.456, t=11.864, p=0.000<0.05), which means there is a positive association between perceived benefits and attitude toward online purchasing. Therefore, findings revealed that there are positive relationships between each of perceived usefulness, perceived ease of use, innovativeness, and perceived benefits with attitudes towards online purchasing.

The results do not support the proposed relationship between privacy concerns and attitude toward online purchasing (β=0.064, t=1.592, p=0.112>0.05). Therefore, H3 was not supported. In addition, the relationship between trust and attitude toward online purchasing was not supported (β=0.056, t=1.374, p=0.170>0.05). Therefore, H4 was not supported.

Finally, the relationship between attitude toward online purchasing and online purchase intention was also tested. Table 5 shows that there is a significant link between attitude toward online purchasing and online purchase intention (β=0.617, t=15.944, p=0.000<0.05). Therefore, H7 was supported.

5 DISCUSSION AND IMPLICATIONS

The online market is struggling to compete and survive in a turbulent global market (Fang, et al., 2011). It is essential to detect the causes behind why consumers choose to visit an online store and in which manner their behaviors differ following their predisposition toward online shopping (Pappas, et al., 2017). The noticeable growth of internet usage and digitalization of society has contributed to a remarkable change in consumers' shopping behaviors and preferences (Han, et al., 2018). Therefore, the current study examines the effect of perceived usefulness, perceived ease of use, privacy concerns, trust, innovativeness, and perceived benefits on attitude toward online purchasing and in turn on online purchase intention in the Lebanese market concerning apparel products. Seven hypotheses were proposed based on a thorough literature review. Following the regression analysis, five hypotheses out of seven were supported.

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Perceived usefulness positively affects attitude toward online purchasing, supporting H1. This finding is in harmony with past studies on the factors affecting the attitude toward online purchasing (e.g., Agag & El-Masry, 2016; Ben Mansour, 2016; Bigne-Alcaniz, et al., 2008; Cass & Fenech, 2003; Crespo & Del Bosque, 2008a; Faqih, 2016). Our findings demonstrated that the more useful the system used to shop online is, the more this could result in a positive attitude toward online purchasing. A system that seems useful and helpful for people could help them to complete their tasks efficiently and effectively so that people are more likely to adopt it (Faqih, 2016).

Perceived ease of use was also found to shape attitude toward online purchasing positively, supporting H2. Previous studies (e.g., Agag & El-Masry, 2016; Ben Mansour, 2016; Bigne-Alcaniz, et al., 2008; Cass & Fenech, 2003; Chiu, et al., 2005; Faqih, 2016) confirmed this result. These studies found that there is a positive relationship between perceived ease of use and attitude toward online purchasing. Therefore, perceived ease of use helps to shape a positive attitude toward purchasing online, so that systems that are characterized by their ease of use and less complexity motivate consumers to shop online. To increase the number of customers, companies have to design a user-friendly website with elements that facilitate the purchasing process (Bigne-Alcaniz, et al., 2008).

Surprisingly, the relationship between privacy concerns and attitude toward online purchasing is not significant, thus H3 was not supported. This result is coherent with the findings of previous studies (Rahman, et al., 2018; Vijayasarathy, 2004). The care of online retailers in developing and sharing privacy policies and statements has reduced the privacy concerns of consumers when operating online (Vijayasarathy, 2004). Besides, consumers will continue to disclose their personal information relying on hypothetical contracts and moral norms despite their privacy concerns (Bandara, et al., 2019). Relying on moral norms, this can weaken the effect of privacy concerns toward consumers' attitude. Some shoppers consider that the online retailer will behave according to the moral norms and will not misuse their personal information or sell it to other companies. When companies implement the necessary protection rules for privacy issues, online users will not worry much concerning their privacy and personal information (Anic, et al., 2019).

Moreover, the positive relationship between trust and attitude toward online purchasing was not realized, so H4 was not supported. Previous studies (e.g., Heijden, et al., 2003; Rahman et al., 2018) confirmed the result of our study. On the other side, the positive relationship between trust and attitude was supported and confirmed in other studies (Akroush & Al-Debei, 2015; Al-Debei, et al., 2015). This finding may be described by the fact that consumers might take the risk of purchasing online without making any consideration to trust the other party. A consumer might buy a computer from a suspicious seller over the internet simply because the computer is highly discounted from its original price without trust or at a minimal level of trust (Kim, et al., 2008). Moreover, the selected respondents of our study have a previous online shopping experience, so they have experienced online environments to purchase apparel products and they tended to trust the party they choose to purchase from beforehand. A past purchasing experience can increase the sense of control over the system and that sense leads to more trust in the system (Weisberg, et al., 2011).

As for the relationship between innovativeness and attitude toward online purchasing, H5 was supported, and the positive relationship is confirmed in this study. The findings are consistent with previous studies (Chiu, et al., 2005; Citrin, et al., 2000). This finding means that the more innovative a consumer is, the more positive the attitude toward online purchasing becomes. The online innovative consumers can aid to promote online shopping over the website (Limayem, et al., 2000). An innovative consumer could spread his experience among other consumers which can push many to take the initiative to try the website to purchase online, so that could explain the positive link between innovativeness and attitude toward purchasing online.

The relationship between perceived benefits and attitude toward online purchasing was also confirmed in this study, thus, H6 was supported. Many previous studies confirmed the positive relationship between perceived benefits and attitude toward online purchasing (Agag & El-Masry, 2016; Akroush & Al-Debei, 2015; Al-Debei, et al., 2015; Wu, 2003). The more benefits the consumer

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gets from online purchasing, the more positive attitudes are generated. The online shopping benefits are many and diverse such as comparing prices, time-saving, low-cost products, delivery service, shopping in the privacy of home, and other benefits. All those benefits could be the reasons behind the positive relationship generated between perceived benefits and attitudes toward online purchasing. The more the benefits are perceived by online consumers, the more likely that a positive attitude is formed toward purchasing online (Al-Debei et al., 2015).

Finally, a positive link between attitude toward online purchasing and online purchase intention was supported in the current study, therefore, H7 was supported. This finding confirms of past studies on the association between attitude toward online purchasing and online purchase intention (e.g., Ben Mansour, 2016; Bigne-Alcaniz, et al., 2008; Chiu, et al., 2005; Crespo & Del Bosque, 2008a). The more positive is the attitude of a consumer, the more this will lead to a high intention of the consumer to purchase online. The positive attitudes push consumers to try online shopping and have serious intentions to purchase online.

The findings of this study could be valuable for companies who would like to have an online presence or to start an e-business in the Lebanese market. Understanding consumer behavior is a major matter especially when the concept of a traditional business is changing to a virtual one. The manager of an online website as well as website developers must work on providing consumers a quick, simple, and secured online shopping experience to remove the barriers of personal contact as well as to provide high service quality. Working on concerns related to the usefulness and ease of use of the system can encourage many consumers to engage in online shopping. Targeting consumers who are more innovative to purchase online could also encourage consumers who are less innovative to try the experience of purchasing apparel products online. Moreover, increasing the benefits of purchasing online such as free delivery fees, fast delivery period, discounted prices, and loyalty programs for online consumers could attract more people to use online shopping.

6 CONCLUSIONS

The purpose of this study was to examine the factors affecting Lebanese attitude toward online purchasing for apparel products, as well as examining the link between attitude toward online purchasing and online purchase intention. Six factors that affect consumers' attitudes toward online purchasing are selected from previous literature: perceived usefulness, privacy concerns, trust, perceived ease of use, innovativeness, and perceived benefits.

Four independent variables are found to affect attitudes toward online purchasing which are perceived usefulness, perceived ease of use, innovativeness, and perceived benefits. However, privacy concerns and trust were found to have an insignificant relationship with attitude toward online purchasing. In addition, the link between attitude and intention in the online purchasing environment was found to be significant and positive.

7 LIMITATIONS AND SUGGESTIONS FOR FUTURE RESEARCH

This study is not limitations free, which provides an opportunity for more research in the future. The study did not specify if the website used to make an online purchase is international or local and this could make a difference in the opinion of respondents considering online shopping. Some might see that buying online from an international website is more secured since it is well known around the world. Therefore, future research could replicate our study and investigate the effect of local vs international websites on the online shopping intention of consumers.

No previous studies have been published before in Lebanon to examine the factors affecting online shoppers' attitude and their intention to shop online. Future research is highly recommended to investigate the attitudes of consumers toward online shopping and to add more factors that might motivate or discourage consumers to shop online. Those studies could be a reference for managerial

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practices in the online setting and to better attract customers as well as developing strong strategies to compete in the e-environment. Besides, the current study examined online purchase intention as the dependent variable, future studies could take our study a step further by exploring if this intention has translated into the actual behavior of purchasing.

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Flow Experience and Consumer Willingness to Pay in Online Mass Customization Processes

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ABSTRACT

This research aims to identify the effects of perceived product value and flow experience during product design on willingness to pay (WTP) and purchase probability in online mass customization. The participants were asked to design shoes to their taste in a custom shoe website. After the design experience, the participants were asked to complete the questionnaire. The analyses suggest that flow experience has a direct effect on consumers' WTP and an indirect effect on purchase probability through the mediating role of perceived value of mass customized products. Furthermore, perceived value of a mass customized product has a significant effect on purchase probability and no effect on WTP found in the analyses.

KEYWORDS

Flow Experience, Mass Customization, Online Shopping Experience, Willingness to Pay

INTRODUCTION AND BACKGROUND

This study's point of departure is that although mass customization is an extremely attractive and popular concept, it has so many obstacles. The co-design process in mass customization has some complex, risky, and uncertain features that may take customers away from participating in customization (Piller et al. 2005:0). For instance, customers may be confused due to the many options offered in the design process or may perceive a risk of being misled by the online presence of the product they want to acquire. This problem may cause customers to give up buying the items that they co-designed or to reject paying more for the products.

When it comes to business, the researchers indicate that customization allows (and forces) businesses to charge higher prices for their co-designed products and services (Dewan et al. 2003; Cavusoğlu et al. 2007; Jiang et al. 2015) due to two reasons. Firstly, customers are frequently ready to pay a premium that reflects the added value of satisfaction that emerges from customization. Secondly, the raised cost of integrating technology with the customization process requires higher price charges or higher unit costs (Cavusoglu et al. 2007). Therefore, proper management of customization processes by the businesses that desire to apply mass customization as a strategic marketing tool is of great importance. Whereas these reasons are not sufficient, there is another aspect that needs to be focused on by the customization-oriented business under heavy competitive market pressures, i.e.,

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online marketing dynamics. It is relatively fast and easy to compare prices in online markets, which supports online customers that are after bargain prices. For instance, the main reason why Turkish consumers prefer online shopping is that they want to access more reasonably priced products (PWC Retail Survey, 2016; Webloyalty Consumer Research, 2016). Similar findings with European sampling were offered by Aichner and Coletti (2013) and found that for 65% of buyers, the price was the primary source of consumers' attraction. However, as mentioned before, prices for mass customized offers are somewhat higher than standard products, which may reduce consumers' intention to buy mass-customized products and their intention to pay more for them.

In sum, online customization is an attractive strategic tool that offers benefits and obstacles to both businesses and customers. When overcoming the barriers, handling the customization process is the first point to focus on. In their work, Jiang et al. (2015) emphasized that as customization activities become more enjoyable, consumers are likely to engage in them more often. They also suggested the concept of "process value" and related it to Csikszentmihalyi's Flow Theory (1975). Nevertheless, their measure of process value was limited to covering "flow experience" fully. However, Novak et al. (2000) have reported that flow experience is an essential variable in understanding online consumer behaviors in the online environment. Franke and Piller (2003;12), whose works are frequently referenced in mass customization studies, identified a research gap and emphasized that flow experience should be examined as a variable in mass customization studies.

Merle et al. (2010) defined the mass customization experience value as "the interaction between the individual and the product design using a specific method of preference revelation." Csikszentmihalyi (1977, p.36) defined the flow as "the holistic sensation present when acting with total involvement; it is what people feel when they enjoy what they are doing." The term resonates with Merle et al.'s (2010) concept of the hedonic value of mass customization experience.

Considering the research gap pointed and the resonations of the terms offered both in mass customization literature and flow theory, the present paper suggests that if mass customization experience is enriched with an ecstatic state of deep concentration (Csikszentmihalyi describes this ecstasy state as flow) while conducting mass customization, it will increase mass-customized product value. Consequently, customers' probability of buying and WTP more for the designed product in an ecstatic state will be positively affected.

As mentioned before, while overcoming online mass customization obstacles, handling pricing is the second point to focus on. It is challenging to sell customized products, which are somewhat higher to the consumers for whom low prices are known as the most significant source of motivation in preferring online shopping. Ferguson et al. (2014) have stated that economic and cognitive limits are essential factors to consider; if these limits are exceeded, a well-customized product will not be purchased.

Offering consumers a process of designing their products may be perceived as a complex, uncertain, and risky process. Besides, there are online shopping and price dynamics to be considered. Also, it is unknown whether the complicated process of mass customization creates a "flow experience" for the customers. Even if it creates a flow experience, this experience will also affect the product's perceived value. This observation makes up the main problem of this study.

Based on the obstacles mentioned above, the objective of this study is; to empirically identify the direct and indirect effects (through the mass-customized product value) of flow experience on purchase probability and WTP.

THEORETICAL BACKGROUND

Mass Customization and Mass Customized Product Value

Stan Davis (1987) used the term "mass customization" for the first time in his work Future Perfect. He defined mass customization as "a process that provides a one-of-a-kind product manufactured

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on a large-scale basis". Following this perspective, Kaplan and Haenlein (2006, p. 177) defined mass customization as "a strategy that creates value by company-customer interaction at the design stage of the operations level to create customized products". Piller (2004, p.314) asserted that there was no common understanding of the concept of mass customization in the literature; therefore, various definitions were proposed. Piller (2004) also emphasized that mass customization cannot be an academic discipline or a business strategy adopted by managers unless a standard definition and understanding is decided. According to Piller, the definition of mass customization should include four elements which are; designing the product together with the customer, meeting the needs of each customer, fixed solution space, fair price, and cost level. Therefore, these four elements are essential in distinguishing mass customization from other concepts of customization.

As Aichner and Coletti (2013) stress, mass customization is necessary to improve a user's personal benefit and demonstrate personal status and strength. Merle et al. (2009) identified two different mass customization value categories as; product and experience value. Similarly, Broekhuizen and Alsem (2002) identified two different kinds of mass customization benefits; instrumental and hedonic benefits. Schreier (2006) identified four different categories of mass customization benefits; perceived uniqueness, functional benefits, pride of authorship and process benefit of self-design (i.e., hedonic, experiential benefit).

Anderson et al. (1992, p.5) define value as "the perceived worth in monetary units of the set of economic, technical, service and social benefits received by a customer in exchange for the price paid for a product, taking into consideration the available suppliers' offerings and prices". In the process of mass customization, the value evaluations of customers who design their very own products via producers' production bases may get a little complicated. When the buyer is involved in the production process, they spend more time and competencies. Simultaneously, Ferguson et al. (2014) assert that customers prefer customized products by the desire to possess their self-designed object without paying the high price of craft manufacturing. However, there is a price premium of the customized product in mass customization compared to the standard one (Piller et al. 2004). At this point, by combining industrial and consumer points of view, there must be an equilibrium between the value evaluations of consumers and the value offers of producers. The equilibrium lies under the assumption that consumers who are willing to buy mass customized products offered by the firms that provide customized products to the masses have an additional WTP more for their self-finished goods. Mass customization "implies that the same large number of customers can be reached as in mass markets of the industrial economy, and simultaneously they can be treated individually as in the customized markets of pre-industrial economies" (Davis, 1987, p.169).

Flow Experience in CMEs and Mass Customization

Mihalyi Csikszentmihalyi (1975), who suggested the flow theory to the psychology literature, defined flow as "a peculiar dynamic state, a holistic sensation, and a source of happiness that people feel when they act with total involvement" (p. 36). Thus, he evolved his terminology and updated the flow definition as "fully immersion, focus, and mental attachment to an activity (Csikszentmihalyi, 1990)". Later, he described the flow state as "a person's experiences at the highest levels of well-being" (Csikszentmihalyi, 2014).

In her paper, Reid (2004) claimed a positive relationship between flow and creativity; nevertheless, her paper relies on case study reports and randomized trials. She also suggests a need for larger trials and more data. Csikszentmihalyi (2014) notes;

"in creativity or optimal experience, I have found that it is always a struggle, and the struggle has to do with essentially opening yourself up and yet delving deeply into yourself… So I see flow as a very important dynamic in the evolution of complexity. It gives you the incentive, the motivation, the reward for going beyond what you have" (p.274).

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Csikszentmihalyi's way of explaining flow and creativity is echoing "mass customization" from consumers' perspectives. Both flow and mass customization theories denote rebelling the standard offers, asking for more, and feeling psychic energy from the creation process (Schreier 2006; Trentin et al. 2014). Besides, the flow concept was also advised as a possible metric of the online consumer experience (Koufaris, 2002, p.207). In their study, Smith and Sivakumar (2004) explored the relationships between online buying behaviors (browsing, buying, and repeat purchasing) and flow experience. Their findings indicate that experienced flow during online shopping can increase the willingness to buy and shop online. Then again, in his critical literature review, Obada (2013) summarized reported relationships between flow and behavioral intentions such as online purchase intentions, revisit intentions or intention to use the Web-based on the works of Agarwal and Karahanna (2000), Luna et al. (2002), Luna et al. (2003), Richard and Chandra (2005), Sanchez-Franco (2006), Bridges and Florsheim (2008).

Flow is a state of rising (ecstasy) that does not emerge when the individual feels very relaxed and takes a rest; on the contrary; it occurs when the individual feels nervous/stressed, has difficulty, and feels good despite all his/her stress and uneasiness (Csikszentmihalyi, 1990). In mass customization, the design process also involves some difficulties for the individual due to its complex, uncertain, and risky structure, as we stated previously. It also requires a certain degree of creativity. However, despite the difficulties, the individual also has the potential to focus on capture the image that he/she has formed the mind while designing the product, be very curious about the product to be obtained as a result of the design process, get the pleasure of creating his/her product, feel that he/she holds the control of the product that will appear in direct proportion to the diversity of the design tools provided by the site in his/her hands.

Besides, the flow may also be a solution for price concerns; in their research, Congwen et al. (2010) exemplified the Taobao website and stated that the site's motto "seek what you like" has become Taobao users' driving force; the joy shoppers found what they like accidentally, its value is much higher than the discount prices. Hinting that if a customer finds pleasure and experiences an ecstatic state of creating and designing his/her product via mass customization, he may buy the product he/she designed at a higher price.

Purchase Probability

One of the most frequently used methods of estimating target consumers' future buying behavior is to use purchase intention measures. Juster et al. (1966) reported that the numerical measurements would give more accurate results than the intention measurements made through verbal expressions (high probability vs. small probability). They proposed a 10 point numerical purchase probability scale. Clawson (1971) stated that the most commonly used purchase intention scales used until the mid-1960s use 5 point scales. The scales offer statements such as "yes, definitely will buy", "yes, probably will", "might, perhaps, it depends", "no", "don't know". After saying "no", some of the respondents change their decisions and buy the product, which leads to numerous errors in the prediction. They stated that this error was mostly due to the verbally expressed scale's inadequate calibration (partitioning).

As a solution, Clawson then in 1971 proposed a scale between 0% and 100% where participants were asked about their possibility of purchase. Dissatisfaction with the accuracy of purchasing behavior estimates based on demographic and socioeconomic variables has made the researchers investigate consumer purchasing intentions and attitudes as an alternative measurement (Juster 1966; Heald 1970; Gabor & Granger 1972). Wright and Macrae (2007) argued that the likelihood of purchasing is also a form of determining the intention of purchasing, but the responses to the two measurements will differ (p.18). Based on these observations, in this study, the probability of purchase is measured with a statement asking the respondents' likelihood of buying their own-designed sneakers in the next 12 months. The participants were asked to mark between 0% and 100% indicating the probability of purchase.

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Willingness to Pay (WTP)

Wertenbroich and Skiera (2002) conceptualized the term WTP as "the maximum price a buyer is willing to pay for a given quantity of a good". Many recent empirical studies revealed that consumers want to pay more for self-designed products. For example, in their paper in which they explored users' WTP for their self-designed watches Franke and Piller (2003) found out that the WTP for a customized watch is higher than the same quality standard WTP. On average, they found an increase of 100% WTP for self-designed watches. Similarly, Schreier (2006) found an increase of 207% in the WTP for self-designed mobile phone covers, an increase of 113% in the WTP for self-designed T-shirts, and an increase of 106% in the WTP for self-designed scarfs. Yu and Yang (2020) found an increase of 41.05% in the WTP for men's ready to wear dress shirt. Kamali and Locker (2002) also reached similar results indicating that customers are ready to pay more to the self-designed products than the standards.

Emphasizing the importance of experience in a design process, Merle et al. (2009) proposed an integrated framework that combines mass customization value components with the WTP for mass customized products. In their model, they divided mass customization value components into two: product value and experience value. They concluded that while the value of the mass customization experience (i.e., experience value) had no direct effect on the WTP, on the other hand, indirectly, the value of the customized product (product value) increases the WTP.

In this paper, the participants were asked their maximum amount of money (in Turkish liras) they would like to pay for a pair of sneakers that are not designed by themselves (i.e., standards) and their maximum amount of money (in Turkish liras) they would like to pay for a pair of sneakers designed by themselves, and their WTP (ΔWTP) is calculated.

HYPOTHESES DEVELOPMENT

When the literature on mass customization is viewed, the empirical findings suggest that consumers tend to buy more customized products than standards (Franke et al., 2009; Kamali and Locker, 2002; Wu et al., 2012; Liao et al., 2013). Also, it is found that buying intention is directly related to the degree of customization (Kamali and Locker, 2002), and the perceived value of the customized product increases buying intention directly (Wu et al., 2012) and indirectly (Liao et al., 2013). When these studies are taken into consideration, there is an insight that the customized product's perceived value will influence the purchase intention. Hence our first research hypothesis is:

H_1: **Mass customized product value positively influences purchase probability**

The perceived value of a mass-customized product, in the literature, was constructed based on the works of Merle et al. (2009) and Merle et al. (2010), and it is considered as a four-dimensional structure; consisting of utilitarian value, uniqueness value, self-expression value, and creative achievement value. One dimension of this four-dimensional structure, creative achievement value, in a way, may relate to the Flow Experience. Thus, in addition to the assumption of the customization of a product's effect on customers' purchase intentions discussed above, the impact of the flow experience during the design of the product on the purchase intention should also be examined. The most important contribution of the Flow Theory to marketing has been the determination of the effect of flow experience on attitudes, intentions, and behaviors (Hoffman and Novak, 2009, p.29), and in various researches, flow experience has been directly (Luna et al., 2003; Pu et al., 2009a; Pu et al., 2009b; Liu et al., 2016) and indirectly (Pu et al., 2015) related to the intention of purchasing. Ozkara et al. (2017) conclude that three flow subdimensions positively influence online purchase intention; online enjoyment, perceived control, and merging. Besides, Korzaan (2003) found that flow experience increased positive attitude towards online shopping. Bridges and Florsheim (2008, p.309)

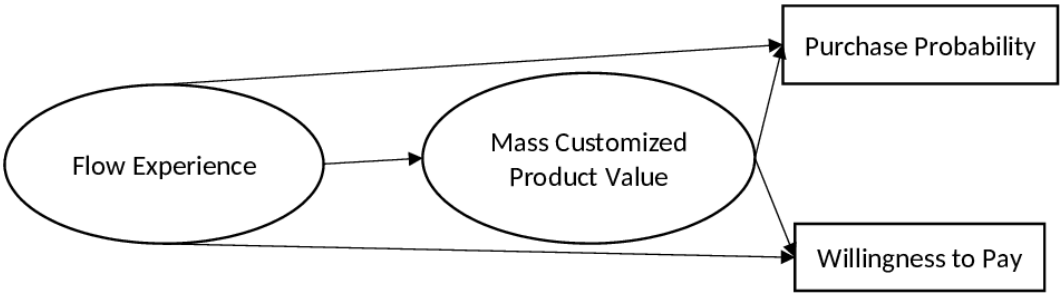
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concluded that online consumers who experience flow buy more and return to the site to repeat the experience. Accordingly, our second hypothesis is:

H_2: **Flow experience positively influences purchase probability**

As regards the previous research that explores the effect of the perceived value of the customized product and flow experience during customization on purchase intention; the first and second hypotheses of the study are constructed as follows:

Figure 1. Research model



WTP, also referred to as the increase in economic value for mass customized product (Franke and Schreier, 2008), has been addressed in many mass customization studies (Kamali and Locker, 2002; Franke and Piller, 2004; Bardakçı and Whitelock, 2003; Franke and Schreier, 2006; Schreier, 2006; Schoder et al., 2006; Franke et al., 2009; Franke et al., 2010a; Franke et al., 2010b; Yu and Yang, 2020) and customer participation study (Wei et al., 2018). Researchers have suggested that consumers are ready to pay higher amounts for mass customized products because they can get more suitable products for their needs via mass customization (Pine, 1993, p. 46; Peppers and Rogers, 1997, p. 178; Piller, 2002, p.11). Here are some more empirical studies that prove these arguments (Piller, 2002; Franke and Piller, 2004; Franke and Schreier, 2006; Schreier, 2006; Franke et al., 2009; Chen et al. 2020). In sum, all these studies predicted that mass customization leads to an increase in the WTP.

Furthermore, if there is a positive change in consumer's WTP for a mass customized product, it is necessary to determine the factors to which this change is attributed. According to Franke and Schreier (2010), the explanation of the increase in WTP for customized products is the perceived entertainment and process effort factors that originate from the process, along with the product preference fit factor.

Merle et al. (2009) examined the effects of product value and experience value on WTP for mass customized products and concluded that experience value increased the WTP via perceived product value. When these research findings are considered, it is inferred that consumers' WTP for customized products is related to their ability to reach a more suitable product to their preferences and experiences in the design process. Within the scope of these assumptions, the third and fourth hypotheses of the study were inferred as follows:

H_3: **Mass customized product value positively influences WTP**
H_4: **Flow experience positively influences WTP**

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Franke and Piller (2003) suggested that the customized product and process notions should be considered independent variables in mass customization research. The empirical investigations of the relationship among them need to be explored. Merle et al. (2009), considering this suggestion, examined the effect of mass customization experience value on the value of the mass-customized product and concluded that mass customization experience value directly affected mass-customized product value. Considering the studies mentioned above and considering that mass-customized product is a direct output of the co-design process, there is a foresight that the co-designing experience will also affect the product value. The assumption that mass customization is not independent of product value and experience value predicts that mass-customized product value can act as a mediator in the relationship between flow experience and WTP and flow experience and purchase probability. Within the scope of these assumptions, the fifth, sixth, and seventh hypotheses of the study were inferred as follows:

H_5: **Flow experience positively influences mass-customized product value**
H_6: **Mass customized product value mediates the effect of flow experience on WTP**
H_7: **Mass customized product value mediates the effect of flow experience on purchase probability**

METHODOLOGY

Sample

This study has a target population of consumers who can make their own purchase decisions living in Turkey. Because of the unavailability of the sampling frame, a non-probability judgmental sampling technique was used. Three hundred thirty completed questionnaires were received at the end of the two weeks.

Procedure

Flow represents a state, a temporary subjective experience of the human-computer interaction, that can be captured more reliably when measured immediately during or after the interaction (Webster et al., 1993, p.420). Therefore in our research, to implement the survey immediately after the design process (i.e., human-computer interaction), the potential participants were invited to surf into a Web Site that includes; 1) an informative text about the research, 2) an introductory teaser video (https://youtu.be/y_kfHHVxKKM), 3) the mass customization Web Site link (www.NIKEiD.com.tr) that they are asked to go through to design their sneakers and 4) the survey link. Thus, all the research processes were conducted in one place. At first, a teaser video was uploaded to a video-sharing site and shared by researchers and survey participants on social media. A pretest of the research instrument was conducted using a small sample of undergraduates to ensure that the format, design, and wording of the questions were clear and easy to be understood.

This study used an existing online mass customization website, "NIKEiD". NIKEiD has one of the most advanced mass customization programs in Turkey that uses a configuration toolkit in a B2C consumer setting at the time of the research (2017). As a product category, sports shoes were selected because of their popularity regardless of the age of shoppers and their appropriateness for customization. Moreover in recent mass customization literature sport shoe category is a key research area (Ribeiro et al., 2017; Runfola et al., 2018, Zhou et al., 2019; Kalantari et al.,2021) After visiting the NIKEiD customization program, participants were asked to create their own sports shoes using the program and imagining that they would have made a purchase. They could select styles, fabrics, colors, and patterns and add text or logo when customizing their shoes. Finally, they completed the questionnaire.

Before testing the hypotheses, descriptive statistics were computed to generate the means, standard deviations, and correlations for the study variables. Since research constructs were previously

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identified, confirmatory factor analysis (CFA) was utilized to confirm how well the measured variable represents the constructs (Hair et al., 2010). Structural equation modeling (SEM) was conducted to explore the relationships among constructs because it allowed us to evaluate the relationships among the latent variables at the same time and guarantee statistical efficiency. Baron and Kenny's (1986) four-step approach was used to test mediation. All these steps were performed within the estimated SEM model, and calculations were performed with SPSS 21.0 and AMOS 22.0. Discriminant validity and internal consistency were conducted through MS Excel.

Measures

The current study relies on the measurements used in the existing literature. Since the scales were initially developed in English, a standard translation and back-translation procedure was conducted.

Mass customized product value was measured by nine items covering three factors: uniqueness value, utilitarian value, and self-expressive value (Merle et al., 2010). Items are rated on a 5-point scale ranging from 1 (totally disagree) to 5 (totally agree).

Flow experience was measured with twelve items covering four factors; control, concentration, curiosity, and intrinsic interest (Webster et al., 1993). Items are rated on a 5-point scale ranging from 1 (totally disagree) to 5 (totally agree).

Purchase probability was measured with one item "What is the probability that you will buy mass-customized sports shoes in the next 12 months? Participants rated from 0 to 100 percent. A five-point verbal purchase intention scale is also used in the study (1: definitely will not buy, 2: probably will not buy, 3: may or may not buy, 4: probably will buy, 5: definitely will buy) to test the reliability and validity of purchase probability measure.

The contingent valuation method was used to measure the WTP for each participant. In the contingent valuation method (CVM), the respondents are directly asked how much they are willing to pay for a product or service (Mitchell and Carson 1989). Precisely, WTP is a ratio-scaled measure of the buyer's subjective value to a product (Wertenbroch and Skiera 2002, p. 228). The difference in ΔWTP between a mass customized product and the most preferred (chosen) standard product yields the value increment of customization (Harzer, 2013).

Equation 1. Formula for calculating willingness to pay change

$$\Delta WTP = \frac{\text{WTPMassCustomizedProduct-WTPStandartProduct}}{\text{WTPStandartProduct}}$$

DATA ANALYSIS AND RESULTS

Demographic Profiles

Table 1 shows all the demographic variables for this study. The majority of the respondents are female and from the age group of 25-34. It is also showed that the respondents with a bachelor's degree or higher make up a significant percentage of the sample. Besides, quite a high number of participants have a personal monthly income below 4500₺. Lastly, a considerable number of them are frequent online shoppers.

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Table 1. Demographic characteristics of participants

		Frequency	Percentage
Gender	Female	217	66
	Male	113	34
Education	Secondary School	6	2
	High School	55	17
	Bachelor Degree	157	48
	PhD - Master	112	33
Age	Less than 25	70	22
	25 - 34	188	56
	35 – 44	46	13
	More than 44	26	8
Personal Monthly Income	1500 ₺ and lower	76	23
	1501 – 3000 ₺	96	30
	3001 – 4500 ₺	94	28
	4501 – 6000 ₺	38	11
	6001 – 7500 ₺	10	3
	7501 ₺ and higher	16	5
Online Shopping Frequency	None	17	5
	Once every two or three years	24	8
	Once or several times in a year	104	32
	Once or several times in a month	153	46
	Once or several times in a week	28	8
	Once or several times in a day	3	1

Results Of Statistical Analyses

Structural equation modeling (SEM) was utilized in this study. SEM helps researchers analyze complex associations among variables and build models. Moreover, two mediation analyses were performed to observe mediation between the variables flow experience - perceived mass customization product value – WTP and flow experience - perceived mass customization product value – purchase probability. Baron and Kenny's (1986) four-step approach was used to provide evidence for mediation. All these steps were applied within the estimated SEM model, and calculations were performed using the IBM SPSS AMOS 24 program.

Because the data were self-reported for all of the variables, it is possible that the common method bias could be present The Harman's one-factor test (Podsakoff and Organ, 1986) was conducted to examine whether common method variance existed in the data set. According to Harman's one-factor test, if common method bias is present, a single factor will explain greater than 50% of the variance in a principle components factor analysis with no rotation. All the items were entered in an exploratory factor analysis using the unrotated solution to a principal components analysis. The analysis finally showed that all items can be categorized into five factors with an eigenvalue greater than one, rather than a single factor The first factor accounted for 36.8% of the variance; the first

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factor did not account for a majority of the variance. The results indicated that common method bias was not a serious concern for the current study(Podsakoff et al., 2003).

Consistent with the two-step approach advocated by Gerbing and Anderson (1988), this study first developed the measurement model by conducting confirmatory factor analysis (CFA). Then, the structural model examined and tested the research hypotheses. Table 2 lists standardized coefficients, critical values, error variances, AVE, and composite reliability values. In order to see how current variables shared their variance and how they differed from other measures, discriminant and convergent validity were conducted (Fornell and Larcker, 1981). Thus, in addition to plausible composite reliability coefficients ($\geq$ .70; Nunnally,1994), the shared variance between factors was examined with the average variance extracted (AVE), which exceeded 0.50 in most measures. Factor loadings for all measurement items are positive and exceed the threshold value of 0.5. All standardized item loadings are significant ($p<0.01$) (Hair et al., 1998). AVE's square root exceeded the correlation with each other factors, suggesting that our measurement scales represent ideal discriminant validity (Table 3). Correlation coefficients suggested that all constructs were correlated positively. Table 4 demonstrates the summary of model evaluations. The goodness of fit statistics were acceptable aside from the GFI. However, a cut-off point of 0.9 is not appropriate under all conditions because both parameter number and sample size can influence it (Shevlin & Miles, 1998:90)

Investigating the Structural Model

As shown in Table 5, the findings showed that flow experience had a significant positive direct effect on purchase probability, WTP, and mass-customized product value providing support for hypotheses 2, 4, and 5. Perceived mass-customized product value had a significant positive direct effect on purchase probability but had no direct effect on WTP. Hence, the findings supported Hypothesis 1 but failed to support Hypothesis 3.

A summary of model evaluations is made in Table 6. The goodness of fit statistics was acceptable aside from the GFI. GFI can be low because both sample size and parameter number can influence it (Sharma et al., 2005; Steiger, 2007).

Mediation Analysis

Two simple mediation analyses were conducted to assess the effect of perceived mass customization product value on the relationship between flow experience and purchase probability and the relationship between flow experience and WTP. In the mediation analysis, the four-step method put forward by Baron and Kenny (1986) and Judd and Kenny (1981) was used to establish partial or complete mediation. According to Baron and Kenny (1986) and Judd and Kenny (1981), a hypothesis of mediation is supported if the following conditions are met:

(1) The causal variable is associated with the dependent variable (see Fig. 2, c'),
(2) the causal variable is related to the potential mediating variable (see Fig. 2, a),
(3) the potential mediating variable is associated with the outcome variable (see Fig. 2, b), and (4) the relationship between the causal variable and the outcome variable is no longer significant when the potential mediator is included in the model (see Fig. 2, c).

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Table 2. Standardized coefficients, critical values, error variances, AVE, and composite reliability values

Factors	Items	Standard Coefficient	Critical value (z-value)	Error Variance	Ave	Composite Reliability
Concentration (α=0.751)	Concentration 3	0,832	-	-	0,505	0,747
	Concentration 2	0,745	10.707	0.099		
	Concentration 1	0,518	8.320	0.087		
Uniqueness value (α=0.780)	Uniqueness value 1	0,65	-	-	0,558	0,79
	Uniqueness value 2	0,816	11.015	0.121		
	Uniqueness value 3	0,765	10.707	0.099		
Self Expressiveness Value (α=0.817)	Self Expressiveness 3	0,797	15.042	0.071	0,604	0,82
	Self Expressiveness 1	0,738	13.773	0.070		
	Self Expressiveness 2	0,794	-	--		
Control (α=0.755)	Control 3	0,555	-	-	0,538	0,772
	Control 2	0,744	9.355	0.154		
	Control 1	0,869	9.773	0.169		
Utilitarian Value (α=0.815)	Utilitarian Value 3	0,851	-	-	0,598	0,816
	Utilitarian Value 2	0,751	15.136	0.058		
	Utilitarian Value 1	0,71	14.044	0.052		
Curiosity (α=0.775)	Curiosity 1	0,628	10.091	0.082	0,564	0,795
	Curiosity 3	0,726	-	-		
	Curiosity 2	0,8	12.154	0.092		
Intristic Interest (α=0.736)	Intristic Interest 1	0,802	10.634	0.109	0,482	0,734
	Intristic Interest 2	0,628	9.169	0.093		
	Intrinsic Interest 3	0,639	-	-		
Flow Experience	Control	0,794			0,606	0,858
	Intrinsic Interest	0,867				
	Concentration	0,583				
	Curiosity	0,837				
MC Product Value	Uniqueness value	0,674			0,597	0,811
	Utilitarian Value	0,964				
	Self Expressiveness Value	0,94				

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Table 3. The square root of AVE (shown as bold at diagonal) and factor correlation coefficients

	Uniqueness Value	Self Expressive Value	Utilitarian Value	Control	Concentration	Intrinsic Interest	Curiosity
Uniqueness Value	**0,746***						
Self Expressive Value	0,535	**0,777***					
Utilitarian Value	0,498	0,753	**0,773***				
Control	0,356	0,484	0,614	**0,733***			
Concentration	0,226	0,302	0,346	0,334	**0,710***		
Intrinsic Interest	0,327	0,388	0,310	0,492	0,310	**0,694***	
Curiosity	0,382	0,415	0,301	0,518	0,301	0,632	**0,750***

*Diagonal elements (in italics) represent the square roots of the AVE's for individual constructs.

Table 4. CFA model evaluations

	Criteria	Model Fit Values
x^2/df	≤3 Excellent * ≤5 Acceptable*	2,713
RMSEA	0≤RMSEA≤0,08 **	0,072
GFI	0,90≤GFI≤1,00 **	0,874
CFI	0,90≤GFI≤1,00 **	0,903
* Hair et al. 2010 ** Schumacker & Lomax,1996		

Table 5. Path coefficients, critical values, and hypotheses results

Hypotheses	Structural Path	Path Coefficients	Critical Value (Z Statistics)	Supported
H_1	MCPV→PP	,303***	3,345	Yes
H_2	FE→PP	,199**	2,131	Yes
H_3	MCPV→WTP	,016	0,161	No
H_4	FE→WTP	,294	2,897	Yes
H_5	FE→MCPV	,719	10,061	Yes
Notes: *** p<0.001; ** p<0.05; FE= flow experience; MCPV= mass customized product value; PP= purchase probability; WTP= willingness to pay				

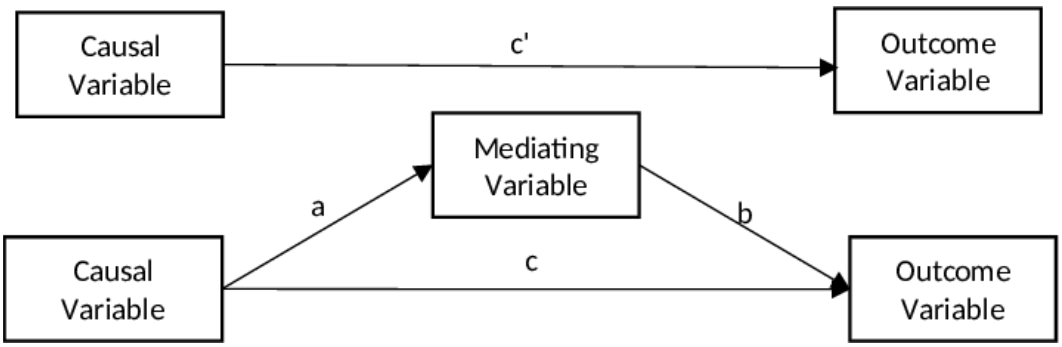
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Table 6. Goodness of fit statistics for the model and critical fit values

Fit Statistics	*Model Fit Values*	*Acceptable Fit*	*Good Fit*	*Excellent Fit*
x^2 / sd	*2,713*	*3-5*	*-*	*≤ 3*
RMSEA	*0,072*	*0.08-0.10*	*0.05 – 0.08*	*≤0.05*
GFI	*0,874*	*-*	*≥0.90*	*≥0.95*
CFI	*0,903*	*-*	*≥0.90*	*≥0.95*

If the criteria for all four of these steps are met, then the data are consistent with the hypothesis that the mediating variable fully mediates the relationship, supporting full mediation. On the other hand, if the first three steps are met but in the fourth step when the potential mediator is included in the model, the relationship between the independent variable and the dependent variable gets lower (in Fig. 2., when c<c') then partial mediation is indicated.

Figure 2.



Flow Experience – Willingness to Pay Mediation Analysis

For the sixth hypothesis, Baron and Kenny's (1986) and Judd and Kenny's (1981) mediation steps were applied within the hypothesized model explained below:

1. In the first step of our mediation analysis, the direct impact of flow experience on WTP was significant and positive, and the path coefficient was 0,289, $p < .01$; Step 1 was approved.
2. In the second step of our mediation analysis, flow experience had a positive and significant impact on perceived mass customization product value, and the path coefficient was 0,718, $p < .01$. Step 2 was approved.
3. In the third step of our mediation analysis, the effect of perceived mass-customized product value on WTP was not significant, $p<.01$. Step 3 was not approved.

Because the third step criterion was not met, hypothesis 6 was rejected.

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Flow Experience – Purchase Probability Mediation Analysis

For the seventh hypothesis, Baron and Kenny's (1986) and Judd and Kenny's (1981) mediation steps were applied within the hypothesized model explained below:

1. In the first step of mediation analysis, the direct impact of flow experience on purchase probability was positive and significant, path coefficient was 0,398, p<.01. Step 1 was approved.
2. In the second step of mediation analysis, flow experience had a positive and significant impact on perceived mass customization product value, and the path coefficient was 0,717, p<.01. Step 2 was approved.
3. In the third step of mediation analysis, perceived mass-customized product value had a positive and significant impact on purchase probability, and the path coefficient was 0,318, p<.01. Step 3 was approved.
4. In the fourth step of mediation analysis, when the potential mediator was included in the model, the relationship between the causal variable flow experience and outcome variable purchase probability was no longer significant. The path coefficient was 0,178, p>.05. Step 4 was approved.

Baron and Kenny (1986) and Judd and Kenny's (1981) four-step approach revealed a full mediation of perceived mass-customized product value in the relationship between flow experience and purchase probability.

A general evaluation of the mediation analysis results, as shown in Table 7, indicates that the perceived mass-customized product value mediates the relationship between flow experience and purchase probability.

Table 7. Hypotheses results for mediations

Hypotheses	Causal Variable	Mediating Variable	Outcome Variable	Supported
H_6	FE	MCPV	WTP	No
H_7	FE	MCPV	PP	Yes

Notes: FE= flow experience; MCPV= mass customized product value; PP= purchase probability; WTP= willingness to pay

CONCLUSION

The goal of this research was to empirically identify the direct and indirect effects of flow experience on purchase probability and WTP in online mass customization processes and introduces a model to achieve this aim.

In this study, it was revealed that consumers' flow experience while designing their products on Websites that offer mass customization service to their customers has a direct effect on the WTP and an indirect effect on the purchase probability through the perceived value of the mass-customized product. We can explain this with two scenarios:

Scenario 1: If customers do not experience a flow state during designing their product on the mass customization website while perceiving the customized product by themselves as a valuable product, they may think that the product reflects their uniqueness, originality, and personality. The customer buys products that meet his tastes. Therefore, in this scenario, the probability of purchasing a customized product will be directly and positively affected; however, customers may not be willing to pay more for it.

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Scenario 2: If customers experience a flow state during their product's designing process on the mass customization website, they will evaluate the self-designed products as more valuable. Since the value of the flow state will be reflected in the product designed during the flow state, the possibility of purchasing and the WTP more for the self-designed products will be positively affected.

This study make several theoretical contributions to the field of marketing. The research is the first study in the literature investigating experiences in the online mass customization process in detail within a scope of the flow experience and its corresponding effects. Hoffman and Novak (2009) state that, from a marketing perspective, the most critical outcome of flow theory is its influence on attitudes, subsequent behavioral intentions, and behaviors. In light of this study's results, it was showed that there is a direct positive effect of the flow experience on the WTP, which may contribute to the marketing literature by highlighting the marketing outcomes of the flow experience.

The findings of the study offer some significant advantages to entrepreneurs, brand managers, and marketing managers who are already applying or who aim to implement online mass customization. It can be concluded that it would be difficult for the practitioners to persuade Turkish customers who stated that the most significant motivational source for choosing the online channels was the lower prices related to online mass customization at higher prices compared to the products of mass production. To this end, it is crucial to determine what factors increase the customers' WTP and their degree of influence. The literature says that consumers are willing to pay more for the products they have designed (Franke et al., 2010), or they are willing to pay more for the products they think they have contributed to their design process (Norton et al., 2011, Wei et al., 2018). WTP varies according to the product category and may reach up to 200% (Schreier, 2006). In the present study, an increase of 25% WTP for the sneakers designed by the respondents was observed. Although the increase in WTP observed in this study seemed relatively low compared to 200% increases in different researches. These previous papers explored WTP for mass customized sneakers made earlier reached some similarities to the results of this study, among which a 30% increase in payment requests was observed (Merle et al., 2008; Merle et al., 2009; Piller, 2002). This result shows the practitioners that the prices they will set for the customized product should consider the type of product they would offer for customization and the standard products' price level.

In other words, the price-sensitive online consumer who designed their sneakers buy these sneakers not because they suit his/her needs better, their originality, nor their ability to reflect individual tastes but for the flow experience he/she had while designing the sneakers (for instance he/she was able to focus, to feel inner interest for it; and feel of personal control over the outcome). He/she, therefore, wants to pay more.

Simultaneously, the flow experience increased the perceived value of the customized product, thereby increasing the likelihood of purchase. When online marketing executives focus only on the value of mass-customized product such as its' uniqueness, originality, individuality, etc. but not the experience that the customer during the customization process, clearly the customer will perceive the product "valuable" since she or he designed it. Therefore she/he will evaluate the possibility of purchasing the self-designed product high. Nevertheless, when it comes to paying more for the customized product, his/her WTP higher prices will suddenly decrease and leave the shopping basket since he/she did not experience a flow experience. Therefore, online marketing managers who want to sell mass customized products that are slightly more expensive than the standard ones are advised to design their sites, and mass customization toolsets in such a way that consumers will experience flow (i.e., curious, interesting, not distracting, and giving control to the customers) is required.

Under these circumstances, it is of great importance for online marketing executives who aim to implement mass customization to design their Websites and mass customization toolsets to arouse curiosity and attract attention. Also, they should not distract the customer, and they need to offer customers a level of customization so that he/she feels that he/she is in control of the design process.

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As a result, it is showed that the variable of "flow experience", which has not been studied in detail before, is of great importance in the process of mass customization in the design process. The present research findings also revealed that managing "the experience" in the self-design process to create flow will contribute significantly to the advancement of businesses.

Limitations and Future Research

Although this study provides fascinating insights into how flow experience influences perceived mass-customized product value, WTP, and purchase probability, it has several limitations. First of all, the study used a convenience sample; therefore, interpretation and generalization of the findings should be made carefully. Future research should include a larger sample of consumers to test the results of this study. Secondly, the product category used in the present study was sports shoes. Therefore, respondents' evaluations of mass customization were only based on a single category of product. Other studies should focus on various types of products. Besides, the Nike ID program is an example of "partial customization" (Duray et al. 2000); it would be worth testing the proposed model on other types of customization programs. Although before starting customization, respondents were asked to imagine that they had actually purchased their design, the experiment remains virtual, hence lacks realism.

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ENDNOTES

This article is extracted from Dr. Ozer-Canarslan's doctoral dissertation supervised by Prof. Barış. If it is possible I would like to point out this information.

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Online Retail Store Community Engagement and Its Impact on Purchase Intentions in the Context of Online Reviews

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ABSTRACT

Consumers want to engage with the retail store community in the shopping context. This study examines how consumers' perceptions of source credibility and perceived usefulness of online reviews impact an intention to engage with the online retail store community and purchase intentions. The study also identifies differences when reviews are posted by customers versus store employees. Results show that the proposed relationships within the structural model are significant, and reviews posted by store employees are seen as more credible, whereas reviews posted by other customers are perceived to be more useful. Theoretical and managerial implications are discussed.

KEYWORDS

Online Reviews, Perceived Usefulness, Purchase Intentions, Retail Store Community Engagement, Source Credibility

INTRODUCTION

Shopping is an inherently social experience (Carter, 2020). Whether it be shopping with friends and family, interacting with other shoppers or salespeople, or chatting with store employees at checkout, customers have reported that shopping in the retail store is communal in nature (Borghini et al., 2009; Jones & Kim, 2011; Peters & Bodkin, 2018). However, as shopping has increasingly moved online, due to convenience factors and the more recent pandemic stay-at-home orders, one may question whether the communal nature of shopping that exists in retail stores applies when customers are shopping online.

Practitioners are advocating that the social nature of shopping still exists when customers move online but in a way that differs from shopping in traditional retail stores (Beaulac, 2019; McIntyre, 2020). "Social shopping" is defined as customers going online to find reviews and recommendations about products and then using those reviews to guide purchase decisions (Widmer, 2020). The concept of "social shopping" is significant, as 93% of consumers have anecdotally reported that online reviews have impacted their purchase decisions (Widmer, 2020).

The purpose of this study is to examine the communal nature of utilizing online reviews for online shopping. There is a growing literature on community in physical stores, but marketers need greater understanding of community within online retailing. In addition, there are few papers that examine source credibility and perceived usefulness as they relate to community engagement in the context of

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online reviews, factors which may be important to social shopping. And while the literature suggests that shoppers view both customers and store employees as members of the retail store community, few studies to date have unpacked differences that may exist when online reviews are provided by these disparate sources. Thus, this study examines the following research questions to address the gaps in the literature: How do consumers' perceptions of source credibility and usefulness of online reviews, impact feelings of community and purchase intentions when posted by customers and store employees? Furthermore, do these relationships differ when the post is provided by other customers versus store employees?

LITERATURE REVIEW AND HYPOTHESES

While there is plethora of studies on community in marketing, the literature on retail store community is just beginning and has been mostly focused on physical stores, as opposed to online shopping (Borghini et al., 2009; Jones & Kim, 2011; Peters & Bodkin, 2018). And while there are numerous papers on online reviews, there are virtually no studies to date examining the impact of reviews on community engagement in the context of online shopping. Although this study is primarily grounded in theory on community, theories related to online reviews were also incorporated into the development of the hypotheses.

Source Credibility

Source credibility's impact on the believability of online reviews (Tahir & Khan, 2020), as well as its impact on purchase intentions in the context of online reviews (Assaker, 2020; Kiran & Srivastava, 2020; Xu et al., 2020), has been well established in the literature. There are no studies to date, however, that have examined the relationship between the source credibility of online reviews and consumers' intention to engage in an online retail community. The literature on virtual community shows that the credibility of information posted within these types of communities is of utmost importance to consumers. For example, Hajli et al. (2015) studied an online health community and found that consumers often questioned whether the other peoples' posts were credible and accurate. Chou et al. (2015) examined a virtual community and found that source credibility predicted the adoption of knowledge from the group. Nair and Harshavardhini (2002) also found that source credibility impacted eWOM adoption through perceived usefulness. Chih et al. (2013) surveyed a fashion community and found that website reputation, source credibility, obtaining buying information, and social orientation influenced perceived review credibility, and in turn attitudes and purchase intentions. Thus, based on the role of source credibility theory in the context of online reviews, as well as studies examining the impact of source credibility within virtual communities, the following hypotheses are posited:

Hypothesis One: In the context of online reviews, source credibility has a positive effect on online retail store community engagement intention.

Hypothesis Two: In the context of online reviews, source credibility has a positive effect on purchase intention.

Perceived Usefulness

Davis (1989) introduced the technology acceptance model (TAM), which identifies ease of use and perceived usefulness as determinants of the use of technology, to the field of information systems. The TAM model was introduced in a work context but has been applied in other consumer contexts, including e-commerce and online reviews (Bernardino et al., 2020; Kiran & Srivastava, 2020; Ventre & Kolbe, 2020). The model has also been utilized in the literature on social commerce and virtual community. Zhang et al. (2016) examined social commerce among Chinese consumers and found that website quality, trust, and eWOM influenced online purchase intentions with perceived ease

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of use and perceived usefulness moderating the relationship between website quality and purchase intentions. More recent studies have also confirmed a direct relationship between perceived usefulness and purchase intentions (Kiran & Srivastava, 2020; Xh et al., 2020; Ventre & Kolbe, 2020; Zhang et al., 2021).

Xie et al. (2020) found that, within social commerce, user satisfaction, IT self-efficacy, user relationship network structure, platform quality, perceived usefulness, and emotional attachment significantly impact users' intentions to adopt information. Similarly, Assaker (2020) found that perceived usefulness, trustworthiness, expertise, and ease of use were significantly related to usage intention of online reviews, with these relationships differing by age and gender. The impact of perceived usefulness on the adoption of information provided in reviews has been reinforced in more recent studies as well (Nair & Harshavardhini, 2020).

The virtual community literature has also identified perceived usefulness as a significant predictor of consumers' intention to engage with the community and adopt information from the group. Corkindale et al. (2018) identified trust and interface design as factors that drove consumers' intention to engage with a firm-hosted online community, with perceived usefulness mediating these relationships. Gadgil et al. (2018) conducted a study of an HIV virtual community and found that perceived usefulness and information quality were the two primary factors that influenced the quality of support and continued user engagement among group members. Other studies in the literature on virtual community have also reinforced perceived usefulness, among other variables, as significantly impacting consumers' intentions to engage with an online community (Chen & Qi, 2015; Wang et al., 2012). Tsai et al. (2011) examined online group buying, and found that perceived usefulness, a sense of virtual community, and trust in the community were determinants of purchase intentions. Thus, based on studies utilizing TAM variables in the context of online reviews, as well as papers identifying the impact of perceived usefulness on intentions to engage and purchase in the contexts of social commerce and virtual communities, the following hypotheses are posited:

Hypothesis Three: In the context of online reviews, perceived usefulness has a positive effect on online retail store community engagement intention.

Hypothesis Four: In the context of online reviews, perceived usefulness has a positive effect on purchase intention.

Online Retail Store Community Engagement Intention

Early on, the literature on brand community found that an intention to engage with the group impacted brand loyalty, which encompasses both attitudinal and behavioral components (McAlexander et al. 2002). As the community literature has grown, this finding has been reinforced and expanded into both the retail store and virtual community contexts. Borghini et al. (2009), identified the American Girl retail store as an active brand community that inspires intense loyalty. Jones and Kim (2011) examined single-brand retail stores and found that consumers' identification with the store was related to brand community, which in turn, impacted brand loyalty. Peters and Bodkin (2021) found that retail store community engagement had a positive impact on store employee trust, store satisfaction, store commitment, and shopping enjoyment.

Although no papers on retail store community engagement have captured its impact on purchase intentions, the larger literature on virtual communities has identified the significance of this relationship. In their study of social media, Ekanayake and Abeysekera (2016) found that brand and brand community engagement predicted both brand equity and purchase intentions. Lyu and Kim (2020) also examined e-commerce activities via social media. They showed that perceived interactivity and perceived sense of community (i.e., engagement and sharing stories) related to self-brand connection and brand commitment, which impacted brand attitudes and purchase intentions. Prentice et al.'s (2019) study showed that identification with a virtual community related to attitudinal

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and behavioural engagement with the group, which predicted purchase intentions. Dos Santos et al. (2018) examined sports sponsorship of a Soccer team and found that trust and a sense of membership within the community related to attitudes toward the team and sponsorship brands, which in turn, impacted purchase intentions. Huang et al. (2012) found that a sense of virtual community moderated the influence of comments posted on attitudes toward the product and purchase intentions. Thus, based on theory about community in marketing and its impact on attitudes and behaviours, as well as studies identifying the impact of engagement with virtual community and its relationship to attitudes and purchase intentions, the following hypotheses are posited:

Hypothesis Five: In the context of online reviews, online retail store community engagement intention has a positive impact on purchase intention.
Hypothesis Six: In the context of online reviews, online retail store community engagement intention has a positive impact on attitude toward the store.
Hypothesis Seven: In the context of online reviews, attitude toward the store has a positive impact on purchase intention.

Customers Versus Store Employees

Early on, the personal selling literature identified salesperson expertise as a factor which impacts purchasing behavior among customers (Woodside & Davenport, 1974). The role of salesperson expertise continues to be studied, but in a more nuanced fashion. Sharma (1990), for example, examined the relationships between salesperson credibility (i.e., expertise and trust), the buyer's product evaluations, and purchase intentions. Sharma showed that salesperson credibility has a positive impact on product evaluations and purchase intentions when buyers have little prior knowledge or low expectations of the brand. Similarly, Belonax et al. (2007) found that salesperson credibility was more impactful in the context of low involvement purchases. And within a retailing context, Friedman et al. (2011) discovered that store employee expertise predicts customer satisfaction and share of wallet. When considering theory about salesperson expertise and its impact on attitudes and behaviours, as well as specific studies examining salesperson credibility and the expertise of store employees impacting satisfaction and purchase intentions, the following hypotheses are posited:

Hypothesis Eight: The impact of source credibility on online retail store community engagement intention will be stronger for store employee reviews, compared to reviews provided by other customers.
Hypothesis Nine: The impact of source credibility on purchase intention will be stronger for store employee reviews, compared to reviews provided by other customers.

There are a few studies which compare differences in consumers' perceptions when the information posted in an online review is provided by other customers versus store employees. Chen et al. (2016) examined source differences in online reviews and found that purchase intentions differed when the source of information was from other customers, a third party, or the manufacturer/retailer. Moreover, the differences varied by consumers' susceptibility to informational influence (Chen et al., 2016). For those consumers with high susceptibility to informational influence, word of mouth provided by other customers was perceived to be the most useful. In contrast, for consumers with low susceptibility to informational influence, reviews provided by all three sources were equally useful. Tata et al. (2020) also examined whether reviews provided by retailers versus a third party had a significant impact on attitudes and purchase intentions when shopping online, but they found no significant difference between these sources. Diwanji and Cortese (2020) also found no difference in perceived usefulness nor consumers' purchase intentions when videos reviews were posted by other users versus the brand.

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The variance in the perceptions of the sources in online reviews, as noted by Chen et al. (2016), may be related to trust and authenticity. Research shows that consumers are living in a postmodern, experiential economy and are actively seeking engaging experiences via their shopping and consumption behaviours (Leigh et al., 2006). Moreover, the literature on online reviews has affirmed the importance of trustworthiness of the review in impacting purchase intentions (Lee & Hong, 2019; Reimer & Benkenstein, 2016). Additionally, Qiao et al. (2020) found that when users are paid to post reviews, the resulting reviews are more positive yet lower quality. Thus, based on theory on authenticity and trust in online reviews, combined with Chen et al.'s (2016) study showing consumers with a high susceptibility to informational influence viewed reviews provided by other customers as more useful than reviews provided by a third party or manufacturer/retailer, the following hypotheses are posited:

Hypothesis Ten: The impact of perceived usefulness on online retail store community engagement intention will be stronger for reviews provided by other customers, compared to store employee reviews.

Hypothesis Eleven: The impact of perceived usefulness on purchase intention will be stronger for reviews provided by other customers, compared to store employee reviews.

The literature on retail store community has found that the characteristics of brand community (moral responsibility, shared consciousness, and rituals/traditions; Muñiz & O'Guinn, 2001) exist among shoppers in the retail store context. Furthermore, this research also shows store employees are an active part of the community, in conjunction with other shoppers. Peters and Bodkin (2018) state that store employees are "brand ambassadors and members of the brand community" (p. 8) and they facilitate community by representing the brand while simultaneously building relationships with customers in the context of the retail store. More recent studies have also found that shoppers' retail store community engagement positively impacts store employee trust, which relates to their overall store satisfaction and commitment (Peters & Bodkin, 2021). Because these findings situate the store employee as part of the retail store community in conjunction with other customers, no differences would be expected in how the relationships between engagement with the online community and its impact on attitudes toward the store and purchase intentions would vary when reviews are provided by other customers versus store employees. Thus, the following hypotheses are posited:

Hypothesis Twelve: There will be no difference in the impact of online retail store community engagement intentions on attitude toward the store, when comparing store employee reviews to reviews provided by other customers.

Hypothesis Thirteen: There will be no difference in the impact of attitude toward the store on purchase intentions, when comparing store employee reviews to reviews provided by other customers.

Hypothesis Fourteen: There will be no difference in the impact of online retail store community engagement intentions on purchase intentions, when comparing store employee reviews to reviews provided by other customers.

METHODOLOGY

Data Collection

Data were collected via an online survey built from five pre-existing scales, with established validity and reliability, along with demographic questions. The scales were modified for an online clothing context. An online clothing store was an appropriate context for studying retail store communities because these retailers often focus on community engagement as part of their missions (https://www.abercrombie.com/shop/us/about-us; https://info.lululemon.com/about/our-story). The survey was administered, via Qualtrics, to students attending a large university in the south-eastern United States.

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Respondents agreed to participate in the study in exchange for a 1% bonus point. Of the 303 initial email invitations sent, 230 responded and were randomly assigned to view one of two scenarios; 113 (49%) were shown a scenario where other customers were the source of communication; 117 (51%) were shown the same scenario with store employees as the source of communication. The scenarios were worded as follows:

You are shopping online at [store they provided in earlier question] for an article of clothing. You notice that the website has posted product reviews that are provided **by other customers [by store employees].** These reviews discuss the quality and craftsmanship of the different styles of clothing that you are considering.

Twenty-five respondents incorrectly answered the filter questions that were embedded to ensure community engagement, resulting in a final sample of 205 (i.e., 67.6% response rate, consistent with Cook et al., 2000; 51% were shown the 'other customers' scenario; 49% were shown the 'store employees' scenario). The majority of respondents were male (55%), white (70%), and employed part-time (53%). These demographics are reflective of the larger student population at the university; thus, non-respondents had similar backgrounds. Approximately 95% of respondents indicated they spend money on clothing in a typical year; 84.4% indicated they spend "a little" to "a moderate" amount. The average response to the question, "How often do you shop for clothing at [a clothing store they named]?" was 3.1 using a scale of 1 (Not very often) to 7 (Very often).

Measurement

Online retail store community engagement intention is defined as "the consumer's intrinsic motivation to interact and cooperate with community members" in an online retail store context (Algesheimer et al., 2005, p. 21). This construct was measured using a scale developed by Lee et al. (2011), which was modified from brand community to retail store. *Source credibility* was defined as the extent to which consumers believe information provided by other people and organizations about products and services is based on expertise and trustworthiness (Pornpitakpan, 2004) and was measured by adapting the Flanagin and Metzger's (2000) scale of reviewer credibility. *Perceived usefulness,* defined as when "people tend to use or not use an application to the extent they believe it will help them" (Davis, 1989, p. 320), was measured by adapting a scale developed by Hassanein and Head (2007). *Attitude toward the store* is defined as the degree to which a customer holds a positive impression of an online store. Muehling et al.'s (1991) scale to assess brand attitude was adapted to ask respondents about their attitude toward the online store. Finally, *purchase intention* was measured by adapting a scale used by Putrevu and Lord (1994) to assess consumers' likelihood to purchase a brand in the future. Table 1 shows the full list of survey questions, along with measurement properties; each of the scales was reliable according to Cronbach's alpha.

Data Analysis

Data analysis began by assessing the validity of the constructs. First, the Kaiser-Meyer-Olkin (KMO) test was conducted to assess how well data were suited to factor analysis. The test score of .91 suggested that the sampling was adequate. Bartlett's test of sphericity yielded a p-value <.001. Subsequently, the researchers followed Fornell and Larcker's (1981) criteria to examine convergent and discriminant validity. First, the researchers examined the factor loadings of the constructs and found that all exceeded .70, except for one item in the online retail store community engagement construct. After validating the factor loadings, the researchers examined the average variance extracted (AVE) for each construct and found that all five constructs had an AVE greater than or equal to .60. The composite reliability value for all five constructs exceeded .70, suggesting good internal consistency (Hair et al., 2010). The greatest percentage of variance extracted from any single factor was 23.8%. To confirm the existence of different constructs in the model, the square root of the AVE of each construct was compared with the squared correlations of the construct itself and other constructs in the model (Fornell & Larcker, 1981; Hair et al., 2010). See Table 2 for details.

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Table 1. Matrix of factor loadings, Cronbach's Alpha, average variance extracted, and composite reliability

Construct		Variable	Factor loading	Cronbach's alpha
Community engagement	(Thinking about your purchase from ___ in the scenario above, please indicate your level of agreement)			
	SC1	I would be supportive of other members of the online store community	0.720	0.880
	SC2	I would provide new information about the online store to other people	0.784	
	SC3	I would actively participate in the online store community's activities	0.599	
	SC4	I would say positive things about the online store community to other people	0.827	
	SC5	I would recommend the online store to anyone who sought advice about the store and its communit	0.868	
	SC6	I would encourage other people to use the online store in the future	0.826	
	SC7	I would not hesitate to refer other people to the online store	0.758	
Perceived usefulness				
	PU1	___ reviews from ___ provide good information about quality and price	0.907	0.940
	PU2	___ reviews from ___ improve my performance in assessing the product quality and price	0.944	
	PU3	___ reviews from ___ increase my effectiveness for product quality and price assessment online	0.906	
	PU4	___ reviews from ___ are useful for evaluating the product quality and price	0.942	
Source credibility				
	Cred1	___ reviews from ___ are credible	0.922	0.940
	Cred2	___ reviews from ___ are believable	0.917	
	Cred3	___ reviews from ___ are accurate	0.924	
	Cred4	___ reviews from ___ are trustworthy	0.929	
Purchase intention		(Based on ___ reviews from ___,)		
	PI1	It is every likely that I would purchase from this online retailer	0.930	0.930
	PI2	I would purchase from this online retailer the next time I need clothing	0.950	
	PI3	I would definitely make a future purchased from this online retailer	0.935	
Attitude toward the store		Thinking about your purchase from ____ in the scenario above, please how you feel about ___		
	Att1	Bad/Good	0.887	0.870
	Att2	Unfavorable/Favorable	0.894	
	Att3	Negative/Positive	0.889	

[a] Scale: 1=very unlikely; 7= very likely
[b] Scale: 1=strongly disagree; 7= strongly agree

Next, the researchers created summary measures of the constructs (Martinez-Lopez et al., 2013). Hair et al. (2010) suggest that single-item measures are appropriate for simple concepts, such as behavioural outcomes, and averaging scales also can reduce model complexity (Martinez-Lopez et al., 2013). Scales used in the present study were reliable (Table 1; Cronbach's alpha ranged from .87-.94) and valid in previous research. Because the scales were reliable and drawn from previous research, Petrescu (2013) suggests the use of single-item indicators. As single items do not affect internal reliability, common method bias is minimized (Bergkvist & Rossiter, 2007), and using common latent factor analysis, only 1.4% common method bias was found (Gaskin, 2012).

Method of Analysis

Based on assessment of the constructs, the analysis proceeded with an examination of the proposed model, which was tested using path analysis. Path analysis allows examination of multiple regression models and is useful when assessing the fit of a priori theoretical models, because it allows examination of the significance of hypothesized relationships (Stage et al., 2004). Path analysis also provides an overall indicator of model fit (i.e., a chi-square statistic) and multiple goodness-of-fit indices. Although path analysis does not allow determination of the impact of individual scale items, the reliabilities of the variables were excellent. Finally, due to the sample size, path analysis provides greater predictive power relative to latent variable analysis. Subsequent to the examination of the proposed model, a group analysis was conducted to assess the influence of customer versus store employee reviews.

RESULTS

Path Analysis (via AMOS 26) was used to test the model and hypotheses shown in Figure 1, Model 1. Multiple indicators, averaged for each construct, were used to represent the construct (Li & Calantone, 1998). The results of the path analysis for the original model indicated a good fit of the data. The fit statistics for the model are as follows: χ^2 =5.501, p=.064, root mean residual=.028, goodness-of-fit=.989, adjusted goodness-of-fit=.921, normed fit index=.989, comparative fit index=.993, Tucker-

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Table 2. Convergent and discriminant validity (Fornell and Larcker)

Construct	Means	Std. Dev.	Composite Reliability (CR)	Community engagement	Perceived Usefulness	Source Credibility	Purchase Intention	Attitude toward the store
Community engagement	4.89	1.23	0.91	***0.60*** [a]				
Perceived usefulness	5.22	1.45	0.96	0.19	***0.86***			
Source credibility	4.64	1.41	0.96	0.25	0.49	***0.85***		
Purchase intention	4.83	1.51	0.96	0.31	0.53	0.61	***0.88***	
Attitude toward the	2.77	0.41	0.92	0.18	0.08	0.12	0.13	***0.79***

[a] Average variances extracted (AVE) are on the diagonal; squared correlations are off-diagonal. The AVEs for each construct are greater than the corresponding inter-construct square correlations which supports

Lewis index=.965 incremental fit index=.993, and root mean square error of approximation=.09. The statistically non-significant χ^2 (p=.064) indicated a good fit of the model to the data, which was confirmed by the other indices. Standardized path coefficients for the model indicated that six of the seven hypotheses were significant (p<.05) (see Table 3). One of the standardized path coefficients for the hypothesis was not significant at p<.05 (H_7, p<=.047).

The R-squared value is the proportion of variance accounted for by its predictors (Arbuckle, 2009). In the original model, approximately 26% of the variance in online retail store community engagement was explained by source credibility and perceived usefulness, and the model explained approximately 18% of the variance in attitude toward the store, and 71% of the variance in purchase intention (see Figure 1, Model 1). Based on the findings and support of the proposed hypotheses, the model appeared to provide insight into the role of online retail store community engagement intention and its ability to impact purchase intentions.

Group Analysis

The results of the path analysis for the group model (customer versus store employee reviews) indicated a good fit of the data. The fit statistics for the model are as follows: χ^2 =7.146, p=.128, root mean residual=.03, goodness-of-fit=.987, adjusted goodness-of-fit=.899, normed fit index=.985, comparative fit index=.993, Tucker-Lewis index=.966 incremental fit index=.993, and root mean square error of approximation=.06. The statistically non-significant χ^2 (p=.128) indicated a good fit of the model to the data, which was confirmed by the other indices. Standardized path coefficients for the model indicated that six of the seven hypotheses were significant (p<.05; see Table 3). One of the standardized path coefficients for the hypothesis was not significant at p<.05 (H_{14}, p<=.076 for customers and p<=.009 for store employees).

For the customers model, approximately 22% of the variance in online retail store community engagement was explained by source credibility and perceived usefulness, and the model explained approximately 17% of the variance in attitude toward the store, and 67% of the variance in purchase intention (see Figure 1, Model 2). For the store employee model, approximately 31% of the variance in online retail store community engagement was explained by source credibility and perceived usefulness, and the model explained approximately 18% of the variance in attitude toward the store, and 73% of the variance in purchase intention (see Figure 1, Model 3). Based on the findings and support of the proposed hypotheses for the grouped analysis, the models appear to provide insight into the role of source credibility and perceived usefulness of online reviews and their ability to impact purchase intentions. Specifically, source credibility of online store employee reviews and perceived usefulness of online customer reviews significantly (p=.007; p=.011, respectively) increases purchase intention.

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Indirect Effects

In addition to the models' direct effects, indirect effects of source credibility, purchase intention, and online retail store community engagement on purchase intention were examined. A mediation analysis was conducted using a bootstrap sample of 200. As seen in Table 3, mediation was identified using variance ratios (Hair et al., 2014). According to Hair et al. (2014), a complete mediating effect occurs if the proportion of the total variance that is systematic variance (VAF)>.80, a partial mediating effect if .20<VAF<.80, and no mediating effect if VAF<.20. For the present study, significant indirect effects (p<=.01) were found between source credibility and purchase intention for the original and store employee models, but no significant mediation effect was found (VAF<.20).

CONCLUSION

In the same way that characteristics of brand community exist within physical stores, community is in play for online retailers with respect to social shopping and reviews. This study found that consumers' perceptions of source credibility and the perceived usefulness of online reviews significantly impact an intention to engage with the online retail store community and shape consumers' attitudes toward the store and purchase intentions. The finding that source credibility relates to an intention to engage with the online retail store community is new, while the impact of credibility on purchase intentions is consistent with previous research (Assaker, 2020; Kiran & Srivastava, 2020; Xu et al., 2020). Moreover, the study's results show that consumers' perceived usefulness significantly impacts an intention to engage with the online retail store community, which is also unique, while the impact on purchase intentions aligns with previous research (Kiran & Srivastava, 2020; Xu et al., 2020; Ventre & Kolbe, 2020; Zhang et al., 2021).

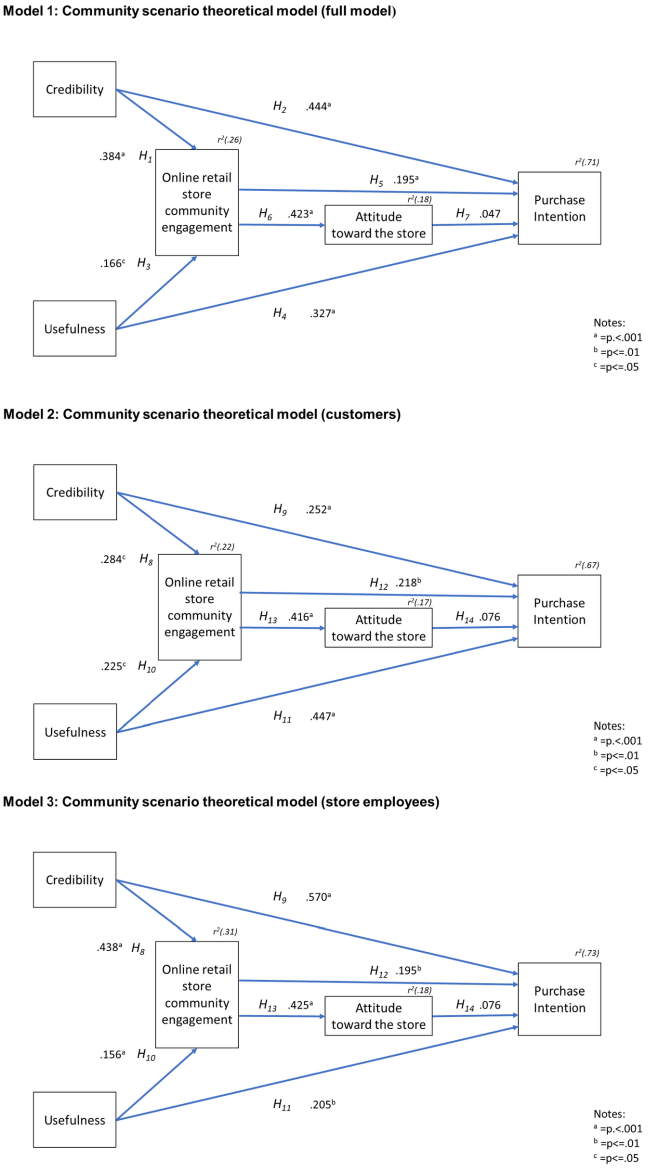
Online retailers may believe that including reviews as part of their websites is enough to increase sales. However, this study's results suggest that there may be a bigger picture to the influence of reviews, one in which the social nature of shopping plays an important role. Like shopping in-stores, consumers want to engage with the online retail store community via reviews. Retailers should consider ways they can increase community among consumers posting reviews and those reading them. Inviting customers to post reviews and providing suggestions for how these reviews can help other shoppers is an example of how an online retailer can encourage moral responsibility among shoppers. Asking shoppers to post reviews on a regular basis can be construed as rituals and traditions that exist within the community. And shared consciousness could be encouraged by explaining to customers that they share common interests with other shoppers and the contents of their reviews are thus helpful to the community.

This study has specific managerial implications for retailers that also extend beyond community into consumer decision making. Models such as Prospect Theory and Means End Chain Theory detail what factors motivate consumers and influence their purchase decisions. The model developed in this study similarly highlights how perceptions of online reviews (with respect to the source, credibility, and usefulness), impact socialization factors such as community engagement, which ultimately influence consumer decision making. The findings also highlight how reviews provide social proof that motivate communal engagement, which ultimately benefits the online retailer. And, in the end, retailers who utilize reviews, whether from existing customers or store employees, are viewed as more trustworthy, helpful, and social, factors which increase sales.

Another contribution of this study is that consumers evaluate reviews from other customers differently from those provided by store employees. Reviews provided by employees are viewed as more credible, most likely due to their perceived level of salesperson expertise. In contrast, reviews provided by other customers are perceived as more useful, potentially due to the authenticity and trust that comes from believing that consumers are not paid to post the reviews and thus are more objective. Some successful online retailers, like Zappos.com, rely on store employees to provide reviews; while others, like Amazon.com, rely more on customer reviews. This study highlights how

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Figure 1. Community scenario models



different sources for the reviews influence community engagement and purchase intentions in unique ways, which extends the literature on online reviews and source differences (Chen et al., 2016; Diwanji & Cortese, 2020; Tata et al., 2020).

Managerially, this study's results also suggest that retailers may want to consider the strategic objectives behind utilizing reviews on their websites. Reviews provided by store employees reinforce expertise, while reviews provided by other customers appear to be more useful for shoppers. And both constructs impact online retail store community engagement and purchase intentions. Thus, depending on the objectives and resources of the online store, a retailer may only want to provide one source for reviews. For example, if the product sold is technical, such as power generators, the

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Table 3. Results of path analysis model

Constructs			Hypotheses	Total model	Hypotheses	Customers	Store Employees	Difference between customers and store employees
				Path Coefficient		Path Coefficient (Std.)	Path Coefficient (Std.)	(p-value)
Source credibility	--->	Community engagement	H_1	**0.384**[a]	H_8	**0.284**[c]	**0.438**[a]	0.625
Source credibility	--->	Purchase intention	H_2	**0.444**[a]	H_9	**0.252**[a]	**0.570**[a]	**0.007**
Perceived usefulness	--->	Community engagement	H_3	**0.166**[c]	H_{10}	**0.225**[c]	**0.156**	0.547
Perceived usefulness	--->	Purchase intention	H_4	**0.327**[a]	H_{11}	**0.477**[a]	**0.205**[b]	**0.011**
Community engagement	--->	Purchase intention	H_5	**0.195**[a]	H_{12}	**0.218**[b]	**0.195**[b]	0.916
Community engagement	--->	Attitude toward the store	H_6	**0.423**[a]	H_{13}	**0.416**[a]	**0.425**[a]	0.817
Attitude toward the store	--->	Purchase intention	H_7	0.047	H_{14}	0.076	0.009	0.510
Indirect Effects:								
Source credibility	--->	Purchase intention		**0.088**[b]		0.071	**0.087**[b]	
Perceived usefulness	--->	Purchase intention		0.037		0.056	0.031	
Community engagement	--->	Purchase intention		0.024		0.032	0.004	
Total Effects:								
Source credibility	--->	Purchase intention		**0.527**[b]		**0.323**[c]	**0.657**[c]	
Perceived usefulness	--->	Purchase intention		**0.363**[c]		**0.533**[b]	**0.235**[c]	
Community engagement	--->	Purchase intention		**0.214**[c]		**0.249**[c]	**0.198**[c]	
Attitude toward the store	--->	Purchase intention		0.047		0.076	0.009	

[a] p<=.001; [b] p<=.01; [c] p<=.05

credibility of the reviewer may be especially important. Alternatively, for products such as a pair of jeans, perceived usefulness may be of greater importance. And some retailers might want to include both types of reviews to maximize the impact on community engagement and purchase intentions among shoppers on their website.

Limitations and Future Research

This study has several limitations, suggesting calls for future research. The study utilized a sample of U. S. college students, who were mostly white males and employed part-time. While college students are an appropriate sample with 91% purchasing online (Lester et al. 2006), an increasing number of older, more diverse, fully employed consumers are shopping online. Previous research also suggests that the effects of this study are likely to exist within other countries and cultures (Dos Santos et al., 2018; Ventre & Kolbe, 2020; Zhang et al., 2016). Thus, future research should examine how various demographic groups in both the U.S. and in other countries utilize online reviews and intend to engage with community as they shop online.

While the original model was assessed using 11 indicators of significance, the root mean square error of approximation (RMSEA) was .09. While this value may be perceived as high, Chen et. al (2008, p. 476) states there "is no empirical evidence for the use of 0.05 or 0.10 as universal cut-off values to determine adequate model fit." Moreover, not all potential antecedents were included in the model, such as ease of use, social influence, cognition, and individual differences in computer use. And the structural model did not capture all possible outcomes, such as brand loyalty. Additionally, within the results, the relationship between attitude toward the store and purchase intentions was not significant. Theory of Reasoned Action (Ajzen & Fishbein, 1999) suggests that attitudes would predict purchase intentions. However, the justification of H7 was based on attitudes toward the product, rather than the store. Because the respondents did not actually interact with the product, know the brand name, nor view a specific website, the absence of information about the online store may have contributed to the lack of significance for this hypothesis.

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Understanding the Intention to Use Virtual Currency in a Gamified E-Commerce Context

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ABSTRACT

Gamification is all about using game features in a serious context to encourage people to use products or services. E-commerce has used gamification for marketing determinations to strengthen intent to use. Technology acceptance model is employed in the current study to investigate the intention to use gamified virtual currency in the Indian context. A survey was used to accumulate records from a random sample of relevant virtual currency users in an e-commerce context using a quantitative approach. Structural equation modeling (SEM) is expended for data analysis in this research. This empirical research shows the incremental impact of the users' perceptions of the ease of use and usefulness on intention to use virtual currency. The variable attitude seems to be not mediating significantly. There are also discussions, restrictions, and future research directions mentioned in this study.

KEYWORDS

E-Commerce, Gamification, TAM, Usage Intention, Virtual Currency

INTRODUCTION

Is there anyone who never played anything in their childhood? Hide and seek, tag, musical chair, pen - paper – scissors, so on, to name a few. Do you think playing remains only for childhood days? Cricket, football, field hockey, etc., to name a few. Playing games have always been a cross-generational activity that piqued interest and fun (Huotari & Hamari, 2017). More than a billion people play games on mobile globally (Joensuu & Ryynänen, 2019). No doubt that the game makes the play more fun and entertaining (Heeter et al., 2011). And many are part of serious games knowingly or unknowingly, which can be considered as something beyond fun and entertainment. Like the concept of serious games, gamification uses game elements but as part of an entertainment game for purposes other than their regular expected use (Buckley et al., 2019). Gamification has been more specifically well-defined as "the use of game design elements in non-game contexts" (Deterding, 2012). Gamification in this context can be defined by employing or embedding the features of game elements like virtual currency, leader boards, points, etc., as a set of any specific behavior and processes applied for getting better engagement of its consumers (Bozkurt & Durak, 2018). Gamification is an efficient design strategy that is still developing in recent years for driving game mechanics into existing real-life

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contexts, especially in e-commerce (Rapp et al., 2019). The interest in gamification in academics and e-commerce is increasing, and it has already been proven. Still, the absence of enough empirical evidence on individual game elements, especially in online commerce(García-Jurado et al., 2019), is something important that needs to be studied. India, being one among the internationally recognized, quick developing, and most significant emerging e-commerce industries that expect to cross 200 billion US dollars by 2026 from 38.5 billion US dollars as of 2017 (*E-Commerce in India| IBEF*, 2021).

Flipkart, along with other minor specialist e-commerce platforms, is presently among the most dominant e-tailers in the Indian e-commerce industry (Prashar et al., 2015; Rajan, 2020). Flipkart Plus is a no-cost successful loyalty initiative for all of its regular customers (Trauboth, 2020). Buyers must earn five hundred SuperCoins to be eligible for this program. Flipkart Plus members earn four SuperCoins per each one hundred rupees invested on a Flipkart purchase, while non-plus customers receive two SuperCoins (Gayathry, 2019). Flipkart SuperCoins is one of the limited virtual currencies used for e-tailing (Moin & Rahman, 2019). The virtual currency reward system that has got some exchange value within the system can be considered virtual currency in the gamification framework (Dicheva et al., 2021). Therefore the virtual currency has got an attribute of private currency that is mainly used in online transactions by using various technological factors. In the case of Flipkart SuperCoins, it can be used for payment for the products, buying coupons, buy exclusive deals and book flight tickets by using the same platform (Flipkart, 2019). The fact of the matter is that virtual currency is slowly replacing real physical currency in the e-commerce context, but nobody is having enough statistical proof on the exact replacement. If we consider the Indian population, there is a tremendous amount of interest invested in virtual currencies (Mohammad Usman, 2020). According to the recent statistics, India being appraised to contribute around ten percent of the overall virtual currency market worldwide. Even though more than one thousand five hundred types of virtual currencies are seen in India (Sackheim & Howell, 2018), we could hardly find the virtual currency in an e-commerce platform. This scenario constantly questions how far the virtual currency is affecting real-life scenarios (Wang & Mainwaring, 2008). This question remains in the mind of both academicians and business people. Therefore this investigation focuses on virtual currency in a gamified e-commerce context. The gamified virtual currency considered for this empirical study is SuperCoins which are stored and transacted only through the Flipkart domain's designated mobile application or website.

The use of game elements like virtual currency and its game mechanics and game dynamics in non-game situations like e-commerce has become one of the fastest emerging practices in marketing-related activities (Yang et al., 2017). Whatsoever the research that provides empirical studies as evidence for game elements' influence on consumer behavior in domains like e-commerce marketing is still deficient (Behl et al., 2020; Deterding et al., 2011). From the existing pieces of literature, it is clearly understood that virtual currency is a gamification element (Helmefalk et al., 2019), and there is a need to identify the relevance of virtual currency concerning consumer behavior perspective in an e-commerce context (Kamboj et al., 2020). There is a clear requirement of empirical research that would examine how each single gamification element like virtual currency individually affect the business environment (Azmi et al., 2021; Xi & Hamari, 2020). Therefore, this study comprises empirical research to bridge the gap on how far this individual game element called virtual currency is affecting the individual's intention to use SuperCoins in the environment of e-business by using the Technology Acceptance Model (TAM).

The key research questions are as follows:

- Determining whether and how much a specific game mechanic termed virtual currency influences e-commerce usage intention?
- To forecast customer behavior in virtual e-commerce environments, should 'Attitude' be used in conjunction with the existing TAM framework, which is in conflict?

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LITERATURE REVIEW

The SuperCoins of Flipkart – Virtual Currency

Whether it's their business practices, product catalogs, web designs, or service offers like rewards, the e-commerce platform should learn what matters effectively for them (Khanna & Sampat, 2015). Flipkart, an e-commerce platform, employs gamification rewards as a promotional and engagement technique to entice more people to visit their platforms on a continuous basis for their success (Raj & Gupta, 2018). "Use of immediate signs of accomplishment through virtual and monetary rewards" is how rewards are defined (Conaway & Garay, 2014). Virtual currency is a kind of reward that is used in gamified context. Virtual currency is something that works like money (Kňažková & Ondrušová, 2019) but definitely, nothing to do with real cash in gamified framework. Virtual currency's existence in a gamified system leads to the formation of a virtual economy. Customers can receive virtual currency called SuperCoins as a reward from Flipkart, and they can also use the same virtual currency as a payment option (Moin & Rahman, 2019). Therefore Flipkart SuperCoins somewhat act as a medium of exchange or function like a currency, and such kind exchange of virtual currency can also be seen in many online gaming websites (Shin, 2008). The main focus of the virtual currency used by Flipkart called SuperCoins is to bring customer loyalty. This supposition is based on the sense that the virtual currency can be used for redeeming tangible goods will be good enough to motivate customers to come back and enhance the buying behavior. But the lack of empirical evidence on the above-stated logic leads this research.

Technology Acceptance Model (TAM)

Theory of Reasoned Action (TRA) (Fishbein & Ajzen, 1975) is the basic inspiration for the formation of the Technology Acceptance Model (TAM) and is used to envisage the acceptance of technology. The Technology Acceptance Model is a widely used model for studying online consumer behavior, especially in e-commerce shopping (Aparicio et al., 2021; Aydin, 2015). One of the worldwide accepted models is Technology Acceptance Model (TAM), especially in an online shopping context in order to predict any kind of adoption of new technologies like Gamification (Raman, 2020). Past studies that used the Technology Acceptance Model have observed the value of Cronbach'sreliability more than 0.9 and considered scales of the Technology Acceptance Model that have got a greater level of convergent validity, discriminant validity, and construct validity(Davis & Venkatesh, 1996). Therefore this study is conducted with an underlying theoretical framework called Technology Acceptance Model (TAM). The two independent variables, perceived usefulness and perceived ease of use, are explored in this research that influences the dependant variable intention to use in the framework of the virtual currency called SuperCoins through user survey research with the help of a mediating variable called attitude.

Perceived Usefulness (PU)

According to (Davis, 1989)the variable perceived usefulness is described by way of "the degree to which an individual believes that using a particular system will improve his or her job performance." The description refers to whether or not anyone considers a piece of technology to be helpful for the task at hand. According to (Lin, 2007), if we consider the virtual community's environment, perceived usefulness specifies the users' confidence in their ability to achieve information and amenities, share their know-how with other people around, and augment their performance in information interchange while using the virtual community. Perceived usefulness is considered an important factor in innovative technology adoption (Kim et al., 2017). The significance and usefulness of technology is still a concern for organisations due to a rise in solutions and technological advancements (McLean & Wilson, 2016). Therefore this study considers perceived usefulness, the one for which a user believes in the existence of a positive attitude-intention relationship.

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H1: Perceived usefulness has a positive influence on attitude toward employing the virtual currency.
H2: Perceived usefulness has a positive influence on the intention to use virtual currency.

Perceived Ease of Use (PEOU)

According to (Davis, 1989) perceived ease of use is well-defined as "the degree to which a person believes that using a particular system would be free of effort." The word ease can be further explained as the freedom from strain or great energy. In this research, the exact definition of perceived ease of use for framing the hypothesis. The study by (R. Hendrickson & R. Collins, 1996)confirmed that there is an existence of a straight connection between perceived ease of use and intention to use.The Technology Acceptance Model's empirical studies have clearly demonstrated that perceived ease of use has a major positive impact on attitude and intention (Khan, 2021; Wong et al., 2021). The gamification scenario, especially in the Indian E-commerce context, is still considered an innovation. Initial acceptance of an invention is aided by the variable perceived ease of use, which is essential for favorable reception and sustained use(Davis et al., 1989). All these contexts are considered for this study.

H3: Perceived ease of use has a positive influence on attitude toward using virtual currency.
H4: Perceived ease of use has a positive influence on the intention to use virtual currency.

Attitude (AT) and Intention to Use (IU)

In the context of this study, the mediating variable attitude is well-defined as "the extent to which a person's positively or negatively evaluate or assess the behavior." The key component and direct antecedent of doing a specific behavior, per the TRA, is "intention" (Aydin, 2015). Both the variables are considered for predicting the intention to use SuperCoins. Initially, research on Technology Acceptance Model (TAM) clearly claims that the attitude on the way to action has an affirmative influence on users' intention, leading to people performing an action. The attitude was eventually eliminated from the Technology Acceptance Model (TAM) as it was shown that it was not directly associated with the use of technology (Teo, 2009). Further studies on Technology Acceptance Model (TAM) also say that the mediating variable attitude may or may not have an impact on a person's intention toward using technology and, it still remains a point of discrepancy among researchers (López-Bonilla & López-Bonilla, 2017). There are also studies using Technology Acceptance Model (TAM) without applying the mediator attitude (R. Hendrickson & R. Collins, 1996). Whatsoever this study is also trying to figure out the relevance of mediation because most of the studies conducted in TAM has got the tremendous contextual difference (Azmi et al., 2021). The validity of the Technology Acceptance Model (TAM) model used in the research is also confirmed through the following hypotheses.

H5: Attitude toward virtual currency has an affirmative influence on the intention to use virtual currency.

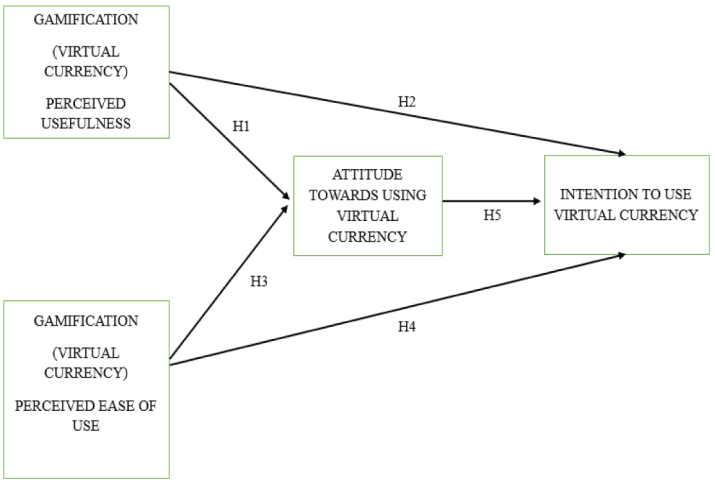
METHODOLOGY

Data Collection and Sample

The primary goal of this research is to understand the intention to use virtual currency in an e-commerce platform called Flipkart. An internet-based questionnaire survey was used and administered for this research from January 2021 to February 2021. Even though we got the responses from 210 participants, we have screened and taken only responses of 165 participants who are Indian nationals relevant for the

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Figure 1. The proposed research framework for testing



study, and therefore, the sample size for this research is N=165. Even now, in e-commerce websites in India, the embedding of virtual currencies is not yet commonplace.; therefore, it was quite necessary to toe the line that the respondents had enough understanding about the virtual currency. Hence, we screened each and every response to determining whether or not users had considerable awareness of this manner of interchange. Considering the respondents' data, most of the participants in the study fall in the age category of 21-30 years and 31-40 years, representing 73% of the total sample. The distribution of the gender-wise data of the selected sample shows that 60% of the participants were males, and 40% of the participants were females. Most of the participants were professionals contributing to 42%, followed by students 25%. About 67% of the respondents have been using Flipkart for more than three years.

Measurement

An online questionnaire survey was conducted to gather the data. The instrument for data collection was developed grounded on the relevant evaluation of pieces of literature and was used to measure each construct. The instrument for the research had 13 items basically representing four different variables as mentioned in the model of the research, and its sources are publicized in the Appendix. The questionnaire was done by means of the 7-point Likert- scale type, which ranges from "strongly disagree" to "strongly agree," and all these were drawn from previously validated instruments. The variable perceived usefulness was finally assessed through 3 items taken from (Davis, 1989), which assessed the magnitude to which a person believes that virtual currency could be used favorably with provided positive, predictable results. The variable perceived ease of use was finally assessed with three items scale taken from (Venkatesh, 2000). The mediating variable called attitude is measured by using three items scale taken from (Van Der Heijden, 2004). The intention to use was the dependent variable that the researcher used, and it was measured by using four items scaled adopted from (Davis, 1989).

Data Analysis

The measuring instruments' reliability and validity had been checked with Partial Least Squares (PLS), and the test was also done using WarpPLS. There are more than a few reasons for using the Partial Least Squares (PLS) regression method. (Reinartz et al., 2009) had already claimed that Partial Least Square is beneficial for a lot of multi-dimensional models of a predictive nature with sample sizes that were comparatively lower, and for this study, the sample size (N) was 165, which matches

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the relatively small size. Therefore the researcher identified that Partial Least Squares (PLS) was the better statistical approach considering the circumstances of this research. Convergent validity, discriminant validity, and tests for reliability were directed grounded on methods commonly used for empirical studies. Initially, the quality of being dependent on the survey questionnaire was checked by calculating Cronbach's α to establish internal consistency. Every single item's reliability was inspected using composite factor analysis. The results obtained from both the analysis satisfy the condition that all of the findings were upper than the suggested standard of 0.7. Each item's outcomes are mentioned in table 1, and the further breakup of each construct is detailed in Table 2.

Table 1. Reliability analysis

Constructs	Cronbach's α	Composite Factor Reliability
Perceived Usefulness	0.945	0.946
Perceived Ease of Use	0.938	0.942
Attitude	0.936	0.941
Intention to Use	0.955	0.955

Table 2. The constructs reliability values

Constructs	Items in the beginning	Final items	Variables	M (SD)	Standard error	Composite factor reliability
Perceived Usefulness	5	3	PU1	5.129 (1.625)	0.083	0.946
			PU2		0.063	
			PU3		0.057	
Perceived Ease of Use	3	3	PEOU1	4.721 (1.680)	0.067	0.942
			PEOU2		0.067	
			PEOU3		0.102	
Attitude	4	3	AT1	5.484 (1.385)	0.08	0.941
			AT2		0.04	
			AT3		0.045	
Intention to Use	4	4	IU1	5.172 (1.634)	0.106	0.955
			IU2		0.058	
			IU3		0.057	
			IU4		0.051	

The method that we employed to check the model's convergent and discriminant validity was suggested by (Fornell & Larcker, 1981). The same procedure was used in this study as well. The item loadings, cross-loadings, and average variance derived in this study were examined carefully and methodically by using exploratory factor analysis (AVE) and also conducted a convergent validity test to confirm consistency across items. Through the test, it was identified that the loadings of each item surpassed the smallest detectable sensation of 0.7 (Chin et al., 2003) and hence indicated

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convergent validity. The shared variance between constructs has been compared with the average variance extracted (AVE) from the individual constructs for examining the discriminant validity of this study. The average variance extracted (AVE) from the individual constructs was less than the shared variance between constructs, confirming the discriminant validity of the constructs used in the research. The inter-construct correlation referring to the construct's diagonal correlations to their latent variables is smaller than the square root of average variance derived (AVE) for all constructs(Loch et al., 2003). Therefore, it proves discriminant validity. In table 3, the specifics of both convergent and discriminant validity are described.

Table 3. Discriminant validity and Convergent validity

Constructs	CR	AVE	MSV	MaxR(H)	4	3	2	1
Intention to Use	0.955	0.841	0.719	0.962	**0.917**			
Attitude	0.941	0.841	0.601	0.952	0.727	**0.917**		
Perceived Usefulness	0.946	0.853	0.719	0.952	0.848	0.776	**0.923**	
Perceived Ease of Use	0.942	0.844	0.707	0.948	0.841	0.733	0.781	**0.919**

RESULTS

The complete model fit had been checked along with the quality indices. The average path coefficient (APC) value is 0.354, the average R-squared (ARS) value is 0.684, and the average adjusted R-squared (AARS) value is 0.679; all the values are shown significance($P<0.001$). The average block VIF (AVIF) is 2.593, and the acceptable if value is ideally less than or equal to 3.3. According to (Kock, 2011) the standards for both average path coefficient (APC) and average R-squared ought to be below 2 (0.05 level significance), whereas the AVIF value ought to be lesser than 5. The average full collinearity VIF (AFVIF)=3.376, and it is acceptable since the value is less than or equal to 5. The models TenenhausGoF (GoF) is 0.781. The Simpson's paradox ratio (SPR) is 1.000, which ideally should be equal to 1. The R-squared contribution ratio (RSCR) is 1.000, which ideally should be equal to 1. The statistical suppression ratio (SSR) is 1.000, which is acceptable if the value is greater than or equal to 0.7. The nonlinear bivariate causality direction ratio (NLBCDR) is 1.000, which is acceptable if the value is greater than or equal to 0.7. The researchers identified the structural equation framework has a good fitness for the data from the above values. The average total collinearity value also infers low overall collinearity, as a result representing trivial method variance.

In order to support the hypothesis, the beta coefficient (β) of affiliation essential be at 0.05 for the significance otherwise underneath the mentioned value (Sarstedt et al., 2014). The adjusted R squared value for each hypothesis is also checked, which specifies the independent variables' explanatory influence on the dependent variable. The respective values are detailed in table 4. As indicated in table 4, the perceived usefulness and ease of use on the importance of virtual currency usage intention shows a significant positive relationship with the dependent variable attitude and describes 61.8% of the variance in attitude. Therefore it supports hypotheses 1 and 3. The independent variables such as perceived usefulness and perceived ease of use, along with the non-significant mediating variable, explain 74.1% of the change in the dependent variable intention to use, thereby supporting hypothesis 2 and hypothesis 4. TAM models are used widely in two different ways: TAM with attitude (TAM-O) and TAM without the attitude (TAM-R). As a result, none of them seems to have the upper hand. There have been studies that show that adjusting one's attitude toward technical usage is insufficient to hold this definition in TAM (López-Bonilla & López-Bonilla, 2017). Whatsoever scenario plays a critical role in deciding the need for the TAM mediation variable for the research. In our indirect

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study effect of attitude towards intention to use is not significant. Considering the factors like this study happens in gamified virtual currency scenario, and the research happens in the Indian context, the direct relationship plays more importance than the indirect relationship. Future researches on this discrepancy can help to realize a better understanding of this subject.

Figure 2. The tested structural framework of intention to use virtual currency

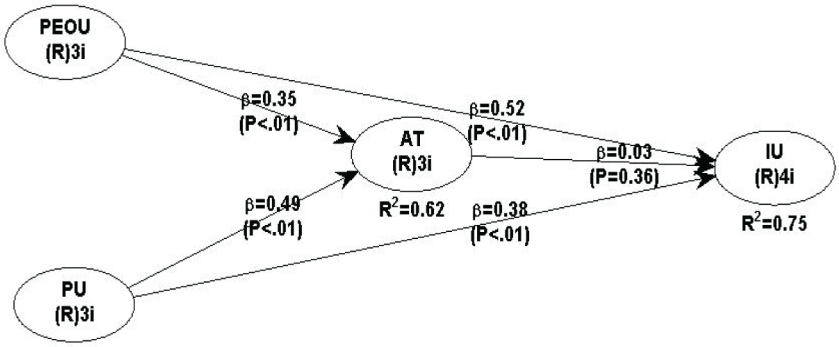


Table 4. PLS-SEM results of the full model

Path	Hypothesis	Path coefficient (β)	Adjusted R2	Significance
PU→AT	H1 is supported	0.49	0.618	P<0.01
PU→IU	H2 is supported	0.38	0.741	P<0.01
PEOU→AT	H3 is supported	0.35	0.618	P<0.01
PEOU→IU	H4 is supported	0.52	0.741	P<0.01
AT→IU	H5 is not supported	0.03	0.741	P<0.36

DISCUSSIONS AND CONCLUSION

Theoretical Implications

The antecedents and outcomes of the usage of virtual currency in the Indian context were investigated using the Technology Acceptance Model (TAM) (Davis et al., 1989). The purpose of this study was to exam empirically in order to understand the nature of people's behavioral intentions to use virtual currency. The calculation and structural model testing findings back up the proposed research model to a greater extend. Overall, all theories demonstrating direct relationships were confirmed, while indirect relationships were not confirmed significantly, according to the results. There have been prior studies that found no mediation of the variable attitude in TAM (Tobbin, 2014), and it is likely that a stronger understanding and forecasting of behavioral intentions can be reached provided attitudes' multifaceted effects are correctly included for mediation. The mediation of attitude in TAM was a subject of discrepancy among the researchers for a long time (López-Bonilla & López-Bonilla, 2017), and another reason probably be the methodology used in this research may have had an effect on the outcome.

According to the empirical evidence, the independent variable perceived ease of use has a positive direct affinity on the intention to use virtual currency, which matches prior studies (Aydin, 2015). It means that the higher the degree to which an individual relies on that using gamified virtual

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currency would be permitted of struggle, the higher the strength of one's intention to perform a specified behavior of virtual currency usage. Similarly, the independent variable perceived usefulness has a significant positive relationship with virtual currency's intention to use, where the result is in accordance with the outcomes of previous research (Aydin, 2018). This means that the higher the degree to which a person believes that using gamified virtual currency would enhance his or her performance, the higher the strength of one's purpose to carry out a particular action of virtual currency usage. Users, on the other hand, are more likely to adopt a virtual currency because it involves reduced work or provides advantages in their operations. In other terms, if consumers can exert less effort when using a virtual currency like SuperCoins, the level of perceived usefulness or perceived ease of use is improved, and these findings are consistent with earlier studies in a similar setting (Tony et al., 2020).

Theoretical contributions to the topic of gamification are made in this paper. Although prior publications have looked into the topic of gamification aspects being adopted, the matter of individual game mechanics like virtual currency being applied to the e-commerce environment is still up for debate. The research utilized the Technology Acceptance Model to show the effects of perceived ease of use, perceived usefulness, and attitude on behavioral intention, with results that are not completely identical to those of earlier studies. As a result, the external validity of the Technology Acceptance Model must be studied again in the gamification environment, taking individual game mechanics into account.

Practical Implications

For gamification consultants and designers, this study has various practical consequences. Certain criteria that need to be considered in order to improve the acceptance of gamified virtual currencies are perceived ease of use and perceived usefulness, which are important factors influencing the desire to use virtual money in an e-commerce setting. For gamification designers, gamification consultants, and customer experience programmers, the characteristics of perceived ease of use and perceived usefulness may provide some inspiration for adopting the same in different e-tailing contexts. People are far more likely to adopt gamified virtual currency if it requires less exertion or has capabilities that meet their needs. Gamification designers and developers could provide a set of tutorials or videos to help customers learn how to utilize gamified virtual currency for the first time. As a result, people can see the significance of rewards like virtual currency in e-commerce. Furthermore, the ease of usage of virtual currency may alleviate the difficulties experienced by early users. As a result, the ease of use of such gamification mechanics will be perceived by the early adopters.

Limitations and the Scope for Future Research

The current investigation has some drawbacks that scholars can discuss in future studies. A non-random sample is used since a randomly selected sample from the study's population is difficult to obtain. As a measure, the PLS-SEM was chosen as the most suitable tool for statistical analysis with regard to this investigation. The research on the same concept with a larger sample size by using different methodology might yield different outcomes. This paper will provide further research opportunities for the already intensely discussed topic of the role of attitude in the technology acceptance model (TAM). Besides, the term gamified virtual currency is not too broad. Additional game features such as scores, badges, and feedbacks may be incorporated into future research to expand the scope of gamification. The analysis was restricted to the Indian background. The virtual currency we chose for the study was SuperCoins, and future research could involve more virtual currencies to extend the findings. Further research can examine the virtual currency's intention to use from both a quantitative and qualitative angle. This can provide a clear blueprint to highlight the association among independent variables such as perceived usefulness, perceived ease of use, and the dependant variable intention to use virtual currency. Current research collected facts from target respondents in India over and done with an online questionnaire survey. As a result, it is suggested that this analysis be carried

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out in a more number developing nation or a proportional investigation between two nations with dissimilar beliefs be used. For further research, a multi-group study based on a social demographic function is also suggested.

Conclusion

Through examining the effects of a single gaming mechanic known as virtual currency on e-commerce customers, the study addressed a gap in the literature. The purpose of this research was to examine if TAM's theoretical assumptions for undermining the impact of virtual currency on its usage intention could be implemented to the scenario of e-commerce among individual customers. According to the findings, there is a significant positive association between perceived ease of use, perceived usefulness, and customer intent to use the the gamified virtual currency. The non mediation of the variable attitude is notable in contrast to the existing TAM model. With the insights obtained from these structural relationships in this study, future researchers are encouraged to build on the findings of this study.

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APPENDIX: ITEMS PRESENT IN THE QUESTIONNAIRE

Perceived Usefulness – PU (Davis, 1989)

PU1: SuperCoins tend to be very useful in my Flipkart transactions.
PU2: I believe SuperCoins will help me improve my Flipkart transactions.
PU3: In my view, SuperCoins would help to increase the quality of Flipkart transactions.

Perceived Ease of Use – PEOU (Venkatesh, 2000)

PEOU1: I have realized that the provision to be straightforward and understandable.
PEOU2: I have realized that the provision does not necessitate a great deal of psychological exertion.
PEOU3: I have realized that the provision to be easy to use.

Attitude – AT (Van Der Heijden, 2004)

AT1: I'd be interested in purchasing a product from this website.
AT2: The idea of purchasing anything from this website appeals to me.
AT3: Purchasing a product from this website is a smart idea.

Intention to Use – IU (Davis, 1989)

IU1: In the future, I hope to transact with SuperCoins.
IU2: As much as possible, I want to use SuperCoins.
IU3: I recommend that others use SuperCoins.
IU4: In the forthcoming, I want to endure using SuperCoins.

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Skepticism Toward Online Advertising: Causes, Consequences, and Remedial Moderators

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ABSTRACT

Despite the enormous spending on digital advertising, consumers are skeptical toward online advertising (STA). The authors integrated advertising value and stimulus-organism-response (SOR) frameworks to develop a model of STA's causes and consequences. Product knowledge and perceived ethics of online seller (ETH) were proposed as moderators. For Study 1, moderated-moderated mediation technique was applied on the time-lagged data of 411 consumers. For Study 2, a between-subject experiment (n = 179) compared the effects of skepticism across video and picture ads. The results indicate that ETH and product knowledge moderated the relationships between stimulus-organism and organism-response states, respectively. Moreover, consumers showed favorable attitudes toward video ads. This study made novel contributions to research on STA by filling multiple voids, integration of advertising value and SOR, infotainment and puffery as predictors, product knowledge and perceived ethics of online seller as moderators, and comparison across advertisement type (video vs. picture).

KEYWORDS

Attitude Toward Online Advertising, Experiment, Moderated Mediation, Online Marketing, Online Seller Ethics, Product Knowledge, Skepticism Toward Online Advertising, Stimulus-Organism-Response Model

1. INTRODUCTION AND BACKGROUND

Online advertising has experienced tremendous growth over the past decade. The amount spent online has even surpassed the amount spent on television advertising. The worldwide spending on Internet advertising in 2020 was 41% of the entire market, while in the same period, the amount spent on TV advertising was 28% of the total market (Guttmann, 2021). Despite the effort and money spent on online advertising, research claims that, in general, consumers do not show trust in advertisements (Amawate & Deb, 2021; Obermiller & Spangenberg, 1998). Accordingly, skepticism toward advertising is a predisposition to distrust the claims made in advertisements (Obermiller & Spangenberg, 1998).

Skepticism is considered an essential component of consumer persuasion knowledge and a generalizable belief about how the marketplace operates (Friestad & Wright, 1994). Although the concept of skepticism toward advertising has been considered necessary, it has not been investigated widely. The existing literature has focused on skepticism toward environmental claims (Cheng, Chang, & Lee, 2020; Yu, 2020) or cause-related marketing (Amawate & Deb, 2021; Bae, 2018). However, skepticism toward online advertising, in general, has received limited attention from researchers.

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The advertising value model (Ducoffe, 1996) laid the foundations for understanding the effects of advertising characteristics on consumer attitudes. Infotainment in advertising is one such ad characteristic that refers to the integration of informativeness and entertainment (Okazaki, 2004). Overall, the research suggests informativeness (Jacobson, Gruzd, & Hernández-García, 2020) and entertainment (Gaber, Wright, & Kooli, 2019; Koshy & Manohar, 2018) influence consumer attitude toward advertising. Therefore, collectively, infotainment should have a positive influence on consumer attitude toward advertising.

The other ad characteristic which could influence consumer attitude toward advertising is ad puffery. Preston (1996) defines advertising puffery as any action by the advertisers which involves using superlatives, subjective views, concealing evidence, or exaggerations in ads to praise a product or service. Research shows that deceit in marketing is one of the trigger points for consumers' lack of trust towards advertising (Xie, 2016). Moreover, when exposed to puffed ads, consumers generate adverse reactions and negative product evaluations (Raziq et al., 2018). In addition, when consumers are less skeptical, advertisers exploit them using puffery (Obermiller & Spangenberg, 1998).

Given the current state of research, this study endeavors to fill several gaps in the domain of consumers' skepticism toward online advertising. First, the existing body of research on STA shows limited attention toward theoretical foundations (Bae, 2018; Obermiller & Spangenberg, 1998). Therefore, the present study integrates the advertising value model (Ducoffe, 1996) and stimulus organism response model (Mehrabian & Russell, 1974) to explain STA. Based on the advertising value model, two ad characteristics, infotainment, and puffery were identified, serving as a stimulus for the organism state. Consumer skepticism toward advertising (STA) has previously been defined as an attitude and consumer's inclination to disbelieve advertising claims (Obermiller, Spangenberg, & MacLachlan, 2005). Therefore, in the present study, skepticism toward online advertising characterized the organism part of the stimulus-organism-response framework. Accordingly, once skepticism has been developed among consumers, it will lead to responses. Bagozzi (1986) referred to the response state as the final state in the SOR paradigm, which comprises the behaviors and attitudes. Therefore, attitude toward online advertising has been proposed to function as a response state in the SOR model. To date, research integrating advertising value and SOR models to explain STA is not known.

Apart from theoretical foundations, the effects of infotainment and puffery as predictors of skepticism toward online advertising have not been investigated so far. Thus, it adds to the novelty of the present study. Additionally, investigating the SOR paradigm in different contexts, researchers have found that moderating factors influence the relationship between stimulus and organism states (Fu, Chen, & Zheng, 2020) and between organism and response states (Indibara, 2017). Furthermore, Hosany, Buzova, and Sanz-Blas (2020) proposed investigating factors that could moderate stimulus-organism and organism-response states. In response to this call, researchers identified online seller's product knowledge and perceived ethics as possible moderators.

The present study also made significant contributions to the subject area on the methodological front—study 1 employed a moderated-moderated mediation approach (Hayes, 2018) to investigate the skepticism phenomenon. The application of this approach is still novel to the subject area. The findings of this study will advance the understanding and methodology of future investigations on skepticism toward online advertising. Whereas study 2 anchored a between-subjects experiment to understand the effects of STA across advertisement types (video vs picture) on Instagram. Thus, study 2 further validated the conceptual model used in study 1 and tested the (Obermiller & Spangenberg, 1998) assumption that consumers may have different skepticism levels depending upon the medium and specific ad.

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2. HYPOTHESES DEVELOPMENT

2.1. Infotainment and STA

The term infotainment is derived from two interrelated and important advertising characteristics: informativeness and entertainment (Okazaki, 2004). The role of information is imperative in online advertising because consumers respond favorably toward relevant and informative ads (Aitken, Gray, & Lawson, 2008; Ducoffe, 1996). Thus, consumers on the internet make decisions after evaluating the information (relevant and factual) they obtain from the advertisements. Jacobson et al. (2020) further added to this vein of the investigation by finding that people use social media to generate and share information. Researchers have a consensus that informativeness induces consumers' motivations to view and process advertisements (Resnik & Stern, 1977).

The other aspect of the infotainment concept is entertainment. In the existing literature, the positive effects of entertainment on consumer attitudes have been emphasized. For example, Koshy and Manohar (2018), H. Lee and Cho (2019), and Gaber et al. (2019) have verified that consumers develop positive attitudes toward entertaining ads on the internet and social media. Thus the overall literature supports that infotainment induces favorable consumer attitudes, whether the internet or mobile advertising (Hongyan & Zhankui, 2017; Okazaki, 2004). In addition, recent research has shown that ads with detailed information and high quality of entertainment are positively embraced (Lin, Paragas, Goh, & Bautista, 2016).

The overarching theory (SOR) for the present study posits that environmental stimuli influence individuals' cognitive and emotional states. These cognitive and affective states, in turn, lead to specific actions (Mehrabian & Russell, 1974). Stimuli can be either controllable or uncontrollable. Advertising characteristics are controllable stimuli as they can be controlled and manipulated by the advertisers/sellers. Aligned with the advertising value model (Ducoffe, 1996), infotainment is an ad characteristic and a controllable stimulus, influencing skepticism toward online advertising. Skepticism toward advertising refers to a specific type of attitude different from the general attitude (Obermiller et al., 2005), defined as "an attitude of doubt or a disposition to incredulity either in general or toward a particular object" (Dictionary, 2019). As a result, for the sake of this study, STA falls into the organism realm of the SOR paradigm.

The studies investigating the direct influence of infotainment on STA are not known. However, some evidence exists for the potential relationship. For example, Hernandez, Chapa, Minor, Maldonado, and Barranzuela (2004) and Tsang, Ho, and Liang (2004) argued that lack of entertainment generates negative attitudes. When an ad fails to entertain consumers or provides irrelevant information, it is deemed boring and results in adverse reactions, including reduced credibility (Kim & Sundar, 2010; Teixeira & Stipp, 2013). Thus, it can be stated that boring and less informative ads will lead to skepticism (since skepticism refers to incredulity/lack of credence). To put it simply, STA being a negative attitude should be inversely related to infotainment. Therefore, this study proposes that infotainment will have a negative influence on skepticism toward online advertising.

2.2. Puffery and STA

Advertising puffery is any action by the advertisers which involves using superlatives, subjective views, concealing evidence, or exaggerations in ads to praise a product or service (Preston, 1996). The use of deceit in marketing is one of the trigger points for customer distrust towards advertising (Xie, 2016). The extant research provides some reference for the relationship between ad puffery and skepticism. Stern and Callister (2020) compared the effects of puffery and hyperbole on consumer attitudes. They found that consumers perceive both types of exaggerations to be deceptive. Moreover, when exposed to puffed ads, consumers develop adverse reactions and offer negative product evaluations (Raziq et al., 2018). Sellers' credibility and perceived truthfulness are at stake when ads contain puffery (Simons, 2017). Obermiller and Spangenberg (1998) added that advertisers exploit them using puffery when consumers are less skeptical.

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Many advertisers use exaggeration to thrive in today's very competitive business climate, such as overstating the advantages of their goods or the overall value proposition of their market offering (Amyx & Lumpkin, 2016). Gao and Scorpio (2011) also supported that advertisers are motivated to use puffery in the wake of a highly competitive culture. Despite the benefits cashed by the marketers, increased use of puffery will erode the credibility of advertisements, eventually leading to customer skepticism.

There is a scarcity of empirical evidence on the relationship between puffery and STA. Amyx and Lumpkin (2016) investigated how the interaction between ad puffery and consumer ad skepticism influence consumer persuasion. The present study relies upon the advertising value model (Ducoffe, 1996) and argues that puffery is a controllable ad characteristic, entirely under the jurisdiction of marketers to use puffery or not. When integrated into the SOR realm, it could be another factor serving as a stimulus for STA. Therefore, it is expected that puffery will have a positive effect on skepticism toward online advertising.

2.3. STA and ATA

Attitude toward online advertising is defined as the aggregation of assessments of online advertising's perceived attributes and benefits (Wang, Sun, Lei, & Toncar, 2009). In the view of Voorveld, Neijens, and Smit (2011), a marketer's understanding of consumers' attitudes about online advertising is critical these days because most companies are shifting to online marketing and advertising. However, marketers' excessive use of online advertising may create a tendency to avoid advertisements on the internet. Because online communications and promotions are mostly pushed toward consumers, they are considered intrusive (Shavitt, Vargas, & Lowrey, 2004). Therefore, consumers tend to believe that they have no control over the content they watch on the Internet (Morimoto & Chang, 2006). A survey by MarketingCharts (2012) confirmed this tendency that two out of every three individuals were afraid of being exposed to excessive advertisements over the internet.

Past literature supports that skepticism leads to negative attitudes toward advertising (Rozendaal, Buijzen, & Valkenburg, 2011). Obermiller and Spangenberg (1998) also believed that skepticism toward advertising might be related to one's general attitude toward advertising. While investigating skepticism in the context of corporate social responsibility, Joireman, Liu, and Kareklas (2018) found that skepticism toward advertising had a negative impact on attitude toward advertising. Similarly, Raziq et al. (2018) investigated the influence of the need for cognition and skepticism toward advertising on attitudes and found that skepticism toward advertising negatively predicted consumer attitudes.

Residing on the SOR paradigm, research has usually opted for behavior as a response. However, According to Bagozzi (1986), response state refers to the final action or outcome in the SOR paradigm, and it includes both attitudes and behaviors. Therefore, ATA being a general attitude, is a response to skepticism toward online advertising. Aligned with the literature review, it can be concluded that skepticism toward online advertising will show a negative relationship with attitude toward online advertising.

2.4. Product Knowledge as a Moderator

Product knowledge, also referred to as product expertise (Obermiller & Spangenberg, 1998), has been studied extensively in consumer behavior research to facilitate information processing (Brucks, 1985). The level of consumers' product-related knowledge (high-knowledge versus low-knowledge) is linked directly to how they process the information. According to Bettman and Sujan (1987), consumers with high knowledge follow an analytical approach in their decision-making since they follow the proper decision-making process using the information already available in memory. Biswas and Sherrell (1993) argued that in comparison to low-knowledge consumers, high-knowledge consumers rely less on heuristic cues for their purchase decisions. Research also suggests that the consumer's knowledge

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about the product also influences how he/she processes the advertisements about that product (G. E. Smith & Wortzel, 1997).

Despite the rich literature on product knowledge and information processing, there is insufficient empirical research on the moderating influence of product knowledge on ad skepticism. Obermiller and Spangenberg (1998) proposed that individual factors such as product knowledge or expertise could influence ad skepticism and its outcomes. Building on Obermiller and Spangerberg's study, Tan and Tan (2007) empirically examined the influence of product knowledge to reduce the negative consequences of skepticism toward health claims. However, they could not find a significant relationship. While the SOR paradigm also supports that the relationship between stimulus and organism could be influenced by the moderating variables (Chang, Eckman, & Yan, 2011; Minton, 2019). Based on the overarching theoretical paradigm of SOR, mix findings from the literature, and little empirical evidence, it is imminent to assume that further investigation is needed on product knowledge and online ad skepticism.

In a recent study, Han, Yu, and Kim (2019) argued that increased product knowledge increases consumer trust and fosters positive attitudes. When consumers are skeptical about claims made in marketing, they use their information and knowledge to access the truthfulness of messages (Bae, 2018). The researchers believe that consumers with more product knowledge evaluate the informativeness and puffery in the advertisements, ultimately affecting the skepticism toward online advertising. In this way, knowledgeable consumers should be less prone to skepticism when exposed to ads containing infotainment. Furthermore, the effects of puffery should be less pronounced for knowledgeable consumers.

Hypothesis 1: Product knowledge moderates the relationship between (1a) infotainment, (1b) puffery, and skepticism toward online advertising, such that the relationship is weaker when product knowledge is high.

2.5. Perceived Ethics of Online Seller a Moderator

Román (2007) identified that the most compelling issues of online buyers include online security, the privacy of financial information, the reliability of the online seller, fraud, and the quality of the products being sold. Recently, a shift has been seen from identifying ethical concerns to how these ethical concerns lead to various perceptions and behaviors. For example, Limbu, Wolf, and Lunsford (2012) studied ethical dimensions on customer satisfaction and loyalty. They found that order fulfillment, security, and non-deception were the imperative dimensions of ethical concerns that ultimately lead to customer satisfaction. Other studies, such as Adam, Aderet, and Sadeh (2008), stated that ethical concerns were related to consumers' purchase intentions, while Ratnasingham (1998) found them related to loyalty.

Signaling theory also comes into play in this regard. The theory states that whenever there is an information gap between a seller and a buyer (i.e., the buyer knows little about the product compared to the seller), the buyer must rely on some signal to overcome this information asymmetry (Spence, 2002). The perceived ethics of the online seller could serve as one such signal. Román (2007) found that consumers' trust is reduced if they perceive deception from the online seller. Once trust is reduced, it will further lead to a reduction in satisfaction and loyalty (Román & Cuestas, 2008). Obermiller et al. (2005) added that skepticism toward advertising creates multiple responses among consumers; one is searching for other information cues. As argued, ethical practices by the seller could be used by consumers as a cue to negate the effects of skepticism toward online advertising.

Although the concepts of ad skepticism and seller ethics are related, the literature does not provide enough empirical evidence for the interaction between online ad skepticism and perceived ethics of online sellers. However, from the SOR perspective, there is evidence for the effects of moderating

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factors between organism and response states (Indibara, 2017; Joireman et al., 2018; Tan & Tan, 2007). Based on the theoretical standpoint of SOR and presented literature, it is proposed that:

Hypothesis 2: The perceived ethics of the online seller moderates the relationship between skepticism toward online advertising and attitude toward online advertising, such that the relationship is weaker when the perceived ethics of the online seller is high.

Based on the first two hypotheses, there is an indication of the moderated-moderated mediation model suggested by Hayes (2018). In the moderated-moderated mediation model, the mediating variable (skepticism toward online advertising) carries the effects of independent variables (infotainment and puffery) to the dependent variable (attitude toward online advertising), and those effects are influenced by the moderating variables (product knowledge and the perceived ethics of the online seller). The conceptual framework in figure 1 presents the graphical depiction of this argument.

Hypothesis 3: Product knowledge and the perceived ethics of the online seller moderate the indirect relationship of (3a) infotainment, (3b) puffery on attitude toward online advertising through skepticism toward online advertising such that the indirect effects are realized when product knowledge and the perceived ethics of the online seller are high.

2.6. Effects Across Advertisement Type

Research on the effects of multimedia content on websites advocates that consumers develop positive attitudes when the content is rich. Thus, websites employing richer multimedia content are deemed more persuasive by consumers than websites with little or no multimedia (Appiah, 2006). Coyle and Thorson (2001) also affirmed that consumers show favorable audio/animated content attitudes than static content. They further added that audio, video, and animated advertisements enhance the consumer experience by increasing vividness. Since web surfers establish a sense of "connection" with characters in video ads (K. M. Lee, 2004), they find them more rich and vivid.

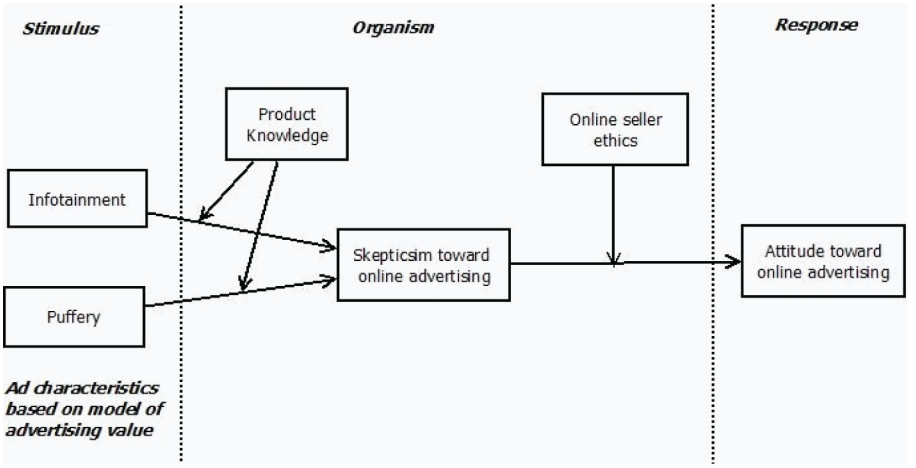
The underlying mechanism for the more substantial effects of video ads compared to picture ads can be attributed to the "social richness," as Short, Williams, and Christie (1976) explained. They argued that social richness depends on communication, i.e., face-to-face communication is the most socially rich mode, followed by video, audio, and written communication. Overall, the research posits that respondents showed a positive attitude toward the information in video form compared to static pictures (Appiah, 2006; Mechling & Gustafson, 2008). Therefore, attitude toward online advertising as a general attitude, and skepticism toward online advertising as a specific attitude (Obermiller & Spangenberg, 1998), should be influenced differently based on ads (video vs picture) shown to the consumers. In addition, Obermiller and Spangenberg assumed that consumers might have different levels of skepticism depending upon the medium and type of ad. Expressly, it is assumed that the effects will be more substantial for video ads aligned with the existing body of knowledge. We, therefore, propose that:

H4a: The effect of infotainment on skepticism toward online advertising is stronger for video ads than for picture ads.
H4b: The effect of puffery on skepticism toward online advertising is stronger for video ads than for picture ads.
H4c: The effect of skepticism toward online advertising on attitude toward online advertising is stronger for video ads than for picture ads.
H4d: The effect of product knowledge on skepticism toward online advertising is stronger for video ads than for picture ads.

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H4e: The effect of perceived ethics of online seller on attitude toward online advertising is stronger for video ads than for picture ads.

Figure 1. Conceptual framework



3. STUDY 1

3.1. Methodology

3.1.1. Sampling and Data Collection

There were roughly 37 million Internet users in Pakistan (17 percent of the total population) (Bank, 2019). An online sample size calculator Raosoft (2004), recommended the minimum sample size of 385 based on population size. To recruit the participants, (a) the study was widely advertised on the internet as widely as possible; (b) asked masters students to promote it among their friends, family, coworkers, and neighbors (c) personal contacts.

Existing studies state that collecting all data in one go can lead to common method variance. Podsakoff, MacKenzie, Lee, and Podsakoff (2003) suggested separating data collection for predictor and criterion variables to overcome this problem. Therefore, data in this study were collected in three phases with a gap of three weeks each. Phase 1 collected the data for advertisement characteristics (infotainment and puffery) and product knowledge. Phase 2 collected data for skepticism toward online advertising and the perceived ethics of the online seller. Phase 3 collected data for attitude toward online advertising.

Respondents who participated in phase 1 were requested to share their e-mail addresses for future correspondence. Subsequently, those e-mail ids were used to request participation in phase 2 and phase 3. As compensation, 15 participants were given USB pen drives that were shared among random participants. At the end of data collection, responses of 411 participants were available, filled for all three data collection stages. Because the collected data were according to the desired size, data analyses were performed.

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3.1.2. Measures

For this study, already established scales were used to elicit subjects' responses. For study 1, a five-category Likert scale was used. Respondents were asked to recall any online ad they have recently watched and then respond to the questions.

Skepticism toward online advertising was operationalized using a 9-item scale adapted from Obermiller and Spangenberg (1998). The original scale was developed for skepticism toward traditional advertising. The scale was modified for the online context. A sample item is *Online advertising is a reliable source of information about the quality and performance of products*. All the items in the scale were reverse coded, with higher values indicating lower skepticism toward online advertising. Consumers' responses regarding online advertising were elicited using five items adopted from Wang et al. (2009). A sample item is *I consider online advertising essential.* A negative item for attention check (*In general, I think that online advertising increases the cost of products*) was reverse-scored before analyses.

Infotainment was assessed using five items adapted from Ducoffe (1996). The scale was modified to fit the online setting. The sample item is *Online advertising is exciting*. Puffery was operationalized using six items adapted from Amyx and Lumpkin (2016). The items were revised for the current study, and sample items are *The online ads are exaggerated,* and *The online ads are overstated*. After factor analyses, four items were retained for study 1.

Product knowledge was measured on four items scale adopted from D. C. Smith and Park (1992). This scale was also adjusted for online context with a sample item *I feel very knowledgeable about the product in the online ad.* The perceived ethics of the online seller was measured using 13 items adopted from Román (2007). The scale taped all four dimensions of perceived ethics of online sellers; security, privacy, fulfillment, and non-deception. A sample item is *The security policy of the company in the online ad is easy to understand.*

3.2. Data Analyses and Results

As far as demographic information is concerned, most respondents belonged to the age group of 20-30 years (65%). Seventy percent of the respondents were male. Most of the respondents were educated, having bachelors and masters degrees, while less than 3% had doctoral or other qualifications.

3.2.1. Summary of Confirmatory Factor Analysis

Factor loadings were estimated as part of the analyses, and AVE (average variance extracted) and CR (composite reliability) values were computed. Fornell and Larcker (1981) argued that the lower threshold value for AVE should be 0.40 only if the CR values are above 0.60. Because all the CR values are above the 0.60 threshold level, AVE values are considered acceptable, indicating appropriate convergent validity. For discriminant validity, Gaskin, Godfrey, and Vance (2018) suggested that HTMT values should be less than 1.00. All the observed values were below the given threshold, indicating adequate discriminant validity (Table 1).

3.2.2. Hypotheses Testing

To test the hypotheses, PROCESS macro based on version 3.5 was used. It was proposed in hypothesis 1 that product knowledge moderates the relationship between infotainment (1a), puffery (1b), and skepticism toward online advertising, such that the relationship will be weaker when product knowledge is high. The results of data analysis in this regard are presented in table 2.

It can be observed from the table that the interaction term for the model (infotainment-skepticism), when moderated by product knowledge, is significant ($\beta = .13$, $p < .01$). The interaction plot for this relationship is presented in figure 2. Skepticism toward advertising was found to be lower when infotainment and product knowledge were high. Hence, H(1a) was supported.

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Table 1. Convergent and discriminant validity

Variables	CR	AVE	ATA	INF	ETH	PK	PUF
Attitude toward online advertising	0.85	0.55					
Infotainment	0.89	0.62	*0.83*				
Perceived ethics of online seller	0.89	0.41	*0.88*	*0.79*			
Product knowledge	0.85	0.59	*0.91*	*0.89*	*0.88*		
Puffery	0.84	0.56	*0.74*	*0.63*	*0.68*	*0.67*	
Skepticism toward online advertising	0.90	0.51	*0.91*	*0.82*	*0.91*	*0.91*	*0.66*

Note. CR = composite reliability; AVE = average variance extracted; STA = skepticism toward online advertising; ETH = perceived ethics of online seller; PK = product knowledge; INF = infotainment; ATA = attitude toward online advertising; PUF = puffery. Italicized values indicate HTMT ratios.

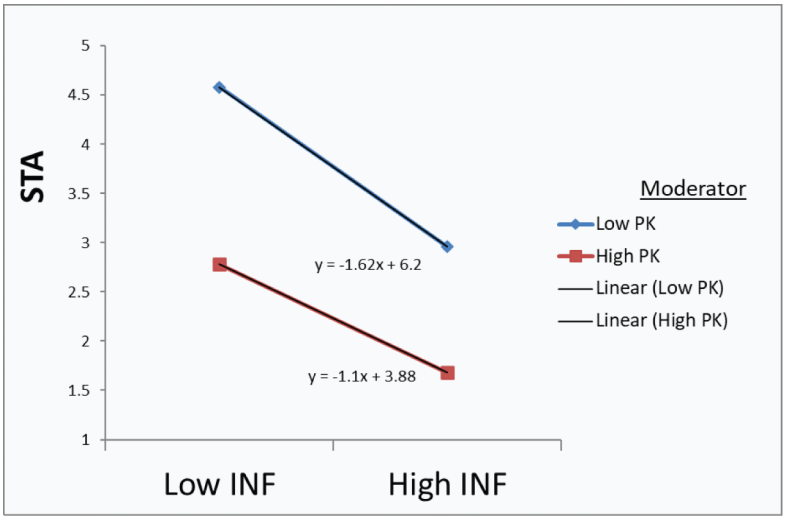
Table 2. Regression results of the overall model

Model	β	SE	T	R^2
Infotainment-Skepticism model				*.65*
Constant	6.19	.18	35.00 ***	
Infotainment	-.68	.08	-8.74***	
Product knowledge	-.75	.07	-10.76***	
Infotainment x PK	.13	.02	5.28***	
Puffery-Skepticism model				*.63*
Constant	2.35	.35	6.80***	
Puffery	.68	.08	8.21***	
Product knowledge	.04	.10	.41	
Puffery x PK	-.17	.03	-6.47***	
Skepticism-Attitude model				*.64*
Constant	5.80	.44	13.14***	
STA	-1.07	.10	-11.00***	
ETH	-.35	.11	-3.22**	
STA x ETH	.18	.03	6.49***	

Note. STA = skepticism toward online advertising; ETH = perceived ethics of online seller; PK = product knowledge. *** $p < .001$; ** $p < .05$.

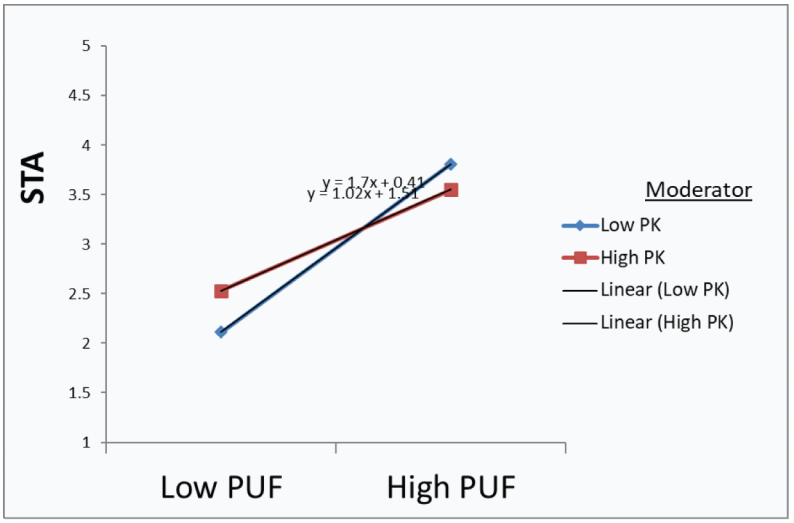
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Figure 2. Interaction plot for infotainment-skepticism model



Similarly, the interaction term for the model (puffery-skepticism) when moderated by product knowledge is significant ($\beta = -.17$, $p < .01$). Figure 3 presents the interaction plot for this relationship. It can be observed from the figure that skepticism is higher in the presence of puffery, but the relationship is weakened in the presence of product knowledge. Hence, H(1b) was supported by the results.

Figure 3. Interaction plot for puffery-skepticism model

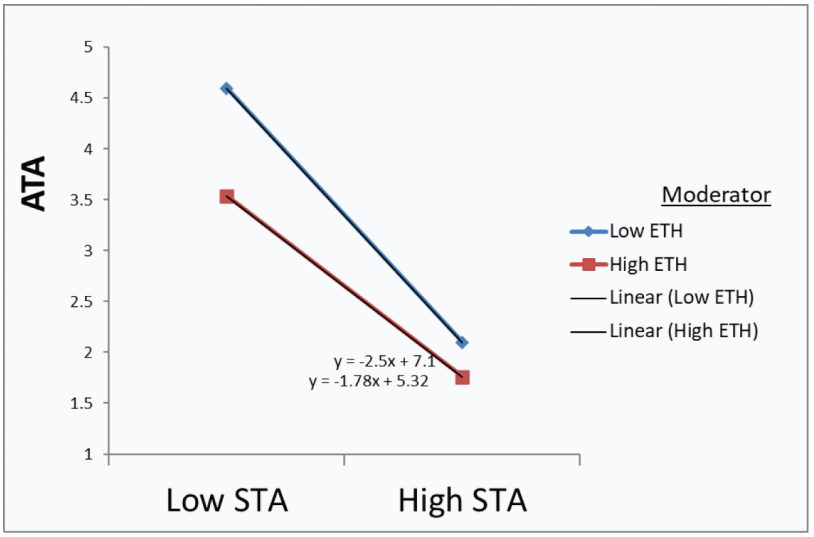


For H2, it was posited that the perceived ethics of the online seller moderates the relationship between skepticism toward online advertising and attitude toward online advertising, such that the relationship is weaker when the perceived ethics of the online seller is high. By observing table 2, it can be found that the interaction term for the model (skepticism-attitude), when moderated by

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perceived ethics of online seller, is significant ($\beta = .18$, $p < .01$). Figure 4 presents the interaction plot for this relationship. It can be observed from the figure that the negative effect of skepticism toward online advertising on attitude toward online advertising is weakened when the perceived ethics of the online seller is higher. Hence, H2 was supported by the results.

Figure 4. Interaction plot for skepticism-attitude model



The third hypothesis stated that product knowledge and the perceived ethics of the online seller moderate the indirect relationship of infotainment (H3a) and puffery (H3b) on attitude toward online advertising through skepticism toward online advertising such that the indirect effects are realized when product knowledge and the perceived ethics of the online seller were high. To test these hypotheses (3a and 3b), two moderated-moderated mediation analyses were performed using the PROCESS macro 3.5 and model 21 in SPSS.

Table 3 presents the results of the conditional process analysis performed for H(3a). The table lists the effect sizes and (upper and lower limit) confidence intervals for the conditional effect of infotainment on attitude toward online advertising through skepticism toward online advertising at four different combinations of product knowledge and the perceived ethics of the online seller. As shown in the table, the indirect effect of infotainment on attitude toward online advertising was

Table 3. Conditional effect of infotainment

Value of moderators		Indirect effect size	Confidence interval	
Product knowledge	Perceived ethics of online seller		Lower	Upper
Low	Low	.19	.13	.25
Low	High	.10	.05	.16
High	Low	.08	.03	.13
High	High	.04	.02	.09

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significant for all four conditions of product knowledge and the perceived ethics of the online seller. Thus, hypothesis 3a was supported.

Table 4 shows moderated-moderated mediation analysis results in a similar vein when puffery was taken as a predictor variable (hypothesis 3b). It can be observed from the table that the indirect effect of puffery on attitude toward online advertising was significant when both the perceived ethics of the online seller and product knowledge were low (β = -.16, LL = -.21 & UL = -.11). The indirect effect was also significant for low product knowledge and high perceived ethics of online seller case (β = -.09, LL = -.14 & UL = -.05). The other two combinations found zero value between their upper and lower limit confidence intervals. Therefore, it can be concluded that there was partial support for hypothesis 3b.

Table 4. Conditional effect of puffery

Value of moderators		Indirect effect size	Confidence interval	
Product knowledge	Perceived ethics of online seller		Lower	Upper
Low	Low	-.16	-.21	-.11
Low	High	-.09	-.14	-.05
High	Low	-.01	-.05	.04
High	High	-.01	-.04	.03

4. STUDY 2

To achieve a deeper understanding of the skepticism phenomenon in the online setting and empirically validate the assumptions of advertising richness and advertising vividness (Ge, Sui, Zhou, & Li, 2020; Short et al., 1976), study 2 tested skepticism across video and photo ads. Moreover, Obermiller and Spangenberg (1998) assumed that consumers' skepticism depends upon the medium and type of ad. Therefore, this study replicates the research model adopted in study 1 (see figure 1) with an experimental design. The experiment served to overcome multiple voids left intact in study 1; (a) test across advertisement type (b) further validate the conceptual framework due to high internal validity (c) test media richness assumptions in context of STA.

4.1. Methodology

4.1.1. Manipulation

A group of masters students (n = 30) were invited to decide on a platform and product for experimental manipulation. The members who participated in the focus group were not invited to the experiment. The participants were asked, "Which online platform they preferred for watching ads?". Most of them responded with Instagram (mode = 19). When asked, "Which product ads did they watch most?" the focus group informed that they preferred to watch ads related to mobile phones when surfing Instagram. Thus, researchers selected Instagram as a platform and mobile phones as a product category. Next, they were inquired about their favorite smartphone brand/model, to which they responded with Vivo V20. Then participants were divided into a group of two. One group was asked to search for their preferred video ad of Vivo V20 on Instagram. The other group was asked to search for three picture ads of the same mobile phone on Instagram (see appendix A). Since videos take longer to watch, the choice of 3 picture ads was made on the recommendation of the focus group to ensure both experimental groups were exposed to stimuli for a similar amount of time. Later, one video ad and

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three picture ads selected by the focus group were randomly shown to the subjects in anticipation of data collection. The participants in this pretest were not included in the main experiment.

4.1.2. Participants

The present study employed a between-subjects experiment based on advertisement type (video vs picture). Data were collected from Internet buyers using an online experiment. Subjects participated voluntarily in the study in response to the survey advertisement on social media, while the participants of study 1 were not invited for the experiment. The participants were randomly assigned to one of two experimental scenarios. An initial screening question ensured that only the Internet buyers participated in the experiment. Next, the participants were shown either video or picture advertisements subject to random assignment. After watching advertisements, respondents completed an online survey on all variables from the study.

A total of 179 Internet buyers participated in the experiment. Regarding gender, 99 (55%) participants were male. The majority had a bachelor's degree (67%) and were less than or equal to 40 years old. In terms of ad distribution, 91 participants watched video ads while 88 watched picture ads.

4.1.3. Measures

In contrast, to study 1, a 7-point Likert scale was used for study 2. The scales for all the variables were reused from study 1. After the factor analysis, all six items for ad puffery were retained for study 2 (in contrast to 4 items for study 1). Similarly, a complete set of 13 elements were used for study 2 (9 items were used for study).

4.2. Data Analysis and Results

Data were analyzed through AMOS 21 using structural equation modeling (SEM) and multigroup analysis. The multigroup analysis was performed using the path coefficient comparison approach suggested by Keil et al. (2000). See Appendix B for calculations of path differences using multigroup analysis.

The overall sample was separated into two groups based on video or picture advertisements. The relationship between infotainment and skepticism toward online advertising was different based on advertisement type: video ($\beta = -.32$, $p < .000$) and picture ($\beta = -.27$, $p < .000$). And difference in path coefficients was significant ($\Delta\beta = .05$, $t = -4.77$). Table 5 presents the summary of path coefficients comparison across advertisement types.

Table 5. Path coefficient comparison

Model	Video		Picture	
	Estimate	Sig.	Estimate	Sig.
INF→STA	-.32	***	-.27	***
PUF→STA	.27	***	.19	**
PK→STA	-.19	*	-.43	***
STA→ATA	-.42	***	-.39	***
ETH→ATA	.18	.136	.37	**

Note. INF = infotainment; STA = skepticism toward online advertising; PUF = puffery; PK = product knowledge; ATA = attitude toward online advertising; ETH = perceived ethics of online seller. *** p < .000; ** p < .05; * p < .10.

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The effect of puffery on skepticism toward online advertising was stronger for video ads (β = .27, $p < .000$) than for the picture ads (β = .19, $p < .05$) and the path difference was significant ($\Delta\beta$ = .08, $t = 6.86$). The effect of skepticism toward online advertising on attitude toward online advertising was different across advertisement type: video (β = -.42, $p < .000$) and picture (β = -.39, $p < .000$). And path coefficient was significant ($\Delta\beta$ = .03, t = -1.84).

5. DISCUSSION

This study integrated advertising value and stimulus-organism- response (SOR) frameworks to develop a model of STA's causes and consequences. Furthermore, product knowledge and perceived ethics of online seller were proposed as moderators. Two separate studies were conducted to test the hypotheses. Study 1 employed a survey technique on a representative sample of Internet users, while study 2 resided on an experimental design.

The first three hypotheses, i.e., H1, H2, H3, were investigated through study 1. It was proposed in H1 that product knowledge will moderate the relationship between infotainment (1a), puffery (1b), and skepticism toward online advertising, such that the relationship will be weaker when product knowledge is high. The data analysis supported the results. Like the previous research, it was found that infotainment as a stimulus predicted STA as an organismic realm (Gaber et al., 2019; Teixeira & Stipp, 2013; Tsang et al., 2004). Similarly, puffery, the other stimulus, positively influenced the organism state (Raziq et al., 2018; Stern & Callister, 2020). In the Asian context, Jamil and Qayyum (2019) also found that various stimuli influence STA (the organism state). The influence of infotainment and puffery on STA further validated the assumptions of the advertising value model (Ducoffe, 1996). Furthermore, the product knowledge interacted with both stimuli (infotainment and puffery) to influence the organism state (STA). Previously, research on SOR has also supported the interaction effects of stimuli and moderators on organism state (Chang et al., 2011; Minton, 2019). Overall, consumers with higher product knowledge pay more attention to infotainment of the online ads, thus less susceptible to STA. In contrast, consumers with high product knowledge could diagnose ad puffery and thus lesser prone to online ad skepticism.

It was proposed in H2 that the perceived ethics of the online seller moderates the relationship between skepticism toward online advertising and attitude toward online advertising, such that the relationship is weaker when the perceived ethics of the online seller is high. The results also support the idea that the perception that an online seller follows ethical considerations leads to a reduction in the negative influence of skepticism and ultimately boosts a positive attitude toward online advertising. Extant literature also validates the fact that the perceived ethical behavior of a seller leads to positive outcomes. For example, Adam et al. (2008), Limbu et al. (2012), and Román (2007) found the ethical behavior of a seller to be related to positive outcomes such as purchase intentions, satisfaction, and loyalty. More recently, Asian consumers also showed that interaction between STA and perceived ethics results in favorable consumer attitudes toward online information (Jamil & Qayyum, 2019). Additionally, Indibara (2017) and Joireman et al. (2018) argued that interaction between organism and moderating variable(s) could influence the outcome of response realms.

The H3 posited that product knowledge and the perceived ethics of the online seller moderate the indirect relationship of infotainment (3a) and puffery (3b) on attitude toward online advertising through skepticism toward online advertising such that the indirect effects are realized when product knowledge and the perceived ethics of the online seller were high. This overall moderated-moderated mediation was supported fully for H3a and partially supported for H3b. These findings have imperative implications for the stimulus-organism-response model, which postulates that the organism state mediates the relationship between stimulus and response states (Fu et al., 2020), and contextual factors influence this indirect effect at both the stimulus-organism (Chang et al., 2011; Minton, 2019) and organism-response (Indibara, 2017; Joireman et al., 2018) states. Existing studies on the SOR paradigm have taken a piecemeal approach to investigate stimulus-organism and organism-response

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with the effects of moderators. However, this study has adopted a holistic approach by simultaneously investigating stimulus-organism-response in the presence of two moderators. This has provided a more comprehensive examination of the SOR paradigm, which is currently lacking in the literature.

Study 2 investigated H4 by comparing the effects across advertisement types. The effects of infotainment and puffery on STA were stronger in video ads than in picture ads. Hence H4a and H4b were supported. Similarly, in line with the literature, the effect of skepticism toward online advertising on attitude toward online advertising was stronger for video ads than for picture ads, lending support for H4c. These findings align with the existing studies such as Appiah (2006) and Mechling and Gustafson (2008). They also found that consumers prefer online content when it is in video form compared to static pictures.

Moreover, Jamil and Qayyum (2021) showed that Asian consumers respond differently to online skepticism when exposed to alternative sources and types of information. Research on advertising richness and advertising vividness explains these outcomes (Ge et al., 2020). Advertising vividness refers to consumer behaviors in response to the sensory stimulation (Weinberger, Swani, Yoon, & Gulas, 2017), i.e., higher sensory stimulation leads to higher vividness. Moreover, video ads are more vivid than picture or textual ads. Therefore, the present study's findings support that consumers find video ads more rich and vivid than picture ads.

In contrast to researchers' expectations, the effect of product knowledge on STA was more substantial for picture ads. Similarly, the effect of perceived ethics of online seller was more substantial on ATA for picture ads. Therefore, both H4d and H4e were not supported. One possible explanation for these outcomes may be that consumers with higher levels of product knowledge do not require more vividness of ads to make decisions. Since product knowledge facilitates information processing (Obermiller & Spangenberg, 1998), pictorial ads are sufficient to shape consumer decisions. Similarly, it can be assumed that the perceived ethics of the online seller serves as a quality signal; consumers may find picture ads sufficient when the sellers are believed to be ethical.

5.1. Implications

This study has implications for both researchers and practitioners. For researchers, it advances the understanding of skepticism toward online advertising by integrating the advertising value model and SOR framework. Residing on the advertising value model, it identified two controllable stimuli (infotainment and puffery) which influence STA. Additionally, product knowledge and perceived ethics of online seller were introduced as moderators. Previously, the researchers relied upon attribution theory to explain skepticism phenomena (Leonidou & Skarmeas, 2017), which is more suitable in general contexts. However, in online advertising, the advertising value SOR models are more specific and relevant, thus opening new avenues of research for future researchers.

Study 1 relied on the Hayes (2018) moderated-moderated mediation approach to simultaneously investigate the effect of the stimulus on the organism, organism to response, and two moderating variables. This approach resulted in a better explanation of the SOR paradigm and the phenomenon of skepticism toward online advertising. Thus, a path has been paved for a much more comprehensive examination of SOR for future investigations.

Two studies were conducted to triangulate and enhance the validity of the proposed conceptual framework. Study 1 followed the survey approach, while study 2 followed the experimental design, promising external and internal validity. In addition, the comparison across ad types (video vs picture) increased the understanding of the researchers that skepticism could be influenced by the type of advertisement and the richness of the content.

As far as practitioners are concerned, this study provides essential prescriptive advice on overcoming the perils of skepticism toward advertising. Because the study found that skepticism toward online advertising results in a negative attitude toward online advertising, researchers introduced product knowledge and the perceived ethics of the online seller as remedial moderating variables. Learning from these findings, practitioners and managers of online businesses should increase the

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positive impact of their advertising efforts by adopting ethical practices to induce positive signals. The ethical practices of online sellers inculcate trust among consumers (Jamil & Qayyum, 2019), resulting in reduced STA.

As far as product knowledge is concerned, online sellers should increase awareness and knowledge of the product(s) in the market. For example, asking independent experts to share/write product reviews would be helpful. Another possible method would be to attract product-relevant influencers, who could review the products and post videos on their pages. Social media influencer reviews have influenced learning and decision-making (Jamil & Qayyum, 2021; J. E. Lee & Watkins, 2016). Therefore, informed consumers would be able to assess, adapt, and recommend online information.

In terms of mode, the findings of this study recommend the use of video ads over picture ads. Although people share more pictorial content on social media, particularly Instagram, the consumers, on the other hand, prefer video content. The video ads are considered socially richer (Ge et al., 2020), developing a connection between the source and recipient. Additionally, Jamil and Qayyum (2021) supported that Asian consumers prefer online information in video format and presented in their native language. Therefore, videos should be the preferred choice of online advertisers when it comes to promoting their products.

The advertisement characteristics (infotainment and puffery) are controllable variables, i.e., managers and practitioners can manipulate them. The relevant and entertaining ads help consumer information processing, resulting in reduced skepticism toward online advertising (Jamil & Qayyum, 2019). Conversely, the exaggerated claims make consumers believe that online sellers are unreliable, further escalating consumer STA. Therefore, online advertisers should mitigate consumers' online skepticism by making the ad content more informative/entertaining and avoiding exaggerated claims.

5.2. Limitations and Future Research Directions

Like every study, this one is not free from limitations. First, only two ad characteristics (infotainment and puffery) were used as stimuli in the study. Other variables could be included in the model that may influence skepticism, such as emotional appeals, confusion, and informativeness. Second, only one response variable has been included; other responses, such as media choice and negative word of mouth, may extend the model further. Third, as a novel contribution, the perceived ethics of online sellers was introduced in the model. Other remedial variables may be introduced to investigate whether they mitigate the perils of skepticism toward online advertising. Fourth, both survey and experimental designs were employed to strengthen the methodology of this study. At the same time, future investigations may opt for longitudinal designs to further understand the skepticism phenomenon.

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APPENDIX A

Link to video ad on Instagram:
https://www.instagram.com/tv/CHfW914gKQH/
Link to picture ad 1 on Instagram:
https://www.instagram.com/p/CLV0M-8lyw2/
Link to picture ad 2 on Instagram:
https://www.instagram.com/p/CLa37l5HpYq/
Link to picture ad 3 on Instagram:
https://www.instagram.com/p/CMAYYb2HWNB/

APPENDIX B

Calculations for multigroup analysis:

$$t = \frac{PC_1 - PC_2}{\left[S_{pooled} \times \sqrt{\left(\frac{1}{N_1} + \frac{1}{N_2} \right)} \right]}$$

where:

Spooled = a pooled estimator of variance
N_1 = Sample size of dataset 1
N_2 = Sample size of dataset 2
SE_1 = Standard error of structural path for group 1
SE_2 = Standard error of structural path for group 2
PC_1 = Path coefficient for group 1
PC_2 = Path coefficient for group 2

$$S_{pooled} = \sqrt{\left\{ \left[\frac{\left(N_1 - 1 \right)}{N_1 + N_2 - 2} \right] \times SE_1^{\,2} + \left[\frac{\left(N_1 - 1 \right)}{N_1 + N_2 - 2} \right] \times SE_2^{\,2} \right\}}$$

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The Impact of Credibility Attributes on Consumer Intention to Adopt Information on Social Networking Sites: Application to Tourism Services

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ABSTRACT

This study examines the effect of consumers' perceived credibility of information on SNS on their attitude and intention to adopt this information in the Arab world. The study adopted a quantitative research approach using a survey questionnaire conducted among 317 individuals. Structural equation modelling was conducted. The results highlighted that there are positive direct effects of authority cues, expertise, trustworthiness, social identity, and argument strength on perceived credibility. Although both authority cues and transparency have non-significant direct effects on perceived credibility, they have positive direct effects on attitude and indirect effects on intention via attitude. Finally, the positive impact of credibility on intention is mediated by attitude. This study contributes to the literature on the credibility of information on SNS and its effect on consumer intention to adopt it.

KEYWORDS

Credibility Attributes, Intention to Adopt Information on SNS, Social Networking Sites (SNS), Tourism Services

INTRODUCTION

Social media plays a critical role for both travelers and tourism companies (Chung & Buhalis, 2008; Ye, Fong, & Luo, 2021). Instead of resorting to official company websites, consumers are increasingly turning to social networking sites (SNS) pages to search for any relevant information, including travel-related tips (Mehraliyev, Choi, King, 2021). Nowadays, suggestions from friends and family, online recommendations, as well as online comments influence travelers' decisions for planning for a trip (Berhanu & Raj, 2020). SNS allow travelers to share their experiences and ideas with others. SNS provide easily accessible forums on which travelers may share their personal information, experiences and ideas with others and leave their comments, reviews, opinions and suggestions about airlines, hotels or restaurants (Hur, Kim, Karatepe & Lee, 2017). As information from SNS comes from a large number of users every day, it can be difficult to quickly differentiate reality from fiction or actual news from mere opinions (Johnson & Kaye, 2014). According to Sunny Tsai and Rita Men

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(2013), to support consumers' engagement with SNS, the credibility of information provided by both brand developers and consumers on SNS pages is an essential element for establishing trusting relationships with consumers (Pop, Săplăcan, Dabija, & Alt, 2021). Similar findings regarding trust in news in social media were also reported recently (Karlsen & Aalberg, 2021). Perceived online credibility is defined as the extent to which one perceives information to be believable and this is considered as a strong predictor of consumer's further action (Li & Suh, 2015) and decision making (Alrubaian, Al-Qurishi, Alamri, Al-Rakhami, Hassan, & Fortino, 2018). Prior research has reported how to judge the information credibility on traditional media or websites (Metzger & Flanagin, 2013). However, information on social media platforms suffers from a lack of professional gatekeepers who can monitor content (Li & Suh, 2015), leading to the presence of some fake news in social media (Abdallah, 2021). It is not unusual to find unverified or false information flow on social media (Di Domenico, Sit, Ishizaka, & Nunan, 2021). In this situation, online consumers are forced to look for new ways to evaluate the credible information.

Although past research has examined the use of social media for travel-related purposes (Varkaris & Neuhofer, 2017; Rahman, Sharmin & Akhter, 2021), scant attention has been paid to how perceived credibility of information on SNS influences consumers' intention to adopt such information in purchasing decisions, specifically in the hospitality marketplace (Li & Suh, 2015). Since consumers can access services indirectly and form a degree of trust of a service provider through online interactions with both a company and other customers (Daugherty, Eastin & Bright, 2013), it is expected that SNS can also play a vital role in consumers' intentions to adopt and act on information on SNS. Although previous research has attempted to investigate antecedents that drive consumers to visit or follow SNS information (Daugherty et al., 2013), more research on a larger and more diverse sample which is applicable to a variety of SNS is needed to aid understanding of consumers' perceived credibility of information concerning SNS and how this affects their intention to adopt it (Li & Suh, 2015).

Importantly, Aluri et al., (2016) argued that despite the popularity of social media usage by travel services, companies still wonder whether social media channels truly enhance the overall experience of travelers, particularly their intention to adopt information on SNS. Research on the perceived credibility of information on SNS is still not very clear or well established, and future research is still needed (Alkhamees et al., 2021). Johnson and Kaye (2014) found that SNS was ranked as less credible than traditional and other online sources of information. Johnson and Kaye (2014) recommended that more research should be conducted to examine the relationship between SNS credibility and the intention to adopt information on SNS for purchasing decisions. This is especially important because according to Inversini and Masiero (2014) and Ayeh, Au and Law (2013), describing and explaining the role of social media in the tourism and travel industry involves cultural elements. Accordingly, further research on a variety of customer attributes is needed in order to increase our understanding of the impact of the perceived credibility of information on SNS on purchasing in tourism services (Song & Yoo, 2016).

Tourism SNS and Credibility

Recently, social media has become an effective channel for the marketing of tourism services (Zeng & Gerritsen, 2014; Hysa, Karasek, & Zdonek, 2021; Mehraliyev et al., 2021). Reports showed that there are 4.38 billion internet users, including 3.48 billion active social media users (Global Web of index, 2019). This may be because tourism customers have become more active and information shared in social media considerably reduces the burden of their decision-making process (Icoz, Kutuk & Icoz, 2018). The Forbes (2017) report highlights that social networking sites have gained remarkable popularity especially among the younger generation of people because they can share pictures, videos and stories. As the travel planning process frequently entails high level of uncertainty due to the characteristics of tourism, the availability of online travel information that can be accessed anytime and anywhere has become increasingly important in order to optimise travel plans (Xiang & Gretzel, 2010, Hsu & Cox, Burgess, Sellitto, & Buultjens, 2009).

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The issue of credibility of online travel information sources continued to be arguable among scholars (Kol et al., 2021). Scholars have not reached a consensus regarding the credibility of tourism applications (Munar & Jacobsen, 2014). Some scholars highlighted that information on tourism applications are credible (Cantallops & Salvi, 2014; Fotis, Buhalis & Rossides, 2012). Consumers trust comments from other travellers on social media prior to their consumption and consider it to be more trustworthy than any information provided by official tourism websites or travel agents as these comments are not intended for commercial purposes (Cantallops & Salvi, 2014; Ye, Law, & Chen, 2011). Furthermore, several methods are being used by several social media websites in order to investigate trustworthiness of reviews (Zelenka et al., 2021). In agreement with this view, Sigala et al. (2012) confirmed that social media represents a key source for potential travellers to understand previous travellers' experiences about destinations, facilities and restaurants (for discussion, see Shankar, 2021). For example, more than 74% of customers use the comments of other consumers on SNS as information sources when planning leisure trips (e.g. choosing a hotel or a restaurant). However, on the other hand, others considered information on tourism SNS as less credible (Tham et al., 2013; Yoo et al., 2009; Zelenka, Azubuike, & Pásková, 2021). Some highlighted that as social media comprises a great deal of user-generated information, users may not consider it as credible source of information (Cox et al., 2009; Fotis et al., 2012).

THEORETICAL FOUNDATION

The current research model for this study was built on the elaboration likelihood model (ELM) developed by Petty and Cacioppo (1986). Although ELM was originally applied to consumer behaviour before the advent of the Internet, it has been employed in different online contexts (Cyr, Head, Lim Stribe, 2018) and specifically for online reviews and to analyse consumer behaviour (Shahab, Ghazali, & Mohtar, 2021). The elaboration likelihood model (ELM) is an information processing theory (Petty & Cacioppo, 1986) that provides a theoretical framework for understanding how people process messages that are intended to be persuasive. According to ELM, there are two major routes by which persuasive messages can be processed: the central and peripheral route. If information recipients view a given message as being important and relevant to the target behaviour, they will carefully consider the issues presented in the message and evaluate the merits of the arguments. Under these circumstances, the recipient will undergo further cognitive processing and exert more effort to evaluate a message. On the other hand, people do not always elaborate on a message's arguments. If they view a message as having little personal relevance, they may not be willing to spend the time and effort in analysing it, but they may rely on what ELM refers to as peripheral cues (Petty & Cacioppo, 1986).

As a central cue, quality and merits of the message presented determines one's attitude towards that message (Petty & Cacioppo, 1986; Chaiken, 1980). Thus, in the context of SNS reviews, it is expected that a review with a stronger message credibility should be perceived as more credible. Unlike central cues, the peripheral route requires less cognitive work. People use simple heuristic cues or informational indicators, such as source credibility, to assess the believability of a message. Peripheral cues are contextually oriented towards the source environment rather than the credibility of the message. Theoretically, people may elaborate a message purely through either the central or peripheral route. For example, social media users who search travel information with high involvement conditions prioritise the central route and critically consider several online reviews (Park, Lee & Han, 2007), while they make judgments using simple peripheral cues under low involvement conditions (Tang, Jang & Morrison, 2012). However, in practice, people elaborate a message at a moderate level employing both routes or may shift processing from the heuristic mode to systematic processing (Oinas-Kukkonen & Harjumaa, 2009).

Based on prior discussion, evaluation of the credibility of information on SNS is expected to be affected not only by central, but also, peripheral cues. Consumers tend to make judgments about the credibility of the information on SNS as well as the credibility of the source prior to deciding to adopt this information (Li & Suh, 2015).

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Measures and Hypotheses

Source Dependency

As consumers increasingly rely on social media as one of their main sources of information (Berhanu & Raj, 2020), source dependency has been included in this study to establish the effect of SNS dependency on consumers' perceptions of SNS information credibility, which may affect their attitude and intention to adopt this information. SNS dependency is included in this study due to users' tendency to depend on SNSs to gain social relationships, information and knowledge acquisition (Ha et al., 2015). According to Johnson and Kaye (2014) dependence on social media is a strong predictor of judgments of high credibility. Johnson and Kaye (2014) found that dependence is a significant indicator of online credibility because the more familiarity users have with a specific media source, the more their ability is to judge its credibility. Recently, Hassan (2020) found that credibility is correlated with dependence on social media sources in Iraq. Social media dependency is important for practitioners and organisations as it allows behaviours that were now possible before the advent of such tools, including visibility, persistence, editability, and association (Treem & Leonardi, 2013). The following hypothesis is therefore proposed:

H1a. Perceived dependency has a positive impact on perceived credibility.

Authority Cues (AC)

According to Sundar (2008), authority cues serve as "a major criterion for assigning credibility to a website" by assessing "whether the source is an official authority or not" (p. 84).

Koh and Sundar (2010) found that information from more authoritative sources was found to have significantly higher trust ratings. According to Lin, Spence and Lachlan (2016) and Westerman, Spence and Heide (2014), individuals have more trust in medical experts or officials, such as tweets from the American Heart Association. As uncertainty is very high on SNS and in many cases, consumers could not guarantee the consistency of peer cues of other consumers, authority cues influenced consumers' decisions (Sundar, Xu, & Oeldorf-Hirsch, 2009). Further, more recent studies have shown that authority cues increase creditability in social media (Lin et al., 2016; Lin & Spence, 2018; Conrad, Leidner,& Schilder, 2008). Accordingly, the following hypothesis is proposed:

H1b. Perceived authority cues have a positive impact on perceived credibility.

Homophily

Humans have a natural need to engage with others who share their way of thinking and interests (McPherson et al., 2001). As social media helps gather people of similar interests together, it encourages the need for homophily (Kang and Chung, 2017; Gu et al., 2014), although other studies have reported different results (Bisgin, Agarwal, & Xu, 2012). According to social identity theory (Ashforth and Mael, 1989), a person's demographic attributes have important influence on others' evaluations about this person (Huang and Kwok, 2013). Reviewers' demographic characteristics of reviewers (such as age or gender) or beliefs and attitudes are expected to have significant impacts on the extent to which others perceive the credibility of reviews. Due to the anonymous nature of the online environment, consumers consider it challenging to determine the credibility of information provided in online reviews as well as the intentions of the information provider (Racherla, Mandviwalla & Connolly, 2012). Wei and Liu (2020) highlighted that homophily fosters users' positive judgements towards similar rather than dissimilar ones (also see Trandoc, 2019). Liu and Park (2015) argued that people consider information as believable if it matches their views, and not believable if it is contradictory to their beliefs, regardless of how this information is well argued or properly sourced.

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Organisations and practitioners should rely on data on homophily by highlighting comments and feedback by existing consumers who show similar traits to potential consumers. In other words, personalised marketing techniques by organisations and practitioners should rely on findings on the impact of homophily on credibility. The following hypothesis is therefore proposed:

H1c. Perceived homophily has a positive impact on perceived credibility.

Social Identity

According to social identity theory, people may decrease uncertainty by communicating with others who share similar social identity (Tajfel & Turner, 1979). Supporting this proposition, Jahng and Littau (2016) have found that in online communities, disclosure of users' profile characteristics such as individual identification (demographics and lifestyle), geographical location, and real name of the reviewer rather than a nickname or screen name have an impact on credibility perceptions. Liu and Park (2015) found that reviewers with a real photo, name, and address are rated more believable than others (for similar results, also see Lee & Shin, 2014). Cyr et al. (2018) considered perceived social presence to be a major effect on online consumers' trust. Recently, Cooley and Parks-Yancy (2019) found that people find reviews by people they know to be more credible.

In the tourism industry, the presence of distinguishable reviewer information such as name and gender improves the perceived credibility of online hotel reviews (Hlee et al., 2018), which amplifies their persuasion and consumers' intention to adopt them (Xie et al., 2011). Like homophily, organisations and practitiones should reveal social identities to increase credibility in online consumers. The following hypothesis is therefore proposed:

H1d. Perceived social identity has a positive impact on perceived credibility.

Source Transparency

Watts and Wyner (2011) asserted that source transparency is the degree to which users can validate the information provided by a source, which in turn increases their perceptions of the information and source credibility. There have several studies in the relationship between source transparency and credibility in online news (Masullo, Curry, Whipple, & Murray, 2021; Curry & Stroud, 2021). Porumbescu (2015) confirmed that one of the most important aspects of online transparency is that information presented should be current, understandable and accessible by online users. Some researchers have suggested that SNS users consider social media as credible as they are independent from media control, which allows users to write in-depth reviews in an open, interactive and transparent way (Arshad & Khurram, 2020; Song & Lee, 2015; Yang & Lim, 2009). This ultimately leads to developing users' trust (Park et al., 2015). Furthermore, Masullo, Curry, Whipple, and Murray (2021) argued that organisations should capitalise on source transparency to increase consumer credibility, although this was not conducted in practice yet. The following hypothesis is therefore proposed:

H1e. Perceived transparency has a positive impact on perceived credibility.

Perceived Expertise

As mentioned earlier, perceived credibility is based on two main factors: trustworthiness and expertise. Perceived expertise is commonly cited as one of the critical factors of perceived credibility (Hovland, Janis & Kelley, 1953). Expertise incorporates whether a source is both qualified and is presenting relevant, accurate information (Hovland et al., 1953), knowledgeable, skilled and competent (Fogg et al., 2002). A message can be considered expert when the source enables easy engagement for both experienced users and beginners (Guido, Prete & Sammarco, 2010). More recent studies in different

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contexts, including tourism and online healthcare, have also emphasised the substantial effects of source trustworthiness on various consumers' attitudes, information adoption intentions (Balouchi, et al., 2017; Ayeh et al., 2013; Thon & Jucks, 2017; Thomas et al., 2019; Lo & Yao, 2019). Perceived expertise is important for organisations as it allows for increased visibility (Treem & Leonardi, 2013), although to our knowledge, this was not incorporated in practice. The following hypothesis is therefore proposed:

H1f. Perceived expertise has a positive impact on perceived credibility.

Perceived Trustworthiness

Ohanian (1991, p. 47) defines trustworthiness as a ''consumer's confidence in the source for providing information in an objective and honest manner.'' A message displays signs of trustworthiness when it gives users an easy and direct idea of the source content and purpose and presents facts in a confident and transparent way (Guido et al., 2010). Low perceptions of the trustworthiness of online information have been considered as a main obstacle to consumers' intentions to embrace this information (Ayeh et al., 2013; Filieri, 2015; Machackova & Smahel, 2018;; Mumuni, O'Reilly, MacMillan, Cowley, & Kelley, 2020). Trustworthiness is critical as consumers usually agree that most online information is not trustworthy (Elsantil, 2020). In this regard, Hensel and Deis (2010) asserted that consumers should be careful when searching information and reading comments on social media sites because SNS users may post fake reviews. Trust is crucial for online tourism marketing as it enhances travellers' intention to purchase (Li, Ong & Ito, 2020; Balouchi, et al. 2017). There is scarcity of studies that tested how organisations and practitioners may incorporate findings on the impact of perceived trustworthiness on credibility. However, Wiencierz, Pöppel, and Röttger, (2015) show that increasing perceived trustworthiness online that help bolster the reputation of the nonprofict organisations, although this was done in tourism organizations. The following hypothesis is therefore proposed:

H1g. Perceived trustworthiness has a positive impact on perceived credibility.

Argument Strength

Argument strength refers to the persuasive strength of arguments embedded in an informational message (Cheung, Sia & Kuan, 2012). Dillard and Shen's (2005) highlighted that there is a strong probability that strong arguments containing real facts and information can reduce uncertainty and increase message credibility as compared to messages that ignore proof and logic (for similar argument, also see Thomas et al., 2019; Xiao, Wang, & Chan-Olmsted, 2018). In the case of SNS, reviews with relevant, persuasive and convincing claims are perceived to be more credible and have a positive effect on consumers' intentions to adopt this information (Cheung et al., 2012; Lee & Shin, 2014). The manner in which a reviewer presents and describes the information in a review continues to impact other reviewers. Information available in a review helps consumers assess the characteristics of an information as well as the source of information (Racherla et al., 2012). Xu and Yao (2015) indicated that argument quality positively affect the consumers' final decision of the adoption of online reviews. Further, Thomas et al. (2019) claimed that arguments can highlight argument strength in their online platforms in order to increase reputation (also see Bee & Dalakas, 2015). The following hypothesis is therefore proposed:

H1h. Perceived argument strength has a positive impact on perceived credibility.

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Perceived Credibility

Credibility of SNS is a critical factor in creating trusting relationships with consumers. Credibility has been broadly examined in the marketing communication literature from both source and message aspects (Metzger & Flanagin, 2013). Previous studies demonstrated disagreement regarding the credibility of information on SNS and its impact on consumer behavior. Travelers may consider SNS as credible sources because posted information comes from other travelers, who are not considered to have any commercial interest. Others argue that any individual can post information on SNS, which may raise several uncertainties about the trustworthiness of social media (Go, You, Jung & Shim, 2016). The perceived credibility of online reviewers significantly influences consumers' attitudes and intentions (Ayeh et al., 2013; Filieri, 2015). The research hypothesis supporting this proposed is as follows:

H2a. Perceived credibility has a positive direct effect on attitude towards information on SNS.

Attitudes Towards Adopting Information on SNS

Attitudes can be defined as an individual's favorable or unfavorable feelings and evaluations about a particular behavior (Fishbein and Ajzen, 1975). The technology acceptance model, the theory of planned behaviour, and other social psychology theories recognise the role of attitudes in shaping behavioral intention. In many studies, attitudes have been considered as a mediator of the effect of consumers' perceptions on behavioral intentions (Ajzen, 1991). Additionally, in the online context, attitudes are considered to be a major cause of the intention to accept online travel information (Casaló, Flavián & Guinalíu, 2010). Boateng and Okoe (2015) and Chu, Kamal and Kim (2013) have also shown that consumers' attitudes toward social media influence their responses toward it. Positive consumers' action, such as adopting information on social media, is the outcome of their attitude toward this information (Mir, 2012). The research hypotheses supporting this are proposed as follows:

H3a. Attitudes towards information have a positive direct effect on intention to adopt information on SNS.
H4a: Attitudes towards information mediate the relation between information credibility and intention to adopt information on SNS.

Intention to Adopt Information on SNS

Intention to adopt information on SNS to purchase tourism services is defined as the probability that consumers will use information on social media to purchase tourism-related services. SNS users are subjected to an overwhelming amount of electronic Word of Mouth (eWOM) information, which is found to be as influential on consumers' decisions (See-To & Ho, 2014). Users perceive the trustworthiness of a source through message and source features which have been shown to have a great influence on their intention to adopt this information (Li and Zhan, 2011). Casaló et al. (2010) highlighted that consumers' intention to use travel information on SNS is significantly affected by their attitudes.

Hypotheses Development

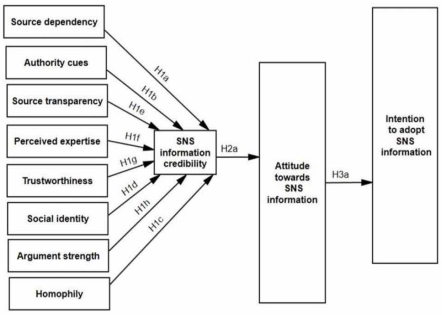
Based on the previous discussion, the anticipated relations of the consumer's intention to adopt SNS information and the study hypotheses are presented in Figure 1.

Methodology

A quantitative research approach was utilized to validate the proposed framework. The questionnaire contained 50 statements measured on a five-point Likert scale ranging from 1 = strongly disagree

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Figure 1. Hypothesized structural model for consumers' intention to adopt SNS information.



to 5 = strongly agree. The questionnaire, distributed using Google forms via messages sent to the participants, entailed two sections: the first included participants' demographic information, such as gender, age, qualifications and income. The second section included statements about each construct. The scales used to measure constructs were adapted from the following sources: homophily from Ayeh et al., (2013), information transparency and argument strength from Li and Suh (2015), authority cues from Choi, Hickerson & Kerstetter (2018), source dependency from Li and Suh (2015), perceived expertise and perceived trustworthiness from Balouchi et al. (2017), perceived credibility from Li and Suh (2015), attitude towards SNS use from Casaló et al. (2010) and Intention to adopt information on SNS from Balouchi et al. (2017).

Participants

A total of 318 individuals residing in different Arab countries completed the survey. The target population of the current study includes consumers who have used information from any SNS to collect travel information during the past two years. The majority (70.10%) of them were between 19 and 25 years old. Women comprised 63.2% of the sample. In terms of level of education, 16% of the participants attended high school, 62% held a bachelor's degree, and 22% had completed a graduate degree. Furthermore, the family income was more than $5,000 per month in 25.00% of the sample; All respondents originated from Arab cultures such as Qatar, Egypt, Jourdan and other Arabic countries. There were no exclusions from the collected data.

Statistical Analysis and Results

The study utilized a two-phase approach (Hair et al., 2010) to analyze the conceptual framework by assessing a measurement model. Structural Equation Modeling (SEM) was then used to examine the hypotheses testing. Descriptive statistics and exploratory factor analyses were conducted using the statistical package SPSS, version 25.0 (IBM Corporation, Armonk, NY, USA). An exploratory factor analysis (EFA) was conducted for the measurement items of the pre-determined constructs. One item for each of homophily and dependency was deleted due to small values of their factor loadings and because it was recommended by reliability analysis. The EFA results for the remaining items showed that Kaiser-Meyer-Olkin (KMO) test = 0.92, and Barlett's test of sphericity p value < 0.05, indicating that the correlation matrix is not an identity matrix and the items sufficiently relate to run a meaningful EFA. Communalities were also above a recommended cut point of 0.4, identifying that each item correlates adequately with all other items. The EFA results revealed that all items loaded on their designed factors, and factor loadings ranged from 0.65 to 0.94, averaging out to greater than 0.70 for each factor. With regard to the reliability test, all Cronbach's alpha coefficients shown in Table 1 achieved the minimum criterion of 0.70, demonstrating satisfactory inter-item reliability.

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A confirmatory factor analysis was used to confirm the factor structure extracted in the EFA and to evaluate the dimensionality of the study variables. Some correlations between error terms were incorporated into the confirmatory factor analysis as suggested by the modification indices, which yielded a significant improvement in the model's goodness of fit. The measurement model, including all the first-order reflective constructs, was tested and the fit indices met the suggested thresholds revealing adequate goodness of fit: Chi-square (2405.514)/degrees of freedom (892) = 2.96, GFI (goodness-of-fit index) = 0.954, AGFI (adjusted goodness of fit index) = 0.915, CFI (incremental fit index) = 0.925, normal fit index (NFI) of 0.950, and RMSEA (root mean square error of approximation) = 0.073.

Factor scores were implemented and the correlations between them do not exceed 0.70 and are significant at the 0.05 and 0.01 levels (see Table 1). The reliability and convergent and discriminant validity of the measurement scales were assessed. The reliability was examined based on composite reliability (CR), whereas a construct is considered to have an adequate composite reliability if composite reliability is greater than 0.70. As shown in Table 1, all composite reliability values ranged from 0.71 to 0.92. In addition, discriminant validity was achieved as the inter-construct correlations did not exceed the square roots of AVE. Finally, all item loadings/standardized path coefficients were equal to or above a recommended cutoff value of 0.60, and the values of AVE were above 0.56, proposing convergent validity of the scales.

The structural model in Figure 1 was conducted to examine the potential relationships between the credibility attributes on customers' attitudes and intention to adopt information on SNS. The Amos software (Arbuckle, Bennett & Speed, 2014) was used to conduct the SEM, estimate the parameters of the prosed structural model, and investigate its goodness of fit. The calculated modification indices recommended that direct effects of perceived source dependency, authority cues and source transparency on attitude towards SNS as well as a regression path between source dependency and intention would improve the goodness of fit of the proposed model in Figure 1. Therefore, regression paths that show these direct effects were successively incorporated into an adjusted model, as shown in Figure 2, and added to the tested hypotheses as H2b- H2d, and H3b.

The results shown in Table 2 revealed an adequate fit for the adjusted model shown in Figure 2, whereas the values of model fit indices meet the suggested thresholds and recommend an adequate goodness of fit for the developed structural model.

Direct Effects

The results shown in Table 3 are consistent with the hypotheses presented in Figure 2. These results reveal that H1a, H1d, and H1f-H1h are supported, indicating that participants' level of perceived source dependency ($b = 0.080$, $p = 0.047$), social identity ($b = 0.150$, $p < 0.01$), expertise ($b = 0.114$, $p = 0.023$), trustworthiness ($b = 0.416$, $p < 0.001$), and argument strength ($b = 0.141$, $p = 0.\ 0.013$) had positive significant direct effects on their perceived information credibility. However, H1b, H1c, and H1e were not supported, demonstrating that level of authority cues ($b = 0.074$, $p > 0.05$), homophily ($b = -0.044$, $p > 0.05$), and source transparency ($b = 0.038$, $p > 0.05$) had insignificant direct effects on information credibility. On the other hand, H2a-H2d were supported, presenting that participants' level of information credibility ($b = 0.288$, $p < 0.01$), source dependency ($b = 0.322$, $p < 0.01$), authority cues ($b = 0.237$, $p < 0.01$), and source transparency ($b = 0.128$, $p = 0.045$) had significant positive direct effects on consumers 'perceived level of attitude towards SNS. Furthermore, the results support H3a and H3b, specifying that attitude towards SNS ($b = 0.128$, $p < 0.01$) and degree of source dependency ($b = 0.249$, $p < 0.01$) had significant positive direct effects on intention to adopt information on SNS. The values of squared multiple correlations () in Table 3 demonstrate that the exogenous predictors of the endogenous factors: information credibility, attitude towards SNS and intention to purchase, explain 57.5%, 35.0% and 63.0%, of their variances, respectively.

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Table 1 Cronbach's alpha, correlation coefficients, coverage reliability (CR), average variances extracted (AVE).

Constructs		AC	SD	H	SI	ST	PE	PT	AS	PC	I	AT
Authority cues	AC	0.798										
Source dependency	SD	.480**	0.874									
Homophily	H	.271**	.275**	0.876								
Social identity	SI	.265**	.335**	.405**	0.806							
Source transparency	ST	.446**	.391**	.442**	.348**	0.745						
Perceived expertise	PE	.323**	.116*	.339**	.167**	.333**	0.846					
Perceived trust	PT	.360**	.201**	.321**	.162**	.271**	.679**	0.807				
Argument strength	AS	.381**	0.335**	.377**	.346**	.452**	.504**	.573**	0.807			
Perceived credibility	PC	.415**	.329**	.322**	.346**	.364**	.559**	.688**	.567**	0.824		
Intention to adopt SNS	I	.400**	.536**	.284**	.343**	.276**	.145**	.320**	.386**	.377**	0.812	
Attitude towards SNS	AT	.450**	.481**	.286**	.335**	.372**	.235**	.342**	.431**	.430**	.671**	0.799
AVE		0.64	0.763	0.76	0.65	0.56	0.716	0.65	0.65	0.68	0.659	0.639
CR		0.920	0.7100	0.71	0.72	0.751	0.881	0.72	0.78	0.78	0.734	0.923
Cronbach's alpha		0.704	0.87	0.775	0.838	0.641	0.849	0.900	0.786	0.849	0.894	0.874

Notes: ns, correlation is insignificant. The diagonals represent the square root of average variance extracted (AVE) and the off diagonals represent the bivariate correlations among the constructs. *, **Significant at the 0.05 and 0.01 levels, respectively.

Indirect Effects (Test of Mediation)

The results shown in Table 4 show that H4a was supported, suggesting there were significant indirect effects (IE) of information credibility (IE = 0.196, $p < 0.01$) on consumers' intention to adopt SNS. In other words, the consumers' attitude mediates the relation between consumers' information credibility and intention to adopt SNS information.

In addition to H4a, the developed model allows us to test the added hypotheses H4b-H4e in Table 4. The results show that there were significant indirect effects of source dependency (IE= 0.219, $p< 0.01$), authority cues (IE = 0.161, $p < 0.01$) and source transparency (IE = 0.087, $p = 0.047$) on consumer intention, which were mediated by attitude. Furthermore, the indirect effect of source dependency (IE = 0.016, $p = 0.031$) expressed via credibility and attitude on consumers' intention was statistically significant.

Discussion

The main purpose of this study was to examine the effect of consumers' perceived credibility of information on SNS on consumers' attitude and intention to adopt this information. Social media users employ specific cues when they evaluate perceived credibility online. Oinas-Kukkonen and

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Figure 2. Hypotheses testing for the adjusted model. *p < 0.05, **p < 0.01, ns not significant.

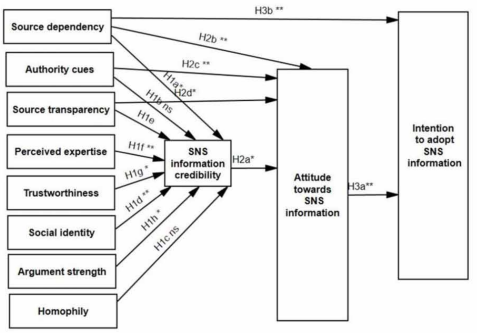


Table 2 Model goodness of fit criteria.

Statistics	Suggested	Obtained
Chi-square significance	<0.05	0.00
Goodness of fit index (GFI)	<0.90	0.983
Adjusted goodness of fit index (AGFI)	<0.80	0.911
Comparative fit index (CFI)	<0.90	0.980
Normal fit index (NFI)	<0.90	0.913
Root mean square residuals (RMSEA)	>0.08	0.068

Harjumaa (2009) reported that users usually assess credibility via content. Similar findings were also reported among Polish and Dutch participants in that similar antecedents as ones used in the current study are also associated with perceived credibility (van Dam, 2013). However, users are also influenced by heuristics such as user names and message topics. As mentioned above, there is a gap in the literature regarding how social media influences consumers' intentions to adopt information on SNS to purchase tourism services, specifically in the hospitality context (Varkaris & Neuhofer, 2017). The current study adds to the literature by investigating antecedents of perceived credibility that affect consumers' perceptions of the credibility of information on SNS and their impact on consumers' attitude and intention to use this information. Our study is the first to investigate the relationship between a large number of antecedents and perceived credibility in Arab context.

Overall, the findings of this study were sometimes consistent with and at other times contradictory to previous research. For instance, the current study suggests that consumers' perceived credibility of information on SNS has an impact on their attitude and intention to adopt this information. The findings of the current study are consistent with previous research that social identity and argument strength have a significant effect on consumers' perceived credibility, attitude and intention to adopt information on SNS. In line with findings by Kusumasondjaja, Shanka and Marchegiani (2012), the current study found that online reviews with an identified source are perceived to be more credible than reviews without an information source. Reviews by users who allow others to access their personal information, such as their name and photo, and allow others to visit their personal pages and see their posts are more credible and have a larger impact on consumers' intention to adopt these reviews when purchasing tourism services. Importantly, one study confirmed the relationship

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Table 3 Estimated direct effects (regression path coefficients) for the adjusted structural model.

Hypotheses	Dependent Variables	Predictor variables	Estimated direct effect (b)	P	Squared multiple correlations ()	Decision
H1a	Information Credibility	Source dependency	0.080	0.047*	0.575	Supported
H1b		Authority cues	0.074 ns	0.128		ns
H1e		Source transparency	.038 ns	0.427		ns
H1d		Social identity	0.150	**		Supported
H1f		Perceived expertise	0.114	0.023*		Supported
H1g		Trustworthiness	0.416	**		Supported
H1h		Argument strength	0.141	0.013*		Supported
H1c		Homophily	-0.044 ns	0.352		ns
H2b	Attitude Towards SNS	Source dependency	0.322	**	0.350	Supported
H2c		Authority cues	0.237	**		Supported
H2d		Source transparency	0.128	0.045*		supported
H2a		Information credibility	0.288	**		Supported
H3b	Intention to adopt SNS	Source dependency	0.249	**	0.630	Supported
H3a		Attitude towards SNS	0.681	**		Supported

*, **significant at the 0.05 and 0.01 levels, respectively, ns not significant.

Table 4 Estimated indirect effects (test of mediation).

Hypo theses	Indirect paths	Estimated indirect effect (IE)	95%CI		P	Decision
			Lower	Upper		
H4a	Information credibility -->Attitude towards SNS --> Intention to adopt SNS information	0.196	.115	.281	.001**	Supported
H4b	Source dependency --> Attitude towards SNS --> Intention to adopt SNS information	0.219	.148	.289	.001**	Supported
H4c	Authority cues -->Attitude towards SNS --> Intention to adopt SNS information	0.161	.089	.246	.001**	Supported
H4d	Source transparency -->Attitude towards SNS --> Intention to adopt SNS information	0.087	.017	.157	.047**	Supported
H4e	Source dependency -->Information credibility --> Attitude towards SNS ->Intention to adopt SNS information	0.016	.004	.033	.031*	Supported

*, **significant at the 0.05 and 0.01 levels, respectively.

between trust and credibility in participants from 31 different countries, vacationing in Indonesia (Kusumasondjaja, Shanka, & Marchegiani, 2011). They have found that revealing social identity of person writing a review is associated with more credibility. In addition, the current study found that the way reviewers build their argument for or against a product or service increases the credibility of other reviewers' perceptions.

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In agreement with Erkan and Evans (2016) and Cheung et al., (2012), the results indicate that as information on SNS is user generated, a review with a stronger argument quality and a high degree of persuasion should be perceived as more credible. This may be also explained by the fact that high uncertainty avoidance cultures, like Arab cultures, consider security issues to be critical, especially online, where there is no personal interaction between the buyer and seller (Elsantiel & Harness, 2011). Accordingly, Arab consumers consider online reviews with identified sources and persuasive and convincing arguments as more credible. Furthermore, similar studies were conducted in other Arab countries, confirming our results, such as Kuwait (Rouibah, Lowry, & Hwang, 2016). However, Rouibah, Lowry, and Hwang (2016) found that there is a relationship between innovativeness, propensity to trust, familiarity and the intention to use an online payment, although credibility was implied implicitly in their study. Similarly, one study has also investigated online credibility in five Arabic countries, including Qatar, Lebanon, Saudi Arabia, Tunisia, and United Arab Emirates (Martin & Hassan, 2020). This study found that there is a relationship between fake media exposure and credibility in all studies countries except Tunisia and Lebanon. Future work should conduct a similar study in the domain of tourism.

Additionally, the results show that dependency, expertise and trustworthiness have a significant effect on consumers' perceived credibility, attitude and intention to adopt information on SNS, in agreement with prior findings (Flanagin, & Metzger, 2013). This finding is consistent with those of Johnson and Kaye (2014), who highlighted that reliance on SNS is a major predictor for SNS credibility perceived by consumers. Loyal SNS users consider SNS as more credible than less loyal users. In addition, the results of the current study align with previous research that expertise and trustworthiness significantly affect users' perception of the credibility of the SNS (Elsantiel & Harness, 2011) as well as their attitudes and intention to adopt information on SNS. The nature of social media interactions is more complex compared to traditional media, and this complexity can have negative impacts due to the great risks and unique features of personal information shared on social media platforms or networks (Miranda, Young, & Yetgin, 2016). Trustworthiness is a major reason for successes in the travel and hospitality sectors (Choi et al., 2018). One of the main reasons for the great importance of expertise and trustworthiness could be the intangibility of tourism products and services. Online users who use information on SNS for travel purposes believe this information is provided by trustworthy tourism consumers who do not have any personal motivations. Similar findings on online credibility were also reported in Bangaldesh (Andaleeb, Rahman, Rajeb, Akter, & Gulshan, 2012) and South Korea (Kim & Johnson, 2009) in that perceived expertise, and revealing social identity do often increase credibility.

However, our research findings reveal that although transparency and authority cues do not have a significant effect on consumers' perceived credibility, and these cues significantly affect their attitude and intention to adopt information on SNS (for similar points, see Sundar, Xu, & Oeldorf-Hirsch, 2009). We attribute the insignificant effect of both antecedents on consumers' perceived credibility to the characteristics of our respondents, who are mainly young people (83%) below 35 years of age, and highly educated (83% were holders of bachelor and postgraduate certificates). These young and educated users have grown up using the internet and social networking through mobile phones, and social media such as Facebook. Our findings are thus in agreement with prior research showing that younger than older people are more likely to rely on using SNS (Ha, Yoon, & Zhang, 2013). This familiarity with the use of social media has made them more risk-averse and less concerned about transparency or authority cues as indicators of credibility. This is especially true for tourism services, where users trust information coming from personal travellers rather than any information coming from official tourism websites. Perceived transparency and authority cues are expected to reduce perceived risk between users and thereby enhance their intention to adopt information on SNS (Arshad & Khurram, 2020; Song & Lee, 2015; Westerman et al., 2014). This appears to be consistent with the high uncertainty avoidance recognized in the current sample that makes them more likely

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to make decisions about online purchasing based on references (reviews) from trusted persons than consumers would do in low uncertainty avoidance cultures.

Previous research by Liu and Park (2015) suggested that reviews from online users who share the same background and attitudes are considered more trustworthy and credible. In contrast, our study found that homophily does not affect consumers' perceived credibility, attitude or intention to adopt information on SNS. The results of the current study might be explained by the widespread use of social media worldwide. Rather than similarities that exist between reviewers on social media, the relationship entails numerous cultural and other considerations such as reviewers' social status, personal reputation, cultural affiliations and experience regarding the topic of the review (Koiranen, Koivula, Keipi & Saarinen, 2019). Because of the great reliance on social media as a key source to gain experience of tourism services, potential travellers are more likely to be exposed to different views and accept reviews from different users from different backgrounds (Koiranen et al., 2019). In addition, as the majority of current research respondents were young users (83%) under 35 years of age, they might be expected to use social media to search for tourism services more often. In addition, younger users are expected to be less careful when evaluating the credibility of information available on SNS.

As expected, the findings from the current study support the elaboration likelihood model (ELM) regarding the mediating role of perceived credibility and attitude. The evaluation of SNS information credibility determines the expected value of the information it provides. Trustworthy Information that imparts significant knowledge has a great influence on consumers' perceived credibility of information and cognitive judgments and enhances their likelihood of adopting this information (Filieri, 2015). It is important to note that model presentation in Figure 1 is a simplified model of the relationship between antecedents (e.g., dependency, expertise, homophily) and credibility. There are possible earlier and intermediate constructs that mediate the relationship between these antecedents and perceived credibility. For example, authority cues and perceived expertise may increase dependence on social media, which in turn may impact credibility. In other words, it is possible that there are some relationships between the antecedents described in Figure 1. However, it is also important to note that it is experimentally difficult to investigate a more complex model. Nonetheless, although our model has limitations, it provides an initial step towards understanding the complex relationship between several factors that may impact credibility in online platforms.

CONCLUSION AND IMPLICATIONS

This study adds to the ever-growing literature on the credibility of information on SNS. It is an appropriate response for the need to more research on information credibility on SNS and has important implications academically, especially, as it broadens our understanding of the impact of consumers' perceived information credibility (considering different antecedents of credibility) on their intention to adopt this information for purchase decisions on a larger and more diverse sample. Being the first study in the Arab world to investigate consumers' credibility perceptions of information on SNS and their intention to adopt this information, this study contributes to the academic research and practice and extend the current concepts of online credibility. Although several scholars have studied the credibility of SNS technical attributes, little work has considered consumers' perspectives regarding the adoption of information on SNS. The main theoretical contribution of this study was to provide a more suitable and comprehensive model for today's dynamic virtual environment, considering antecedents of SNS information credibility and how they impact consumers' intention to adopt information. This model can be applicable to assess online consumers' perceived credibility on various SNS such as Twitter or Facebook. Credible information on SNS can play a significant role in the believability of social media. Credible sources are seen as likely to produce credible messages and credible messages are seen as likely to have originated from credible sources.

The findings of this study improve the growing body of literature by targeting new societies, including the Arab culture, whose information providers could benefit significantly from the insights

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offered by this research. Our research shows that managers or operators of social media platforms should maintain a high level of source dependency, trustworthiness and expertise in order to ensure the credibility of the information disseminated on their sites. They should also ensure source transparency and authority cues, which affect consumers' attitude and intention to adopt information on SNS. On the other hand, users should pay more attention to disclosing their social identity and enhancing the strength of their arguments, which will improve the perceived credibility of online reviews.

The current study also has very important implications for organizations, practice, and society. For example, organizations should attempt to increase credibility in their online platforms. Specifically, organizations should attempt to encourage users to reveal their social identity, increase argument strength, trustworthiness, source dependency, and highlight expertise of users, in online platforms. For example, organizations should highlight reviews by experts who are not related to the organization itself (König, & Jucks, 2019). As suggested by Kusumasondjaja, Shanka, and Marchegiani (2011), the identity of the information posted online should be made available for potential customers, as this can increase credibility. Importantly, managers should strive to address negative reviews online. This can increase trust in the organization, as potential customers will know that organisers are addressing negative reviews, as reported in prior studies (Babin, Lee, Kim, 2005; Kau, & Loh, 2006; Wong, 2004). Organizations should conduct surveys to study how customers make decisions in their online platforms. Doing so will increase trust in the organization and likely increase attitude and intention to adopt SNS information. This will help investigate whether factors mentioned here contribute to their decision making. Such surveys can also help investigate whether homophily impacts intention to adopt SNS information. As for impact on society, our study helps better connect online platforms with customers. As more activities are being conducted online, it is important for future research to understand factors underlying trust and credibility in virtual environments in general.

LIMITATIONS AND FUTURE RESEARCH

However, some limitations need to be addressed for future research. First, data were collected for this study from various Arabic countries without consideration of the cultural differences among these cultures. Future examination may replicate this study in different Arab cultures to find out whether there will be any differences in consumers' perceptions of information credibility. Additionally, conducting a cross-culture comparison of consumers' perceptions of information credibility on SNS and its effect on their intention to adopt this information in purchasing tourism services is an important area to investigate in the future. Another limitation relates to the fact that this study has focused only on the tourism sector. Future research could replicate the same model in different sectors and identify how the type of service affects consumers' perceptions of information on SNS. A future study may also replicate the current study to investigate whether homophily has any effect on consumers' perceived credibility of information on SNS or their intention to adopt this information for a different sample or context. In addition, further studies could examine whether there would be any significant effect for information transparency and authority cues on consumers' perceived credibility. Further research also may examine additional factors that influence SNS credibility. Importantly, future research should also test additional factors, such as interactivity. Research found that more interaction online with potential users increase credibility (Jahng, & Littau, 2016; Johnson & Kaye, 2016; Kaye & Johnson, 2017).

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Can Perceived Ease of Use Improve M-Commerce Adoption?

Role of Mobile Network Service Quality

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ABSTRACT

In India, the e-commerce sector is rising at an enormous pace, but an even larger share has been in the hands of offline retailers until now. A unique type of consumer behaviour has been seen because of the simple accessibility of the internet, where the customer searches for products and its information on the internet and then buys them offline. With this, the researcher is here analysing the mediating effect of perceived ease of use on adoption of m-commerce services with the usage of advanced mobile network generations. Data were collected from 616 respondents by making stratas of respondents in such a way that different telecom users who are doing online transactions were taken. Data were analysed by using structural equation modeling technique, and it was examined that the mediator (perceived ease of use) has a great impact on m-commerce adoption.

KEYWORDS

Convenience, M-Commerce, Network, Perceived Ease of Use, SEM, Service Quality

1. INTRODUCTION

M-commerce has evolved from a trend to reality. Customer demand trends and technological advancement have mitigated challenges and narrowed the line between offline and online trade. E-commerce is part of our shopping habits today, from purchasing groceries to appliances (Tam, Loureiro & Oliveira, 2019). Nowadays, online retailers are using the Internet to convey, communicate, and disseminate information, sell the product, receive feedback, and conduct satisfactory customer surveys. On the other hand, customers used the Internet to purchase the product online and compare prices, product features, and after-sales service facilities that they would receive if they purchased the product from a particular store (Sarkar, Mukherjee & Lahiri, 2020).

Furthermore, as customers have become more tech-savvy, so e-commerce executives need to comprehensively consider the needs and factors of customers that affect their preference to choose and shop on e-commerce websites. There are usually two points of understanding the behaviour of online consumption. The first phase involves how to persuade customers to shop online, and the second phase is to encourage them to buy back, which is vital to the success of an M-commerce portal (Chiu & Cho, 2019).

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Developing countries have seen an incredible and impressive increase in infrastructure production for telecommunications technology over the past decade. In terms of improving access to wireless connectivity, which has a tremendous effect on the lives of billions of people worldwide, development is remarkable. Internet access has not only improved the simplicity and speed of communication, but has also provided marketers with the ability to redefine the art of communicating with potential consumers. A good mobile telecommunications service network is also favourably correlated with a country's economic growth (Yadav, Sharma & Tarhini, 2016). Data services are insightfully changing our lives. Anywhere and anytime, cellular providers make internet connectivity available. This enables instant access to social networks, intranet employment, shopping, internet browsing, entertainment, etc. Cellular companies are striving to improve service and provide their users with a better experience. The growth and advancement of cellular broadband services have been enabled by research and development in different areas of cellular technology (Alomary & Kostanic, 2013).

This study investigates the impact of mobile network service quality on the adoption of mobile commerce services with the help of one mediator, i.e. perceived ease of use (PEOU). First, we will examine whether the determinants of the mobile network service quality, which are: speed, trust, satisfaction, security, privacy, and cost, have an impact on M-commerce services or not. Second, we will examine that, with the mediator's help, i.e., perceived ease of use (PEOU), its adoption will increase or not. Perceived ease of use also has their determinants that are: convenience, and website quality.

2. LITERATURE REVIEW

2.1 Mobile Network Service Quality

The mobile network industry can be considered a typical service industry. In a world country like India, where the market is increasing dramatically, the rapidly evolving characteristics of both network networks and handsets and the restrictions imposed by regulators make the market more challenging and unpredictable. In such a scenario, estimating customer satisfaction poses real challenges for researchers. Customer satisfaction, however, has been hypothesized by many researchers as a result of perceived service quality. Therefore, it is important to estimate the perceived quality of service to assess customer satisfaction (Chakraborty & Sengupta, 2013). The conventional word of mouth (WOM) type has become obsolete and converted into electronic word of mouth due to technological advances (eWOM). eWOM is the type of online open comments about a product or business made by potential, current, or former customers.Meanwhile, the social network is a new forum that allows users to share views and experiences with their friends and acquaintances about goods or services in an online platform (Abedi, Ghorbanzadeh&Rahehagh, 2020).

There are some determinants of mobile network service quality which are covered in this study are elaborated with their supporting studies in the following table:

2.2 Perceived Ease of Use

The Technology Acceptance Model (TAM) is one of the most popular model used by researchers in the study of technology adoption by individuals. TAM indicated that it is possible to use perceived ease of use to predict the attitude towards using new technology, which in turn influences the behavioral intention actually to use the actual device. The perceived ease of use is "the extent to which the prospective adopter expects the adopted new technology to be a free effort to transfer and use it" (Davis, 1989). Therefore, if users believe that online transactions are easy to use and hustle-free, they would be more likely to use that system (Chong, Ooi, Lin & Tan, 2010). A key factor in the technology acceptance model is perceived ease of use;it is the degree to which a person assumes that using a specific technology will be free of cognitive effort. An e-service that appears to be easy to read, understand, and use can reduce customer confusion and overall risk during pre-purchase

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Table 1. Literature Review of determinants of mobile network service quality

Variable	Authors with year	Findings
Speed	Frimpon & Adaku (2018)	Speed refers to the distribution rate of the distributed service, which is the data in this case. Speed is an important metric for customers to determine their broadband choices and decide how they plan to use the Internet. Technically, speed is divided into two categories i.e., download speed and upload speed.
Trust	Tam, Loureiro & Oliveira (2020)	To research the role of trust in e-commerce and its effect on individual performance outcomes, we use total trust, one dimension calculated by these three trust dimensions: competence, benevolence and integrity. Trust is also depends upon impulse buying and previous customers'review.
Satisfaction	Iqbal (2020)	Customer satisfaction is derived from quality service, contributing to long-term customer loyalty. If the service quality is satisfactory or meets customers' expectations, then customers become loyal towards that particular portal.
Security	Ho & Oh (2009)	Online customers cannot physically verify the quality of goods in advance or to test the safety and security of personal information (credit card numbers, ID numbers, etc.) sent through the Internet. The fear of customers would repeatedly be generated without the existence of visible trustworthiness criteria for consumers to differentiate the legitimacy and reputation of e-vendors. For this purpose, e-security seal programmes have been introduced, thus creating standards or principles that allow consumers to differentiate trustworthy e-sellers from others.
Privacy	Zhu, Srivastava & Sutanto (2020)	Privacy of information may be described as the right of an individual to monitor the circumstances under which his or her personal information is accessed and used. Where the Internet is concerned, the ability of the user to monitor how their knowledge is acquired, disseminated and used has been diminished.
Cost	Chakraborty & Sengupta (2013)	Providers of mobile networks should be expected to provide good voice quality, the ability to connect easily to other networks and good service quality at a fair price.

assessments (Featherman, Miyazaki & Sprott, 2010). In TAM, ease of use has a direct effect on behavioural intention to use.

Furthermore, the perceived ease of use also has a direct impact on usability and approach. TAM generally interprets and predicts individual acceptance of information technology (Yeh & Li., 2009). Researchers worldwide have turned their attention to internet banking and e-commerce and similar technology to adopt new technologies. Technology Acceptance Model (TAM) and its later versions are the most commonly used models that helped the users to move their intention towards new technologies acceptance (Goyal, Maity, Thakur & Srivastava.,2013). TAM is founded on the premise that people's perceptions of usefulness and ease of use influence how technology is implemented,

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but the authors expand their distribution into long and near-term perceptions (Snowden, Spafford, Michaelides & Hopkins., 2006).

In this study, two dimensions of perceived ease of use (PEOU) have also taken and these were convenience and website quality.

Table 2. Literature Review of determinants of Perceived Ease of Use

Variable	Authors with Year	Supported Findings
Convenience	Jiang, Yang & Jun (2013)	It was identified that convenience has five determinants that play an important role: ease access, search, evaluation, transaction, and possession/post-purchase convenience.
Website Quality	**Akram, Hui, Khan, Tanveer, Mehmood & Ahmad (2018)**	**The quality of the website has a positive effect on the Online Impulse Buying Behaviour (OIBB); second, sales promotion greatly affects the OIBB and serves as a good moderator in the relationship between the quality of the website and the purchasing of online impulses. Third, the use of credit cards has a positive impact on online impulse purchases. In addition, the use of credit cards strengthens the relationship between the quality of websites and the online impulse purchasing.**

2.3 M-commerce Adoption

The advancement of 4th generation mobile telecommunications technologies in recent years has fueled the rapid expansion of mobile commerce (M-commerce) worldwide. M-commerce uses wireless devices and data connections to conduct transactions that result in the exchange of value for information, services, and goods because of its effect on businesses worldwide, M-commerce has attracted the attention of everyone, offering consumers faster access than e-commerce (Marriot, Williams & Dwivedi, 2017; Sun & Chi, 2019). M-commerce is an extension of the e-commerce in which business activities are carried out in a wireless environment via mobile devices. (Tarhini, Alalwan, Shammout & Al-Badi, 2019).

3. RESEARCH METHODOLOGY

3.1 Sampling

Data were collected from those who are doing online transactions sothat different telecom users were included by using stratified random sampling (example: Airtel, Vodafone Idea, Reliance Jio, and Other). In others BSNL users were taken. For each network users, stratas were constructed. Data were collected through online social media platforms. Duplicacy of data were eliminated by doing proper screening of repeat email addresses and then removed them. At the beginning of the survey, we have

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one data-item (Are you doing any type of online monetary transaction from your mobile devices?). If any respondent tick "NO"that person was excluded from the study. In this way, a questionnaire was sent to 1000 persons only 730 were responded. But from those 730, only 616 respondents were taken as only 616 persons were those do are doing monetary transaction from their mobile devices.

3.2 Sample Size

According to Hair et al., 2006, researcher can take the sample size according to data-items in the study. As in this study, there are 35 data-items of three variables (mobile network service quality, PEOU, and M-commerce adoption), do ten times of it, the number we obtained is the minimum size of the study. So, here in this research, 350 is the minimum sample size. But, for more accuracy of the study, researcher here took data from 616 respondents. During the initial phase when researcher did a pilot testing, a questionnaire was sent to 130 respondents, of whom 121 responded. A pilot investigation was conducted in the preliminary stage to examine the accuracy of the research instrument and was carried out on a small number of participants who were part of the sampling universe. At that time, data were collected from those online customers who were doing job in the telecom sector and also doing online transactions from their mobile devices as they have proper knowledge about the service quality of the mobile network. So, after doing the pilot study 2 data-items were removed from the study,and these data-items were:

- Extra charges for delivery makes you think not to order online next time.
- Lack of touch and feel facility in M-commerce gives you a negative feeling towards online buying.

Among respondents, 49% of respondents were female, and 51% were male. Stratas of different telecom users were shown in the following table:

3.2 Conceptual Framework

Table 3. Stratas of users of mobile network

Telecom Operator	Airtel	Vodafone Idea	Reliance Jio	Other (BSNL)
Number of Users	257	132	218	9

3.3 Hypotheses Formulated

Once developing the conceptual framework, these hypotheses were formulated:

H_{01}: Mobile network service quality not positively influences perceived ease of use.
H_{02}: Perceived ease of use not positively influences M-commerce adoption.

3.4 Reliability Statistics

The most prevalent method for estimating the reliability value is the Cronbach Alpha method. This informs the objects' relationship and says whether or not they belong to the same domain. If the value of the Cronbach's alpha exceeds 0.70 for the questionnaire, this indicates that the items have strong internal consistency (Gliem & Gliem, 2003).

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Figure 1. Conceptual Framework of the study

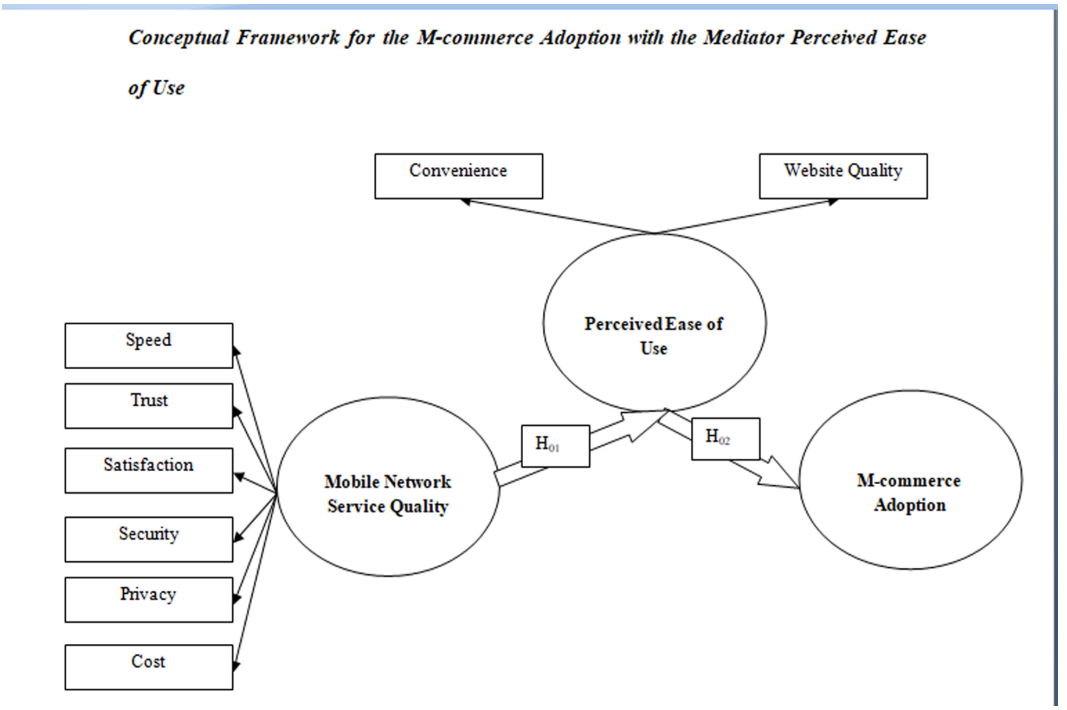


Table 4. *Table showing reliability statistics*

Cronbach's Alpha	Number of Items
.944	35

4. RESULTS

Descriptive Demographic Details

Intrinsic Results

Data were gathered and analysed using structural equation modeling (SEM), a multivariate second-generation method for investigating causal models that comprised an estimate of the two causal model components: the measurement and the structural models, to test the proposed hypotheses. In recent years, such analysis methods have been widely used. AMOS Graphics software was used in our research to test the measurement and the structural models (Hsu & Lin, 2020). In CFA, we are analysing how reliable the observed variables are about the latent constructs, which is represented by squared multiple correlations (SMC), and SEM model is used for determining how much of the variance is accounted for by endogenous variables (Schreiber et al., 2006).

Measurement Model

Before testing the structural model, the measurement model analysis is used to ensure that the scales used accurately represent the latent constructs. The scales were tested for three latent constructs

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Table 5. Demographic Details of Respondents

Measure	Items	Frequency
Gender	Male Female	317 299
Age	16 to 25 26 to 35 36 to 45 More than 45	239 285 54 38
Educational Qualification	Intermediate Graduate Post-Graduate Others	77 168 305 66
M-commerce Platform Users	Flipkart Amazon Myntra Snapdeal Meesho LimeRoad	252 225 78 31 14 16

to ensure that they were reliable and that the hypotheses test results could be supported by valid measurements (Xiao, 2016). The measurement model was shown as follows:

The fitness measures of the measurement model are shown in the following table:

The resulting χ^2 fit statistic of the three constructs model was 1,924.760 with 484 degrees of freedom ($p < .05$). The CFI, GFI, AGFI, NFI, AMSEA and RMR were .902, .829, .802, .874, .070, and.079 respectively. All factors were highly significant ($p < .05$). All fitness measurements were acceptable. As a result, all of the measures indicated that the model fits the data as all the values are under the acceptable range of fitness (Byrne., 2016).

Assessing the Convergent Validity

Convergent validity refers to the degree to which responses to a test or instrument have a close correlation with responses to other tests or instruments that are conceptually identical. To determine convergent validity, we need to know the Average Variance Extracted (AVE) value. It is a measure of the amount of variance reported by a construct in comparison to the amount of variance due to measurement error. An AVE value must be 0.5 or greater than 0.5 (Hair et al, 2019).

Structural Model

This SEM model investigates that the mobile network service quality and PEOU have an immense impact on M-commerce service adoption. Their factor loadings are seen in the figure above, which demonstrates that they have a positive impact on M-commerce adoption as most of the factor loadings demonstrates a good correlation. Let's take a look at the following figure for a brief view of the direct and indirect values of the path.

When we calculate the value of the direct path coefficient, it is .44 and if we see the indirect path coefficient, it is (.77*.91), i.e., 70. It demonstrates that the mediator (perceived ease of use) hasan important role in the relationship between the mobile network service quality and the adoption of M-commerce services. The researchers used the full mediation model (their proposed model) in this study because a buyer is unlikely to adopt M-commerce services unless they demonstrate an acceptable level of perceived ease of use in online transactions (e.g., convenience and website quality). As a result, perceived ease of use completely mediates the effects of mobile network service quality components on M-commerce adoption.The mediation is used because the inference of mediation in

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Figure 2. Measurement Model of the Study

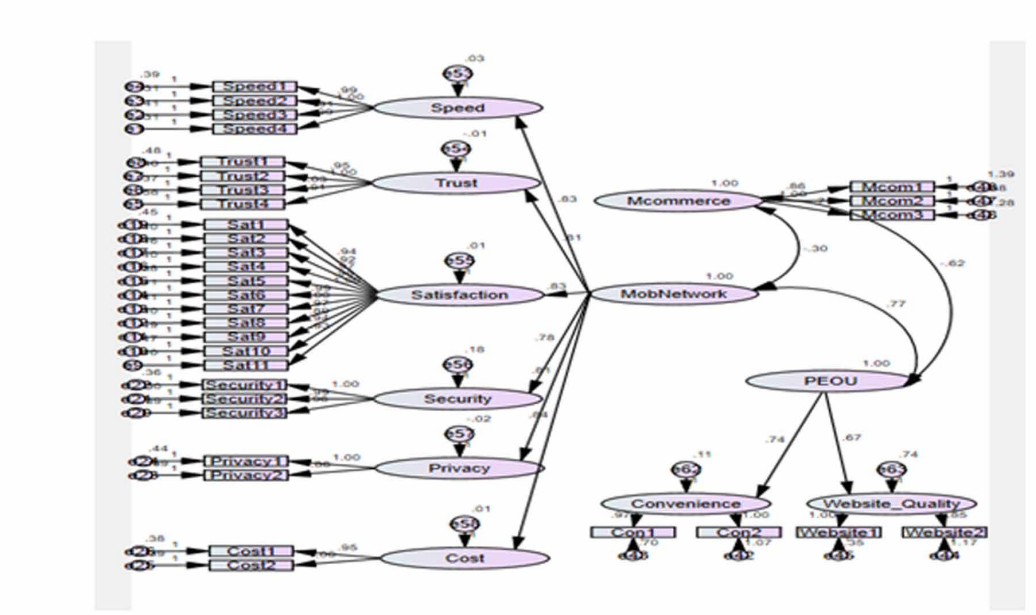


Table 6. Model fit statistics of measurement model

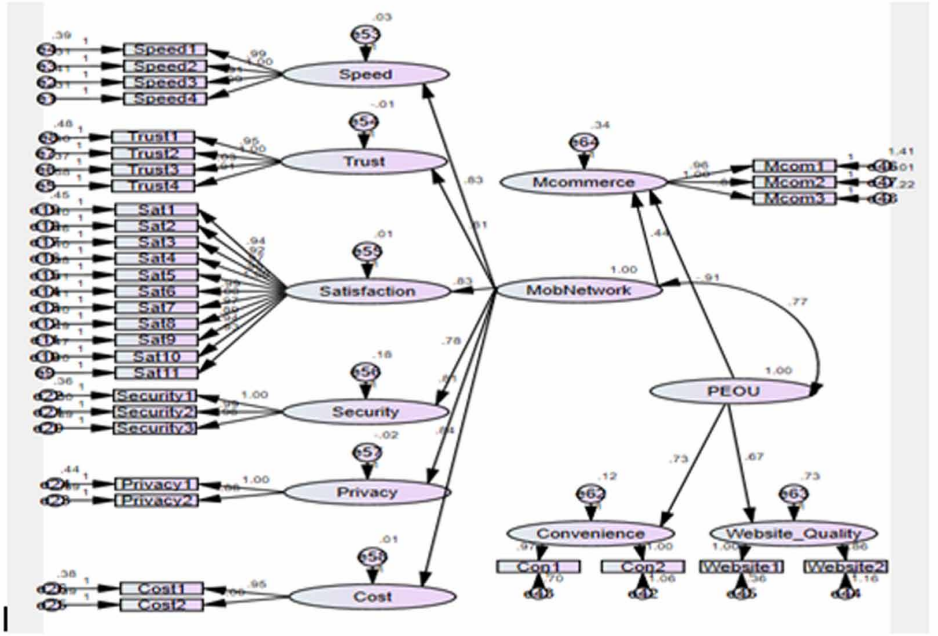
Measures	Recommended Criteria	Obtained Value	Suggested by Authors
CMIN/df	<3 good; <5 acceptable	3.947	Barbara M. Byrne., (2016); Hair et al., (1998)
GFI	Value close to 1	.829	
AGFI	Value near to 1	.802	
NFI	Value near to 1	.874	
CFI	Value close to 1	.902	
RMR	Less than .08	.079	
RMSEA	.05-.08 good	.070	

Table 7. Table showing AVE values

Latent Variables	Average Variance Extracted Value
Mobile Network Service Quality	0.68676
Perceived Ease of Use	0.5198
M-commerce Adoption	0.580209

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Figure 3. SEM Model of the Study



SCM is known as causal mediation analysis. It is based on the mediation formula which was given by Pearl (2014) that it enables the interaction between the causal and the mediator variable. Secondly, he defines mediation consistently for linear and nonlinear models, as well as for continuous versus binary mediators or outcomes (Klien, 2016).

The values of their SEM models are illustrated in the following table:

Regression Weights

The above table depicts that p-values of both the hypotheses are significant as both are less than .005, which interprets that mobile network service quality positively influences perceived ease of use and perceived ease of use also positively influences the adoption of M-commerce services. It means that H_{01} and H_{02} both are rejected.

5. CONCLUSION

This study was conducted to provide a deep understanding of the key factors that could form the intention of customers to accept M-commerce.One of the main determinants of online business success is the perceived online shopping comfort of consumers. If customers feel convenient to use any M-commerce portal and if the website is user-friendly, then customers are inclined towards those m-portals.

Mobile technology is increasingly appealing because it provides versatile, ubiquitous access to the Internet, transforming conventional electronic commerce (e-commerce) into M-commerce. M-commerce has unique characteristics that provide customers with values that are not available in conventional wired e-commerce, including usability, customization, identity, and perceived enjoyment. Mobile devices have become a marketing tool with great advertising potential, encouraging user interactivity and conducting via purchases. The spread of mobile telephony worldwide, the steady

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Figure 4. Direct and Indirect Path Coefficient Value of the Variables

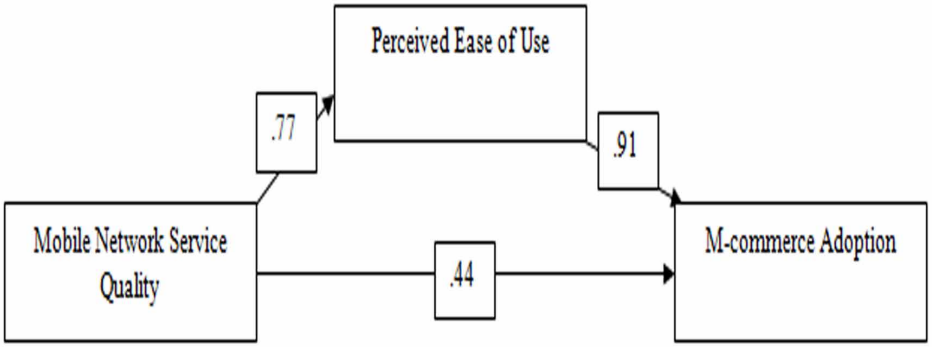


Table 8. Model fit statistics values of the structural model

Measures	Recommended Criteria	Obtained Value	Suggested by Authors
CMIN/df	<3 good; <5 acceptable	3.977	Barbara M. Byrne., (2016); Hair et al., (1998)
GFI	Value close to 1	.830	
AGFI	Value near to 1	.803	
NFI	Value near to 1	.874	
CFI	Value close to 1	.903	
RMR	Less than .08	.077	
RMSEA	.05-.08 good	.070	

The CMIN/df value is 3.977. The CFI, GFI, AGFI, NFI, RMSEA and RMR values were .903, .830, .803, .874, .070, and .077 respectively. All factors were highly significant ($p < .05$).All values are within the appropriate range that interprets the SEM results as acceptable for the model or good results indicator (Byrne., 2016).

Table 9. Regression weights table showing p-values

	Estimate SE C.R. P
MobileNetworkServiceQuality to PerceivedEaseofUse	.772 .037 20.770 ***
PerceivedEaseofUse to Mcommerce	-.625 .054 -11.650 ***

*** $p < 0.001$; S.E means Standard Error; whereas C.R means Critical Ratio

rise in the number of mobile users worldwide, and a projection for the future of m-commerce are evidence of its marketing appeal in both academic and business contexts.

To meet revenue targets, managers of M-service providers are constantly under pressure to distinguish their offerings from those of their rivals as today customers are more up-to-date towards new technology and the service quality aspects. The companies who give better services atlow cost

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will be inclined towards those companies as today they have lots of options. Most of the customers prefer Flipkart and Amazon M-commerce portal for doing their online transactions.

6. IMPLICATIONS FOR RESEARCH

The rapid expansion of global e-commerce initiatives reflects its appealing benefits, which include improved government performance, lower cost structure, greater flexibility and range of services, greater transparency, greater accountability, and faster transactions. In developing countries, the growth of information and communication technologies, or e-commerce, is significant.

There is substantial growth in information and technology, or rather, in e-commerce. For the success of e-commerce, technological efficiency is essential. Human, economic, and other problems, however, also need to be taken into account. In most nations, the current e-commerce situation happens to be "stay organized for the situation." Evaluating the current situation reveals opportunities for organizations to take seriously to survive the effects of globalization and market opening.

Today's financial firms can earn a fair living by embracing the pioneered practices by themselves or others in electronic commerce. They should, just as importantly, provide innovative value-added services to promote electronic commerce, just as their predecessors did in the emergence of modern commerce some centuries ago. They need to get out of their rent looking for attitude, speculative habit, and the mind-set of doing more of the same to do so, i.e., less of imaginary rocket science mathematics, but more of an in-depth understanding of how the supply chain functions Internet.

7. PRACTICAL IMPLICATIONS

The results are useful for M-commerce providers designing marketing campaigns that emphasize promoting the mobility aspect of M-commerce, particularly its use for consumers and its security. M's business needs to be developed and modernised to meet customers' specific needs and requirements better. Marketers need to focus on key factors such as social impact, perceived enjoyment, perceived risk, and anticipated efforts to increase the adoption of M-commerce services to target groups of young and innovative customers.

Data reflects the perception and expectations of customers when using mobile devices. Services are used for quality assurance to identify areas that required improvements. Mobile communications should focus on providing greater trustworthiness, accuracy, and stability in mobile services. An extended focus on the objects of mobile network dimensions like transmission quality and network coverage is also useful for improving the perceived quality of mobile services for their customers.

8. MANAGERIAL IMPLICATIONS

Modern models of technology adoption are based on theoretical differences, like innovation Diffusion Theory (Rogers 1983), which may be expected to influence the customer's adoption of latest technologies. Researchers have investigated several connection between the Diffusion model (Roger theory and Bass model) and acceptance model. This study presents an outline of the various theoretical perspectives through which we came to know that previous studies had addressed individual acceptance of information communication technology.

To make shopping more convenient, e-commerce sellers should have to work on various aspects of service delivery, such as delivering goods, timeliness of the order, tracking facility, and orderly accuracy (Collier & Bienstock, 2006). To achieve these objectives of providing good quality of services, (1) The supplier must meet the expectations of buyers. (2) Online retailers should deliver the product to the buyer's preferred address. (3) Deliver the products on or before the promised date. (4) To assure the buyer that the product was matched with the description that was posted on the online portal.All of these factors have an impact on the perceived transaction value of purchasers, which further enhance the satisfaction of customers (Xiao., 2016).

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9. LIMITATION AND FUTURE RESEARCH

Only those customers who transact from their mobile devices have been included in this study. The study excludes individuals who do not carry out some form of the online transaction. Furthermore, the technological aspect of the advanced network is not included in this study. Researchers have only used the service quality element of mobile networks, which affects the adoption of M-commerce services.

For future research, attempts should be made to understand whether or not customers' reactions to adopting new technologies shift over time and to check their way of shopping over various periods. Furthermore, in various contexts and fields, this research can evaluate and validate some additional constructs with some mediating and moderating effects for different M-commerce services. Additionally, instead of taking the TAM model, researchers can also take the Unified Theory of Acceptance and Use of Technology (UTAUT) model.

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